

ECO200 Microeconomics

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1 Supply and demand

1.1 The model and the market

Definition(Supply): The combined amount of a good that all producers in a market are willing to sell.

Definition(Demand): The combined amount of a good that all consumers in a market are willing to buy.

Definition(Market): In the strictest sense, a market is defined by the specific product being bought and sold, a particular location, and a point in time. In this market, potential buyers meet with potential customers, and vice versa.

Definition(Commodities): Goods traded in markets where consumers view different varieties of the good as essentially interchangeable, which means the goods are homogeneous.

Key assumptions of demand and supply model

- We restrict our focus to supply and demand in a single market.
- All goods bought and sold in the market are commodities.
- All goods sold in the market sell for the same price, and everyone has the same information about prices, the quality of the goods being sold, and so on.
- There are many buyers and sellers in the market, which means no one consumer or producer has a noticeable impact on anything that occurs in the market and on the price level in particular. This is usually quite realistic for buyers, but not always for sellers. For example, the market for cell phones, there are only a few companies and producers in this market, and their actions can impact the market substantially.

1.2 Demand

Definition(Substitutes): A good that can be used in place of another good.

Definition(Complement): A good that is purchased and used in combination with another good.

Factors affecting the demand

1. The price of the good itself is the most fundamental determinant of demand.
2. The number of consumers in the market.
3. The price of goods (in other markets) which are substitutes or complements of the good.
 - When the price of a substitute good falls, consumers will want to buy more of that substitute and less of the initial good.
 - When the price of a complement falls, consumers will want to buy more of it and also more of the initial good.
4. The wealth of the consumers in the market.
5. The tastes of the consumers in the market.

Definition(Demand curve): Demand curves show the relationship between quantity demanded for a good and its price, holding constant "everything else". Visually, it looks like this:

Definition(Inverse demand curve): A demand curve written in the form of price as a function of quantity demanded.

Definition(Demand choke price): The price at which quantity demanded is zero; the vertical intercept of the inverse demand curve.

Mathematically, demand curve can also be represented with functions: The demand curve has equation D_1

$$Q = 1000 - 200P$$

The inverse demand curve, which is actually plotted similarly as above, has equation

$$\begin{aligned} Q &= 1000 - 200P \rightarrow 200P = 1000 - Q \\ \rightarrow P &= \frac{1000}{200} - \frac{1}{200}Q \rightarrow P = 5 - .005Q \end{aligned}$$

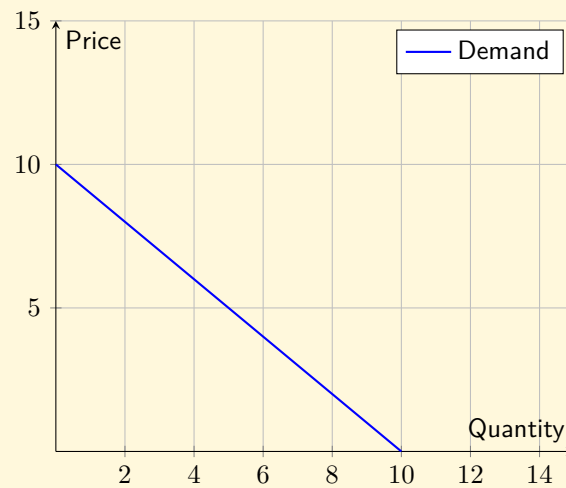


Figure 1: General demand curve

1.2.1 Shifts in demand curve

We have two distinctions in shifts in demand curve:

1. Changes in quantity demanded: A movement along the demand curve that occurs as a result of a change in the good's price.
2. Changes in demand: A shift of the entire demand curve caused by a change in a conditions of demand other than the good's own price. Changes in demand conditions can also affect the slope of demand curve.

Changes in demand that causes the shift in demand curve can be represented graphically and mathematically. Consider changes in consumer preferences of a product that shifts every level of demand, where D_1 shifts to D_2 , and D_1 can also shift to D_3 :

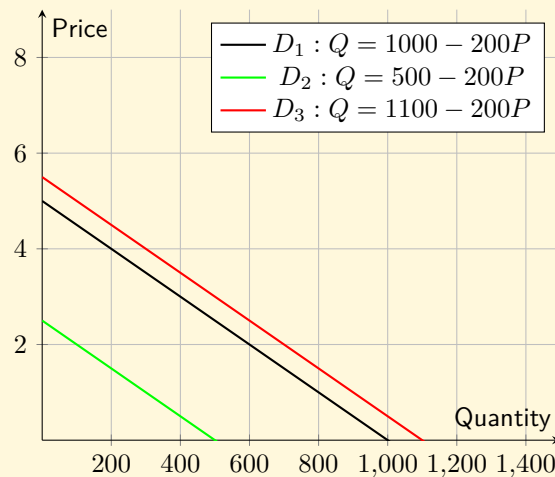


Figure 2: Shifts of demand curves

Mathematically: D_1 , has equation

$$Q = 1000 - 200P$$

The inverse demand curve D_1 has equation

$$\begin{aligned} Q &= 1000 - 200P \rightarrow 200P = 1000 - Q \\ \rightarrow P &= \frac{1000}{200} - \frac{1}{200}Q \rightarrow P = 5 - .005Q \end{aligned}$$

Meanwhile, demand and inverse demand curve D_2 on has equation

$$Q = 500 - 200P \rightarrow P = 2.5 - .005Q$$

Demand and inverse demand curve D_3 has equation

$$Q = 1100 - 200P \rightarrow P = 5.5 - .005Q$$

1.3 Supply

There are several factors that can influence supply.

Factors influencing supply

- Price (Obviously)
- Supplier's cost of production and state of production technology
- The number of sellers in the market
- The sellers' outside options (an opportunity cost of production)

Definition(Supply curve): Supply curves show the relationship between supply a good in a given market and its price, holding constant "everything else". Visually, it looks like:

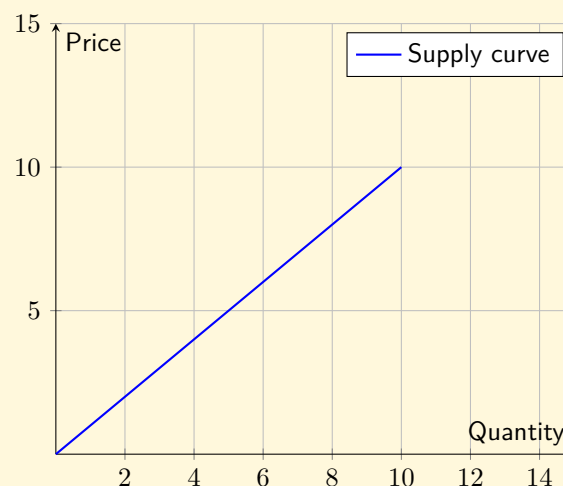


Figure 3: General supply curve

Definition(Inverse supply curve): A supply curve written in the form of price as a function of quantity supplied. (p. 19)

Definition(Supply choke price): The price at which no firm is willing to produce a good and quantity supplied is zero; the vertical intercept of the inverse supply curve.

Mathematically, supply curve can be represented. The supply curve has equation

$$Q = -200 + 200P$$

The inverse supply curve, which is actually plotted in the form above, is:

$$\begin{aligned} Q &= -200 + 200P \rightarrow 200P = 200 + Q \\ \rightarrow P &= \frac{200}{200} + \frac{1}{200}Q \rightarrow P = 1 + .005Q \end{aligned}$$

1.3.1 Shifts of supply curve

There are two distinctions when it comes to changes in the supply curve:

1. Changes in quantity supplied: A movement along the supply curve that occurs as a result of a change in the good's price.
2. Changes in supply: A shift of the entire supply curve caused by a change in a determinant of supply conditions other than the good's own price.

Changes can be represented both visually and mathematically, where S_1 shifts to S_2 , or S_1 can shift to S_3 due to potential changes in production technology:

The supply curve, s_1 has equation

$$Q = -200 + 200P$$

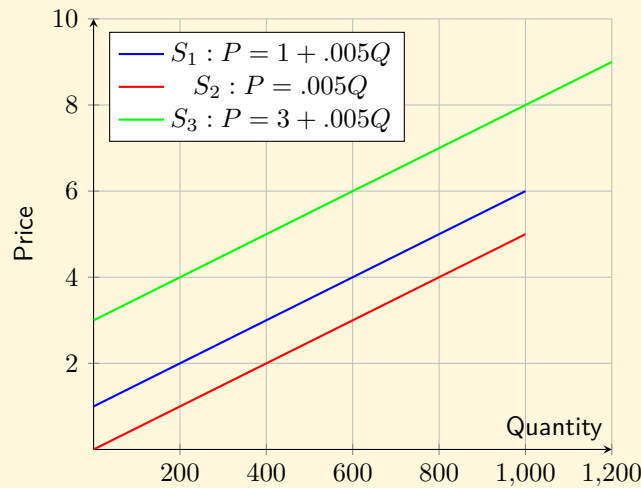


Figure 4: Shift of supply curves

The inverse supply curve, which is actually plotted has equation

$$Q = -200 + 200P \rightarrow 200P = 200 + Q$$

$$\rightarrow P = \frac{200}{200} + \frac{1}{200}Q \rightarrow P = 1 + .005Q$$

Meanwhile, supply and inverse supply curve S_2 has equation

$$Q = 200P \rightarrow P = .005Q$$

supply and inverse supply curve S_3 has equation

$$Q = -600 + 200P \rightarrow P = 3 + .005Q$$

1.4 Market equilibrium

Definition(Market equilibrium): The point where supply and demand cross, which only occurs under the assumption of our demand and supply model.
Visually, the equilibrium looks like:

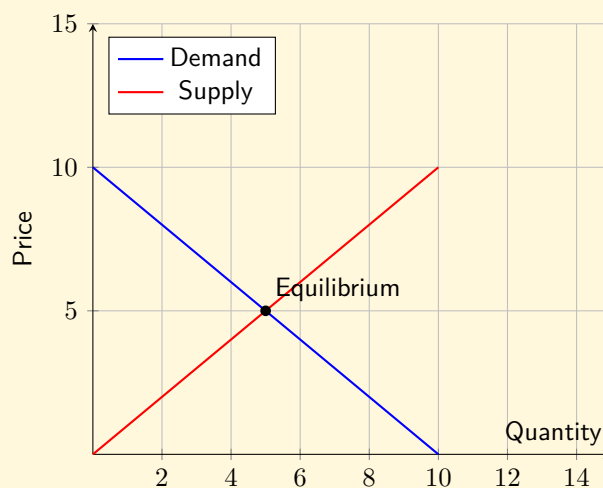


Figure 5: Market equilibrium as intersection

Definition(Equilibrium price): The only price P_e at which quantity supplied equals quantity demanded.

Definition(Equilibrium quantity): The only quantity Q_e at which quantity supplied equals quantity demanded.

We can calculate equilibrium point mathematically:

Letting Q^D stand for quantity demanded and Q^S stand for quantity supplied, we set $Q^D = Q^S$ at price P_e :

$$\begin{aligned}1000 - 200P_e &= -200 + 200P_e \\ \rightarrow 1200 &= 400P_e \\ \rightarrow 3 &= P_e\end{aligned}$$

To get Q_e , plug P_e back into either the demand or the supply curve equation:

$$1000 - 200(3) = -200 + 200(3) = 400 = Q_e$$

The method also works for inverse demand and supply curve:

To find the equilibrium using the inverse demand and supply curves, we'll first express price as a function of quantity. Given the demand equation $Q^D = 1000 - 200P$, we can rearrange it to express P in terms of Q^D . Similarly, for the supply equation $Q^S = -200 + 200P$, we can express P in terms of Q^S .

Let's find the inverse demand and supply curves:

For the demand curve:

$$\begin{aligned}Q^D &= 1000 - 200P \\ P &= 5 - 0.005Q^D\end{aligned}$$

This is the inverse demand curve.

For the supply curve:

$$\begin{aligned}Q^S &= -200 + 200P \\ P &= 0.005Q^S + 1\end{aligned}$$

This is the inverse supply curve.

To find the equilibrium, set the inverse demand equal to the inverse supply:

$$5 - 0.005Q^D = 0.005Q^S + 1$$

Given that $Q^D = Q^S = Q_e$ at equilibrium:

$$5 - 0.005Q_e = 0.005Q_e + 1$$

Combining like terms:

$$\begin{aligned}5 - 1 &= 0.01Q_e \\ 4 &= 0.01Q_e \\ Q_e &= 400\end{aligned}$$

Plugging Q_e into either the inverse demand or supply curve will give us P_e :

Using the inverse demand curve:

$$\begin{aligned}P_e &= 5 - 0.005(400) \\ P_e &= 5 - 2 \\ P_e &= 3\end{aligned}$$

So, the equilibrium price P_e is 3, and the equilibrium quantity Q_e is 400.

1.4.1 Adjusting to equilibrium in economics

There are two distinctions when it comes to disequilibrium:

- Excess demand
- Excess supply

Visually, both scenario looks like:

In both scenarios, the curve will adjust the disequilibrium and move back to the equilibrium point:

- When the market has excess supply, Producers will supply more at this price than consumers are willing to buy, so to move their excess product, producers will lower the price toward P_e .
- When the market has excess demand, consumers will demand more than producers have available to sell, so individual producers will realize they can raise the price and still sell their whole stock.

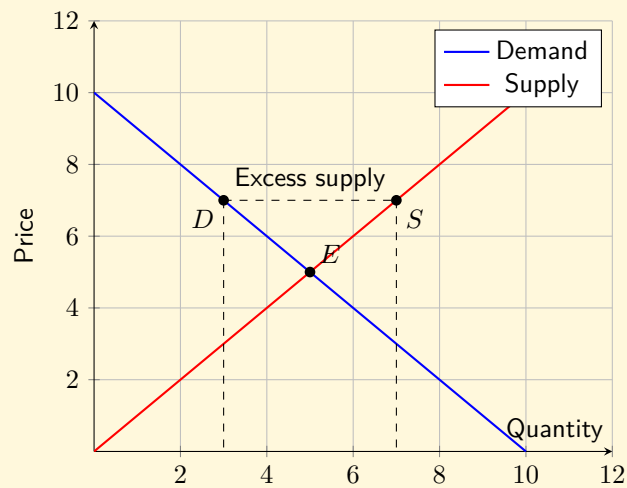


Figure 6: Excess supply

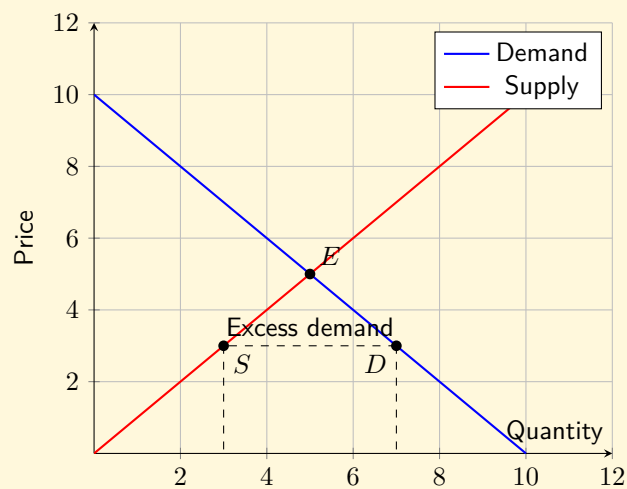


Figure 7: Excess demand

1.4.2 Creating new equilibrium

When there are supply shocks and demand shocks that cause the curve to shift, their effect on equilibrium can be summarized as:

		Impact on Equilibrium	
Curve that Shifts	Direction of Shift	Price	Quantity
Demand Curve	Out (increase in D)	↑	↑
	In (decrease in D)	↓	↓
Supply Curve	Out (increase in S)	↓	↑
	In (decrease in S)	↑	↓

Visually, they can be visualized as:

Shifts can also be presented using math:

The outward shift in supply corresponds to a change in the supply equation from

$$Q_1^S = -200 + 200P \quad \text{to} \quad Q_2^S = 200 + 200P$$

From before, we know that initially $P_e = 3$ dollars and $Q_e = 400$ lbs

But now, at $P = 3$, more tomatoes will be supplied (800lbs) than demanded (400 lbs)

This excess supply is resolved by lowering the price. Re-solving the system given our initial demand equation (which has not changed) and new supply equation gives:

$$P_e = 2; \quad Q_e = 600$$

In some scenarios such as natural disasters and war, both curves can shift:

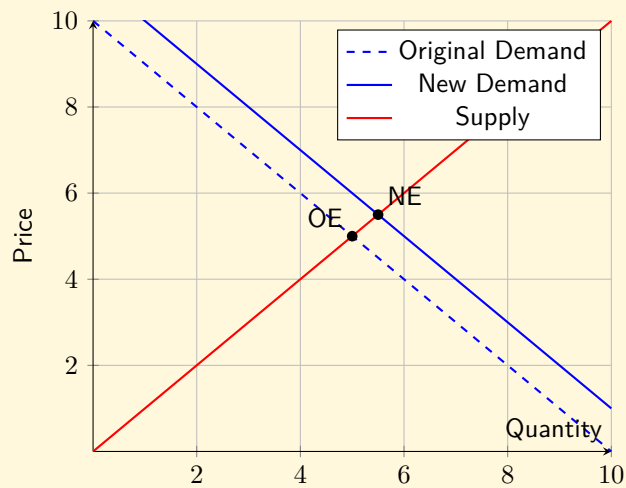


Figure 8: Increase D

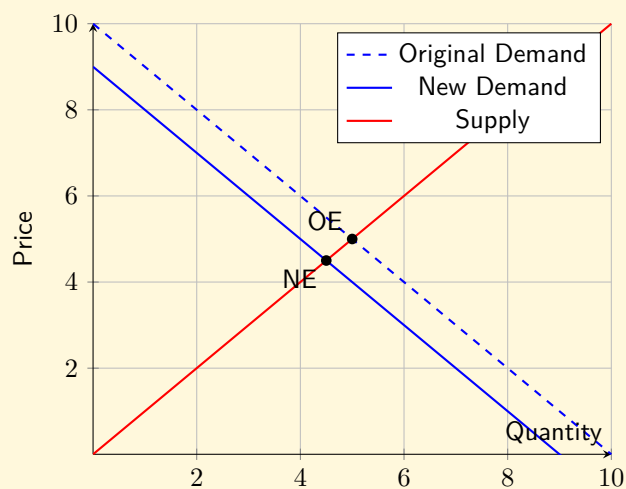


Figure 9: Decrease D

1.5 Comparative statistics

Definition(Exogenous parameter): An exogenous parameter is one whose measure is determined outside the model.

Definition(Comparative statistics): comparative statics is the comparison of two different economic outcomes, before and after a change in some underlying exogenous parameter.

Comparative statics is commonly used to study changes in supply and demand when analyzing a single market.

Consider a demand equation:

$$Q^D = 1000 - 200P$$

The intercept of the demand equation "contains" many different factors that influence the location of the demand curve in P/Q space.

By making these factors (or in this case, one of them) explicit, we can analyze how demand and equilibrium outcomes change when this factor changes

One factor that influence the demand curve is the average consumer income I , and it can be taken into account of demand equation explicitly:

$$Q^D = 400 + 0.01I - 200P$$

Using calculus, it's very simple to see that

$$\frac{\partial Q^D}{\partial I} = 0.01 > 0$$

In words: when average consumer income among shoppers at the farmers market rises by one dollar, total demand for this good at any price rises by 0.01 per unit of goods.

With this understanding, we can introduce the definition of normal goods:

Definition(Normal goods): A type of a good which experiences an increase in demand due to an increase in income.

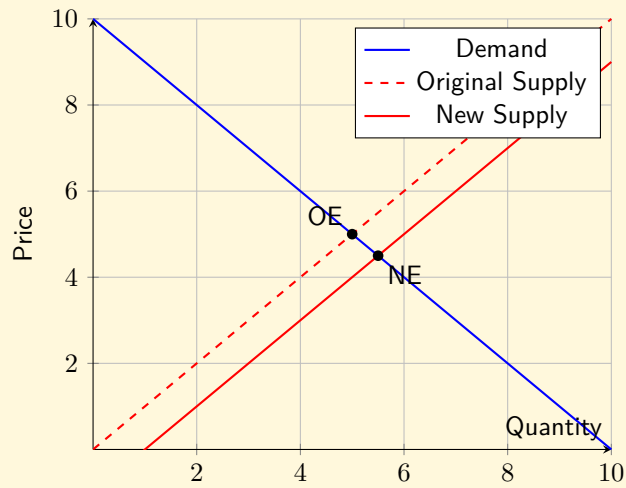


Figure 10: Increase S

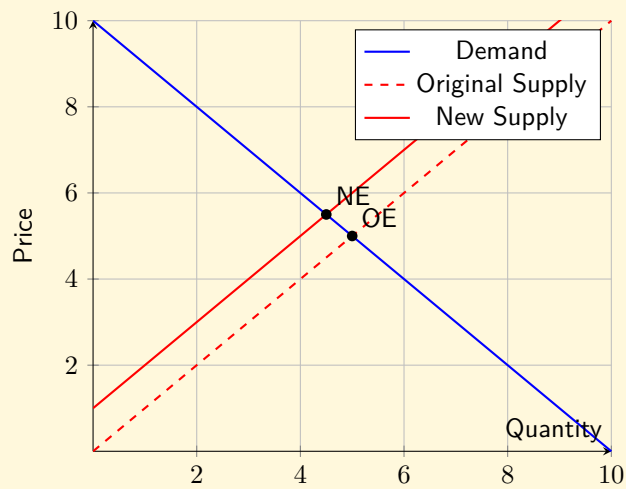


Figure 11: Decrease S

The effect of average personal income can be used to find its effects on equilibrium price and equilibrium quantity: Using some manipulations in calculus, we have the following result:

$$\frac{dP_e}{dI} = \frac{\frac{\partial Q^D}{\partial I}}{\frac{\partial Q^S}{\partial P_e} - \frac{\partial Q^D}{\partial P_e}}$$

$$\frac{dQ_e}{dI} = \frac{\partial Q^D}{\partial I} + \frac{\partial Q^D}{\partial P_e} \left(\frac{\frac{\partial Q^D}{\partial I}}{\frac{\partial Q^S}{\partial P_e} - \frac{\partial Q^D}{\partial P_e}} \right)$$

1.6 Non-linear demand and supply curve

Definition(Exponential demand function):

$$Q = e^{(c_0 - c_1 P)}$$

is an exponential demand function

Inverting this function would mean:

First, we take the natural log of both sides of the exponential demand function

$$\ln(Q) = \ln \left(\exp^{(c_0 - c_1 P)} \right)$$

The natural log of $\exp(x) = x$ for any variable or expression x , so we have:

$$\ln(Q) = c_0 - c_1 P$$

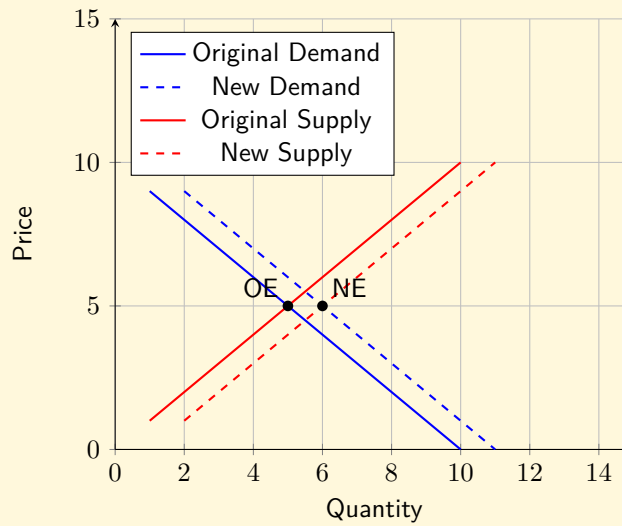


Figure 12: Both curves shift

And now we just rearrange for P so we can plot in P/Q space:

$$P = \frac{c_0}{c_1} - \frac{1}{c_1} \ln(Q)$$

And then let $a_0 = \frac{c_0}{c_1}$ and $c_1 = \frac{1}{c_1}$, so we get:

$$P = a_0 - a_1 \ln(Q)$$

Visually, this demand curve looks like:

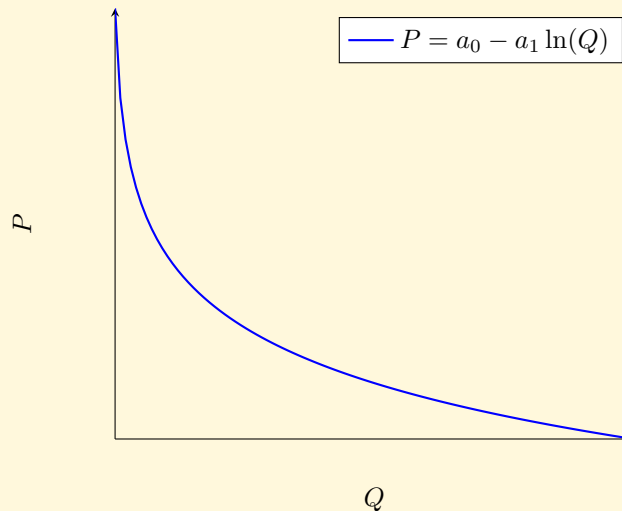


Figure 13: Non-linear demand curve

Our linear-log demand function is:

$$P = a_0 - a_1 \ln Q_d$$

where $a_0 = 300$ and $a_1 = 20$

Suppose we also have linear-log supply function:

$$P = b_0 + b_1 \ln Q_s$$

Also, let's suppose $b_0 = 30$ and $b_1 = 10$

Their equilibrium looks like:

First, we want to solve for Q_e and P_e in equilibrium. Since we have the inverse demand and supply curves in this case, we set these equal to each other and solve for Q_e first

$$\begin{aligned} 300 - 20 \ln Q_e &= 30 + 10 \ln Q_e \rightarrow 270 = 30 \ln Q_e \\ \rightarrow \ln Q_e &= 9 \rightarrow Q_e = 8,103 \end{aligned}$$

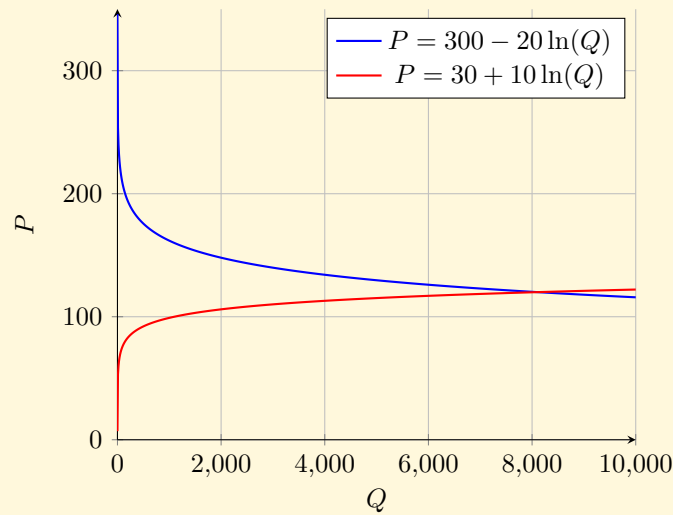


Figure 14: Equilibrium

We can get P_e from either the (inverse) supply or demand equation:

$$P_e = 30 + 10 \times 9 = 300 - 20 \times 9 = \$120$$

1.7 Elasticity

Definition(Elasticity): The ratio of the percentage change in one value to the percentage change in another.

- A **steeper** curve in P/Q space means that a given change in P is associated with a smaller change in Q .
- A **flatter** curve in P/Q space means that a given change in P is associated with a larger change in Q .
- When Q is more responsive to P (a flatter curve), we say the relationship being plotted is (relatively) elastic.
- When Q is less responsive to P (a steeper curve), we say the relationship being plotted is (relatively) inelastic.

Generally, a elasticity is the ratio of the percentage change in one value to the percentage change in another. Formally,

- The price elasticity of demand is:

$$E^D = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in } P} \equiv \frac{\% \Delta Q^D}{\% \Delta P}$$

- The price elasticity of supply is:

$$E^S = \frac{\% \text{ change in quantity supplied}}{\% \text{ change in } P} \equiv \frac{\% \Delta Q^S}{\% \Delta P}$$

We can also write elasticities in terms of partial derivatives, by replacing Δ with ∂

$$E^D = \frac{\partial Q^D}{\partial P} \frac{P}{Q^D}$$

$$E^S = \frac{\partial Q^S}{\partial P} \frac{P}{Q^S}$$

We have the following observations for price elasticity of demand:

- Because demand curves slope downward, the price elasticity of demand is always nonpositive; in special cases, it can be zero.
- A price elasticity can also be thought of as the percentage change in quantity demanded for a 1% increase in price.

We have the following observations for price elasticity of supply:

- The price elasticity of supply is always positive (or again more precisely, always nonnegative) because quantity supplied increases when a good's price rises.
- Supply elasticities can be regarded as the percentage change in quantity in response to a 1% increase in price.

Essentially, price elasticity of demand and supply measure how sensitive demand and supply relative to change in prices.

- Markets with large price elasticities of demand - where quantity demanded is sensitive to price differences - are those where consumers have a lot of ability to substitute away from or toward the good in question.

Example: If the price of one brand or model of smartphone goes up significantly, consumers can easily switch to another brand or model. There are many alternative products available, and many consumers see them as being substitutable. As a result, the quantity demanded for a particular smartphone can be quite sensitive to its price.

- Markets with less price-responsive elasticities of demand - where quantity demanded is fairly unresponsive to price changes - are those where consumers have fewer options for substitution.

Example: Diabetics who require insulin do not have the luxury of choosing a substitute if the price of insulin goes up. Their health depends on it. As such, even if the price rises, the quantity demanded may not decrease by a proportionate amount because there's no true substitute for insulin for those who need it.

- Markets with large price elasticities of supply - where the quantity supplied is sensitive to price differences are those where it is easy for suppliers to vary their amount of production as price changes.

Example: If there's a sudden surge in demand (and price) for a certain type of toy because it becomes a popular trend, manufacturers can often quickly ramp up production to produce more of that toy. This is because making more of an already-designed toy might simply involve running the factory for longer hours or using more materials, which can be adjusted relatively quickly.

- Markets with low price elasticities of supply - where the quantity supplied is fairly unresponsive to price changes - are those where it is difficult to increase production.

Example: No matter how high the demand (and price) for beachfront property, the supply cannot be easily increased. There's a limited amount of land directly on the beach, and once it's developed, there isn't an easy way to "produce" more beachfront land. As a result, the quantity supplied is fairly unresponsive to price changes.

We can classify elasticities by magnitudes:

- **Elastic:** a price elasticity with an absolute value greater than 1.
- **Inelastic:** a price elasticity with an absolute value less than 1.
- **Unit elastic:** a price elasticity with an absolute value equal to 1.
- **Perfectly inelastic:** a price elasticity that is equal to zero; there is no change in quantity demanded or supplied for any change in price.
- **Perfectly elastic:** a price elasticity that is infinite; any change in price leads to an infinite change in quantity demanded or supplied.

For linear demand and supply curve, the calculation of elasticity can be simplified as:

$$E = \frac{\% \Delta Q}{\% \Delta P} = \frac{\Delta Q / Q}{\Delta P / P} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q} = \frac{1}{\text{slope}} \times \frac{P}{Q}$$

At different sections of the demand and supply curve, it has different price elasticity.
For price elasticity of demand, we have:

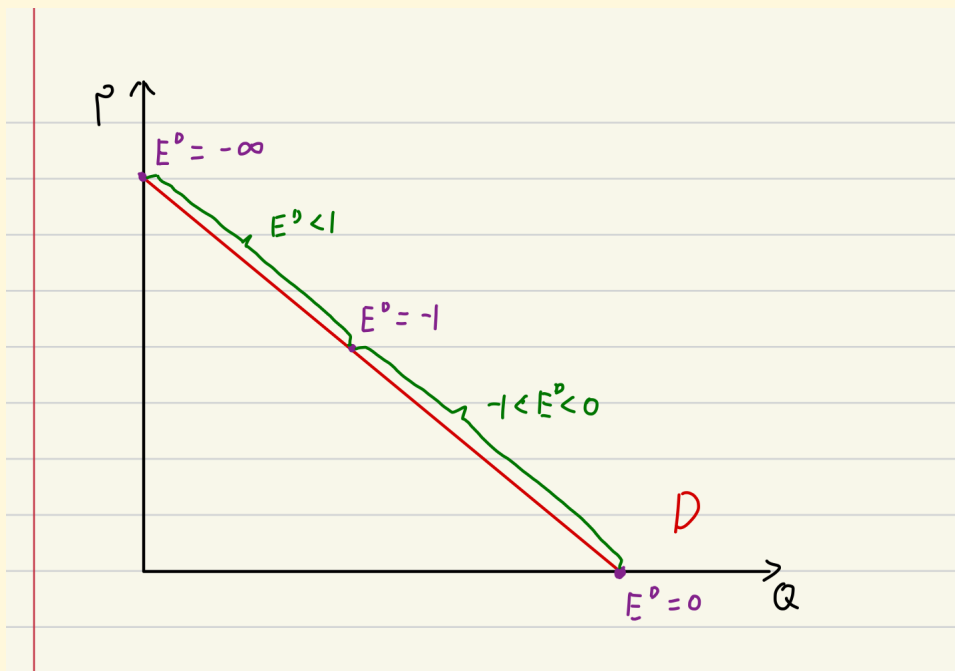


Figure 15: Elasticity along linear demand curve

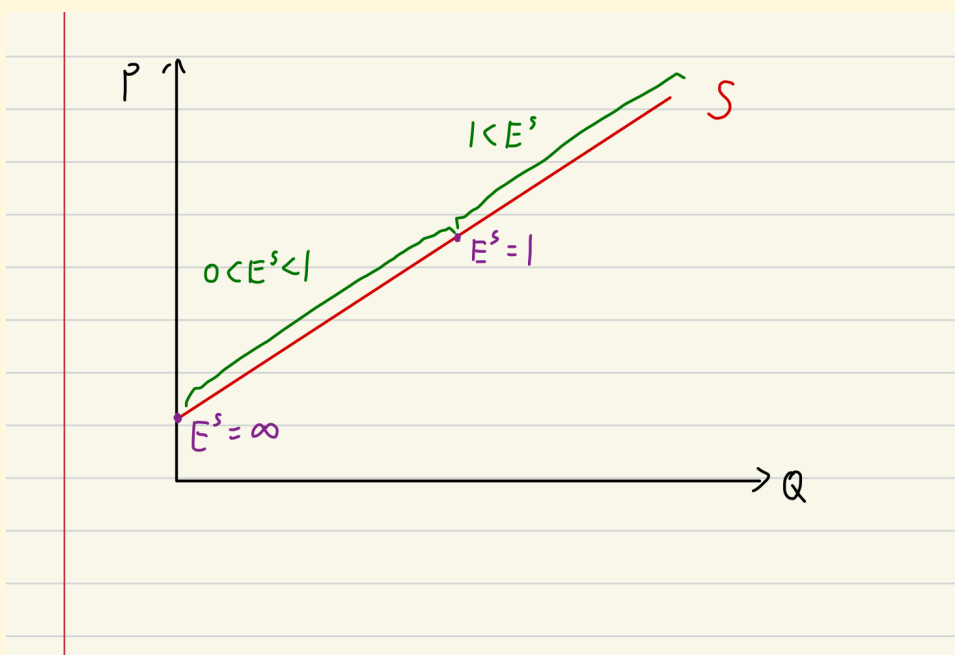


Figure 16: Elasticity along linear supply curve

Perfectly elastic and inelastic demand and supply can be visualized as:

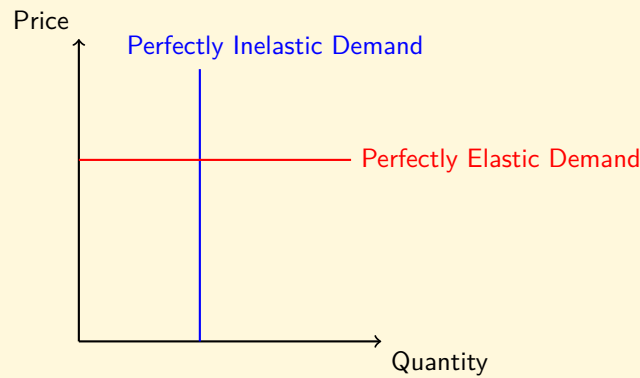


Figure 17: Perfectly inelastic and elastic demand

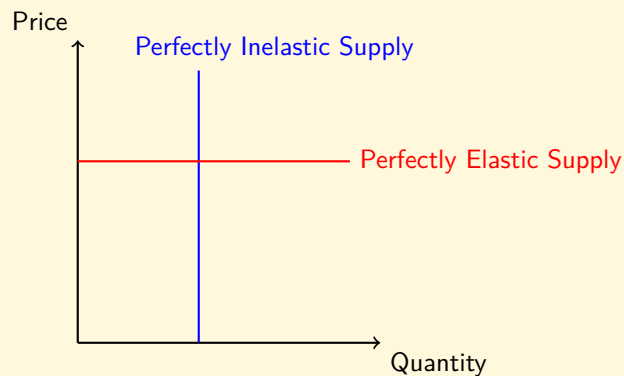


Figure 18: Perfectly inelastic and elastic supply

For instance, if demand for a good is perfectly inelastic, then a shift in supply can change only the equilibrium P , not the equilibrium Q .

Similarly, if supply of a good is perfectly inelastic, then a shift in demand can change only the equilibrium P , not the equilibrium Q .

Perfectly inelastic demand and supply curves are examples of limit cases: unlikely to ever occur in real life but is useful for demonstrating concepts. In contrast to perfectly inelastic demand or supply curves, perfectly elastic demand and supply curves may actually exist under special circumstances. Specifically: perfectly competitive firms face (essentially) horizontal, or perfectly elastic, demand curves at the market price. That is: given what they can produce given their technology, they can sell as much as they want at P_e but essentially nothing above P_e .

Definition (Income elasticity of demand): The ratio of the percentage change in quantity demanded to the corresponding percentage change in consumer income.

$$E' = \frac{\% \Delta Q^D}{\% \Delta I} = \frac{\Delta Q^D}{\Delta I} \cdot \frac{I}{Q^D}$$

- Goods for which E' is positive are **normal goods**.
- Goods for which E' is negative are **inferior goods**.
- Goods for which $E' > 1$ is luxury good.

Definition (The cross-price elasticity of demand for good X): The cross-price elasticity of demand for good X is given by

$$E_{XY}^D = \frac{\Delta Q_X^D}{\Delta P_Y} \frac{P_Y}{Q_X^D}$$

Here, P_Y is the price of a different good, Y

- When E_{XY}^D is positive, goods X and Y are **substitutes** in demand.
- When E_{XY}^D is negative, goods X and Y are **complements** in demand.

2 Analyze the market with supply and demand

2.1 Surplus

Definition(Consumer surplus): For an individual consumer, consumer surplus (CS) is the difference between the price he would be willing to pay for a good and the price he actually has to pay: P_e . Aggregated to a market, CS is the difference between the total amount of money consumers in the market would collectively be willing to pay for the amount of the good transacted, Q_e and the amount of money they actually collectively pay for Q_e .

Graphically, CS in a market can be measured as the area under the demand curve, above the market price, out to Q_e :

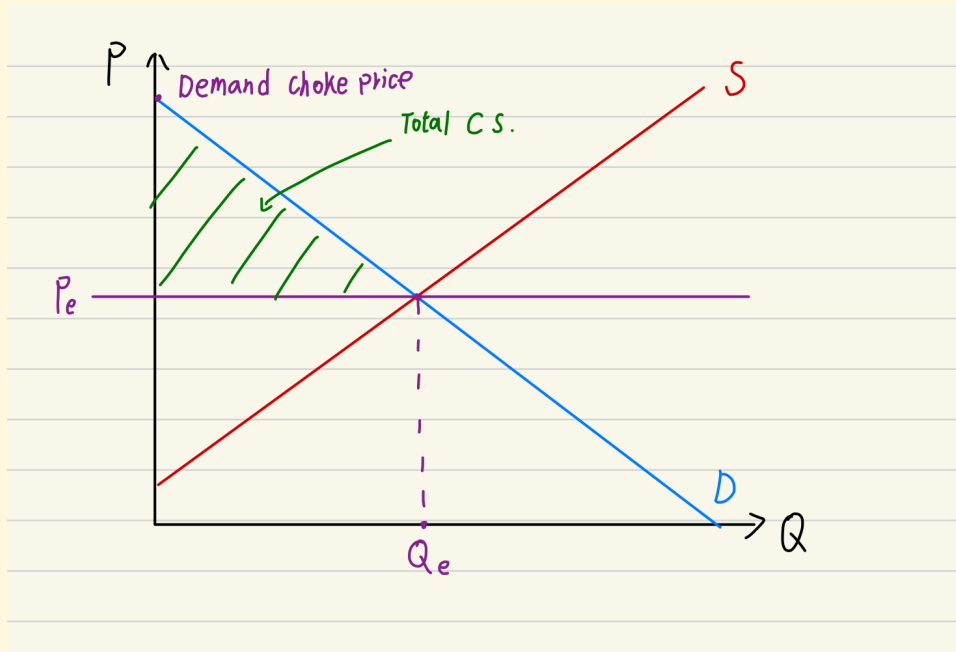


Figure 19: Aggregate consumer surplus

From the visualization above, we can deduce that:

$$\text{Total CS} = \frac{(\text{demand choke price} - P_e)}{2} \times Q_e$$

Definition(Producer surplus): For an individual producer (or firm), producer surplus (PS) is the difference between the price actually received from consumers for some amount of a good, P_e , and the price at which they would just be willing to offer this amount of the good for sale. Aggregated to a market, PS is the difference between the total amount of money producers in the market receive from selling Q_e and the amount of money they would have to receive to collectively offer Q_e .

Graphically, producer surplus in a market can be measured as the area above the supply curve, below the market price, out to Q_e :

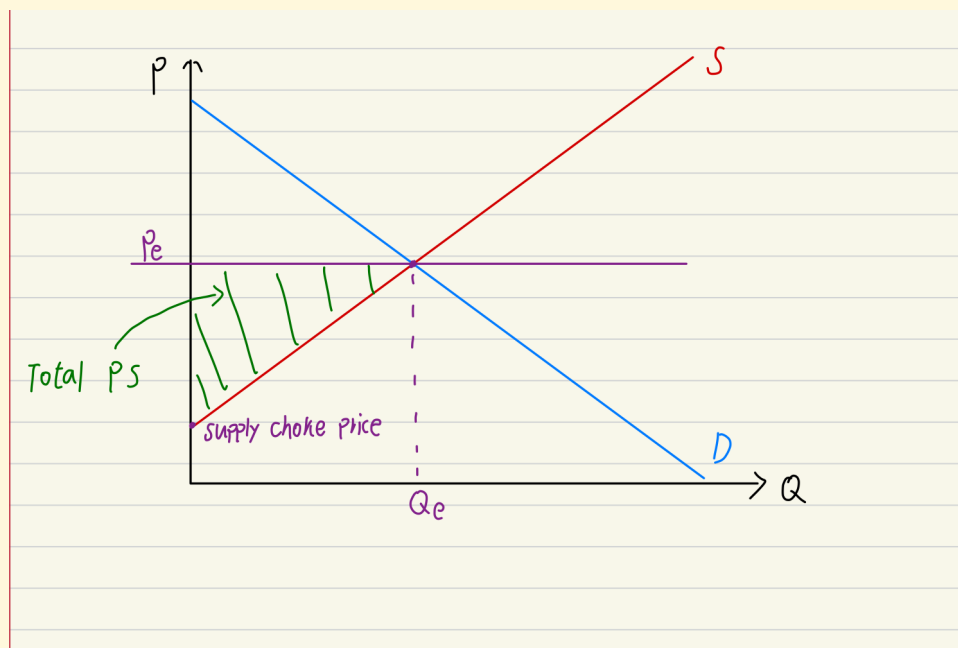


Figure 20: Aggregate producer surplus

From the graph, we can deduce that:

$$\text{Total PS} = \frac{(P_e - \text{supply choke price})}{2} \times Q_e$$

Definition(Total surplus): The total surplus would equal to total producer surplus plus total customer surplus, which equals to the social surplus or the net benefits of the whole market.

Example: The demand and supply curves for candy bars in a midwestern city are given by

$$Q^D = 151 - 20P$$

$$Q^S = 188P - 4$$

where Q is measured in thousands of candy bars per day and P in dollars per candy bar.

- Find the equilibrium price and quantity.
- Calculate the consumer and producer surplus at the equilibrium price.

Equilibrium occurs where $Q^D = Q^S$. Therefore, we can solve for equilibrium by equating the demand and supply curves:

$$\begin{aligned} Q^D &= Q^S \\ 152 - 20P &= 188P - 4 \\ 20P &= 208P \\ P &= \$0.75 \end{aligned}$$

Therefore, the equilibrium price of a candy bar is \$0.75. To find the equilibrium quantity, we need to plug the equilibrium price into either the demand or supply curve:

$$\begin{aligned} Q^D &= 152 - 20P & Q^S &= 188P - 4 \\ &= 152 - 20(0.75) & &= 188(0.75) - 4 \\ &= 152 - 15 & &= 141 - 4 \\ &= 137 & &= 137 \end{aligned}$$

Remember that Q is measured in terms of thousands of candy bars per day, so the equilibrium quantity is 137,000 bars per day.

To calculate consumer and producer surplus, it is easiest to use a graph. First, we need to plot the demand and supply curves. For each curve, we can identify two points. The first point is the equilibrium, given by the combination

of equilibrium price (\$0.75) and equilibrium quantity (137). The second point we can identify is the choke price. The choke prices for demand and supply can be determined by setting Q^D and Q^S equal to zero and solving for P :

$$\begin{array}{rcl} Q^D & = & 152 - 20P \\ 0 & = & 152 - 20P \\ 20P & = & 152 \\ P & = & \$7.6 \end{array} \quad \begin{array}{rcl} Q^S & = & 188P - 4 \\ 0 & = & 188P - 4 \\ 4 & = & 188P \\ P & = & \$0.02 \end{array}$$

So, the demand choke price is \$7.60 and the supply choke price is \$0.02.

The demand and supply curves are graphed in the figure below. Consumer surplus is the area below demand and above the price (area A). Its area can be calculated as

$$\begin{aligned} CS = \text{area } A &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= (0.5) \times (137,000 - 0) \times (\$7.60 - \$0.75) \\ &= (0.5) \times 137,000 \times \$6.85 = \$469,225 \end{aligned}$$

Producer surplus is the area below price and above supply (area B) :

$$\begin{aligned} PS = \text{area } B &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= (0.5) \times (137,000 - 0) \times (\$0.75 - \$0.02) \\ &= 0.5 \times 137,000 \times \$0.73 = \$50,005 \end{aligned}$$

Elasticity also impact consumer surplus:

- If the demand for a good is elastic (consumers are price-sensitive), a price increase leads to a large reduction in quantity demanded. As a result, consumer surplus decreases significantly.
- If the demand for a good is inelastic (consumers are not very price-sensitive), a price increase leads to a small reduction in quantity demanded. As a result, consumer surplus is usually greater than the case which the demand is elastic.

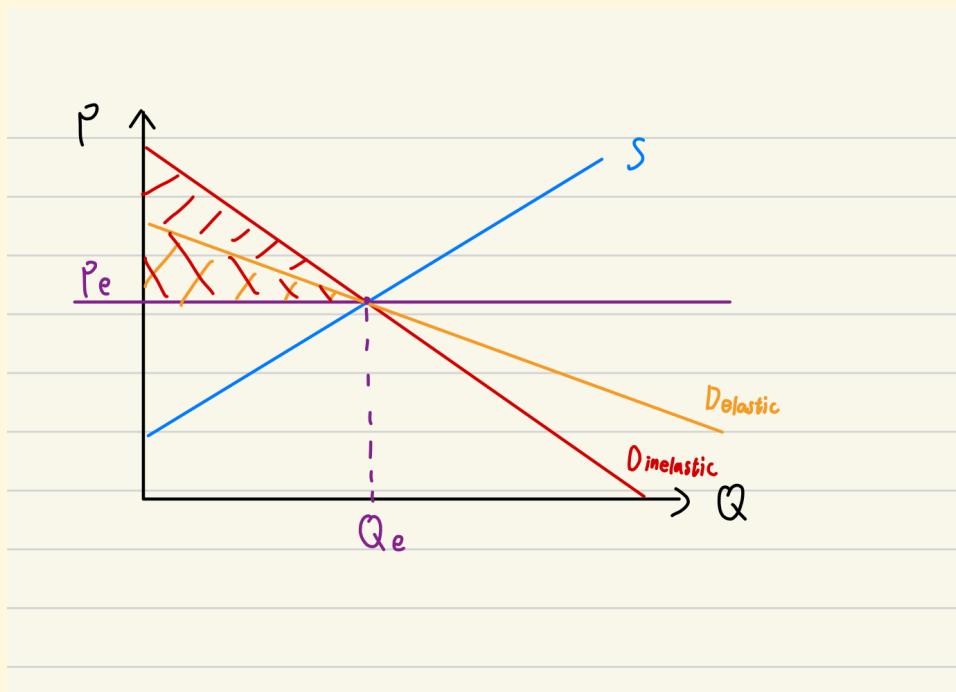


Figure 21: Two different CS

Similarly, for producer surplus, we have:

- If the supply for a good is elastic (consumers are price-sensitive), a price increase leads to a large reduction in quantity demanded. As a result, producer surplus decreases significantly.

- If the supply for a good is inelastic (consumers are not very price-sensitive), a price increase leads to a small reduction in quantity demanded. As a result, producer surplus is usually greater than the case which the supply is elastic.

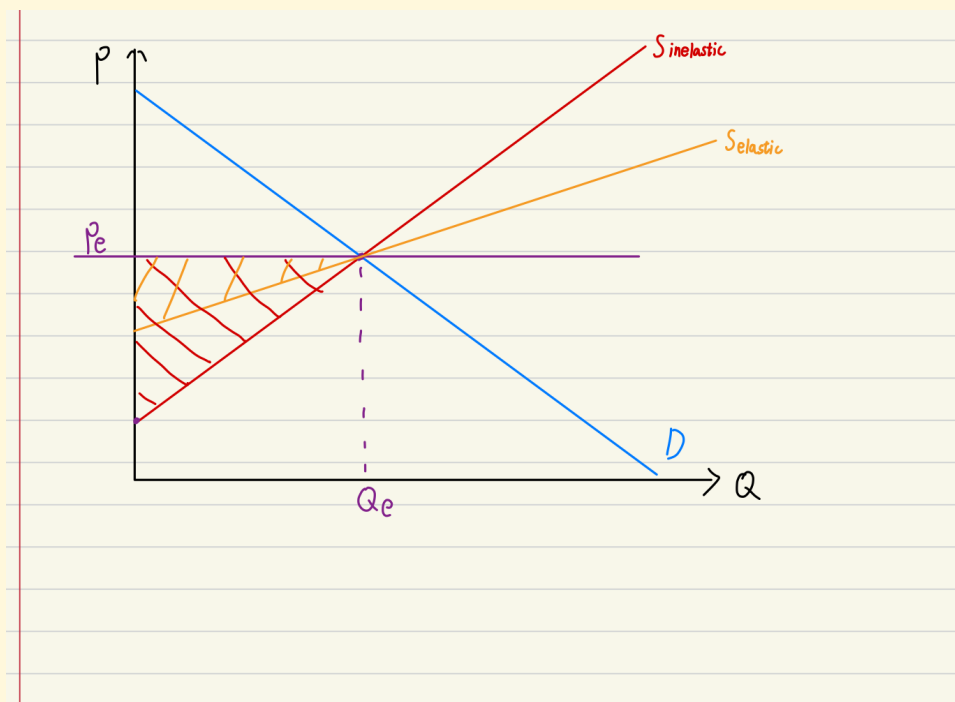


Figure 22: Two different PS

Therefore we know anything that impact the elasticity of the curves also impact the surplus.

Changes in the market conditions that cause the shift in demand and supply curve also changes the surplus and create losses.

Consider the case when supply moves to the left:

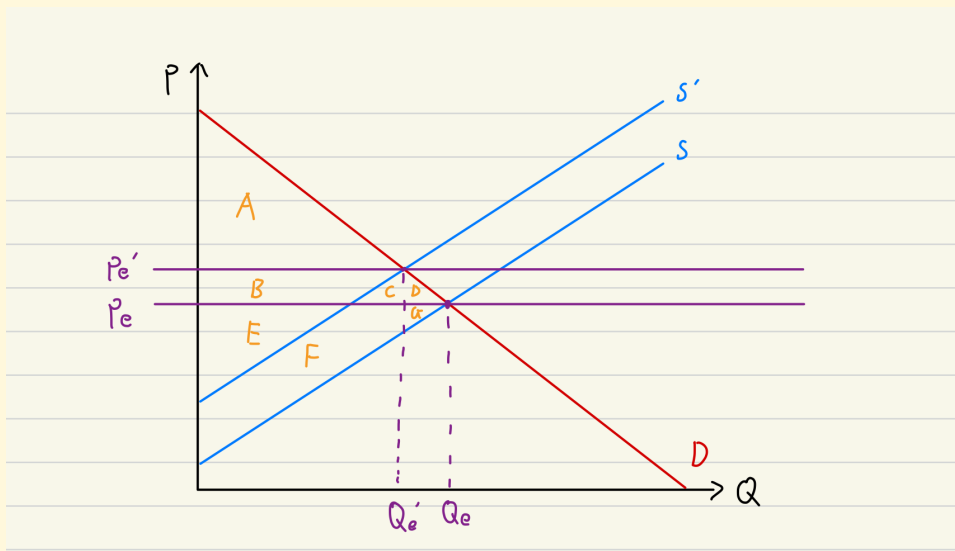


Figure 23: Supply move to the left case

- The customer surplus before the shift is $A + B + C + D$.
- The customer surplus after the shift is A
- The producer surplus before the shift is $E + F + G$
- The producer surplus after the shift is $B + E$

Consider the case when supply moves to the right:

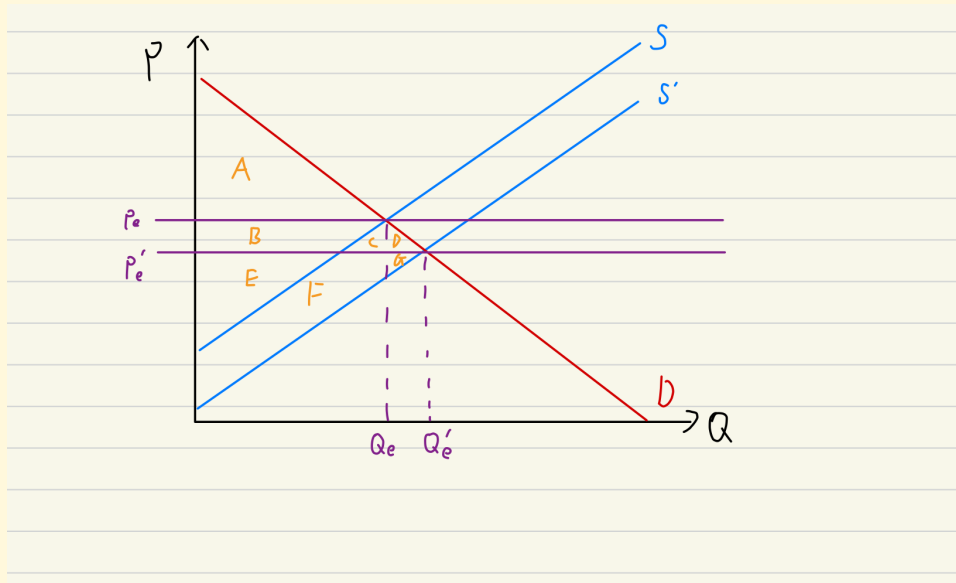


Figure 24: Supply move to the right

- The customer surplus before the shift is A .
- The customer surplus after the shift is $A + B + C + D$
- The producer surplus before the shift is $B + E$
- The producer surplus after the shift is $E + F + G$

Consider the case when demand moves to the right:

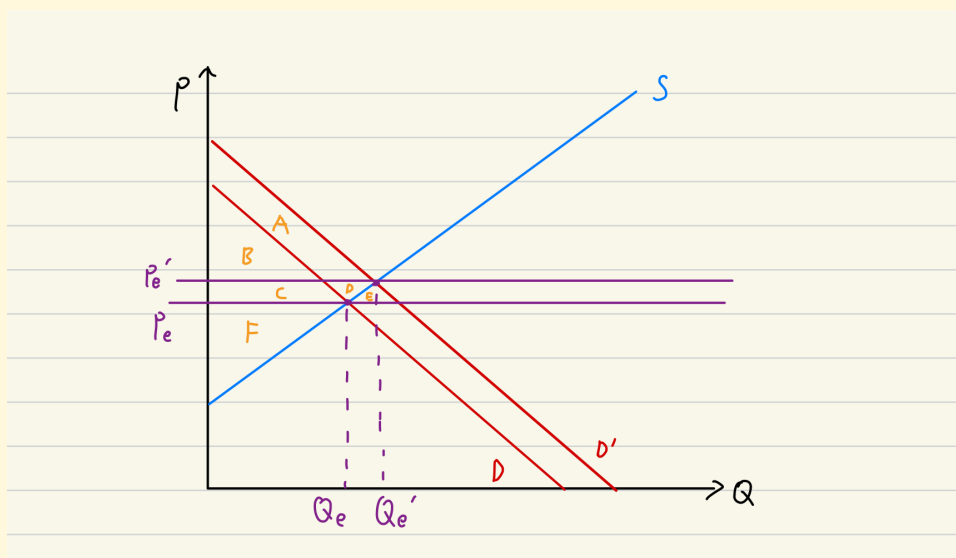


Figure 25: Demand move to the right

- The customer surplus before the shift is $B + C$.
- The customer surplus after the shift is $A + B$
- The producer surplus before the shift is F
- The producer surplus after the shift is $C + D + F$

Consider the case when demand moves to the left:

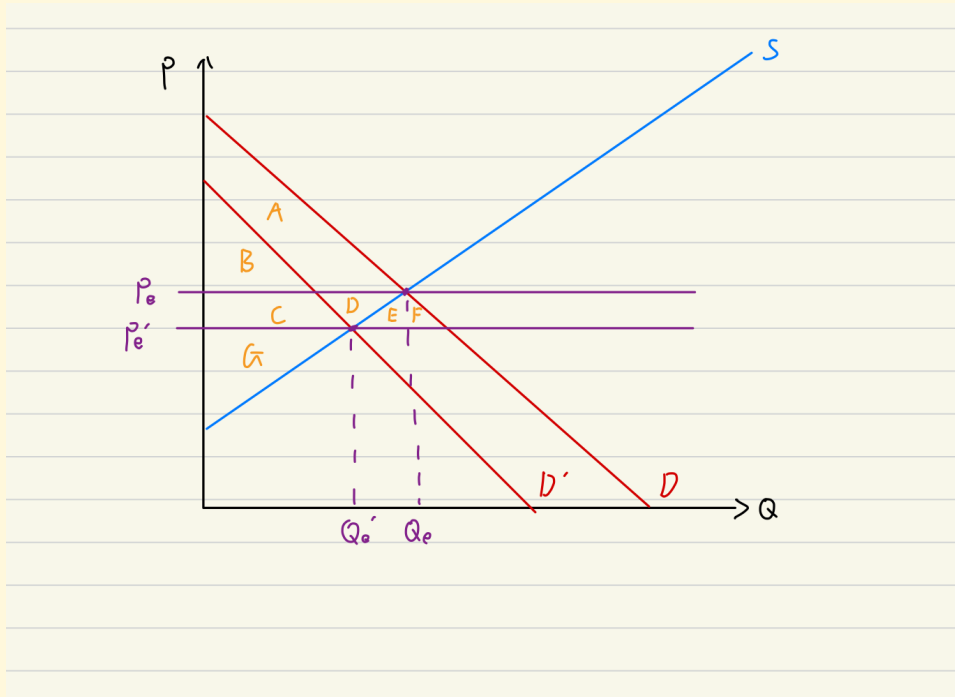


Figure 26: Demand move to the left

- The customer surplus before the shift is $A + B$.
- The customer surplus after the shift is $B + C$
- The producer surplus before the shift is $C + D + G$
- The producer surplus after the shift is G

Example: A local tire market is represented by the following equations and in the diagram offered below:

$$Q^D = 3,200 - 25P$$

$$Q^S = 15P - 800$$

where Q is the number of tires sold weekly and P is the price per tire. The equilibrium price is \$100 per tire, and 700 tires are sold each week.

Suppose an improvement in the technology of tire production makes them cheaper to produce so that sellers are willing to sell more tires at every price. Specifically, suppose that quantity supplied rises by 200 at each price.

- What is the new supply curve?
- What is the new equilibrium?
- What happen to the customer and producer surplus?

Quantity supplied rises by 200 units at every price, so we simply add 200 to the equation for Q^S :

$$Q_2^S = 15P - 800 + 200 = 15P - 600$$

The new equilibrium occurs where $Q^D = Q_2^S$:

$$3,200 - 25P = 15P - 600$$

$$3,800 = 40P$$

$$P = \$95$$

We can find the equilibrium quantity by substituting the equilibrium price into either the supply or demand equation (or both):

$$\begin{aligned} Q^D &= 3,200 - 25(95) & Q_2^S &= 15(95) - 600 \\ &= 8,200 - 2,375 & &= 1,425 - 600 \\ &= 825 & &= 825 \end{aligned}$$

The new equilibrium quantity is 825 tires per week. Notice that because supply increased, the equilibrium price fell and the equilibrium quantity rose just as we would predict.

The easiest way to determine the changes in consumer and producer surplus is to use a graph such as the one shown here. To calculate all the areas involved, we need to make sure we calculate the demand choke price and the supply choke prices before and after the increase in supply. The demand choke price is the price at which quantity demanded is zero:

$$\begin{aligned}Q^D = 0 &= 3,200 - 25P \\25P &= 3,200 \\P &= \$128\end{aligned}$$

The demand choke price is \$128.

The supply choke price is the price at which quantity supplied is zero. Because supply is shifting, we need to calculate the supply choke price for each supply curve:

$$\begin{array}{ll}Q_1^S = 0 = 15P - 800 & Q_2^S = 0 = 15P - 600 \\15P = 800 & 15P = 600 \\P = \$53.33 & P = \$40\end{array}$$

The initial supply choke price is \$53.33 but falls to \$40 when supply increases.

Consumer surplus: The initial consumer surplus is the area of the triangle below the demand curve but above the initial equilibrium price (\$100) :

$$\begin{aligned}CS_{\text{initial}} &= \frac{1}{2} \times \text{base} \times \text{height} \\&= \frac{1}{2} \times (700 - 0) \times (\$128 - \$100) \\&= (0.5)(700)(\$28) = \$9,800\end{aligned}$$

The new consumer surplus is the area of the triangle below the demand curve and above the new equilibrium price (\$95):

$$\begin{aligned}CS_{\text{new}} &= \frac{1}{2} \times \text{base} \times \text{height} \\&= \frac{1}{2} \times (825 - 0) \times (\$128 - \$95) = (0.5)(825)(\$33) \\&= \$13,612.50\end{aligned}$$

So, after the outward shift in supply, consumer surplus rises by \$3,812.50.

Producer surplus: The initial producer surplus is the area of the triangle below the initial equilibrium price and above the initial supply curve (S_1) :

$$\begin{aligned}PS_{\text{initial}} &= \frac{1}{2} \times \text{base} \times \text{height} \\&= \frac{1}{2} \times (700 - 0) \times (\$100 - \$53.33) = (0.5)(700)(\$46.67) \\&= \$16,334.50\end{aligned}$$

The new producer surplus is the area of the triangle below the new equilibrium price and above the new supply curve (S_2) :

$$\begin{aligned}PS_{\text{new}} &= \frac{1}{2} \times \text{base} \times \text{height} \\&= \frac{1}{2} \times (825 - 0) \times (\$95 - \$40) = (0.5)(825)(\$55)! \\&= \$22,687.50\end{aligned}$$

The increase in supply also led to a rise in producer surplus by \$6,353.

2.2 Market interventions

2.2.1 Price ceilings

Definition(Price ceilings): A form of government intervention by setting the highest price that can be paid for a good or service legally.

If the price ceiling is above market equilibrium price, it is nonbinding and will not hinder the free-market outcome. On the other hand, if the price ceiling is below the equilibrium price, it is binding and will impact the market.

Consider the following binding situation:

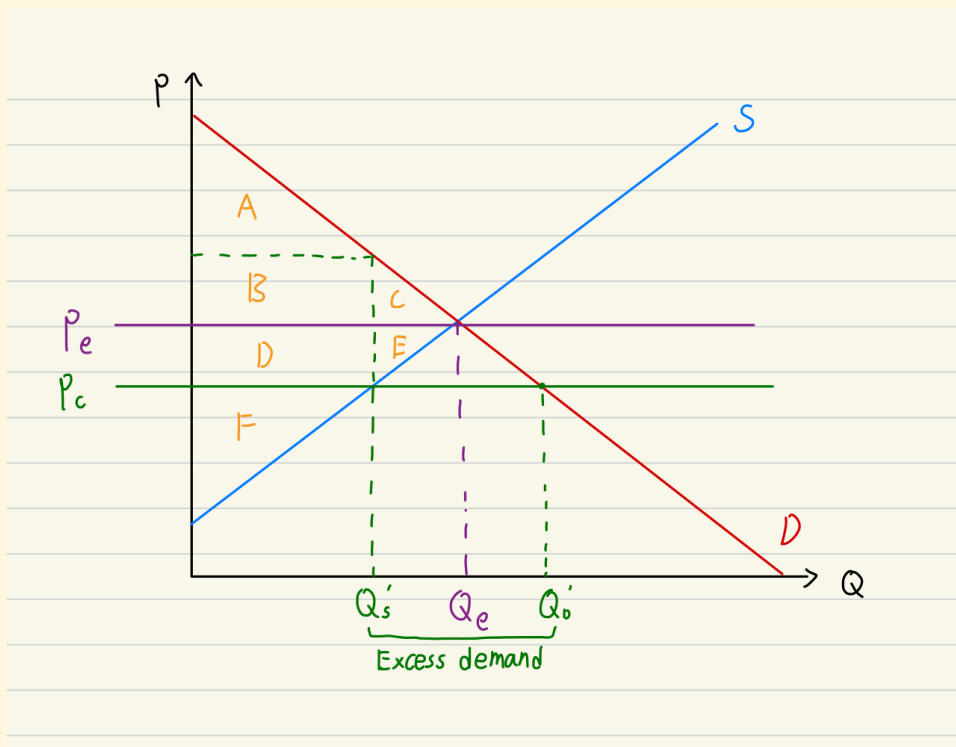


Figure 27: Binding price ceiling

Definition(Deadweight loss (DWL)): The difference between the maximum total surplus that consumers and producers could gain from a market and the combined gains they actually total surplus after a price regulation.

Definition(Excess demand): The difference between the quantity demanded and the quantity supplied at a price ceiling.

Definition(Transfer): Surplus that moves from producer to consumer, or vice versa, as a result of a price regulation.
In general:

A change in quantity from the equilibrium value is the only thing that causes a DWL. Changes in price will cause transfers.

We can identify the quantities visually from figure 27:

- Consumer surplus before $A + B + C$
- Consumer surplus after $A + B + D$
- Producer surplus before $D + E + F$
- Producer surplus after F
- Deadweight loss $C + E$
- Transfer from PS to CS D

The DWL tells us how much money gets burned up in the process of transferring surplus through the regulation: it can be seen as the cost of the transfer.

Area C is the deadweight loss suffered by consumers; area E is lost by producers.

Deadweight loss occurs because the price-control regulation transfers income by changing prices, and price changes affect incentives and lead to inefficiency relative to the market equilibrium outcome.

The goal of the price ceiling was to help the customer by, effectively, transferring surplus to them through lower prices. However, there is a better way for the government without intervening the free market.

Suppose instead that the government simply required producers to transfer $D - C$ to customers without intervening in the market.

Customers are equally well off as under the price ceiling because $A + B + C + D - C = A + B + D$.

The producers would be better off, rather than being left with just F , they would have their free-market producer surplus of $D + E + F$ minus their payment of $D - C$. The areas D in these two values cancel, leaving the pizzerias $E + F + C$, which is larger than the F in producer surplus they get under the price ceiling law.

Price elasticity also has impact on the size of DWL and transfer.

- For Deadweight loss:

1. If both demand and supply are **elastic**, the price ceiling causes a significant DWL due to a large mismatch between quantity demanded and supplied.
2. With **inelastic demand and elastic supply**, the DWL is smaller, as consumers aren't as responsive to the price change, despite producers being able to adjust supply quickly.
3. If demand is **elastic and supply inelastic**, the DWL can be larger since the quantity demanded changes substantially, but producers can't adjust supply accordingly.

- For transfer:

1. With **elastic demand**, a price ceiling leads to a considerable increase in consumer surplus at the expense of producers, as consumers benefit more from lower prices and have alternative choices.
2. In the case of **inelastic demand**, the gain in consumer surplus is smaller, meaning a less significant loss for producers.
3. If supply is **elastic**, producers face a significant loss of surplus due to their ability to adjust production. This loss transfers largely to consumers if demand is also elastic.
4. For **inelastic supply**, producers can't adjust their production much, resulting in a smaller transfer of surplus to consumers.

In general, we can make the following conclusion:

For a more elastic market a price change causes a greater decrease in quantity therefore a policy in a more elastic market will cause a greater deadweight loss with respect to transfer.

2.2.2 Price floor

Definition(Price floor): It is a form of government intervention that sets a price regulation that sets the lowest price that can be paid legally for a good or service.

A price floor is binding if it is above equilibrium price. Otherwise it is not binding.

Consider the following binding situation:

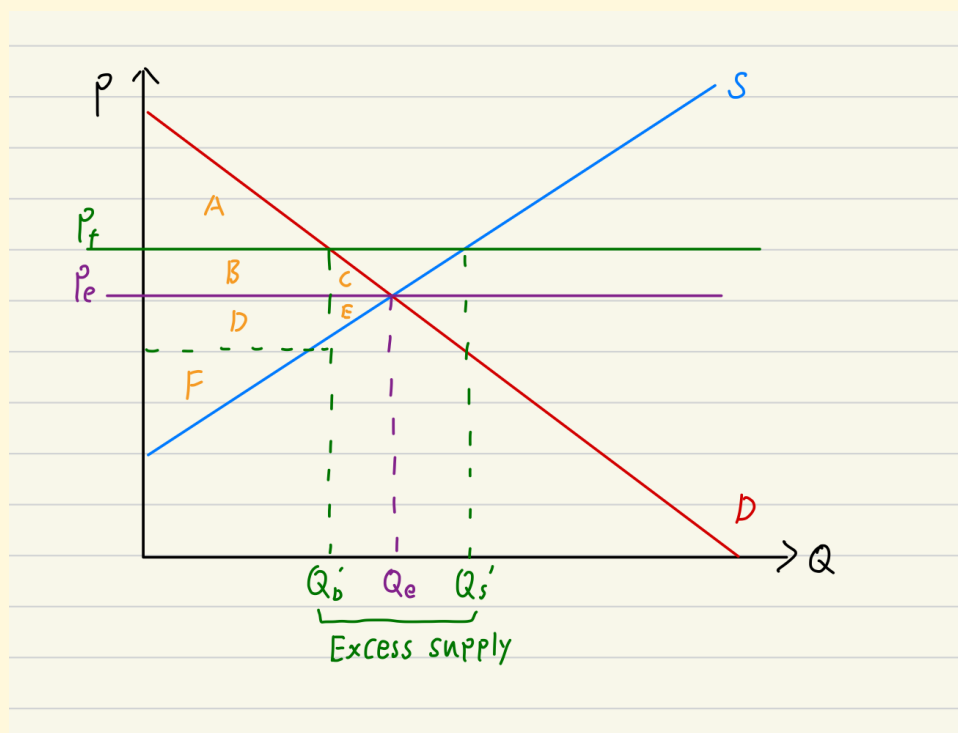


Figure 28: Binding price floor

Definition(Excess supply): The difference between the quantity supplied and the quantity demanded at a price floor.

We can identify the following quantities:

- Consumer surplus before $A + B + C$
- Consumer surplus after A
- Producer surplus before $D + E + F$
- Producer surplus after $B + D + F$
- Deadweight loss $C + E$
- Transfer from CS to PS B

2.2.3 Quota

Definition(Quota): It is a regulation by the government that sets the quantity of a good or service provided.

A quota is binding unless it is at the position of the equilibrium quantity. Otherwise, it is not binding.

Consider the following binding situation:

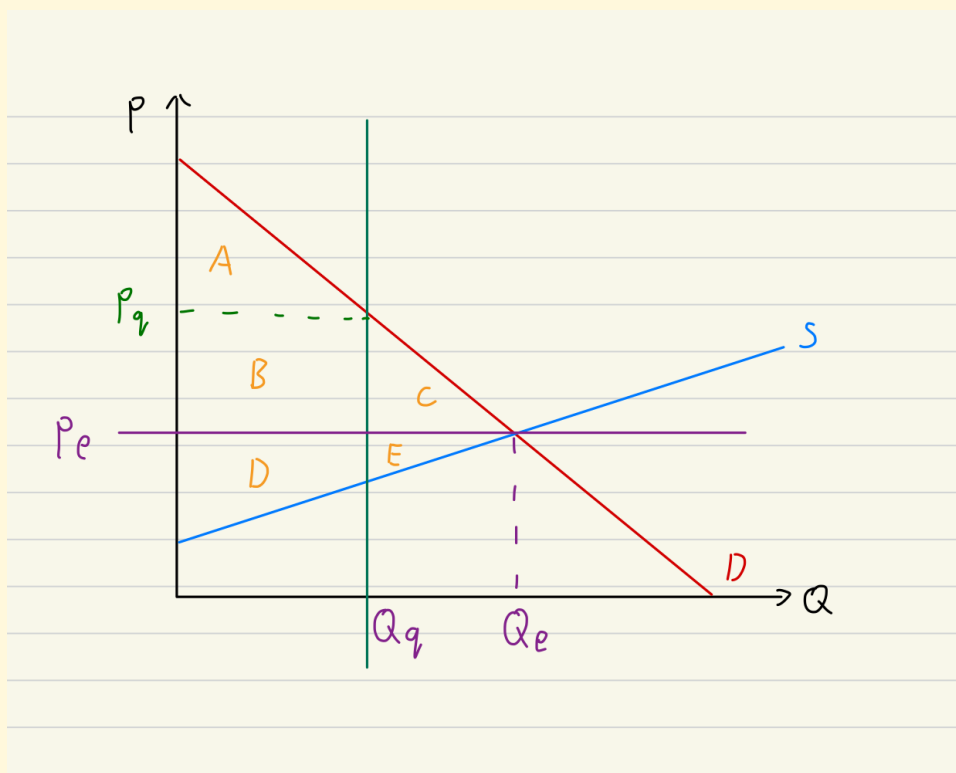


Figure 29: Binding when quota is lower than equilibrium with P_q being the market price

From the graph, we have the following observations:

- Consumer surplus before $A + B + C$
- Consumer surplus after A
- Producer surplus before $D + E$
- Producer surplus after $B + D$
- Deadweight loss $C + E$
- Transfer from CS to PS B

2.2.4 Taxes

There are two types of tax incidence or tax burden created by a tax: the statutory incidence of a tax and the economic incidence of a tax. Typically, a general reference to "tax incidence" refers to the economic incidence of a tax.

Definition(Statutory incidence): The statutory incidence of a tax falls on the party, producers or consumers, that has to physically send a check to the government in the amount of a tax.

Definition(Economic incidence): The economic incidence of a tax falls on the party that bears the actual cost of the tax. Put another way, economic incidence reflects the actual change in an individual's or firm's resources due to the tax. The statutory incidence of the tax is irrelevant to the economic incidence of the tax.

The one who legally bears the tax doesn't determine its real economic impact. For instance, a gas tax on producers may lead to higher prices for consumers, and a consumer tax can reduce demand and impact producer earnings. The true effect, or economic tax incidence, depends on the market's reaction to price changes, not on the initial tax assignment.

We will consider two methods of viewing taxes: **shifting curves** and **wedge method**.

For shift curves, we have two scenarios, shifting supply curves and shifting demand curves.

The supply curve is based on a firm's Marginal Cost (MC), and any factor increasing the MC causes a shift in the supply curve, taxes included. Consider the following graphic representation:

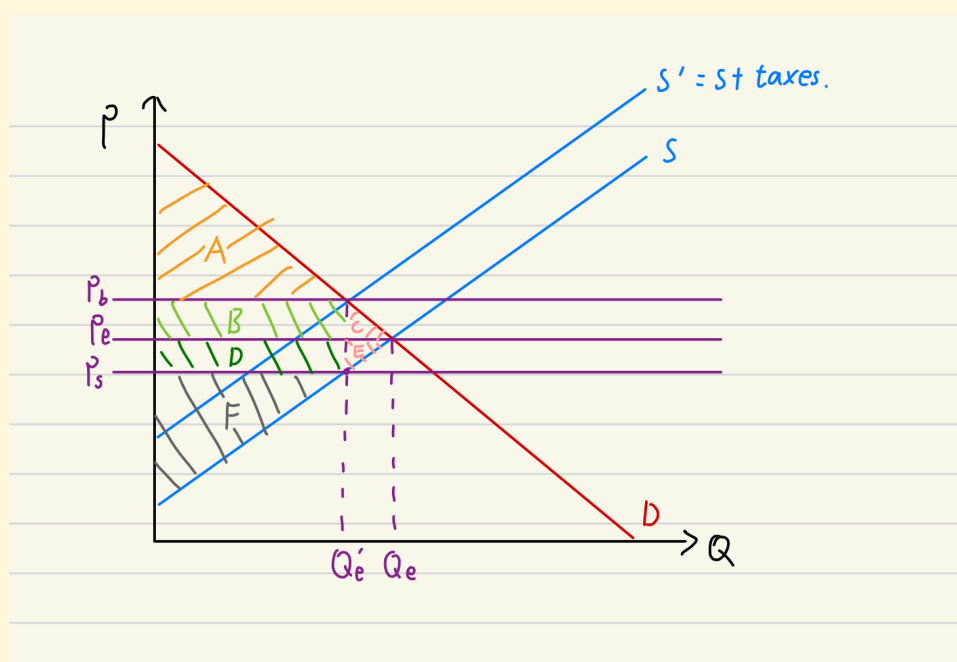


Figure 30: Shifting supply due to tax on supply

From the graph above, we have the following:

- P_b is the market price, or the price actually paid by the customer.
- P_s is the price received by the seller.
- The tax received by the government per unit good or service is $P_b - P_s$.
- Tax revenue will be the sum of tax paid by the sellers (Area D) and tax paid by the buyers (Area B).

We also observe that:

- Consumer surplus before $A + B + C$
- Consumer surplus after A
- Producer surplus before $D + E + F$
- Producer surplus after F
- Government revenue from the tax $B + D$

We then know the total surplus:

$$TS = CS + PS + \text{tax revenue} = A + F + B + D$$

And the dead-weight loss resulted from this tax is:

$$DWL = C + E$$

Since the demand curve represents the consumers' willingness to pay, the demand curve will shift down as a result of the tax. Graphically, it looks like:

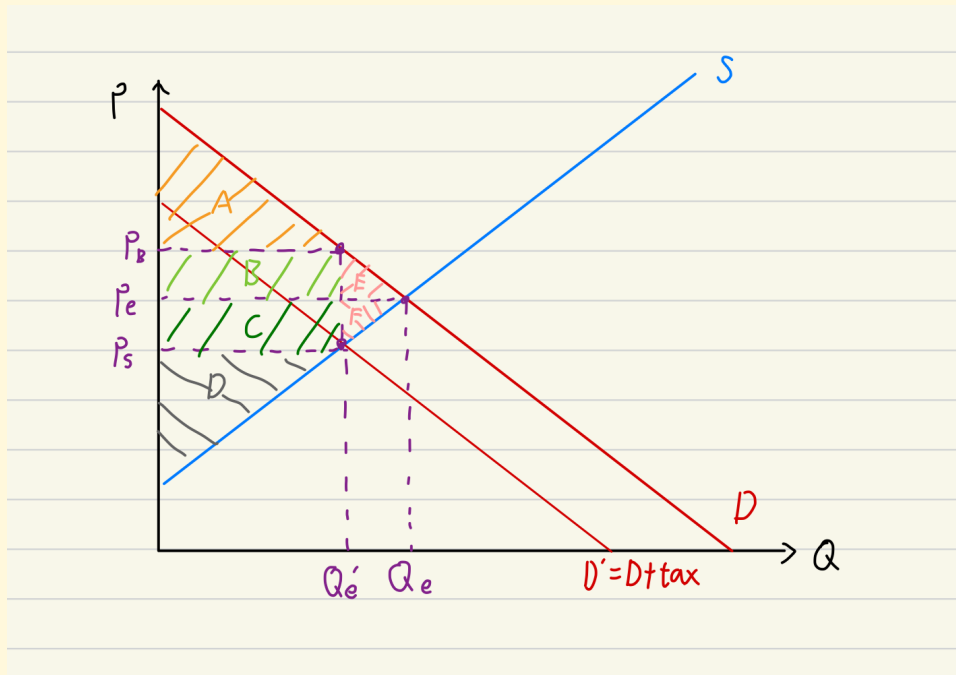


Figure 31: Shifting demand due to tax on buyers

We have the following observations:

- P_b is the market price, or the price actually paid by the customer.
- P_s is the price received by the seller.
- The tax received by the government per unit good or service is $P_b - P_s$.
- Tax revenue will be the sum of tax paid by the sellers (Area C) and tax paid by the buyers (Area B).

We also observe that:

- Consumer surplus before $A + B + E$
- Consumer surplus after A
- Producer surplus before $C + D + F$
- Producer surplus after D
- Government revenue from the tax $B + C$

We then know the total surplus:

$$TS = CS + PS + \text{tax revenue} = A + B + C + D$$

And the dead-weight loss resulted from this tax is:

$$DWL = E + F$$

From the analysis above, we can see that the statutory burden of a tax (who legally pays the tax) doesn't necessarily dictate the economic burden of the tax (who ultimately bears the cost). In many markets, the economic burden is shared between consumers and producers irrespective of the statutory burden.

It turns out there is a general formula that will approximate the share of the tax borne by the consumer and the share borne by the producer:

- Share borne by consumer = $\frac{E^S}{E^S + |E^D|}$
- Share borne by producer = $\frac{|E^D|}{E^S + |E^D|}$

If the price elasticity of supply (E^S) is infinite, the consumers' share is equal to 1 ; that is, consumers bear the whole burden when supply is perfectly elastic.

If the absolute value of the price elasticity of demand ($|E^D|$) is infinite, the consumers' share of the tax burden is zero, and suppliers bear the whole burden of the tax.

Example: Consider the demand and supply for cola in a market represented by the following equations:

$$Q^D = 15 - 10P$$

$$Q^S = 40P - 50$$

where Q is millions of bottles per year and P measures dollars per bottle. The equilibrium price of cola is \$1.30 per bottle, and 2 million bottles are sold each year.

1. Calculate the price elasticity of demand and the price elasticity of supply at the equilibrium price and quantity.
2. Calculate the share of a tax that would be borne by consumers and the share borne by producers.
3. If a tax of \$0.15 per bottle is imposed, what would be the expected price buyers will have to pay? What price will sellers receive after the tax?

The formula for price elasticity of demand is

$$E^D = \frac{\Delta Q^D}{\Delta P} \times \frac{P}{Q^D}$$

From the demand curve, we can calculate $\frac{\Delta Q^D}{\Delta P}$. Each time P changes by 1 unit, Q^D falls by 10 . Therefore,

$$\frac{\Delta Q^D}{\Delta P} = -10$$

Substituting into the formula for elasticity, we get

$$E^D = \frac{\Delta P}{\Delta Q} \times \frac{P}{Q} = -10 \times \frac{1.3}{2} = \frac{-13}{2} = -6.5$$

The formula for price elasticity of supply is

$$E^S = \frac{\Delta Q^S}{\Delta P} \times \frac{P}{Q^S}$$

From the supply curve, we can see that $\frac{\Delta Q^S}{\Delta P} = 40$. Note that each time P increases by 1 unit, Q^S rises by 40. Thus, the price elasticity of supply is

$$E^S = \frac{\Delta Q^S}{\Delta P} \times \frac{P}{Q^S} = 40 \times \frac{1.3}{2} = \frac{52}{2} = 26$$

The proportion of the tax borne by buyers will be

$$\frac{E^S}{E^S + |E^D|} = \frac{26}{26 + |-6.5|} = \frac{26}{32.5} = 0.8$$

The proportion of the tax borne by sellers will be

$$\frac{|E^D|}{E^S + |E^D|} = \frac{|-6.5|}{26 + |-6.5|} = \frac{6.5}{32.5} = 0.2$$

So, buyers will bear 80% of the tax and sellers will bear only 20% of the tax.

$$\text{Increase in } P_b = (0.80)(\$0.15) = \$0.12$$

The price buyers pay will rise from \$1.30 per bottle (the original equilibrium price) to \$1.42.
Sellers will bear the other 20% of the tax:

$$\text{Decrease in } P_s = (0.2)(\$0.15) = \$0.03$$

The price sellers receive will fall from \$1.30 per bottle to \$1.27.

Another method to view taxes is through the wedge method. This method recognizes that who pays the tax is ultimately irrelevant. Instead, the wedge method illustrates that a tax drives a wedge between the price consumers pay and the revenue producers receive, equal to the size of the tax levied.

Because of the tax, consumers pay more for the good (P_D) than they did before the tax, and suppliers receive less for the good (P_s) than they did before the tax.

The tax wedge is the difference between the price consumers pay and the value producers receive (net of tax) from a transaction.

Following the Law of Supply and Demand, as the price to consumers increases, and the price received by suppliers decreases, the quantity that each wishes to trade will decrease. After a tax is introduced, a new equilibrium is reached, where consumers pay more ($P^* \rightarrow P_D$), suppliers receive less ($P^* \rightarrow P_s$), and the quantity exchanged falls ($Q^* \rightarrow Q_t$). The difference between P_D and P_s will be equivalent to the size of the per-unit tax.

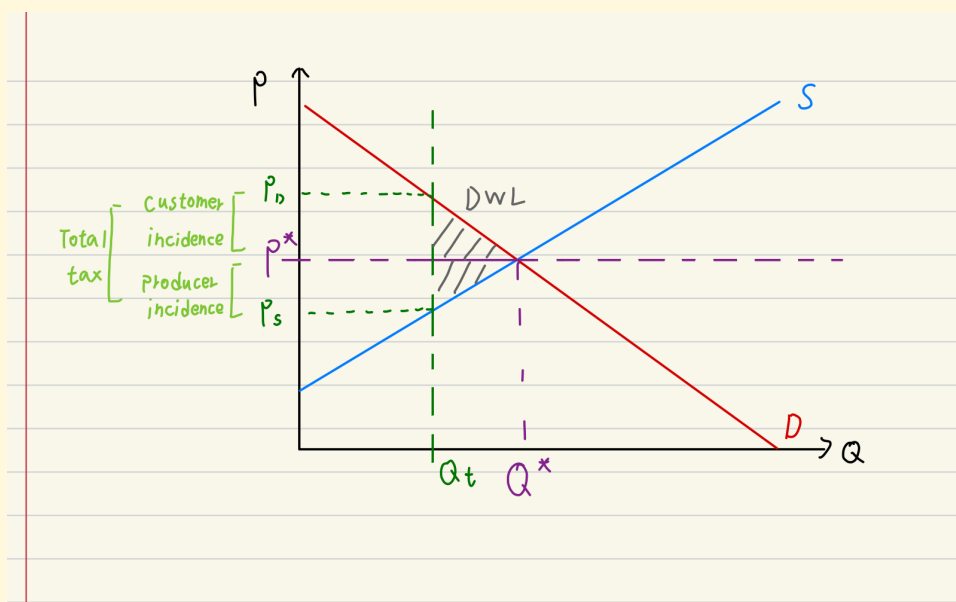


Figure 32: Wedge method

Deadweight loss occurs with a tax because a higher price for consumers, and a lower price received by suppliers, reduces the quantity of the good sold. Thus, the equilibrium quantity of a taxed good is lower than the equilibrium quantity when the same good is not taxed. The deadweight loss created by the tax is equal to $\frac{1}{2} \times \text{Tax per unit} \times (Q^* - Q_t)$ represented by the shaded triangle in the figure.

The party with the more inelastic (steeper) curve bears more of the tax:

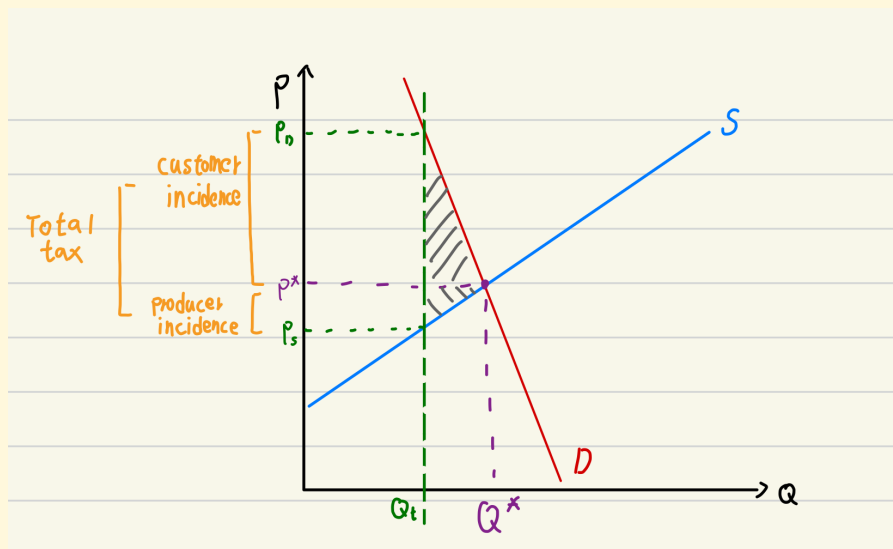


Figure 33: Steeper demand curve

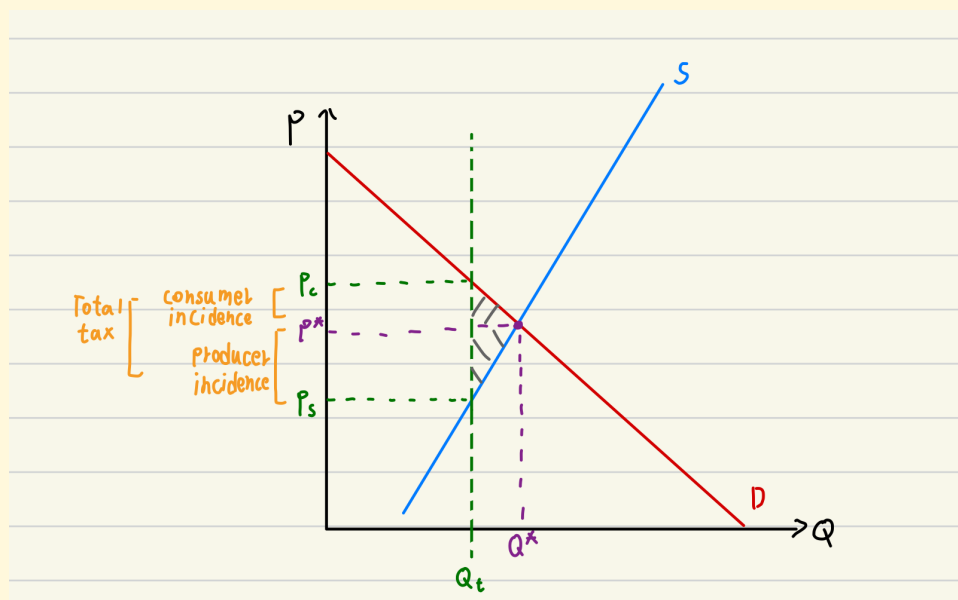


Figure 34: Steeper supply curve

3 Customer behavior

3.1 The Consumer's preferences and the concept of utility

The foundation of market demand is rooted in customer behavior.

And one of the major component in customer behavior model is customer preferences.

Consumers' preferences underlie every decision they make. Economists think of consumers as making rational choices about what they like best, given the constraints they face.

Consumers make many choices every day about what to buy and what not to buy. These choices involve many different goods.

Economists broadly make four (and only four) assumptions about consumer preferences.

The first three assumptions constitute what are often referred to as the **axioms of rational preferences** because consumers whose preferences satisfy these assumptions make "rational", and therefore predictable, choices.

The fourth assumption is a little bit more restrictive and imposes the idea that, generally speaking, consumers have a **taste for variety**.

The four assumptions are:

1. **Completeness and rankability:** A consumer can compare and rank any two bundles of goods. For any two bundles A and B , a consumer can determine one of the following: A is preferred to B , B is preferred to A , or A is just as good as B .
2. **Transitivity:** If a consumer prefers bundle A over bundle B and prefers bundle B over bundle C , then the consumer must also prefer bundle A over bundle C . This ensures that preferences are logically consistent.
3. **Non-Satiation (or "More is Better"):** All else being equal, consumers prefer more of a good to less of it. This assumption ensures that, given two bundles with the same amount of one good, the bundle with more of the other good is preferred.
4. **Convexity (taste of variety):** This assumes that consumers prefer a diversified or balanced bundle of goods. Or the more a consumer has of a particular good, the less she is willing to give up of something else to get even more of that good.

Definition(Utility): Utility is a measure of how satisfied a consumer is. It is closely related to, although not exactly, a measure of happiness or well-being.

Definition(Utility function): A utility function u mathematically describes the relationship between what consumers actually consume and their level of well-being (i.e. utility). It is a function because each combination of goods consumed (i.e. each bundle) maps into a single level of utility.

Definition(Marginal utility function): Marginal utility of a particular good is how much does total utility increase when the amount of that good increases by a small amount, with the amounts of all the other goods in utility held constant.

Definition(Ordinal measure): An ordinal measure classifies data into categories that can be ranked or ordered. However, the actual differences between the categories are not specified or uniform.

Definition(Cardinal measure): A cardinal measure not only categorizes and ranks data, but it also specifies the exact intervals or differences between values. This allows for meaningful mathematical operations (like addition or subtraction) to be performed on the data.

Remark: Preference ranking is ordinal measure, and utility function is cardinal measure

Remark: Because utility functions are just representations of preferences, and are somewhat arbitrary, we can't use utility to make **interpersonal comparisons** among consumers. For example, we can't say a person with utility function $U_1^*(J, K)$ is "better off" when $J = K = 2$ than a person with utility function $U_1(J, K)$, even though $U_1^*(2, 2) = 36 > U_1(2, 2) = 18$

3.2 Indifference curve

Definition(Preference map): A preference map refers to a graphical representation of a consumer's preferences over various combinations or bundles of goods. The most common graphical tool used to represent consumer preferences is an indifference curve diagram.

Definition(Indifferent): It is a special case in which a consumer derives the same level of utility from each of two or more consumption bundles.

Definition(Indifference curve): An indifference curve represents a set of bundles or combinations of goods that give a consumer the same level of satisfaction or utility. They are like contours of a graph. Points yielding different utility levels are each associated with distinct indifference curves and these indifference.

Definition(Indifference map): A graph of indifference curves for several utility levels of an individual consumer.

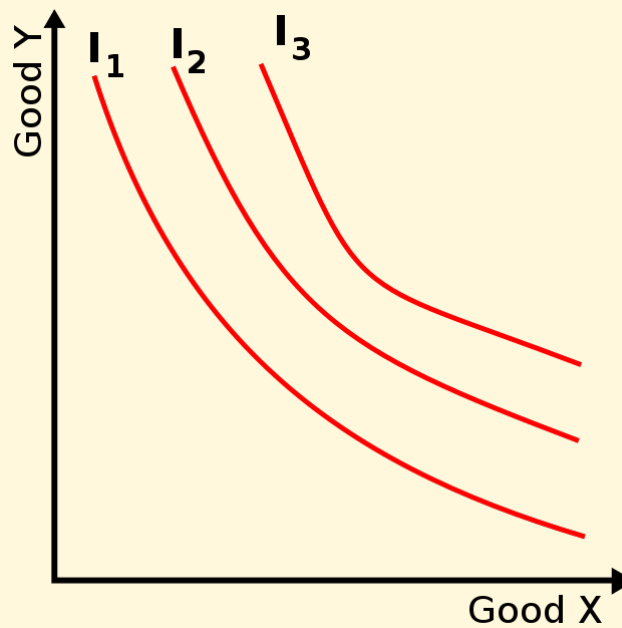


Figure 35: Indifference curves and indifference map

Characteristic of indifference curve

1. Defined only in the non-negative quadrant of commodity quantities (i.e. the possibility of having negative quantities of any good is ignored).
2. Existence of Indifference Curves: Due to the assumption of completeness and rankability, we can always draw indifference curves since all bundles have a distinct utility level and can be ranked.
3. Direction and Slope: The "more is better" principle means that indifference curves representing higher utility levels are located farther out. It also ensures that indifference curves don't slope upwards because as quantity consumed of one good (X) increases, total satisfaction would increase if not offset by a decrease in the quantity consumed of the other good (Y). So, a bundle with more of both goods must be preferred to one with less of both, thus the first bundle must yield a higher utility, and lie on a different indifference curve at a higher utility level.
4. Non-Crossing Property: Transitivity ensures that indifference curves don't intersect. If they did, it would suggest inconsistencies in preferences, implying that a bundle can offer two different utility levels, which is a contradiction.
5. Shape: The diminishing marginal rate of substitution causes indifference curves to be convex to the origin. At high quantities of good Y and low quantities of good X , a consumer is willing to give up a relatively large amount of Y for a small increase in X because the marginal utility of Y is low and the marginal utility of X is high. As the quantity of X increases (and Y decreases), the willingness to substitute Y for more X decreases; hence, the consumer demands more Y for each additional unit of X given up. This changing willingness creates the convex shape.

Definition (Marginal rate of substitution of x): The rate at which a consumer is willing to trade off one good (the good on the horizontal axis, X) for another (the good on the vertical axis, Y) and still be left equally well off.

$$MRS_{XY} = -\frac{\Delta Y}{\Delta X}$$

If the quantity of good x increases by one unit while holding the quantity of good y constant, the utility changes by MU_x . This is the definition of marginal utility for good x . For any change Δx in the quantity of good x , the resulting change in utility is $\Delta x MU_x$.

On an indifference curve, any change in good x (Δx) is compensated by a change in good y (Δy) such that the consumer remains indifferent (or equally satisfied). The change in utility resulting from the change Δy in good y is $\Delta y MU_y$, as per the definition of marginal utility for good y .

As the consumer moves along the indifference curve, the total change in utility has to be zero because the consumer remains equally satisfied at every point on the curve. Hence, $\Delta x MU_x + \Delta y MU_y = 0$.

Rearranging the equation above, we get $\frac{\Delta y}{\Delta x} = -\frac{MU_x}{MU_y}$.

This equation highlights that the slope of the indifference curve, which represents the Marginal Rate of Substitution between x and y (i.e., MRS_{xy}), is equivalent to the negative ratio of the marginal utilities of the two goods.

We have another calculus approaches:

We start by writing down the "identity" equation for a given indifference curve:

$$U(x, y) = \bar{U}$$

where $\bar{U}$ is some utility value Now take the total derivative of the identity as x and y change:

$$\frac{\partial U}{\partial x} dx + \frac{\partial U}{\partial y} dy = \frac{\partial \bar{U}}{\partial x} dx + \frac{\partial \bar{U}}{\partial y} dy \rightarrow MU_x dx + MU_y dy = 0$$

Note that since $\bar{U}$ is just a number, its value doesn't change with x or y Rearranging:

$$\frac{dy}{dx} = -\frac{MU_x}{MU_y}$$

This is the same expression we got before except our d replaces Δ , representing continuous small changes in x and y

Steepness of Indifference Curves

Because the MRS measures the willingness of the consumer to trade one good for another, we can tell a great deal about preferences by examining the shapes of indifference curves.

1. Indifference curves that are relatively steep indicate the consumer is willing to give up a large quantity of the good on the vertical axis to get another unit of the good on the horizontal axis.
2. Relatively flat indifference curves imply that the consumer would require a large increase in the good on the horizontal axis to give up a unit of the good on the vertical axis.

Curvature of Indifference Curves

The curvature of indifference curves reflects information about the consumer's preferences between two goods, just as its steepness does.

- Goods that are highly substitutable are likely to produce indifference curves that are relatively straight. This means the MRS does not change much as the consumer moves from one point to another along the indifference curve.
- Goods that are complementary will generally have indifference curves with more curvature.

For an individual consumer, indifference curves might be flat in some areas of the preference map and very curved in others.

Definition(Perfect substitute): A good that a consumer can trade for another good, in fixed units, and receive the same level of utility.

These kinds of preferences produce linear indifference curves, and utility functions for perfect substitutes take on the general form $U = aX + bY$, where a indicates the marginal utility of consuming one more unit of X and b indicates the marginal utility of consuming one more unit of Y .

Definition(Perfect complement): A good from which the consumer receives utility dependent on its being used in a fixed proportion with another good.

They usually have L-shaped indifference curve. Perfect complements lead to distinctive L-shaped indifference curves. Mathematically, this can be represented as $U = \min\{aX, bY\}$, where a indicates the marginal utility of consuming one more unit of X and b indicates the marginal utility of consuming one more unit of Y .

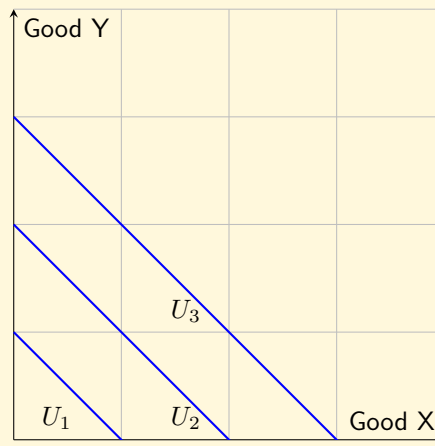


Figure 36: Perfect substitute indifference map

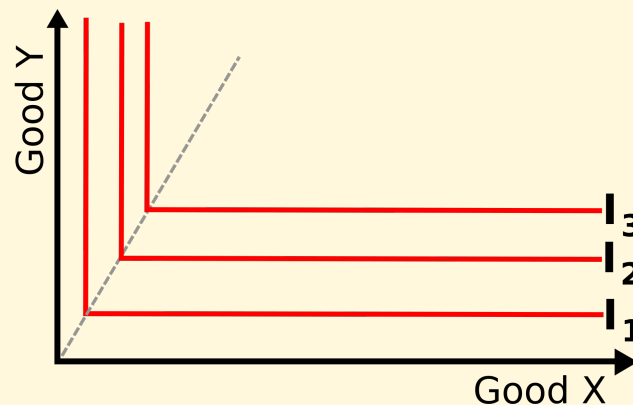


Figure 37: Perfect complement indifference map

3.3 The consumer's constraints

Because consumers have limited money to spend and goods are not free, consumers must make tradeoffs when deciding how much of each good to consume.

These tradeoffs depend not only on the utility consumers get from each good, but also:

- how much money they have to spend on all goods
- The prices of the goods.

These two phenomena characterize the **consumer's budget constraint**.

We must start with some assumptions (again, to keep things simple, we focus on a model with only two goods). We assume three things:

- Goods have fixed prices and how much an individual consumer buys of a good has no effect on this price.
- Consumers have a fixed and exogenously given income to spend on goods.
- The consumer's problem is static: utility from consumption is only realized once so there is no role for saving and borrowing is not allowed.

Definition(Budget constraint): A curve that describes the entire set of consumption bundles a consumer can purchase when spending all income.

The mathematical formula for a budget constraint is

$$\text{income} = p_x x + p_y y$$

where p_x and p_y are the prices for 1 unit of goods X and Y , and x and y are the quantities of the two goods.

We can write the equation of the budget constraint as:

$$p_x x + p_y y = \text{Income} \rightarrow y = \frac{\text{Income}}{p_y} - \frac{p_x}{p_y} x$$

- So an increase in p_x rotates the budget line inward along the x -axis, making it steeper.
- An increase in p_y also rotates the budget line inward, but along the y -axis.
- An increase in income leaves the slope unchanged, but shifts the budget line outward, parallel to itself.
- Similarly, a decrease in income leaves the slope unchanged, but shifts the budget line inward, parallel to itself.

Definition(Feasible bundle): Any combination of goods on or below the budget constraint that the consumer has the income to purchase. (p. 111)

Any combination of goods on or below the budget constraint that the consumer has the income to purchase (i.e., any point between the origin on the graph and the budget constraint, including those on the constraint itself) is a feasible bundle.

Definition(Infeasible bundle): Any combination of goods above or to the right of the budget line that the consumer cannot afford to purchase.

Any combination of goods above or to the right of the budget line that the consumer cannot afford to purchase is an infeasible bundle.

Realistically, there is a form of budget constraint calls quantity discount:

Another form of budget constraint in the real world is also called quantity limits:

3.4 Budget constraint

Definition(Budget constraint): A curve that describes the entire set of consumption bundles a consumer can purchase when spending all income.

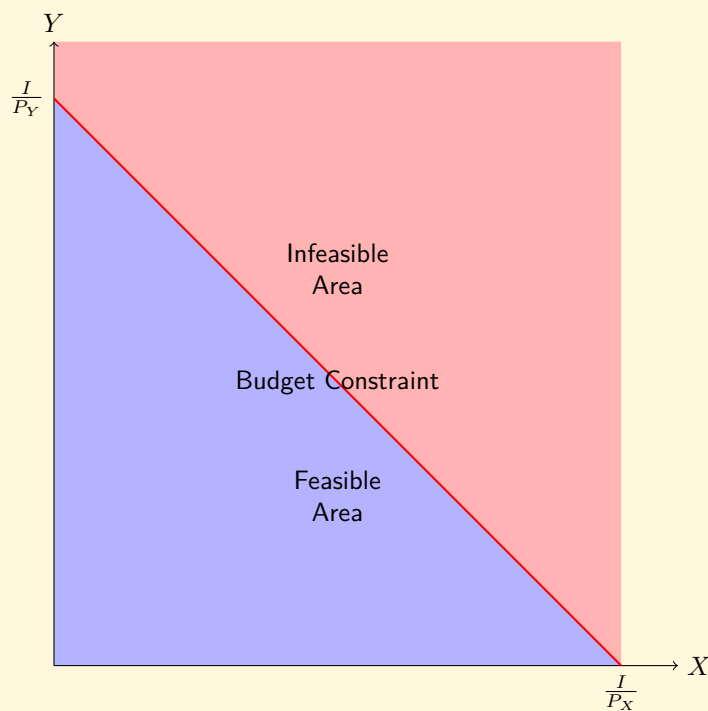


Figure 38: Budget constraint

Definition(Feasible bundle): Any combination of goods on or below the budget constraint that the consumer has the income to purchase.

Definition(Infeasible bundle): Any combination of goods above or to the right of the budget line that the consumer cannot afford to purchase.

The general equation to the budget constraint is:

$$p_x x + p_y y = I \rightarrow y = \frac{1}{p_y} - \frac{p_x}{p_y} x$$

The slope of the budget constraint in y/x space is therefore $-\frac{p_x}{p_y}$.

Shifts and changes to the budget constraint

1. An increase in p_x rotates the budget line inward along the x -axis, making it steeper.
2. An increase in p_y also rotates the budget line inward, but along the y -axis, making it flatter.
3. An increase in I leaves the slope unchanged, but shifts the budget line outward, parallel to itself.
4. Similarly, a decrease in I leaves the slope unchanged, but shifts the budget line inward, parallel to itself.

The budget constraint has been a straight line so far. But there are some cases in which the budget constraint would be other form instead with kinks.

For example, quantity discounts introduce non-linearity and non-convexity to the budget constraint.

Example: Suppose Alex spends his \$100 weekly income on pizzas and data usage on his phone. A pizza costs \$10; if he spends everything on pizzas, he can buy 10. If the price of phone data is constant at 10 cents per MB, he can buy up to 1 GB (1,000MBs) of data. The budget constraint in the case where data use costs a constant 10 cents per MB is given by the solid section of the line running from zero MBs and 10 pizzas up to 1,000MBs and zero pizzas. Now, with the quantity discount: **the first segment** runs from the point where Alex buys only pizzas (10 pizzas, 0MBs) to the point where he buys 600MBs of data and 4 pizzas. This segment reflects the constant price of 10 cents per MB for the first 600MBs. **The second segment** starts from the kink at 600MBs and 4 pizzas and runs to the point where Alex spends all his money on data, which gives him 1400 MBs (0 pizzas, 1400 MBs). This segment reflects the discounted price of 5 cents per MB after the first 600MBs.

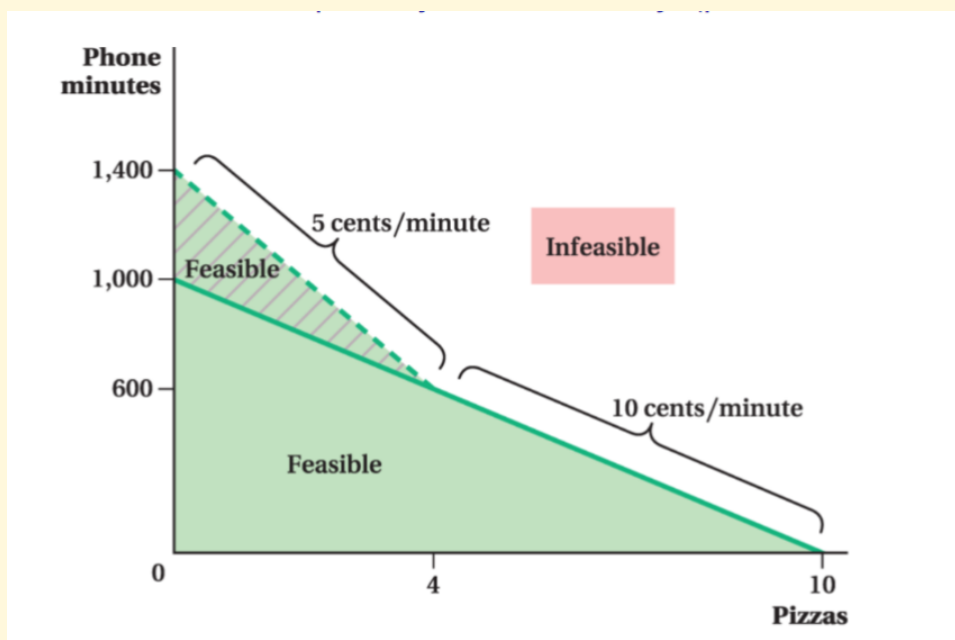


Figure 39: Budget line with quantity discount

Another way a budget constraint can be kinked is if there is a limit on how much of one good can be consumed.

Example: Alex's budget constraint is affected by a cap of 600 MBs on his weekly data usage, which is imposed by either his carrier, the government, or his parents. Despite his weekly income and the prices of pizzas and data remaining the same, Alex is unable to purchase more than 600 MBs of data. Consequently, the budget constraint, which originally extended from 10 pizzas with no data to no pizzas with 1000 MBs of data, now becomes horizontal at the 600 MBs mark. This horizontal line indicates that any additional income cannot be used to purchase more data, rendering any point beyond 600 MBs infeasible. Therefore, if Alex chooses a bundle on this horizontal segment, he will have leftover unspent income. The graph representing this situation would show a kink at 600 MBs and 4 pizzas, with a solid line indicating feasible combinations up to this point and a dashed line beyond, which is labeled as infeasible, highlighting that Alex is restricted from using his income beyond the imposed data cap.

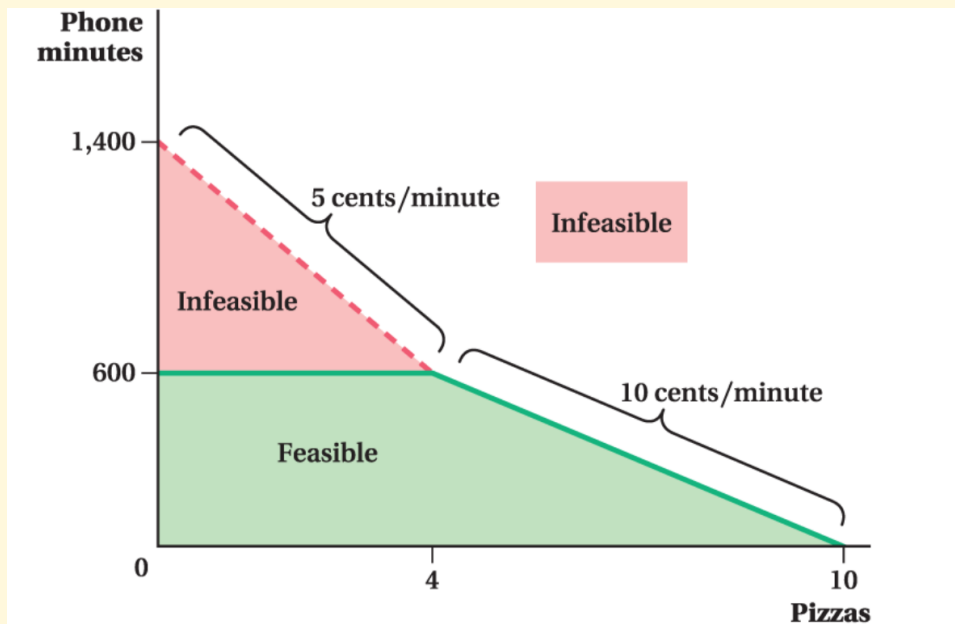


Figure 40: Budget constraint with quantity limit

3.5 Introduction to customer constrained optimization problem

The consumer wants to maximize his utility, but it is limited in his ability to do so by the budget constraint.

This is just an individual-level version of the general economic problem, which exists for individuals, families, organizations, and societies: resources are scarce and we must choose how to deploy them to make ourselves as (materially) well off as possible.

Based on our theory developed to date, however, the general solution to the consumer's optimization problem is intuitively obvious:

Get to the highest attainable indifference curve given the budget constraint.

The general solution to the optimization problem is viewed graphically:

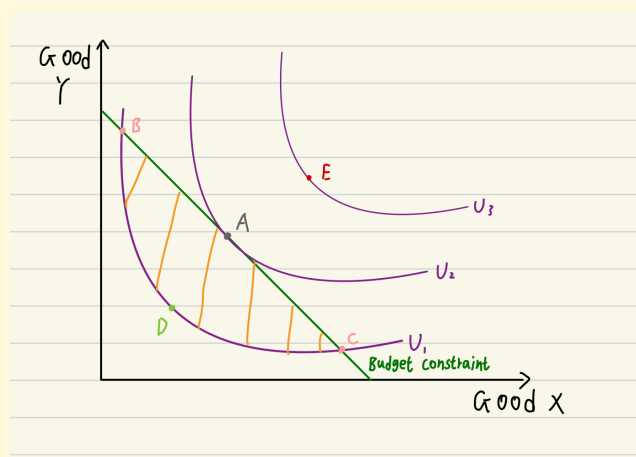


Figure 41: General solution

Definition (Interior solution): A utility-maximizing bundle that contains positive quantities of both goods.

The consumer's optimal consumption bundle occurs at the point of tangency between her budget constraint and her indifference curve, shown here at point A.

The consumer can afford the consumption bundles represented by points B, C, and D, but these are on a lower indifference curve (U_1) than is point A (U_2).

Point E is on a higher indifference curve (U_3), but it lies outside the consumer's budget constraint and is thus infeasible.

The optimal point can be shown mathematically:

The point of tangency between the (highest attainable) indifference curve and the budget constraint is characterized by the fact that their slopes are equal at this point.

The slopes are (in absolute value):

- Indifference curve: $MRS = \frac{MU_x}{MU_y}$
- Budget constraint: $\frac{p_x}{p_y}$

So the optimality condition at an interior solution is

$$\frac{MU_x}{MU_y} = \frac{p_x}{p_y}$$

$$\frac{MU_x}{p_x} = \frac{MU_y}{p_y}$$

In words: spend such that the last dollar on each good generates the same extra utility.

However, in some cases, the highest attainable indifference curve may not be characterized by this point of tangency.

Specifically, if we have a **corner solution**, the highest attainable indifference curve will lie on one of the goods and takes an optimal value of zero and is not consumed at all.

Definition (Corner solution): A utility-maximizing bundle located at the "corner" of the budget constraint where the consumer purchases only one of two goods.



Figure 42: Corner solution

Clearly, the highest attainable indifference curve provides utility U_2 and is reached at point A where zero mystery novels are purchased.

In this case, the indifference curves are everywhere flatter than the budget line so $|MRS_{xy}| < \frac{p_x}{p_y}$ everywhere on the preference map.

There are general two methods for solving a consumer's optimization problem mathematically:

1. Using the tangency condition
2. Using Lagrangian optimization

3.5.1 Using tangency condition and Lagrangian optimization on Cobb-Douglas

If the solution is **interior**, where the consumer optimally consumes positive amounts of both goods, we already know the equation of the solution:

$$|MRS_{XY}| = \frac{MU_X}{MU_Y} = \frac{p_X}{p_Y}$$

We can use this condition, and the budget constraint, to solve for the optimal values of X and Y called X^* and Y^* .

But to do so, we need to know MU_X and MU_Y . These are marginal utilities, which must be derived from utility itself, so we need a specific utility function, i.e. we must choose a functional form for utility.

An extremely common utility function we encounter in economics is the Cobb-Douglas utility function.

Definition(Cobb-Douglas utility function):

$$U(X, Y) = X^\alpha Y^{1-\alpha}, \quad \alpha \in (0, 1)$$

Typically, α is called the share parameter for X , for reasons that will become apparent

We have:

$$|MRS_{XY}| = \frac{MU_X}{MU_Y} = \frac{\alpha X^{\alpha-1} Y^{1-\alpha}}{(1-\alpha) X^\alpha Y^{-\alpha}} = \left(\frac{\alpha}{1-\alpha} \right) \frac{Y}{X} = \frac{p_X}{p_Y}$$

So, solving for Y :

$$Y = \left(\frac{1-\alpha}{\alpha} \right) \left(\frac{p_X}{p_Y} \right) X$$

Plug this into the budget constraint: ¹

$$\begin{aligned} p_X X + p_Y \left(\frac{p_X}{p_Y} \right) \left(\frac{1-\alpha}{\alpha} \right) X &= I \\ \alpha p_X X + (1-\alpha) p_X X &= \alpha I \\ X &= \frac{\alpha I}{p_X} \end{aligned}$$

We have:

$$X^* = \frac{\alpha l}{p_X}$$

As well:

$$Y = \left(\frac{1-\alpha}{\alpha} \right) \left(\frac{p_X}{p_Y} \right) X$$

which reduces to

$$Y^* = \frac{(1-\alpha) l}{p_Y}$$

This solution to the optimization problem and holds for any values of α, l, p_X , and p_Y . Note we can also write:

$$p_X X^* = \alpha l; \quad p_Y Y^* = (1-\alpha) l$$

The optimal values of X and Y for Cobb-Douglas utility are:

1. $Y^* = \frac{(1-\alpha) l}{p_Y}$
2. $X^* = \frac{\alpha l}{p_X}$

For simple problems like the one above, we can use the optimality condition at an interior solution directly. In some cases though, we may want to use an alternative, more general, method called Lagrangian analysis.

The Lagrangian method has one major advantage: it gives us a Lagrangian multiplier, which has a very useful interpretation.

The general form of the Lagrangian is

$$\mathcal{L}(X, Y, \lambda) = f(X, Y) + \lambda[g(X, Y)]$$

- $f(X, Y)$ is the objective function to be optimized (in our case, utility to be maximized).
- $g(X, Y)$ is the constraint: in our case, the income constraint.
- λ is the Lagrangian multiplier for the budget constraint.

Now we will use Lagrangian analysis on Cobb-Douglas. Start by writing out the Lagrangian for our specific Cobb-Douglas utility maximization problem:

$$\mathcal{L}(X, Y, \lambda) = X^\alpha Y^{1-\alpha} + \lambda [I - p_X X - p_Y Y]$$

We maximize the Lagrangian by taking the partial derivatives with respect to X, Y , and λ and setting them equal to zero.

Using our Cobb Douglas utility function:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial X} &= \frac{\partial f(X, Y)}{\partial X} + \lambda \frac{\partial g(X, Y)}{\partial X} = \alpha X^{\alpha-1} Y^{1-\alpha} - \lambda p_X = 0 \\ \frac{\partial \mathcal{L}}{\partial Y} &= \frac{\partial f(X, Y)}{\partial Y} + \lambda \frac{\partial g(X, Y)}{\partial Y} = (1-\alpha) X^{\alpha} Y^{-\alpha} - \lambda p_Y = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= I - p_X X - p_Y Y = 0\end{aligned}$$

The Lagrange multiplier λ is in the first two first order ODE. So, solve each of these equations for λ :

$$\lambda = \frac{\alpha X^{\alpha-1} Y^{1-\alpha}}{p_X} = \frac{(1-\alpha) X^{\alpha} Y^{-\alpha}}{p_Y}$$

We start finding Y as a function of X using the equality from the first two conditions:

$$\begin{aligned}\frac{\alpha X^{\alpha-1} Y^{1-\alpha}}{p_X} &= \frac{(1-\alpha) X^{\alpha} Y^{-\alpha}}{p_Y} \\ \frac{Y^{1-\alpha}}{Y^{-\alpha}} &= \frac{(1-\alpha) p_X}{\alpha p_Y} \frac{X^{\alpha}}{X^{(\alpha-1)}} \\ Y &= \frac{(1-\alpha) p_X}{\alpha p_Y} X\end{aligned}$$

Using the last condition, we can plug this value for Y into the budget constraint:

$$\begin{aligned}I - p_X X - p_Y Y &= 0 \\ I &= p_X X + p_Y \frac{(1-\alpha) p_X}{\alpha p_Y} X \\ I &= p_X X \left[1 + \frac{(1-\alpha)}{\alpha} \right] \\ I &= p_X X \left[\frac{\alpha}{\alpha} + \frac{(1-\alpha)}{\alpha} \right] \\ I &= \frac{p_X}{\alpha} X \\ X^* &= \frac{\alpha I}{p_X} \\ Y^* &= \frac{(1-\alpha) p_X}{\alpha p_Y} \left(\frac{\alpha I}{p_X} \right) = \frac{(1-\alpha) I}{p_Y}\end{aligned}$$

The optimal values of X and Y for Cobb-Douglas utility are same as the method use tangency condition:

1. $Y^* = \frac{(1-\alpha)I}{p_Y}$
2. $X^* = \frac{\alpha I}{p_X}$

Note we also have an expression for λ :

$$\lambda = \frac{MU_X}{p_X} = \frac{MU_Y}{p_Y}$$

λ is the **marginal utility of wealth**, which measures how much does utility rise if I rises by one dollar. And this identity only occurs at interior solution.

3.5.2 More examples of customer constrained optimization problem

Example: Let utility be:

$$U(X, Y) = X^{0.25} Y^{0.75}$$

Assume that $p_X = 5, p_Y = 5, I = 200$:

1. What is MRS_{XY} ?
2. What is X^* and Y^* using the method from the previous question?

3. What is the consumer's solution?

Solution:

1. Marginal Rate of Substitution (MRS_{XY}):

The MRS_{XY} is given by the negative ratio of the marginal utilities of X and Y:

$$MRS_{XY} = -\frac{\partial U/\partial X}{\partial U/\partial Y}$$

Calculating the partial derivatives, we get:

$$\frac{\partial U}{\partial X} = 0.25X^{-0.75}Y^{0.75}, \quad \frac{\partial U}{\partial Y} = 0.75X^{0.25}Y^{-0.25}$$

Thus,

$$MRS_{XY} = -\frac{0.25X^{-0.75}Y^{0.75}}{0.75X^{0.25}Y^{-0.25}} = -\frac{1}{3} \frac{Y}{X}$$

2. Optimal Bundle (X^* and Y^*):

The budget constraint is:

$$5X + 5Y = 200$$

We have equality:

$$|MRS_{XY}| = \frac{Y}{3X} = \frac{p_Y}{p_X} = \frac{5}{5}$$

So we have:

$$Y = 3X$$

If we substitute this identity in the budget constraint, we have:

$$20X = 200$$

Solving for X , we find:

$$X^* = 10$$

Substituting X^* back into the budget constraint, we find Y^* :

$$Y^* = 30$$

3. Consumer's Solution:

The consumer's solution to maximize utility with a budget of \$200, with prices for goods X and Y both at \$5, is to purchase 10 units of good X and 30 units of good Y. And this is a interior solution.

Example: Let utility be:

$$U(X, Y) = X + 10Y^{0.5}$$

Assume that $p_X = 5, p_Y = 2, I = 100$

1. What is MRS_{XY} ?
2. What is X^* and Y^* using the method from the previous question?
3. What is the consumer's solution?

Solution:

1. Given the utility function:

$$U(X, Y) = X + 10Y^{0.5}$$

and the prices and income as $p_X = 5$, $p_Y = 2$, and $I = 100$, we want to find the Marginal Rate of Substitution (MRS_{XY}).

The Marginal Rate of Substitution is the rate at which a consumer can give up some amount of one good in exchange for another good while maintaining the same level of utility. It is calculated as the negative of the ratio of the marginal utilities of the two goods. For two goods X and Y , it is given by:

$$MRS_{XY} = -\frac{\partial U / \partial X}{\partial U / \partial Y}$$

First, we find the marginal utility of each good:

$$MU_X = \frac{\partial U}{\partial X} = 1$$

$$MU_Y = \frac{\partial U}{\partial Y} = 5Y^{-0.5}$$

Then, the MRS between X and Y is:

$$MRS_{XY} = -\frac{MU_X}{MU_Y} = -\frac{1}{5Y^{-0.5}} = -0.2Y^{0.5}$$

Therefore, the Marginal Rate of Substitution of X for Y (MRS_{XY}) is:

$$MRS_{XY} = -0.2Y^{0.5}$$

2. We have equality:

$$|MRS_{XY}| = 0.2Y^{0.5} = \frac{p_Y}{p_X} = \frac{2}{5}$$

So, we have $Y^* = \left(\frac{2}{5 \cdot 0.2}\right)^2 = \frac{625}{4}$

Substitute this into the budget constraint:

$$5X + 2 \cdot \frac{625}{4} = 100$$

Solving for X^* , we have $X^* = -42.5$

3. The quantity $X^* = -42.5$ and $Y^* = \frac{625}{4}$ cannot be the customer solution because the quantity of X cannot be negative and quantity of Y is exceeding the budget constraint. Therefore we conclude there is no interior solution. However, consider the corner solution, because quantity of X is negative, we can set $X = 0$, so we have the corner solution $X^* = 0$ and $Y^* = 50$.

From the example above, we have the following important remark:

Remark: If either X^* or Y^* are negative, we set either X^* or Y^* equal to zero to get the closed corner solution to the optimization problem. If either X^* or Y^* are equal to 0, we naturally have the corner solution.

Example: Let $U(x, y) = \ln(x) + 2 \ln(y)$ Also, let $I = 100$, $p_x = 1$ and $p_y = 2$

1. Can we say for certain the solution is interior?
2. What are x^* and y^* ?

Solution:

1. As $x \rightarrow 0, \ln(x) \rightarrow -\infty$

Also, As $y \rightarrow 0, \ln(y) \rightarrow -\infty$ So $x = 0$ or $y = 0$ cannot be optimal values Another way to see this is by looking at the marginal utilities. We have:

$$MU_x = \frac{1}{x}; \quad MU_y = \frac{2}{y}$$

As x or y go to zero, their marginal utility goes to infinity.

Therefore it is not possible that indifference curves are everywhere flatter or everywhere steeper than the budget constraint, so long as prices are finite.

We can be sure we have an interior solution in this case.

2. Since we know our solution is interior, we can solve via the short-cut method. Start with

$$\begin{aligned} \frac{MU_x}{MU_y} &= \frac{p_x}{p_y} \\ \frac{1/x}{2/y} &= \frac{1}{2} \\ \rightarrow \frac{y}{2x} &= \frac{1}{2} \\ \rightarrow y &= x \end{aligned}$$

Subbing into the budget constraint:

$$x + 2x = 100 \rightarrow x^* = \frac{100}{3}$$

And therefore

$$y^* = \frac{100}{3}$$

Example: Here we have a example of optimizing with more than 2 goods. Let $U(x, y, z) = x^{0.25}y^{0.25}z^{0.50}$, assuming Cobb-Douglas preferences. Also, let $I = 500, p_x = 1, p_y = 2$ and $p_z = 2$. What are x^*, y^* , and z^* in this case?

Solution: When $n > 2$, it is useful to use a Lagrangian to derive the optimality conditions. We write

$$\mathcal{L}(x, y, z, \lambda) = u(x, y, z) + \lambda [I - p_x x - p_y y - p_z z]$$

Here, our objective function $f(\cdot)$ is utility and we have only one constraint: the budget constraint

We know Cobb-Douglas gives rise to interior solutions

Plugging in our specific parameter values we have:

$$\mathcal{L}(x, y, z, \lambda) = x^{0.25}y^{0.25}z^{0.50} + \lambda[500 - x - 2y - 2z]$$

$$\frac{\partial \mathcal{L}}{\partial x} = 0.25x^{-0.75}y^{0.25}z^{0.50} - \lambda = 0 \quad (1)$$

$$\frac{\partial \mathcal{L}}{\partial y} = 0.25x^{0.25}y^{-0.75}z^{0.50} - 2\lambda = 0 \quad (2)$$

$$\frac{\partial \mathcal{L}}{\partial z} = 0.50x^{0.25}y^{0.25}z^{-0.50} - 2\lambda = 0 \quad (3)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 500 - x - 2y - 2z = 0 \quad (4)$$

Then combine (1) with (2) and (2) with (3) to eliminate λ :

$$\begin{aligned} \frac{0.25x^{-0.75}y^{0.25}z^{0.50}}{0.25x^{0.25}y^{-0.75}z^{0.50}} &= \frac{y}{x} = \frac{1}{2} \\ \frac{0.25x^{0.25}y^{-0.75}z^{0.50}}{0.50x^{0.25}y^{0.25}z^{-0.50}} &= \frac{z}{2y} = 1 \\ 500 - x - 2y - 2z &= 0 \end{aligned}$$

From above, we get

$$y = \frac{x}{2}; \quad z = 2y; \quad 500 = x - 2y - 2z$$

We have three equations in three unknowns:

- Two optimally conditions,
- One constraint.

Subbing for y and z in terms of x in the budget constraint gives:

$$x + 2(0.5)x + 2x = 500 \rightarrow x^* = \frac{500}{4} = 125$$

Then subbing for z in terms of y in the budget constraint, taking out expenditure on x :

$$2(y) + 2(2y) = 500 - 125 = 375 \rightarrow y^* = 62.5$$

And solving for z :

$$z^* = \frac{500 - 125 - 62.5}{2} = 125$$

Note that:

$$x^* = 125 = \frac{0.25(500)}{1} = \frac{0.25I}{p_x}$$

As well:

$$y^* = 62.5 = \frac{0.25(500)}{2} = \frac{0.25I}{p_y}$$

And:

$$z^* = 125 = \frac{0.50(500)}{2} = \frac{0.5I}{p_z}$$

We have the following observation about the solutions to this question:

Remark: The general solution when we have Cobb-Douglas preferences is:

$$\mathbf{i}^* = \frac{\alpha_i \mathbf{I}}{\mathbf{p}_i}$$

for any good i with share parameter α_i

Example: In this example we have the following non-standard budget constraint. Let $U(x, y) = 2x^{0.5}y^{0.5}$. Also, let $I = 100$, $p_x = 2$ and $p_y = 2$ up to 10 units of y then $p_y = 1$ beyond 10 units purchased. What are x^* and y^* ?

Solution: Start with the fact that we, again, know that the solution will be interior

As well, we already know the general form the solution will take if $y < 10$:

$$x^* = \frac{0.5I}{p_x}; \quad y^* = \frac{0.5I}{p_y^1}$$

What if the optimal value of y is greater than 10 ? Then the problem becomes:

$$\max_{x,y} (2x^{0.5}y^{0.5}) \quad \text{s.t.} \quad p_x x + p_y^1(10) + p_y^2(y - 10) = 100$$

Or:

$$\max_{x,y} (2x^{0.5}y^{0.5}) \quad \text{s.t.} \quad 2x + y = 90$$

Let $\tilde{I} = 90$. If $y^* > 10$:

$$x^* = \frac{0.5\tilde{I}}{p_x}; \quad y^* = \frac{0.5\tilde{I}}{p_y^2}$$

Suppose the solution holds where $y < 10$. Then:

$$x^* = \frac{0.5(100)}{2} = 25 \quad y^* = \frac{0.5 \cdot 100}{2} = 25$$

Suppose the solution holds where $y > 10$. Then:

$$x^* = \frac{0.5(90)}{2} = 22.5 \quad y^* = \frac{0.5(90)}{1} = 45$$

The actual solution must be the second, since in the first case $y^* > 10$ and so p_y is 1, not 2 at the optimum.

We can rule the first possible solution out. The actual solution is:

$$x^* = 22.5 \quad y^* = 45$$

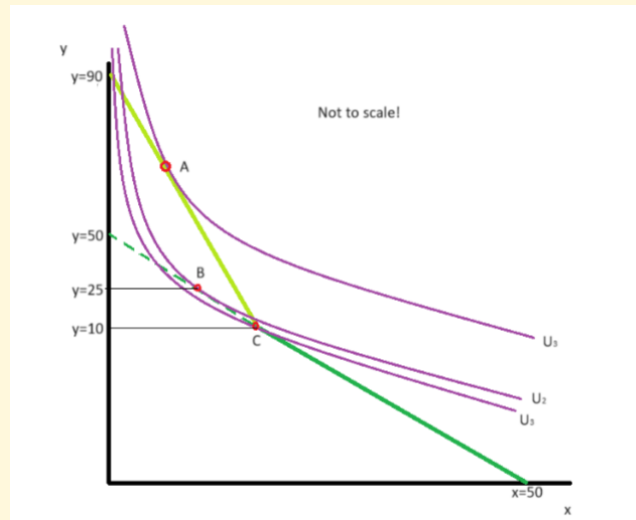


Figure 43: Different solutions

At a kink, the interior optimality condition doesn't hold since the budget constraint is not differentiable at this point.

To test the kink solution, look to see utility at the kink compares to utility at $\{x^*, y^*\}$:

$$U\left(x = \frac{100 - 2(10)}{2}, y = 10\right) = 40; U(x^*, y^*) = 63.64$$

The latter solution is clearly preferable, so the consumer optimally chooses 22.5 units of x and 45 units of y . So in conclusion:

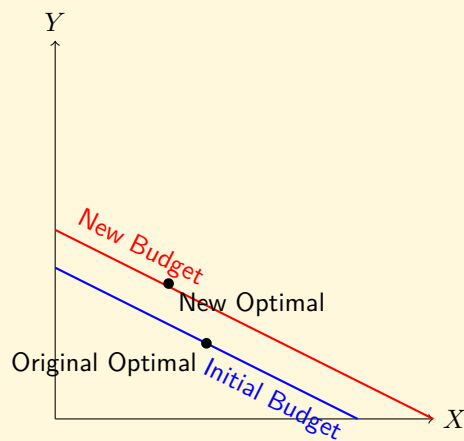
- Bundle A is the solution.
- Bundle B , where $p_y = p_y^1 = 2$, is clearly sub-optimal as the budget constraint doesn't hold with equality (this is the solution we ruled out).
- Bundle C is the bundle at the kink, which here generates the least utility.

4 Individual and market demand

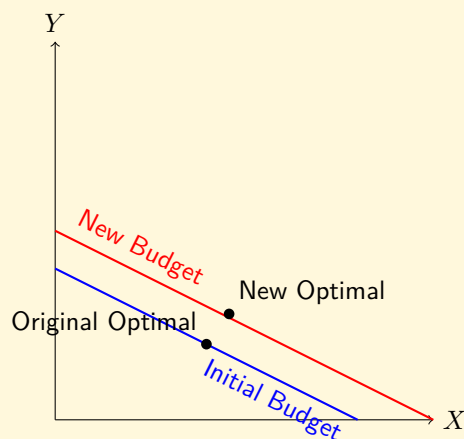
4.1 Change of income and customer choices and optimal

Definition (Income effect): The change in a consumer's consumption choices that results from a change in the purchasing power of the consumer's income.

- If optimal consumption of both goods rises when purchasing power rises, then both items X and Y are normal goods.
- If optimal consumption of one goods decreases when purchasing power rises, then both items X and Y are inferior goods.



(a) Inferior goods



(b) Normal goods

Figure 44: Consumer choices for different types of goods

Recall we defined the income elasticity of demand:

$$E_l^D = \frac{\% \Delta Q}{\% \Delta I} = \frac{\Delta Q}{\Delta I} \frac{I}{Q}$$

Here we think about this elasticity at the individual rather than the market level: Q here is the optimal value of a good from the consumer optimization problem, i.e. Q^* .

It is useful to categorize income effects in terms of their associated elasticities at different levels of I and Q^* :

1. **Inferior goods** have $E_l^D < 0$.
2. **Necessity goods** (which are normal) have $E_l^D \in (0, 1)$.

3. **Luxury goods** (which are also normal) have $E_t^D > 1$.

Definition(Income expansion curve): A curve that connects a consumer's optimal bundles at each income level.

Definition(Engle curve): A curve that shows the relationship between the quantity of a good consumed and a consumer's income.

Consider the following income expansion curve:

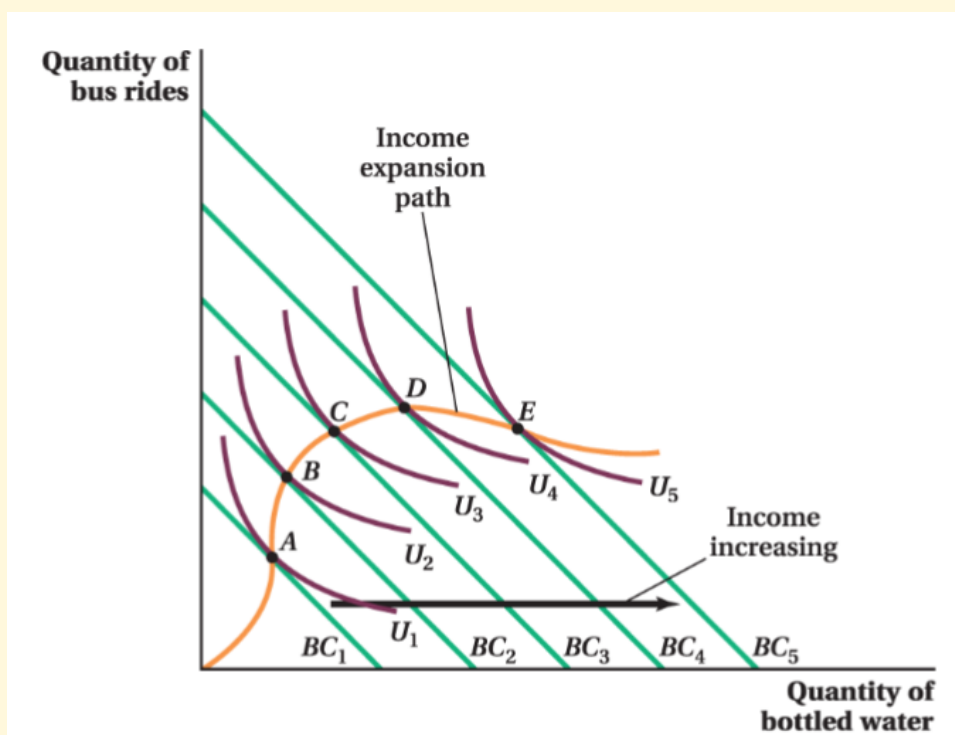


Figure 45: Example income expansion curve

Points A, B, C, D , and E are the optimal consumption bundles at five different income levels that correspond to the budget constraints shown. Note that the indifference curves themselves come from individual utility function.

Consider the following engle curves:

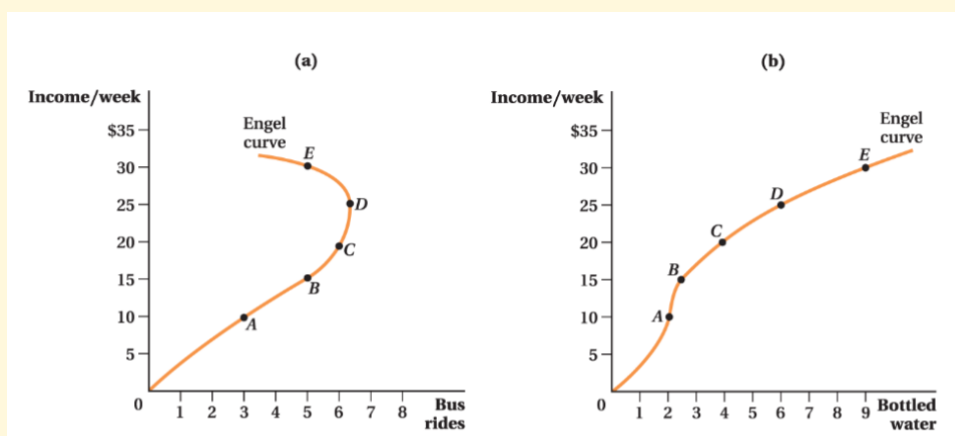


Figure 46: Example engle curves

In the engle curves we can see that bus rides are inferior goods and bottle water is normal good.

4.2 Prices changes and customer's choices

Analyzing the response of a consumer to changes in p_X or p_Y , all else constant, allows us to derive a consumer's individual demand curve.

The specific individual demand curve for good X tells us:

As the price of good X changes (while holding all else constant), how does the quantity of good X in the utility-maximizing bundle change?

To build the demand curve, we start by figuring out the consumer's utility-maximizing consumption bundle at some price for item X . By taking many different prices and find the utility-maximizing consumption bundle for X , we can graph a curve with the horizontal axis being the quantity of X and vertical axis being different prices for Y :

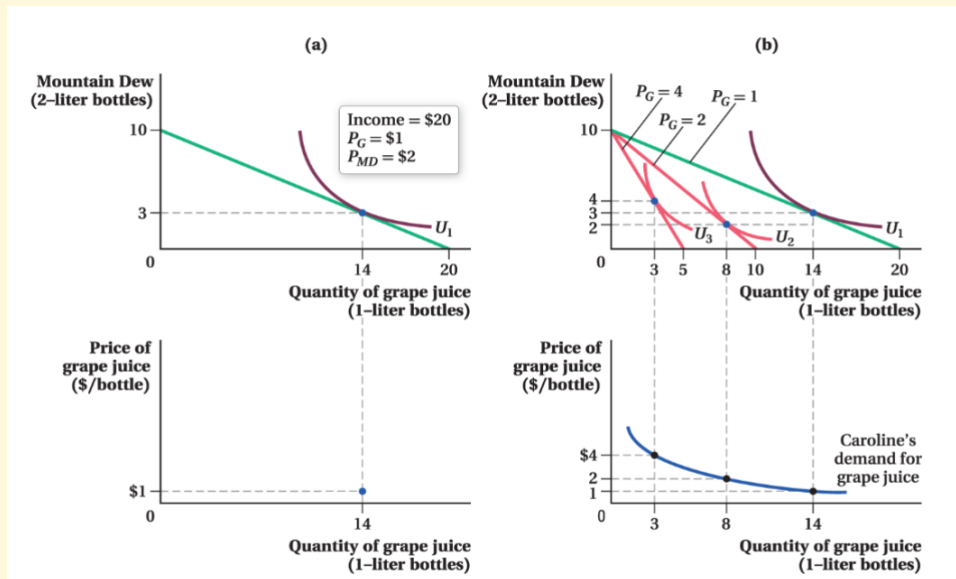


Figure 47: individual demand curve for X

Example: Let $U(x, y) = x + 2 \ln(y)$. Find and draw the demand curve for x . Find and draw the demand curve for y .

Solution: Start by setting $\frac{MU_x}{MU_y} = \frac{p_x}{p_y}$ to find the interior optimum (if it exists), and let $D_X(p_X, p_Y, I) = X^*$ and $D_Y(p_X, p_Y, I) = Y^*$.

We get:

$$\frac{y}{2} = \frac{p_x}{p_y} \rightarrow y^* = \frac{2p_x}{p_y}$$

If we're at an interior solution then:

$$x^* = \frac{I - p_y \left(\frac{2p_x}{p_y} \right)}{p_x} = \frac{I}{p_x} - 2$$

Of course, if we're not at an interior solution, then $y = \frac{1}{p_y}$ and $x = 0$ So we have:

$$D_x(p_x; I, p_y) = \begin{cases} 0 & p_x \geq \frac{I}{2} \\ \frac{I}{p_x} - 2, & p_x < \frac{I}{2} \end{cases}$$

$$D_y(p_y; I, p_x) = \frac{\min[I, 2p_x]}{p_y}$$

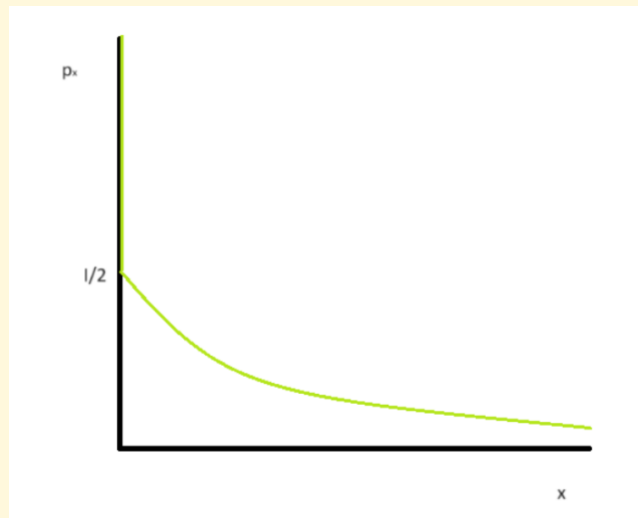


Figure 48: Demand curve for X

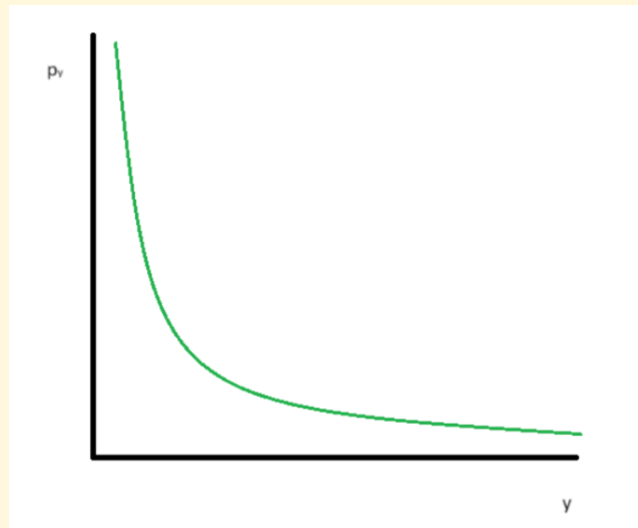


Figure 49: Demand curve for Y

We know that if a consumer's preferences or income change, or the prices of other goods change, then the demand curve shifts. We can see how by tracing out the demand curve under these new conditions.

Suppose the consumer in the previous diagram thinks grape juice is a "healthy alternative" to Mountain Dew. At a party, a nutritionist friend of a friend explains that this is not really the case: drinking grape juice is pretty much like drinking soda. The consumer's preference for grape juice change in response to this information: she now likes it less relative to Mountain Dew.

As a result, she found grape juice less desirable, this means her indifference curves flatten in Dew / Grape Juice space. Another way to think about this is as follows: Because the marginal rate of substitution (MRS) equals $-MU_G/MU_{MD}$, this preference shift shrinks marginal utility of grape juice at any quantity, reducing her MRS - that is, flattening her indifference curves.

We can built the individual demand curve just like before:

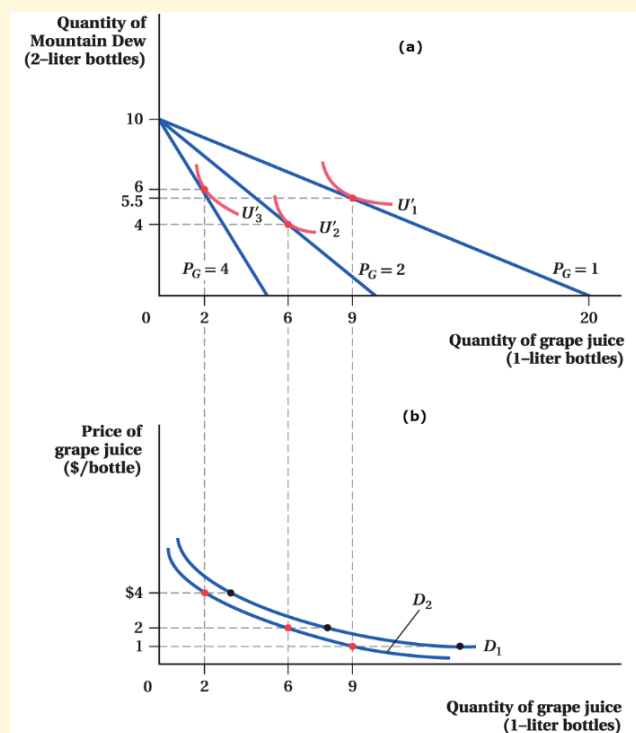


Figure 50: Shift in demand curve

With the flatter indifference curves (labeled U_1' , U_2' , and U_3'), Caroline's utility maximizing consumption bundles have changed. Now her optimal consumption levels of grape juice at prices of \$1, \$2, and \$4 per bottle are 9, 6, and 2 bottles, respectively. At each price level, she now consumes fewer bottles of grape juice.

Because she purchases fewer bottles of grape juice at each price point, Caroline's demand curve for grape juice shifts inward from D_1 to D_2 .

So, she consumes less grape juice at every price of grape juice: a shift in demand.

4.3 Conceptual foundations for income and substitution effects from price changes

Definition(Substitution effect): The change in a consumer's consumption choices that results from a change in the relative prices of two goods $\frac{p_Y}{p_X}$.

Definition(Income effect): The change in a consumer's consumption choices that results from a change in the purchasing power of the consumer's income.

Definition(Total effect): The total change effect in a consumer's optimal consumption bundle as a result of a price change. And we have:

$$\text{Total Effect} = \text{Substitution Effect} + \text{Income Effect}$$

4.3.1 Conceptual example with a drop of a price

Consider the following example shows how one consumer, Carlos, who spends his income on attending music concerts and basketball games, reacts to a fall in the price of basketball tickets.

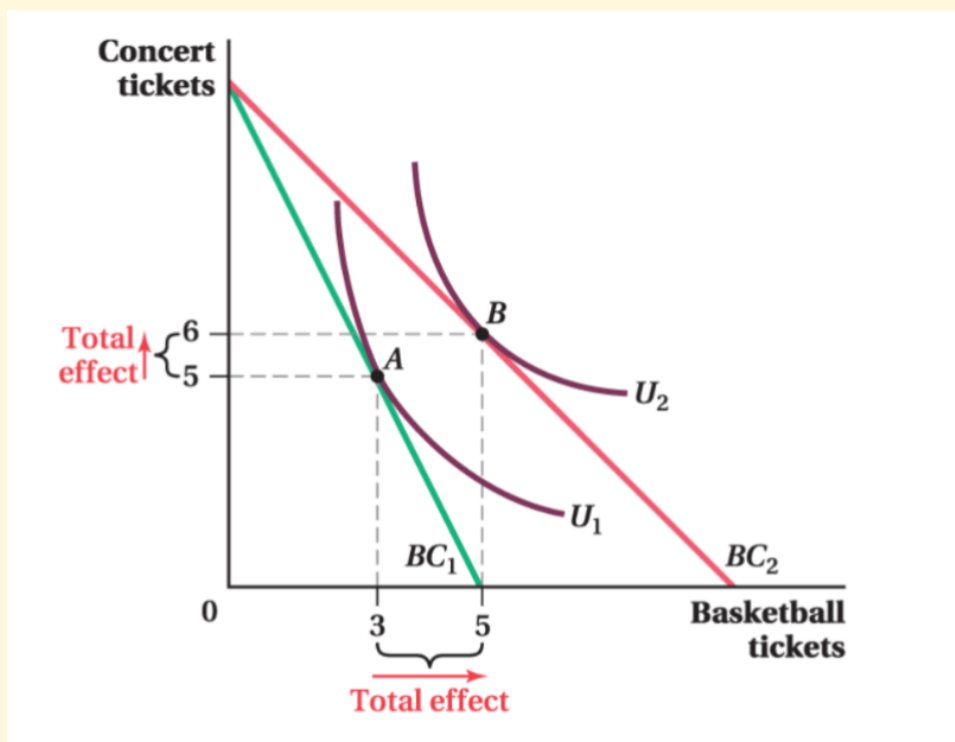


Figure 51: Before the after the price fall

When the price of basketball game tickets decreases, Carlos's budget constraint rotates outward from BC_1 to BC_2 because Carlos can now purchase more basketball tickets with his income.

As a result, the optimal consumption bundle shifts from A (the point of tangency between indifference curve U_1 and budget constraint BC_1) to B (the point of tangency between indifference curve U_2 and budget constraint BC_2).

Because of the fall in the price of basketball tickets, the quantity of concert tickets consumed increases from 5 to 6, and the number of basketball tickets Carlos purchases rises from 3 to 5.

These overall changes in quantities consumed between bundles A and B are the total effect. We want to study how the total effect can break into substitution and income effect. So, we want to isolate substitution and income effect.

Isolating substitution effect

Continue with the example above, let's begin by isolating the substitution effect. This part of the change in quantities demanded is due only to the change in relative prices, not to the change in Carlos's buying power. To isolate the substitution effect, we need to figure out how many concert and basketball tickets Carlos would want to buy if, after the price change, there was no income effect, that is, if he had the same purchasing power as before the price change and felt neither richer nor poorer.

For Carlos to feel neither richer nor poorer, the bundle he consumes after the price change must provide him with the same utility he was receiving before the price change; that is, **the new bundle must be on the initial indifference curve U_1** .

The substitution-effect-only bundle on U_1 has to reflect the fact that the goods' relative prices have changed. Those new relative prices are seen in the slope of the new budget line BC_2 . The problem is that a point of tangency doesn't exist between U_1 and BC_2 . However, there is a point of tangency between U_1 and a budget line with the same slope (i.e., the same relative prices) as BC_2 . The budget line we're talking about is BC' , and the point of tangency is point A' .

Bundle A' is what Carlos would buy if the relative prices of concert and basketball tickets changed the same way, but Carlos experienced no change in purchasing power. This is the definition of the substitution effect. Thus, the substitution effect, in isolation, moves Carlos's demanded bundle from A to A' .

So in general, what we did is we have to shift the post-price-change budget line BC_2 back in parallel (to preserve the new relative prices) until it is tangent to the preprice-change indifference curve U_1 (to keep the original utility level the same). To isolate the substitution effect when a price falls, we need Carlos's income to fall a little bit to erase how much richer he feels because the new lower price increases his effective purchasing power.

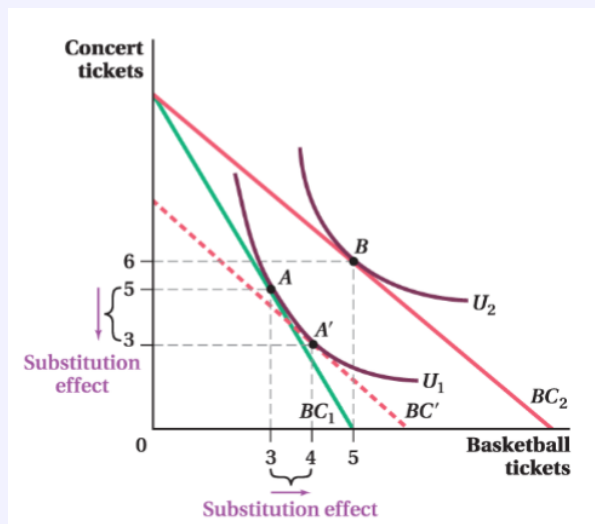


Figure 52: Isolating substitution effect

Isolating income effect

The income effect is the part of the total change in quantities consumed that is due to the change in Carlos's buying power after the price change.

When the price of a good falls, Carlos becomes richer overall. The reduction in a good's price means there's a whole new set of bundles Carlos can now buy that he couldn't afford before because he would have some money left over. In a sense, that money left over is the increase in his income from the price change.

This increase in buying power allows Carlos to achieve a higher level of utility than he did before. The income effect is the change in Carlos's choices driven by this shift in buying power while holding relative prices fixed at their new level. So, it is the shift in consumption quantities (from bundle A' to B) due to Carlos's ability to reach a higher indifference curve (U_2 instead of U_1) while holding relative prices fixed (BC_2 and BC' have the same slope).

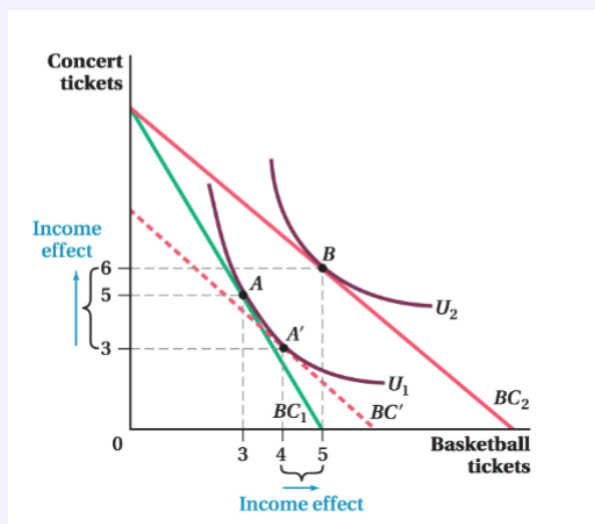


Figure 53: Isolating income effect

4.3.2 Similar example but rise in price

Here we consider a similar example but with rise in basketball ticket price. They follow similar logic:

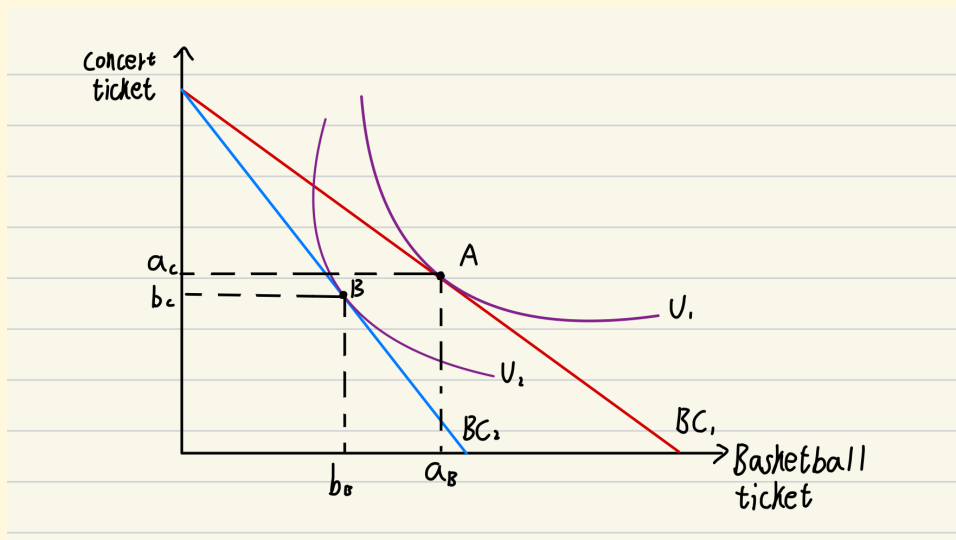


Figure 54: changes due to increase in basketball price

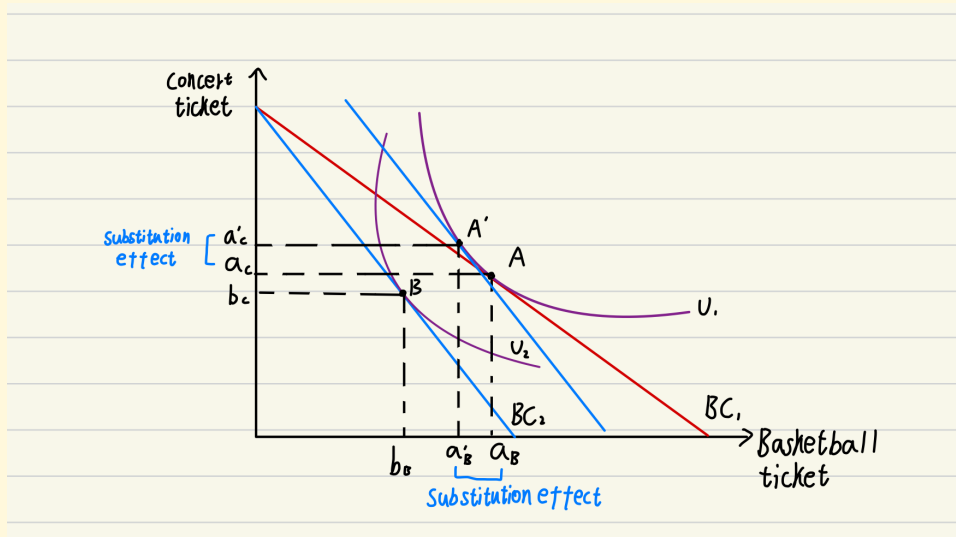


Figure 55: Isolating substitution effect

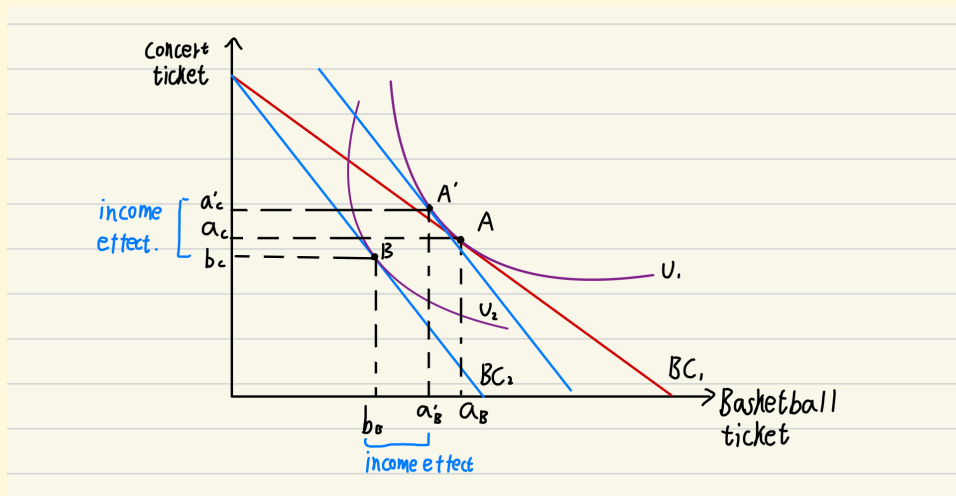


Figure 56: Isolating income effect

Remark: Substitution effects are always negative for price increases so long as preferences are rational and demonstrate taste for variety

4.3.3 Summary from the examples

There are three basic steps to analyzing substitution and income effects. We start with the consumer at a point of maximum utility (Point A) where his indifference curve is tangent to his budget constraint.

1. When prices change, draw the new budget constraint (a price change rotates the budget constraint, altering its slope). Then find the optimal quantity at the point (Point B) where this new budget constraint is tangent to a new indifference curve.
2. Draw a new line that is parallel to the new budget constraint from Step 1 and tangent to the original indifference curve at Point A' . The movement along the original indifference curve from Point A (the original, pre-price-change bundle) to this new tangency (point A') is the substitution effect. This movement shows how quantities change when relative prices change, even when the purchasing power of income is constant.
3. The income effect of the price change is seen in the movement from point A' on the initial indifference curve to point B on the indifference curve at the new price. Here, relative prices are held constant (the budget lines are parallel) but the purchasing power of income changes.

4.3.4 Sizes of income and substitution effect

Substitution effect

- When the indifference curve is highly curved, the MRS changes quickly along the curve. Thus, any given price change will not change consumption choices by much. So, there is less substitution effect.
- When the indifference curve is less curved, the MRS does not change quickly along the curve. Thus, any given price change affects consumption choices more strongly. So, there is more substitution effect.

Income effect

- If a consumer spends a significant portion of their budget on a good, then a price change for that good will have a substantial impact on the consumer's remaining income. So more income effect.
- On the other hand, if a consumer spends less significant portion of their budget on a good, then a price change of that good will not have a substantial impact on the consumer's remaining income. So less income effect.
- In the extreme case where a consumer does not purchase an item at all, a price change in that item will not affect the consumer's income or purchasing behavior. So there is no income effect.

4.3.5 Income and substitution effect on inferior goods

Suppose there is a drop in the price of ramen (inferior good).

The optimal bundle at the original prices is shown at point A . When the price of ramen noodles drops, Judy's budget constraint rotates outward from BC_1 to BC_2 . The total effect of this price change is represented by the increase in quantities consumed from the original utility maximizing bundle A to bundle B . Overall, Judy can reach a higher level of utility U_2 and her consumption of both ramen noodles and steak increases.

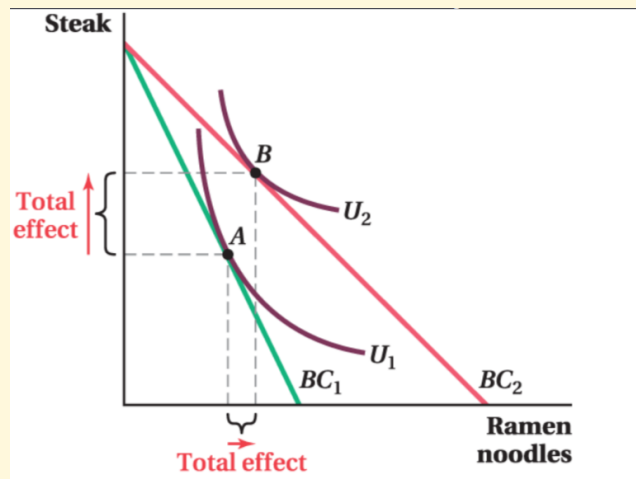


Figure 57: Example of inferior goods

Using the procedure, we can isolate the substitution and income effect:

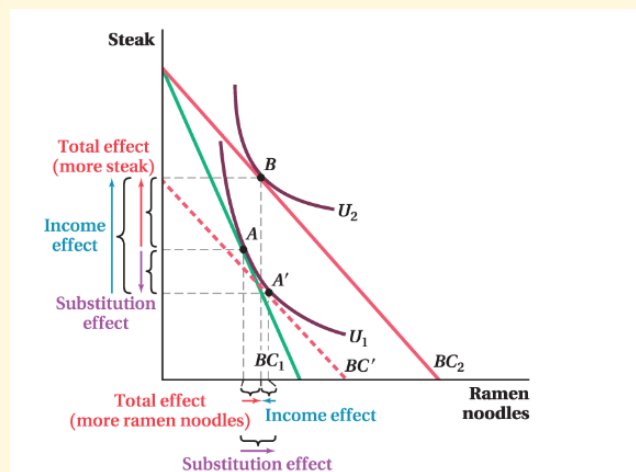


Figure 58: Isolate the substitution and income effect

With the decrease in price of inferior good, the inferior good becomes relatively cheaper compared to other goods. This leads consumers to substitute away from the relatively more expensive goods towards the now cheaper inferior good, even if their income hasn't changed.

The other good, which is now relatively more expensive, experiences a negative substitution effect because consumers are substituting away from it in favor of the now cheaper inferior good. They buy less of the other good as they substitute it with the inferior good.

Because the decrease in the price of the inferior good effectively increases the consumer's purchasing power – they can now buy the same quantity of the good for less money, or more of the good for the same money. For normal goods, this would lead to an increase in consumption as consumers feel wealthier. However, for inferior goods, the income effect works in the opposite direction. As the consumer's purchasing power increases, they actually buy less of the inferior good because they can now afford to purchase more of the preferable substitutes. This is the negative income effect for the inferior good.

For the other good in the consumer's basket, which we assume is a normal good, the increase in purchasing power from the price decrease of the inferior good means that the consumer can now afford to buy more of this normal good, leading to a stronger positive income effect.

4.3.6 A change in price of a substitute good

Consider a example of a pepsi and coke, with a increase in price of pepsi:

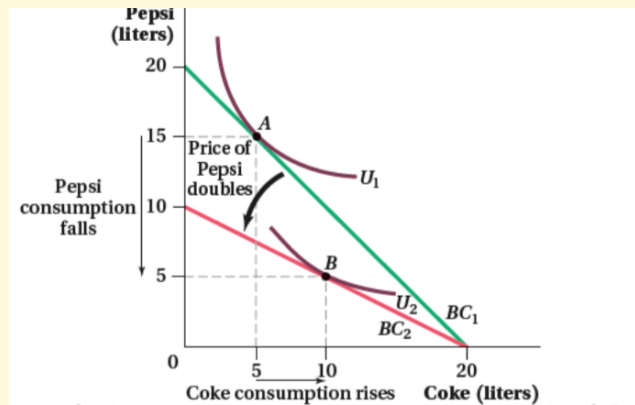


Figure 59: Pepsi and coke

Changes in the prices of a good's substitutes lead to shifts in the good's demand curve. When a substitute for a good becomes more expensive, this raises the quantity demanded of that good at any given price level. As a result, the demand curve for the good shifts out (the demand for that good increases). When a good's substitutes become cheaper, the quantity demanded at any given price falls, and the good's demand curve shifts in.

4.3.7 A change in price of a complement good

Consider an example with an increase in the price of a complement good, in this case the ice cream:

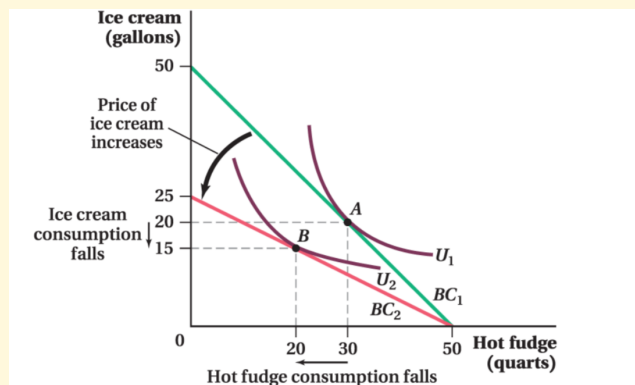


Figure 60: Ice cream and hot fudge

If the price of a complement of a good increases, the quantity demanded of that good at every price decreases, and its demand curve shifts in. If the price of a complement of a good falls, the quantity demanded of that good rises at all prices and the demand curve shifts out. Changes in the price of a complementary good shift the demand curve for the other good. Changes in a good's own price cause a move along the same demand curve.

4.3.8 Complement and substitute

Here is a summary of how a good can experience a shift in demand if price changes for complement or substitute.

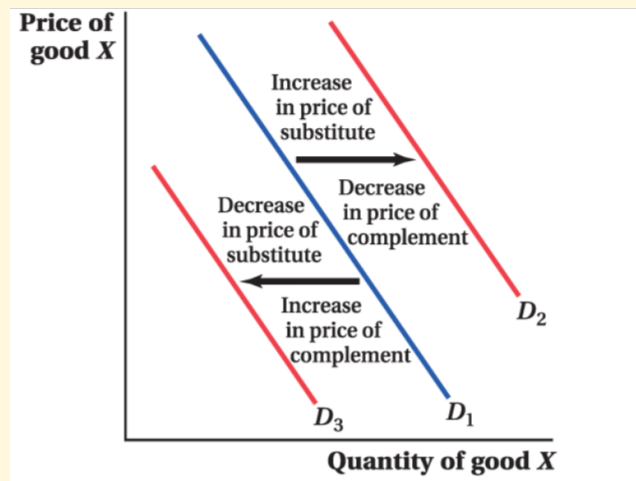


Figure 61: Complement and substitute and demand curves

4.3.9 Summaries about income and substitute effect

Substitution Effects	Income Effects
Involve comparisons of bundles that lie on the same indifference curve.	Involve comparisons of bundles that lie on two different indifference curves.
The direction of the effect on quantity consumed for a given change in the relative price of the good is unambiguous.	The direction of the effect on quantity consumed for a given change in the relative price of the good is ambiguous and depends on whether the good is normal or inferior.
If the good's relative price falls, the substitution effect causes the consumer to want more of it.	If the good is normal, then a fall in either its price or the price of the other good will cause the consumer to want more of it. (A drop in any price, even of another good, increases the effective income of the consumer.) If the good is inferior, then a price drop will cause the consumer to want less of it.
If the good's relative price rises, the substitution effect causes the consumer to want less of it.	If the good is normal, then a rise in either its price or the price of the other good will cause the consumer to want less of it. If the good is inferior, then a rise in either price will cause the consumer to want more of it.

4.4 The math for income and substitute effect

4.4.1 The demand curve from Cobb-Douglas utility

The Cobb-Douglas utility function for goods X and Y is:

$$U(X, Y) = X^\alpha Y^{1-\alpha}$$

Also, we already know the general solution for the optimal values of X and Y , which we derived:

$$X^* = \frac{\alpha I}{p_X}; \quad Y^* = \frac{(1-\alpha)I}{p_Y}$$

Let $D_X(p_X, p_Y, I) = X^*$. The slope of the demand curve D_X with respect to p_X tells us how X^* changes when p_X increases by a small amount, all else constant.

In other words, it's the partial derivative of X^* with respect to p_X :

$$\frac{\partial D_X(\cdot)}{\partial p_X} = \frac{\partial X^*}{\partial p_X} = -\frac{\alpha I}{p_X^2}$$

4.4.2 Decomposing effects in Cobb-Douglas utility function

Consider the Cobb-Douglas utility function:

$$U(X, Y) = X^\alpha Y^{1-\alpha}, \alpha \in (0, 1)$$

Consider a consumer with $\alpha = 0.25$ and initial budget constraint $200 = 5X + 10Y$.

Suppose the consumer above experiences an increase in p_X from 5 to 10.

Let's start by finding the total change in demand, which is the sum of the income and substitution effects:

$$D_X(p_X = 5, p_Y, l) = \frac{0.25(200)}{5} = 10; \quad D_X(p_X = 10, p_Y, l) = \frac{0.25(200)}{10} = 5$$

$$\Delta X^* = 5 - 10 = -5$$

To find the substitution effect graphically, we found the new (imaginary) budget constraint just tangent to the original indifference curve at the new prices.

Mathematically, this means finding a value of I , call it $\tilde{I}$, at which utility is the same at the new p_X as it was at the old p_X when the consumer is optimizing.

First, we need to find the original level of utility: Initially, $X^* = 10$ and $Y^* = \frac{0.75(200)}{10} = 15$ So we have:

$$U^1(X, Y) = 10^{0.25} 15^{0.75} = 13.55$$

This is the original level of utility, i.e. the utility on the original indifference curve before the change in p_X .

Next, let's find the value of $\tilde{I}$ such that:

$$U(X^*, Y^*) = 13.55$$

when income is $\tilde{I}$, $p_X = 10$ and $p_Y = 10$. That is:

$$U\left(\frac{0.25(\tilde{I})}{10}, \frac{0.75(\tilde{I})}{10}\right) = 13.55$$

$$\left(\frac{0.25(\tilde{I})}{10}\right)^\alpha \left(\frac{0.75(\tilde{I})}{10}\right)^{1-\alpha} = 13.55$$

$$\tilde{I} \left(\frac{0.25}{10}\right)^{0.25} \left(\frac{0.75}{10}\right)^{0.75} = 13.55$$

$$\tilde{I} = 237.8 > I$$

Now we decompose the substitution and income effects for X by computing the X^* at the optimum given $I = \tilde{I}$ and $p_X = p_Y = 10$:

1. This is called the compensated value of X .
2. It holds purchasing power constant, and isolates the substitution effect.

We have:

$$X^* = \frac{\alpha(\tilde{I})}{\text{new } p_X}$$

$$X^* = \frac{(0.25)237.8}{10} = 5.95$$

Therefore the substitution effect for X is:

$$\text{compensated } X - \text{original } X = 5.95 - 10 = -4.05$$

And the income effect for X is:

$$\text{final } X - \text{compensated } X = 5 - 5.95 = -0.95$$

The substitution and income effects sum to the total change in X : $-4.05 + -0.95 = -5$.

Now what is the total change in Y ? and What are the income and substitution effects for Y ?

We have the same $\tilde{I} = 237.8 > I$, and the compensated Y is $Y^* = \frac{(1-0.25)\cdot\tilde{I}}{\text{new } p_Y} = 17.835$

Therefore the substitution effect for Y is:

$$\text{compensated } Y - \text{original } Y = 2.83$$

And the income effect for Y is:

$$\text{final } Y - \text{compensated } Y = -2.83$$

The substitution and income effects sum to the total change in Y : $-2.83 + 2.83 = 0$.

4.4.3 More example on the decomposition

Example: Given $U(x, y) = x + 2 \ln(y)$, let $I = 100$ and $p_y = 5$. Suppose p_x falls from 5 to 2. Find the income and substitution effects, and the total change in demand, for x .

Solution: Using the fact we already have from the previous examples in the demand section:

$$D_x(p_x; I, p_y) = \begin{cases} 0 & p_x \geq \frac{I}{2} \\ \frac{I}{p_x} - 2, & p_x < \frac{I}{2} \end{cases}$$

$$D_y(p_y; I, p_x) = \frac{\min[I, 2p_x]}{p_y}$$

- The initial $y^* = \frac{\min[100, 2 \times 5]}{5} = 2$
- The initial $x^* = \frac{100}{5} - 2 = 18$
- After the price change, $x^* = \frac{100}{2} - 2 = 48$
- The total change in demand for x is

$$48 - 18 = +30$$

Here we will follow the step introduced previously:

1. Find initial utility: $U_1 = 18 + 2 \ln(2) = 19.38$.
2. Find $\tilde{I}$ that makes the consumer indifferent between $\{p_x = 5, p_y = 5, I = 100\}$ and $\{p_x = 2, p_y = 5, I = \tilde{I}\}$.
Since we're clearly still at an interior solution after the price of x falls, this is where

$$\frac{\tilde{I}}{2} - 2 + 2 \ln\left(\frac{4}{5}\right) = U_1, \rightarrow \tilde{I} = 43.65$$

3. Find x^* and y^* at the compensated bundle (call these x_c^* and y_c^*)

$$x_c^* = \frac{43.65}{2} - 2 = 19.8, \quad y_c^* = \frac{4}{5} = 0.8$$

Using the information above, the substitution effect for x is $19.8 - 18 = +1.8$ while the income effect is $48 - 19.8 = +28.2$

Example: Given $U(x, y) = x + 2 \ln(y)$, let $I = 100$ and $p_y = 5$. Suppose p_x falls from 5 to 2. Find the income and substitution effects, and the total change in demand, for y .

Solution: Using the fact we already have from the previous examples in the demand section:

$$D_x(p_x; I, p_y) = \begin{cases} 0 & p_x \geq \frac{I}{2} \\ \frac{I}{p_x} - 2, & p_x < \frac{I}{2} \end{cases}$$

$$D_y(p_y; I, p_x) = \frac{\min[I, 2p_x]}{p_y}$$

We have already done most of the work to solve this problem:

- The initial $y^* = \frac{\min[100, 2 \times 5]}{5} = 2$ and $x^* = \frac{100}{5} - 2 = 18$.
- After the price change, $y^* = \frac{4}{5} = 0.8$.
- The total change in demand (the cross-price total effect) for y is

$$0.8 - 2 = -1.2$$

So we see that y is a substitute in demand for x . Because we are working with quasi-linear rather than Cobb-Douglas preferences, cross price effects are not zero.

1. Find initial utility: $U_1 = 18 + 2 \ln(2) = 19.38$.
2. Find $\tilde{I}$ that makes the consumer indifferent between $\{p_x = 5, p_y = 5, I = 100\}$ and $\{p_x = 2, p_y = 5, I = \tilde{I}\}$, which we already saw was 43.65.
3. The x^* and y^* at the compensated bundle are again

$$x_c^* = \frac{43.65}{2} - 2 = 19.8, \quad y_c^* = \frac{4}{5} = 0.8$$

Using the information above, the substitution effect for y is $0.8 - 2 = -1.2$ while the income effect is $0.8 - 0.8 = 0$.

The reason of having zero income effect is because in this case, once the optimally condition of y is reached, all additional resources are devoted to x .

4.5 From individual to market demand

When studying consumer demand, we're often more interested in the combined demand of all consumers, rather than just one consumer's demand.

The market demand for a good is simply the horizontal sum of all the individual demand curves of consumers in the market.

4.5.1 Graphic example

Consider this graphic example:

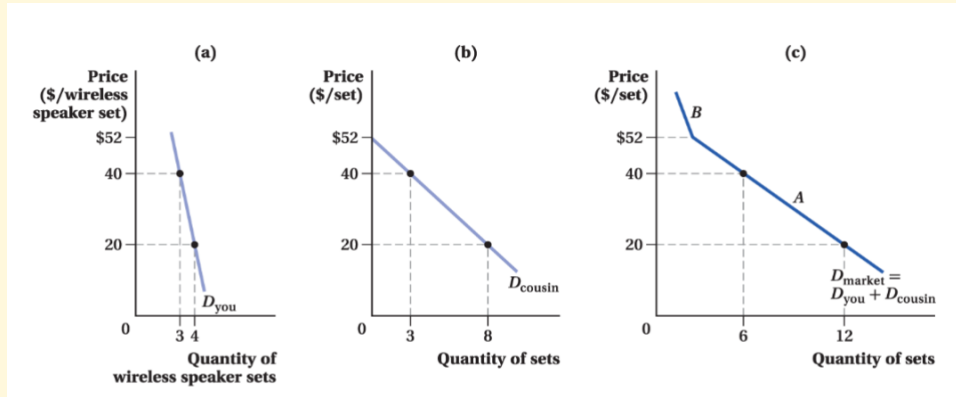


Figure 62: Example individual to market demand

1. Your individual demand curve D_{you} shows the number of sets of wireless speakers you would demand at each price. At a price of \$40 per set, you demand 3 sets; at \$20 per set, you demand 4 sets.
2. Your cousin's individual demand curve D_{cousin} shows that he is more sensitive to price changes in speakers than you are. At a price of \$40 per set, he similarly demands 3 sets, but when the price of a set of speakers is \$20, he demands 8 sets.
3. In a market consisting only of you and your cousin, the total market demand curve D_{market} is the sum of your demand curve D_{you} and your cousin's demand curve D_{cousin} . At a price of \$40 per set of speakers, you and your cousin each demand 3 sets, summing to 6 sets on D_{market} . When speaker sets cost \$20 each, you demand 4 sets and your cousin demands 8, summing to 12 sets of speakers on D_{market} . At prices above \$52, your cousin's quantity demanded is zero, so D_{market} overlaps with D_{you} and therefore D_{market} is kinked at a price of \$52. Market demand will always be flatter and will never be to the left of the individual demand curves.

4.5.2 Algebraic example

From the previous graphic example we have:

$$Q_1 = 5 - 0.05P$$

$$Q_2 = 13 - 0.25P$$

The first demand curve is "yours" and the second is "your cousin's".

Now we want to calculate choke prices for each curve:

$$P_1^{\text{choke}} = \frac{5}{0.05} = 100$$

$$P_2^{\text{choke}} = \frac{13}{0.25} = 52$$

We solve this by writing

$$Q_1 = \max[5 - 0.05P, 0]$$

$$Q_2 = \max[13 - 0.25P, 0]$$

Then the equation for the market demand curve is:

$$Q = \begin{cases} 0 & P \geq 100 \\ 5 - 0.05P & 100 > P \geq 52 \\ 18 - 0.30P & 52 > P \end{cases}$$

Where $Q_1 + Q_2 = 5 - 0.05P + 13 - 0.25P = 18 - 0.30P$

5 Producer behavior

5.1 Production basic

Definition(Production): The process by which a person, company, government, or non-profit agency uses inputs to create a good or service for which others are willing to pay.

Definition(Final goods): A good that is bought by a consumer.

Definition(Intermediate goods): A good that is used to produce another good.

Definition(Firms): Firms are organizations dedicated to production, typically owned and/or organized by producers. We assume production is carried out by economic firms. The eight assumptions listed are the those typically underlying the neoclassical firm:

Assumptions for productions and neoclassical firms

1. The firm produces only one good and had already chosen which good that will be.
2. Firms want to maximize the surplus they make from production, which always involves minimizing the costs required to produce any level of output.
3. The firm uses only two inputs to make its product: capital (which includes buildings and land) and labor.
4. In the short run, a firm can choose to employ as much or as little labor as it wants, but it cannot change how much capital it uses. In the long run, the firm can freely choose the amounts of both labor and the capital it employs.
5. The more inputs the firm uses, the more output it makes: inputs are productive.
6. A firm experiences diminishing marginal returns to labor and capital in production. That is: holding the amount of labour in production constant, increasing the amount of capital yields less and less additional output, and vice versa.
7. The firm can buy as many capital or labor inputs as it wants at fixed market prices. In other words, firms are price takers in the markets for inputs.
8. The firm is able to borrow however much it needs to purchase inputs, with the debts paid back out of the revenue from output. Therefore, the firm does not have a budget constraint. So, it means firm has access to perfect credit markets. In such markets, the firm can finance the purchase of any level of inputs it requires without being restricted by its current financial resources.

Definition(Production function): A mathematical relationship that describes how much output can be made from different combinations of inputs.

So, a production function is just a formula that tells you how much output (which we label Q for quantity) you get as a function of the two inputs, capital (K) and labor (L) :

$$Q = f(K, L)$$

In the context of production functions, concavity is a property that reflects diminishing marginal returns to a factor of production. A production function $Q(K, L)$ is described as concave with respect to both inputs, capital (K) and labor (L), it means that as you keep adding more of either capital or labor while holding the other factor constant, the additional output produced from each additional unit of the input will eventually decrease.

Example: A very common production function, that satisfies concavity on both inputs, is the Cobb-Douglas function:

$$Q = K^\beta L^{1-\beta}, \quad \beta \in (0, 1)$$

5.2 Production in short run

Definition(Marginal product): The additional output that a firm can produce by using an additional unit of an input (holding use of the other input constant).

Definition(Marginal product of labour): Consider short run production function $Q = f(\bar{K}, L)$ with $\bar{K}$ fixed. The marginal product of labor (MP_L) is the change in quantity (ΔQ) resulting from a 1-unit change in labor inputs (ΔL) :

$$MP_L = \frac{\Delta Q}{\Delta L} = \frac{\partial Q}{\partial L} = \frac{f(\bar{K}, L + \Delta L) - f(\bar{K}, L)}{\Delta L}$$

The marginal product of labor at a given level of labor input is the slope of the production function at that point.

Definition(The diminishing marginal product of labor): It is the reduction in the incremental output obtained from adding more and more labor, it is embodied in our assumption.

The figure below shows marginal product of labor as slope of the production function and its diminishing returns shown as decreasing slope.

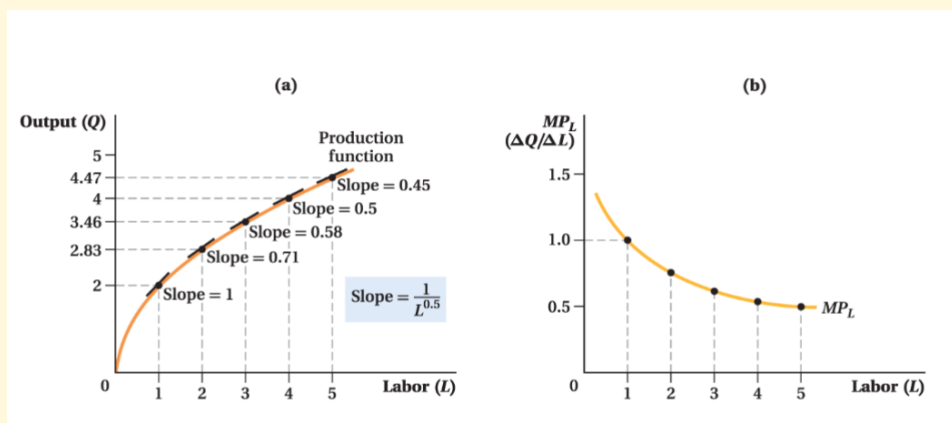


Figure 63: MP of labor as slope and diminishing return

We have two additional notes on diminishing returns:

- K also has diminishing marginal returns, with L fixed. Since K only varies in the "long run", diminishing returns to K are not as notable as diminishing returns to L . However, it's not possible to increase K forever, with L constant, and not experience a diminishing rate of return.
- Diminishing marginal returns to L occur only eventually. It's possible, for low enough L , marginal returns are increasing due, say, to specialization. But eventually, labour runs out of capital to work with.

Definition(Average product): The total quantity of output divided by the number of units of input used to produce it.

Definition(Average product of labor): The average product of labor (AP_L), for example, is the quantity produced, Q , divided by the amount of labor L used to produce it:

$$AP_L = Q/L$$

The average product of labour is of course distinct from the marginal product of labor, but they are connected:

- Whenever the MP_L lies below the AP_L , the AP_L is falling.
- Whenever the MP_L lies above the AP_L , the AP_L is rising.

5.3 Production in long run

In the short run, the diminishing marginal product limits a firm's ability to produce additional output by using more and more labor.

But in the long run, the firm can change its inputs of both capital and labor.

There are no diminishing marginal to returns in long run production function because you are not fixing anything, on the other hand, you have return to scale.

Definition(Return to scale): It describes a change in the amount of output in response to a proportional increase in all the inputs.

Consider the firm with production function

$$F(K, L) = K^\beta L^{1-\beta}$$

Suppose we double the inputs of both K and L . We get **constant returns to scale**:

$$F(2K, 2L) = (2K)^\beta (2L)^{1-\beta} = 2K^\beta L^{1-\beta} = 2F(K, L)$$

Let

$$F(K, L) = K^{0.6} L^{0.6}$$

Then:

$$F(2K, 2L) = (2K)^{0.6} (2L)^{0.6} = 2^{1.2} K^{0.6} L^{0.6} > 2F(K, L)$$

A production function has **decreasing returns to scale** if $F(2K, 2L) < 2F(K, L)$.

5.4 Firm optimization: Cost minimization

Definition(Cost minimization): A firm's goal of producing a specific quantity of output at minimum cost.

We call the challenge of producing a specific amount of a particular good as inexpensively as possible the firm's cost-minimization problem. And the firm's production decision is another constrained optimization problem.

The firm wants to minimize the total costs of its production. However, it must hold to a constraint in doing so: It must produce a particular quantity of output. That is, it can't minimize its costs just by refusing to produce as much as it would like (or refuse to produce anything, for that matter).

5.4.1 Isoquants

Definition(Isoquants): A curve representing all the combinations of inputs that allow a firm to make a particular quantity of output.

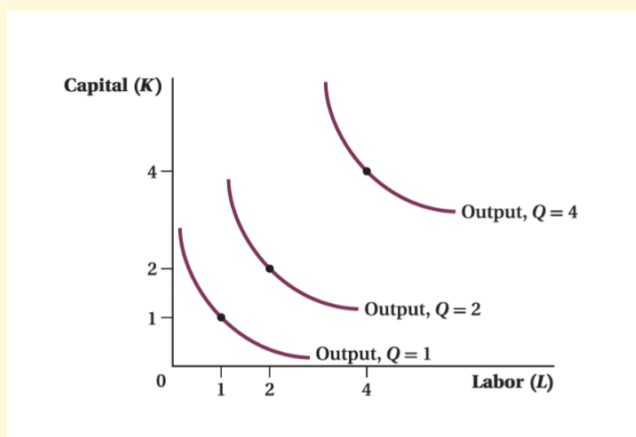


Figure 64: Example isoquants

Just as with indifference curves:

1. Isoquants further from the origin correspond to higher output levels (because more capital and labor lead to higher output).
2. Isoquants cannot cross (because if they did, the same quantities of inputs would yield two different quantities of output).
3. Isoquants are convex to the origin (because using a mix of inputs generally lets a firm produce a greater quantity than it could by using an extreme amount of one input and a tiny amount of the other).

Definition(Marginal rate of technical substitution $MRTS_{XY}$): The rate at which the firm can trade input X for input Y , holding output constant.

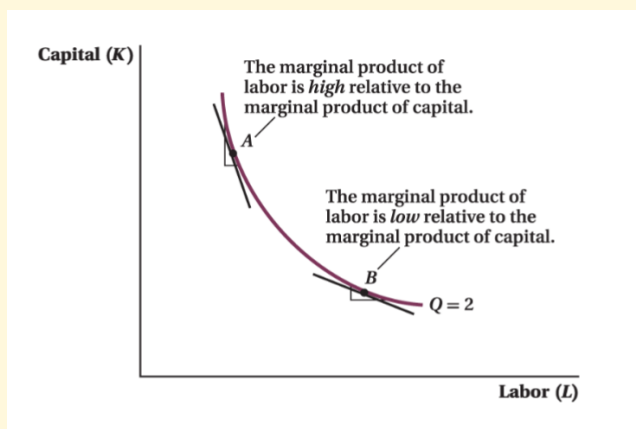


Figure 65: MRTS

The negative slope of the isoquant is the marginal rate of technical substitution of labor L for capital K .

At point A , the marginal product of labor is high relative to the marginal product of capital, and a relatively small decrease in labor would require a large quantity of capital to hold output constant.

At point B , the marginal product of labor is low relative to the marginal product of capital, and a relatively small decrease in capital would require a large quantity of labor to hold output constant.

The curve demonstrates a diminishing rate of technical substitution as the indifference curve.

Like we did for indifference curves, we can derive the slope of the isoquant in terms of the marginal products MP_L And MP_K .

Assume that production function $F(K, L)$ is smooth and differentiable, and let an isoquant have equation $F(K, L) = \bar{Q}$ where $\bar{Q}$ is some amount of output.

Then we take the total derivative of the isoquant identity with respect to K and L

$$\frac{\partial F}{\partial K} dK + \frac{\partial F}{\partial L} dL = \frac{\partial \bar{Q}}{\partial K} dK + \frac{\partial \bar{Q}}{\partial L} dL = 0$$

Rearranging we get:

$$\frac{dK}{dL} = -\frac{\frac{\partial F}{\partial L}}{\frac{\partial F}{\partial K}} = -\frac{MP_L}{MP_K}$$

1. Relatively straight isoquants indicate that $MRTS_{LK}$ does not vary much along the curve. Therefore, labor and capital are close substitutes for each other.
2. Relatively curved isoquants indicate that $MRTS_{LK}$ varies greatly along the curve. Therefore, labor and capital are not close substitutes for each other.

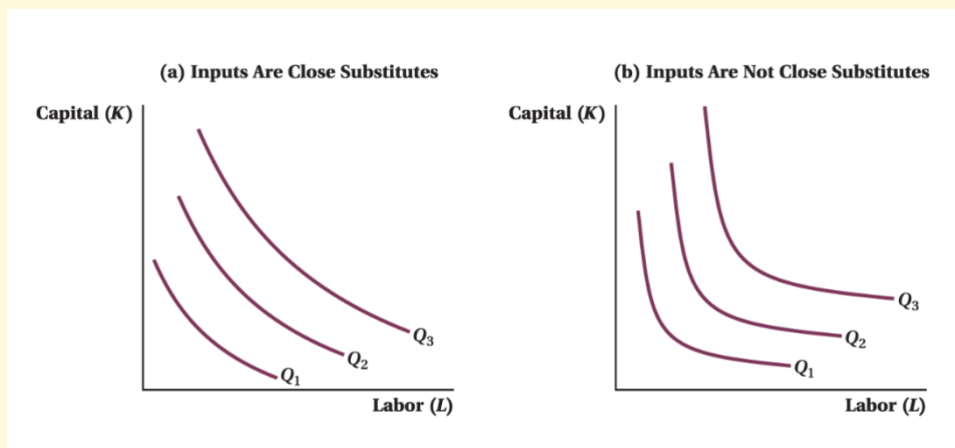


Figure 66: Substitutes and complements in production

We will illustrate perfect substitutes and complements using example:

- Robots K and labor L are perfect substitutes. The isoquants are straight lines, and the $MRTS_{LK}$ does not change along the isoquant. In this case, 2 humans can substitute for 1 robot. And the equation for the curve is a linear combination of these two variables.
- Cabs K and drivers L are perfect complements. The isoquants are L -shaped, and the optimal quantity (K, L) for each output Q is the corner of the iso quant. In this case, 1 cab with 1 driver produces $Q = 1$, while 2 cabs with 2 drivers produce $Q = 2$. And the equation for this curve is $Q = \min(L, K)$.

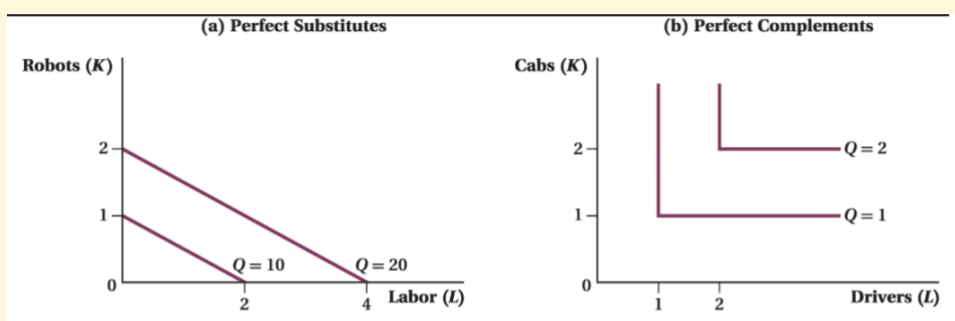


Figure 67: Perfect substitutes and perfect complements

5.4.2 Isocost line

Definition(Isocost line): A curve that shows all the input combinations that yield the same cost. Mathematically, the isocost line corresponding to a total expenditure level of C is given by

$$C = RK + WL \Leftrightarrow K = \frac{C}{R} - \frac{W}{R}L$$

- R is the price (the rental rate) per unit of capital, which is the economic user cost that includes initial purchase price and further depreciations and related opportunity costs.
- W is the price (the wage) per unit of labor.
- K and L are the number of units of capital and labor that the firm hires.

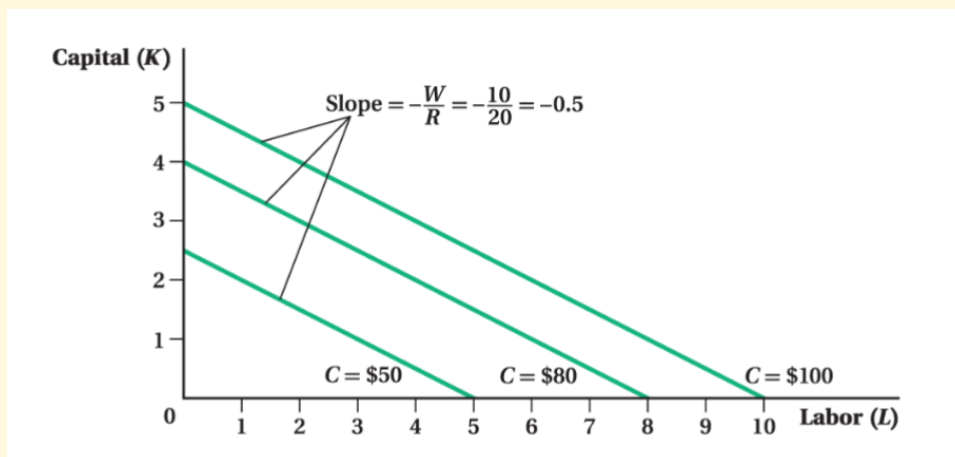


Figure 68: isocost lines

- Isocost lines for higher total expenditure levels are further from the origin. This reflects the fact that, as a firm uses more inputs, total expenditure on those inputs increases.
- Here, W/R is the rate at which I can change labour into capital as I move up the isocost line, without changing total expenditure. And the curves all have the same slope.
- All lines are parallel because firms can infinitely buy as much labor and capital as it wants, and the relative trade offs between inputs remain the same.

Just like budget lines for consumers, when relative prices change, the isocost line rotates. So, it all depends on the change in slope of the curves.

Example:

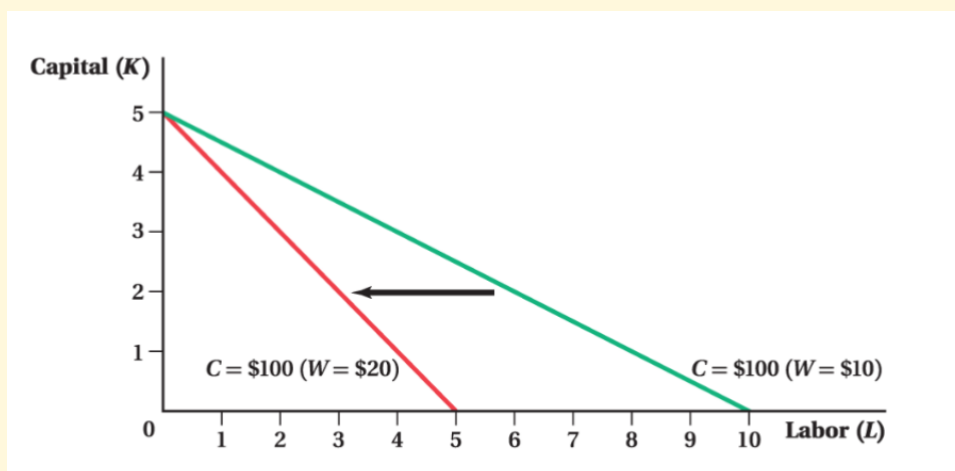


Figure 69: Rise wages and constant capital

When the price of labor increases from $W = \$10$ to $W = \$20$ and the price of capital stays constant at $R = \$20$, the slope of the isocost changes from $-\frac{1}{2}$ to $-\frac{2}{2}$, or -1 . The isocost line therefore becomes steeper, and the quantity of inputs the firm can buy for \$100 decreases.

Example:

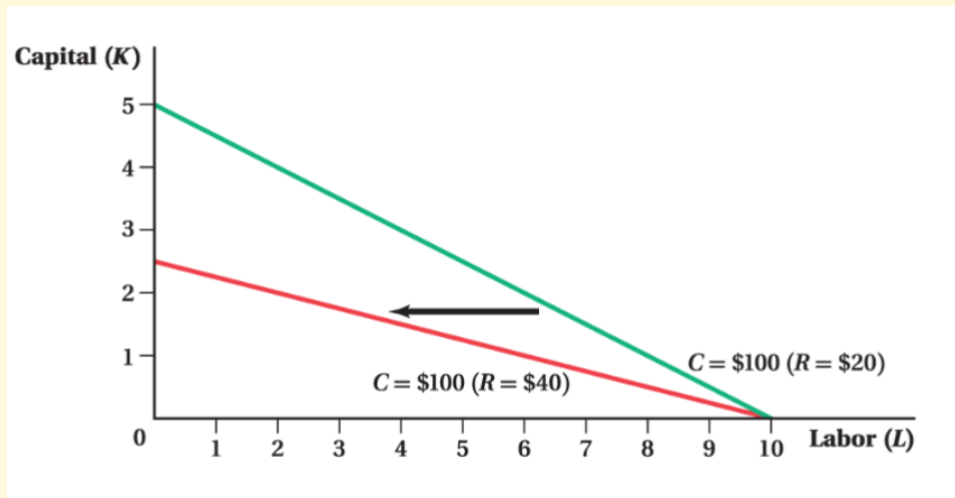


Figure 70: Rise capital and constant labor

When the price of capital increases from $R = \$20$ to $R = \$40$ and the price of labor stays constant at $W = \$10$, the slope of the isocost changes from $-\frac{1}{2}$ to $-\frac{1}{4}$. The quantity of inputs the firm can buy for \$100 decreases, and the isocost line becomes flatter.

5.4.3 Finding cost minimization graphically

Isoquants and isocost curves are the two components of the firm's cost minimization problem

- Isoquants measure production opportunities.
- Isocost curves measure the costs of production.

To solve its cost minimization problem, a firm follows the following general strategy:

Get to the lowest attainable isocost curve for a given isoquant (i.e. for a given level of production)

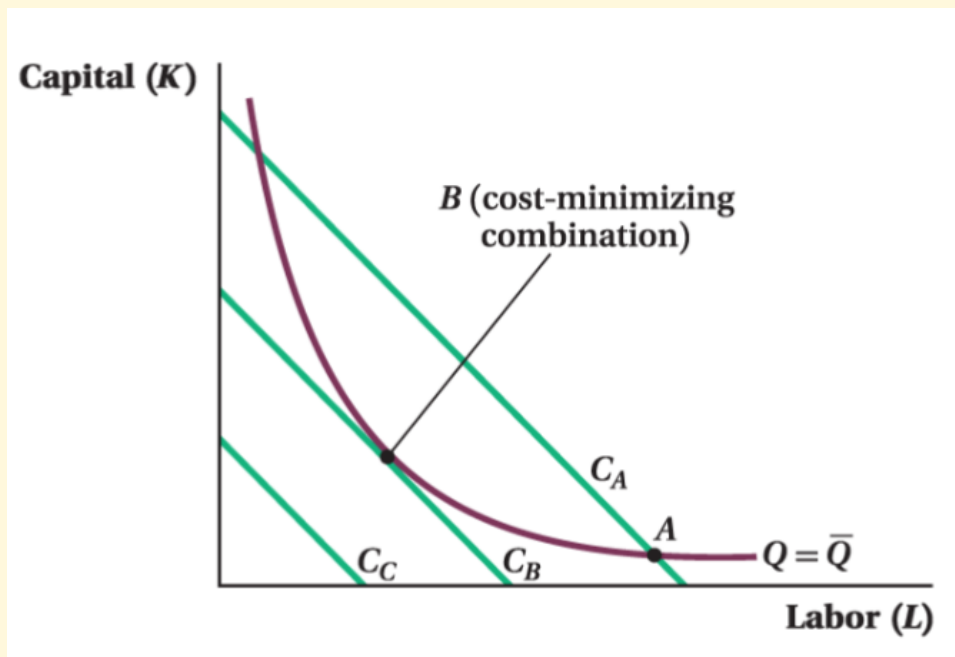


Figure 71: Graphic solution to cost minimize

The firm wants to minimize the cost to produce the quantity $Q = \bar{Q}$. Because A is on the isoquant, the firm can choose to use input combination A to produce $\bar{Q}$. However, A is not cost-minimizing because the firm can produce $\bar{Q}$ at a lower cost at any point below and to the left of the isocost C_A . Point B , located at the tangency between isocost C_B and the isoquant, is the firm's cost-minimizing capital and labor combination. Input combinations on C_C cost less than those on C_B , but are too small to allow the firm to produce $\bar{Q}$.

5.4.4 Computing cost minimization

The optimality condition for an interior solution for the firm is characterized by a tangency condition.

Here, the tangency is between the isoquant corresponding to the desired output Q and the lowest attainable isocost line:

$$\frac{MP_L}{MP_K} = \frac{W}{R}$$

Or:

$$\frac{MP_L}{W} = \frac{MP_K}{R}$$

We can use Lagrangian analysis to get the optimality condition seen above. In this case, we are solving a minimization problem, not a maximization problem as we did for the consumer

Define:

$$\mathcal{L}(K, L, \lambda) = RK + WL + \lambda[\bar{Q} - F(K, L)]$$

The FOCs are:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial K} &= R - \lambda MP_K = 0 \\ \frac{\partial \mathcal{L}}{\partial L} &= W - \lambda MP_L = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= \bar{Q} - F(K, L) = 0 \end{aligned}$$

Consider this example:

Example: Let $F(K, L) = 50K^{0.25}L^{0.75}$. What is the firm's cost-minimizing combination of inputs if it must produce $Q = 10$ units given a wage W of \$20 per unit of L and a rental rate R of \$15 per unit of K .

Solution: I will solve this by the "short-cut method" of simply writing down the optimality condition and subbing it into the production constraint. We know that, at the minimum:

$$\frac{W}{R} = \frac{20}{15} = \frac{4}{3} = \frac{MP_L}{MP_K} = \frac{37.5K^{0.25}L^{-0.25}}{12.5K^{-0.75}L^{0.75}} = \frac{3K}{L}$$

Rearranging for L :

$$L = \frac{9}{4}K$$

The firm has to produce 10 units of output. So we need

$$F(K, L) = 10 \rightarrow 50(K)^{0.25} \left(\frac{9K}{4}\right)^{0.75} = 10 \rightarrow 50K \left(\frac{9}{4}\right)^{0.75} = 10 \rightarrow K^* \approx 0.109$$

And then, by optimality, $L^* \approx 0.245$.

5.4.5 Optimal under price change

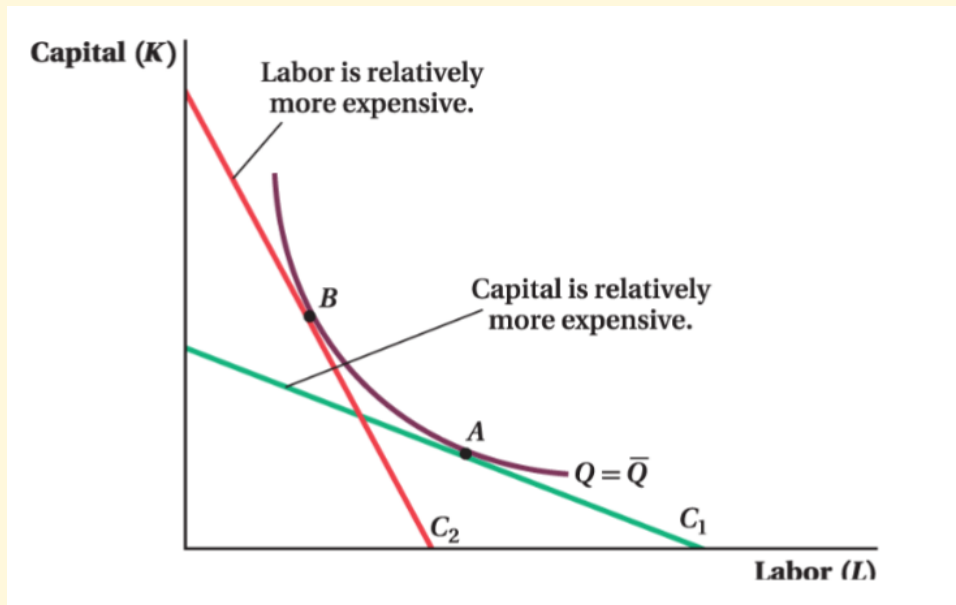


Figure 72: Labor becomes relative expensive

The initial input price ratio gives the slope of isocost line C_1 . The firm wants to produce the quantity $\bar{Q}$, so initially the cost-minimizing combination of inputs occurs at point A .

Now suppose that labor becomes relatively more expensive (or equivalently, capital becomes relatively less expensive). This change causes the isocost lines to become steeper. With the steeper isocost lines, the point of tangency shifts to point B .

Therefore, the increase in the relative cost of labor causes the firm to shift to an input mix that has more capital and less labor than before.

Example: Let $F(K, L) = 50K^{0.25}L^{0.75}$, it must produce $Q = 10$ units given a wage W of \$20 per unit of L and a rental rate R of \$15 per unit of K . Now suppose R falls from \$15 to \$5 per unit of K . What is the new cost-minimizing bundle of inputs given the firm still wants to produce $Q = 10$ as before?

Solution: K has become cheaper relative to L so we know before solving that K^* should rise and L^* should fall along the isoquant for $Q = 10$. We simply re-solve at the new prices:

$$\frac{W}{R} = \frac{20}{5} = 4 = \frac{MP_L}{MP_K} = \frac{37.5K^{0.25}L^{-0.25}}{12.5K^{-0.75}L^{0.75}} = \frac{3K}{L}$$

Rearranging for L :

$$L = \frac{3}{4}K$$

The firm still has to produce 10 units of output. So we need

$$F(K, L) = 10 \rightarrow 50(K)^{0.25} \left(\frac{3K}{4}\right)^{0.75} = 10 \rightarrow 50K \left(\frac{3}{4}\right)^{0.75} = 10 \rightarrow K^* \approx 0.248$$

And then, by optimality, $L^* \approx 0.186$.

5.5 The expansion path and the total cost curve

Definition(Expansion path): A curve that illustrates how the optimal mix of inputs varies with total output.

Definition(Total cost curve): A curve that shows a firm's cost of producing particular quantities.

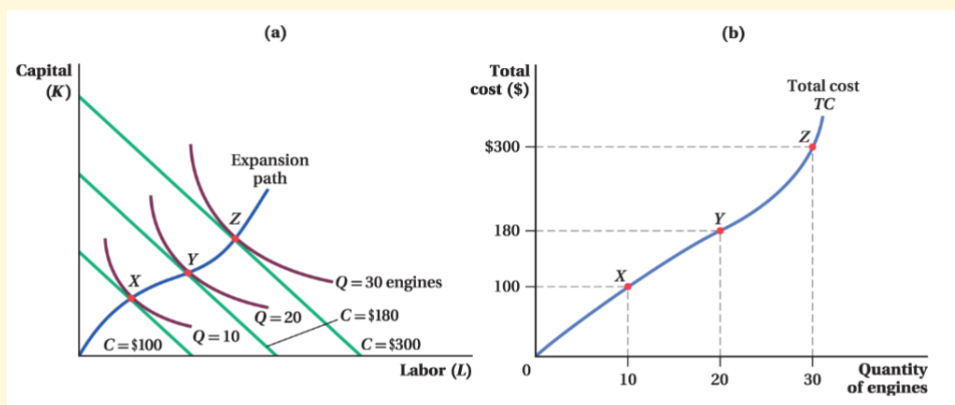


Figure 73: Expansion path and total cost curve

6 Firm costs

6.1 Opportunity cost

Definition(Accounting cost): The direct costs of operating a business, including costs for raw materials, wages paid to workers, rent paid for office or retail space, and the like.

Definition(Opportunity cost): The value of what a producer gives up by using an input.

Definition(Economic cost): The sum of a producer's accounting and opportunity costs.

Definition(Economic profit): A firm's total revenue minus its economic cost.

Definition(Accounting profit): A firm's total revenue minus its accounting cost.

6.2 Sunk cost

Definition(Sunk cost): A cost that, once paid, cannot be recovered.

Definition(Special capital): Capital that cannot be used outside of its original application. And it is a important determinant of sunk cost, and making some sunk cost avoidable.

Definition(Operating revenue): The money a firm earns from selling its output.

Definition(Operating cost): The cost a firm incurs in producing its output.

The business should stay open when operating revenue is bigger than operating cost. And in the consideration, we exclude sunk cost.

Definition(Sunk cost fallacy): The mistake of letting sunk costs affect forward-looking decisions. And in rational decision, we exclude the consideration of sunk cost.

6.3 Types of cost and total cost

Definition(Fixed cost FC): An input cost that does not vary with the amount of output, even if it is zero.

Definition(Variable cost VC): The cost of inputs that change as the firm changes its quantity of output.

Definition(Total cost TC): The sum of a firm's fixed and variable costs:

$$TC = FC + VC$$

To increase output, a firm must increase its variable costs by hiring more variable inputs More correctly, therefore:

$$TC(Q) = FC + VC(Q)$$

We assume that firms operate in only two time-frames: the short run and the long run. In short run, we assume capital is fixed with variable labor. In long run, we assume both capital and labor are variable.

Various institutional factors determine how quickly inputs become variable for a firm, for example:

1. **Active capital rental and resale markets:** If a firm can rent the capital it has on hand to other firms or re-sell it, then the "long run" for capital can be quite short. If active capital markets don't exist for the firm's specific capital inputs, then capital costs are necessarily sunk costs until the capital is used up in production, which may be a long time.
2. **Labour contracts:** In some labour markets, workers expect to receive contracts that require that pay them specified amount regardless of how much time they spend producing output. These type of contracts can arise when the firm requires workers to remain with the firm for a long time to become productive, serving as a commitment device, or they can arise in "closed shop" union firms. Labour contracts can make labour into a fixed cost over a fairly long time horizon

Definition(Cost curve): The mathematical relationship between a firm's production costs and its output.

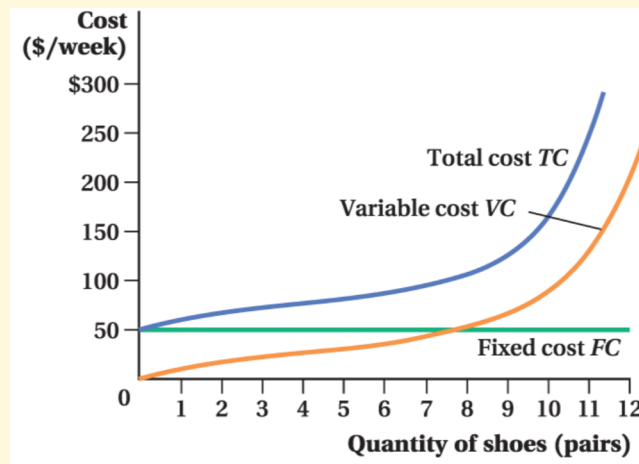


Figure 74: Sample cost curve

6.4 Average cost

Definition(Average fixed cost AFC): A firm's fixed cost per unit of output.

$$AFC = \frac{FC}{Q}$$

Definition(Average variable cost AVC): A firm's per-unit variable cost of production.

$$AVC = \frac{VC(Q)}{Q}$$

Definition(Average total cost ATC): A firm's total cost per unit of output.

$$ATC = \frac{TC(Q)}{Q}$$

And we have the relation:

$$ATC = \frac{TC(Q)}{Q} = \frac{FC}{Q} + \frac{VC(Q)}{Q} = AFC + AVC$$

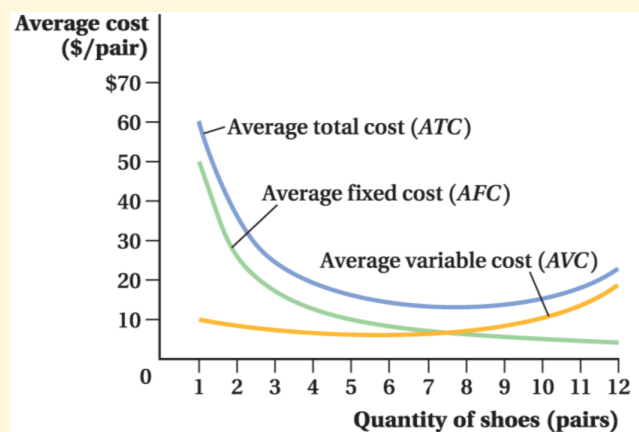


Figure 75: Average cost curves

6.5 Marginal cost

Definition(Marginal cost MC): A firm's cost of producing one more unit of output.

$$MC = \frac{\Delta TC(Q)}{\Delta Q} \equiv \frac{\partial TC(Q)}{\partial Q}$$

Since fixed cost does not change with Q , we can also write

$$MC = \frac{\Delta VC(Q)}{\Delta Q} \equiv \frac{\partial VC(Q)}{\partial Q}$$

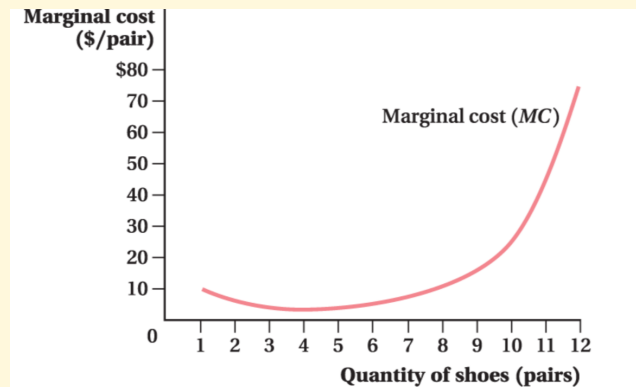


Figure 76: Graphic of marginal cost

From the graph, we see that output levels, it rises steeply. Why? Marginal cost may initially fall at low quantities because of complications in producing the first few units that can be fixed fairly quickly, or because having more scale allows workers to specialize in those tasks that they are best at performing.

As output continues to increase, however, these marginal cost reductions stop and marginal cost begins to increase with the quantity produced.

There are many reasons why it becomes more and more expensive to make another unit as output rises: Capacity constraints may occur, inputs may become more expensive as the firm uses more of them, it may be more difficult for the company to coordinate its operations, and so on.

The AC and the MC are both derived from total cost, so they are directly related.

1. If the MC of output is less than the AC at a particular Q , producing an additional unit of Q will reduce the AC .
2. By the same token, if the MC of output is greater than the AC at a particular Q , producing an additional unit of Q will increase the AC .
3. This argument holds regardless of whether AC refers to ATC or AVC because marginal costs are always increments to variable costs.

Consequently, the MC curve intersects the AVC and ATC curves at their minimums:

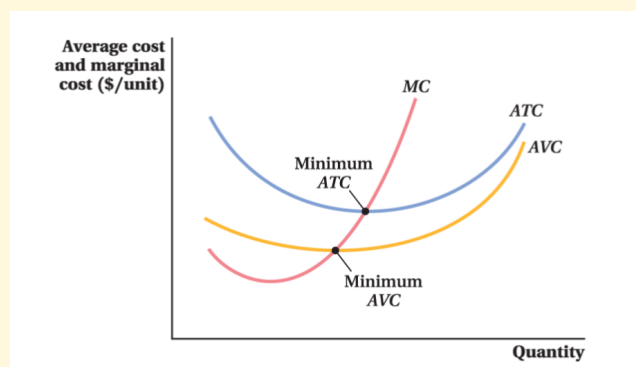


Figure 77: minimum AVC and ATC

We can also see the relationship between the MC and AVC curves mathematically Recall that:

$$AVC = \frac{VC(Q)}{Q}$$

Taking the derivative of this expression with respect to Q and setting it to zero, we get:

$$\frac{VC'(Q)}{Q} - \frac{VC(Q)}{Q^2} = 0 \rightarrow VC'(Q) = \frac{VC(Q)}{Q} \rightarrow MC(Q) = AVC(Q)$$

This result tells us that, when the AVC curve is at its minimum (at the critical value for Q), we must have $AVC = MC$.

Now we do the same exercise for the ATC curve:

$$ATC = \frac{TC(Q)}{Q} = \frac{FC}{Q} + \frac{VC(Q)}{Q}$$

Taking the derivative of this expression with respect to Q and setting it to zero, we get:

$$\begin{aligned} -\frac{FC}{Q^2} + \frac{VC'(Q)}{Q} - \frac{VC(Q)}{Q^2} &= 0 \\ \rightarrow VC'(Q) &= \frac{VC(Q)}{Q} + \frac{FC}{Q} \\ \rightarrow MC(Q) &= AVC(Q) + AFC(Q) = ATC(Q) \end{aligned}$$

This again tells us that, when the ATC curve is at its minimum (at the critical value for Q), we must have $ATC = MC$.

Example: Suppose a firm's total cost curve is

$$TC(Q) = 15Q^2 + 8Q + 45$$

Start by finding the firm's marginal cost, average total cost, average fixed cost, and average variable cost. Clearly, $FC = 45$ since 45 does not vary with Q , while the remaining terms are VC . So:

$$\begin{aligned} MC(Q) &= \frac{dTC(Q)}{dQ} = 30Q + 8 \\ ATC(Q) &= \frac{TC(Q)}{Q} = 15Q + 8 + \frac{45}{Q} \\ AFC(Q) &= \frac{FC}{Q} = \frac{45}{Q} \\ AVC(Q) &= \frac{VC(Q)}{Q} = 15Q + 8 \end{aligned}$$

Next, find the value of Q for which ATC is minimized and show that $ATC(Q) = MC(Q)$ at this value

Again we have:

$$ATC(Q) = \frac{TC(Q)}{Q} = 15Q + 8 + \frac{45}{Q}$$

Taking the derivative and setting it equal to zero:

$$\frac{dATC(Q)}{dQ} = 15 - \frac{45}{Q^2} = 0 \rightarrow Q^2 = 3 \rightarrow Q = \sqrt{3}$$

When $Q = \sqrt{3}$, we have $ATC(Q) = 15(\sqrt{3}) + 8 + \frac{45}{\sqrt{3}} \approx 60$

And we have $MC(Q) = 30(\sqrt{3}) + 8 \approx 60$

So, indeed, $MC(Q) = ATC(Q)$ at its minimum.

6.6 Cost curves in the short run and long run

Definition(Short run cost curve): The mathematical representation of a firm's total cost of producing different quantities of output at a fixed level of capital.

Consider the following short run and long run expansion path:

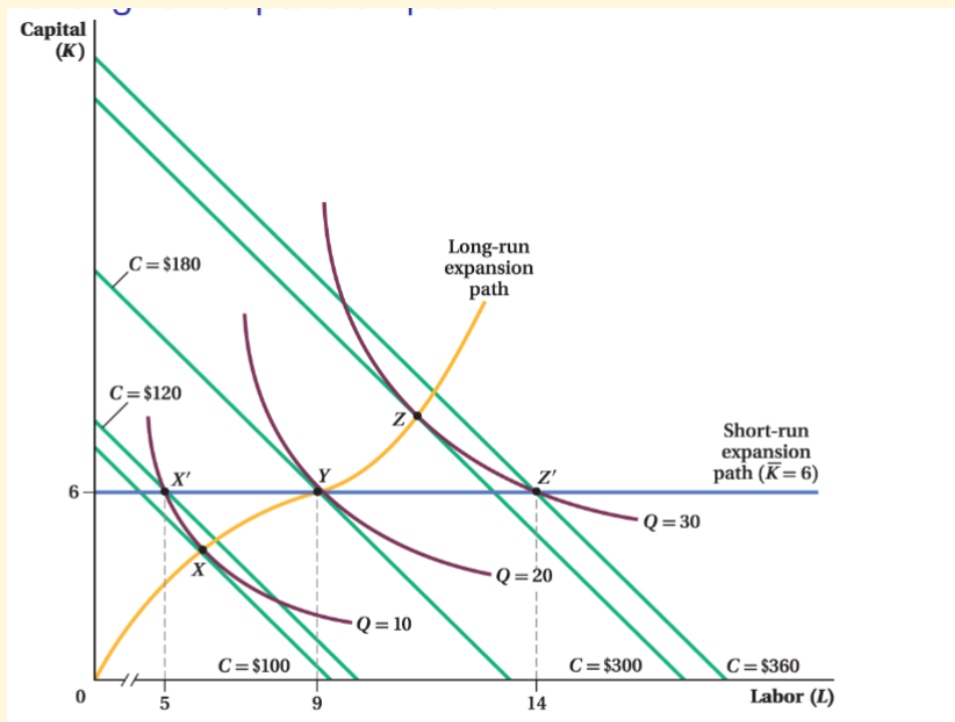


Figure 78: Short run and long run expansion path

The short run expansion path is a straight line at $K = \bar{K}$ (here $K = 6$).

The long run expansion path is upward sloping as L and K both vary.

Increasing only L to reach a higher level of Q could only be optimal in the short run.

In the long run, if our firm wanted to produce more than $Q = 20$ engines, it would take advantage of its greater flexibility in the long run to increase both L and K in the most economically efficient (cost minimizing) way.

Both curves intersect because when $Q = 20$, $\bar{K} = 6$ is the cost-minimizing capital input and $L = 9$ is the cost-minimizing labor input.

The short run cost curve lies above the long run cost curve except at $Q = 20$.

And we have the total cost curves:

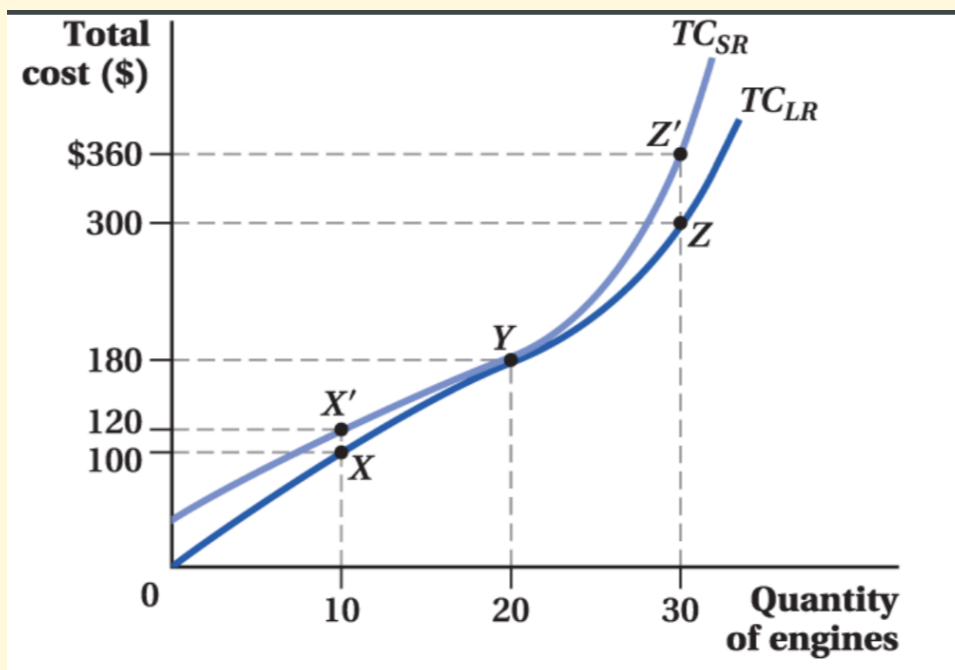


Figure 79: Short run and long run total cost curves

Some points on total cost short run curve is higher than long run because we are producing at a fixed capital,

and we can only achieve the same level of production by increasing labor instead of increasing both, so it is more expensive.

To get these average cost curves, divide total costs of Q in the SR and LR by Q :

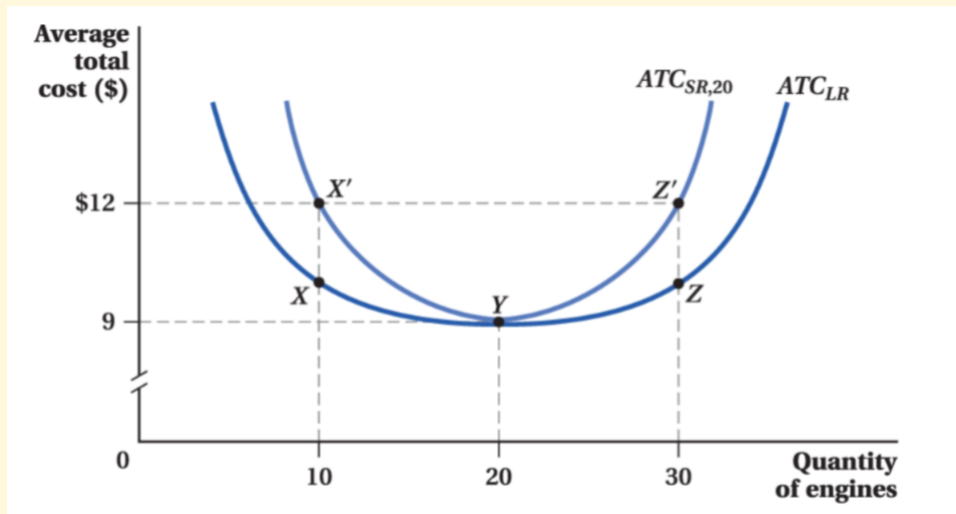


Figure 80: Average total cost curves in the long and short runs

The point where each short run curve touches the long run curve is where the long run and short run expansion paths at $\bar{K}$ intersect.

The Average total cost curve in short run is U-shaped because initially at low number of outputs, the fixed costs and variable costs averaged on each item is quit high, and as the output increases, the costs are averaged out on more items. So, the curve started at a high point, an decreases. Eventually, due to diminishing returns to scale, it is becoming more expensive becuase we need more resources to increase output, so, the curve goes up again after the cost minimizing point.

For the long run average cost curve, the production experience economies of scale because capital becomes more productive and allows the firm to take advantage of the relative cheapness of capital compared to labor at higher levels of scale, even though we have decreasing returns to scale. But eventually, as the production scale up, the diminishing scale to returns will act and causes diseconomies of scale.

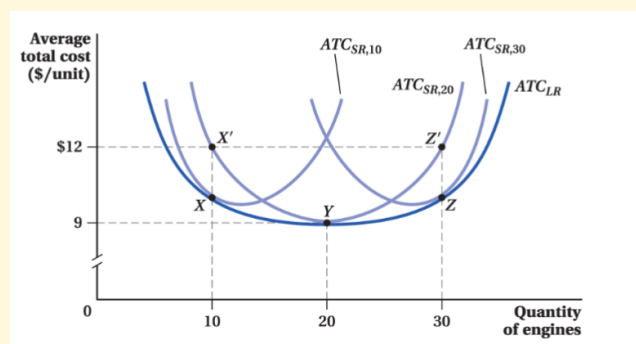


Figure 81: Average total cost curves in the long and short runs

$ATC_{SR,10}$ and $ATC_{SR,30}$ do not touch ATC_{LR} at their lowest points. That's because even the output levels that minimize average total costs in the short run when capital is fixed (the low points on $ATC_{SR,10}$ and $ATC_{SR,30}$) can be produced more cheaply if capital inputs were flexible.

The short run and long run marginal cost curve is shown as:

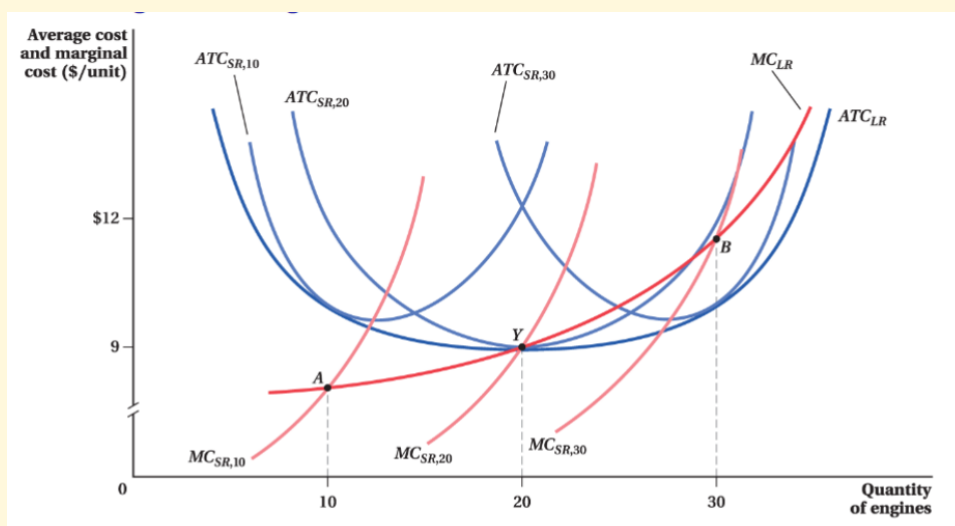


Figure 82: SR and LR marginal cost curves

The long run MC curve is flatter than any of the short run MC curves because in the short run, at least one factor of production (like capital) is fixed. Firms can only adjust variable inputs (like labor). This constraint often leads to increasing marginal costs as production increases, because variable inputs become less efficient when they are used intensively with the fixed input.

The short run and long run MC curve coincide at the same quantity as SR ATC and LR ATC because at this point they have the same level of capital and labor, therefore they have the same level of costs.

6.7 Economies of scale

Definition (Economies of scale): Total cost rises at a slower rate than output rises.

Definition (Diseconomies of scale): Total cost rises at a faster rate than output rises.

Definition (Constant economies of scale): Total cost rises at the same rate as output rises.

- A firm that faces increasing returns to scale necessarily experiences economies of scale: doubling output does not require doubling costs.
- For a firm that faces decreasing returns to scale may or may not experience diseconomies of scale. Obviously, however, strongly decreasing returns to scale will generally lead to diseconomies of scale.

Example: Now consider a firm with Quasi-linear production function:

$$Q = F(K, L) = K + 20L^{0.5}$$

This function has decreasing returns to scale since:

$$F(2K, 2L) = 2K + (2^{0.5}) 20L^{0.5} < 2F(K, L)$$

We can easily show, using methods we've seen before that, at an interior solution for the two factors:

$$L^* = \frac{100R^2}{W^2}$$

And all the rest of production is handled by K . Suppose $W = 10$ and $R = 1$

1. Let $Q = 25$:

Then $L^* = 1$, $K^* = 5$, $TC(Q) = 15$, $ATC(Q) = 0.6$.

2. Let $Q = 30$:

Then $L^* = 1$, $K^* = 10$, $TC(Q) = 20$, $ATC(Q) \approx 0.67$.

So we have decreasing returns to scale and we have diseconomies of scale.

Example: We now derive the long run cost curve for our previous example $Q = F(K, L) = 10K^\beta L^{1-\beta}$.

Start with the cost function itself in terms of K and L :

$$C = RK + WL$$

We know that at the firm's cost-minimizing combination of inputs

$$\frac{MP_L}{MP_K} = \frac{W}{R}$$

With our production function, this is:

$$\left(\frac{1-\beta}{\beta}\right) \frac{K}{L} = \frac{W}{R}$$

Or:

$$K = \left(\frac{\beta}{1-\beta}\right) \frac{W}{R} L$$

The optimal K can be written as a function of L so we can write:

$$Q = 10 \left(\frac{\beta}{1-\beta} \frac{W}{R} L\right)^{\beta} L^{1-\beta} = 10 \left(\frac{\beta}{1-\beta} \frac{W}{R}\right)^{\beta} L$$

Therefore:

$$L = \frac{1}{10} \left(\frac{\beta}{1-\beta} \frac{W}{R}\right)^{-\beta} Q$$

And therefore, from before:

$$K = \frac{1}{10} \left(\frac{\beta}{1-\beta} \frac{W}{R}\right)^{1-\beta} Q$$

Now we are ready to write our cost function in terms of Q :

$$\begin{aligned} C(Q) &= R \left(\frac{1}{10} \left(\frac{\beta}{1-\beta} \frac{W}{R}\right)^{1-\beta} Q\right) + W \left(\frac{1}{10} \left(\frac{\beta}{1-\beta} \frac{W}{R}\right)^{-\beta} Q\right) \\ C(Q) &= \frac{Q}{10} \left(R \left(\frac{\beta}{1-\beta} \frac{W}{R}\right)^{1-\beta} + W \left(\frac{\beta}{1-\beta} \frac{W}{R}\right)^{-\beta}\right) \\ C(Q) &= \frac{Q}{10} \left(\frac{R}{\beta}\right)^{\beta} \left(\frac{W}{1-\beta}\right)^{1-\beta} \end{aligned}$$

So we can write

$$C(Q) = \frac{\theta}{10} Q, \quad \theta = \left(\frac{R}{\beta}\right)^{\beta} \left(\frac{W}{1-\beta}\right)^{1-\beta}$$

And as well $ATC(Q) = \frac{\theta}{10}$: constant average costs in the long run.

Example: Consider the production function

$$F(K, L) = \begin{cases} 10K^{0.2}L^{0.7} & \text{if } Q \leq 20 \\ 10K^{0.7}L^{0.2} & \text{if } Q > 20 \end{cases}$$

Let $W = 10$ and $R = 1$. Consider the case when the firm is producing $Q = 10$. Optimality says

$$\frac{7}{2} \frac{K}{L} = 10 \quad \text{and} \quad 10K^{0.2}L^{0.7} = 10$$

Combining these expressions like usual to isolate for L , we get

$$L^{0.9} = \left(\frac{7}{20}\right)^{0.2} \rightarrow L^* \approx 0.792 \quad \text{and} \quad K^* \approx 2.26$$

$$ATC(Q) = \frac{2.26 + 10 \times 0.79}{10} \approx 1.02$$

Now consider the case when the firm is producing $Q = 30$. Optimality says:

$$\frac{2}{7} \frac{K}{L} = 10 \quad \text{and} \quad 10K^{0.7}L^{0.2} = 30$$

Combining these expressions like usual to isolate for L , we get

$$L^{0.9} = \left(\frac{3}{35^{0.7}}\right) \rightarrow L^* \approx 0.213 \quad \text{and} \quad K^* \approx 7.47$$

$$ATC(Q) = \frac{7.47 + 10 \times 0.21}{30} \approx 0.32$$

Average total cost has fallen, so the firm experiences economies of scale. But this production function clearly has decreasing returns to scale everywhere: doubling inputs always less than doubles output. Why do we have economies of scale? As the firm scales up, capital becomes more productive in its production function. (Here, for simplicity, the increase in the relative productivity of capital is sudden at $Q = 20$, but it could be more gradual.) This allows the firm to take advantage of the relative cheapness of capital compared to labor at higher levels of scale, even though we have decreasing returns to scale.

However, this doesn't work forever. Consider producing $Q = 50$. Again, optimality says:

$$\frac{2}{7} \frac{K}{L} = 10 \quad \text{and} \quad 10K^{0.7}L^{0.2} = 50$$

Combining these expressions like usual to isolate for L , we get

$$L^{0.9} = \left(\frac{5}{35^{0.7}} \right) \rightarrow L^* \approx 0.376 \quad \text{and} \quad K^* \approx 13.17$$

$$ATC(Q) = \frac{13.17 + 10 \times 0.38}{Q} \approx 0.34.$$

Now, decreasing returns to scale lead to diseconomies of scale.

7 Supply under competition

7.1 Market structures and perfect competition in the short run

Definition(Market structure): The competitive environment in which firms operate.

We categorize a market or industry using three primary characteristics:

1. Number of firms. Generally, the more companies in the market, the more competitive it is.
2. Whether the consumer cares which company made the good. In general, the more indistinguishable or identical the products are, the more competitive the market is.
3. Barriers to entry. If new firms can enter a market easily, the market is more competitive.

We have four basic market structure:

	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
Number of Firms	Many	Many	Few	One
Type of Products Sold	Identical	Differentiated	Identical or differentiated	Unique
Barriers to Entry	None	None	Some	Many

Table 1: Four basic market structure

To elaborate on the characteristics of perfect competition market:

1. **Number of Firms:** There needs to be a large number of firms so that no one firm has any impact on the market equilibrium price by itself. Thus, any one firm can change its behavior without changing the overall market equilibrium.
2. **Types of Products Sold:** All firms produce an identical product. By identical, we don't just mean that all the firms make televisions or all the firms make smoothies. We mean the consumers view the output of the different producers as perfect substitutes and do not care who made it. This assumption is more accurate for nails or gasoline or bananas, but less accurate for smartphones or automobiles.
3. **Barriers to Entry:** There are no barriers to entry. If someone decides she wants to start selling nails tomorrow, nothing prevents her from doing it.

A price taking firm (here, an urban chicken farmer named Ty) can sell as much output (as many eggs) as it wants at the market clearing price but nothing above this price. Therefore, the competitive firm's individual demand curve is perfectly elastic (horizontal) at the market price, which is \$2.25 in this case.

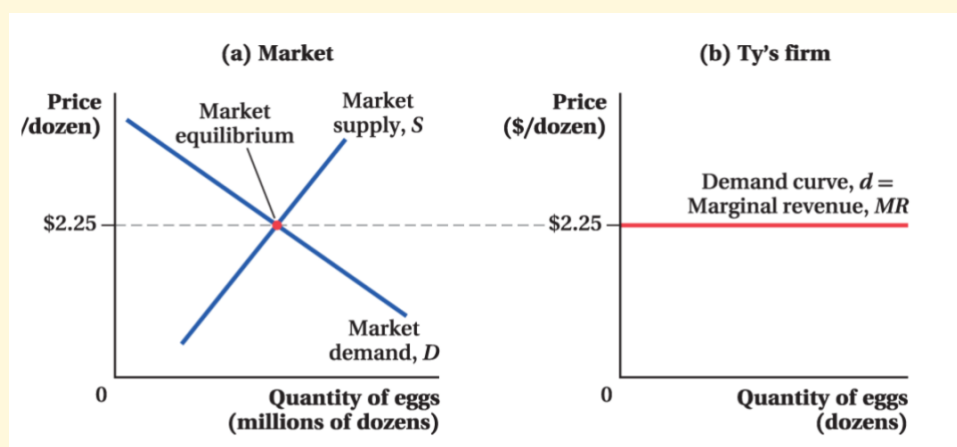


Figure 83: The price-taking firm's individual demand curve

7.2 Profit maximization in perfect competition

As we already know, producers in economics are the owners and operators of firms. As a producer, your income is the profit of your firm.

To maximize profits, a firm must decide how much output it wants to produce: this is the output decision. It chooses output to maximize its profit, i.e. the difference between its revenue from production and its cost of production.

If a firm is operating in a perfectly (or nearly perfectly) competitive market, its profit-maximizing output decision is actually quite easy to understand.

Let profit be π . Then:

$$\pi(Q) = TR(Q) - TC(Q)$$

where $TR(Q)$ is the total revenue from producing and selling output (of whatever good or service the firm makes)

Let's suppose that TR and TC are continuous and differentiable functions of Q . Then we look for a critical value for Q :

$$\frac{\partial \pi}{\partial Q} = 0 \rightarrow \frac{\partial TR}{\partial Q} = \frac{\partial TC}{\partial Q} \rightarrow MR(Q) = MC(Q)$$

We already know about $MC(Q)$. $MR(Q)$ is the marginal revenue of output.

The firm profit maximizes at the level of Q where

$$MR(Q) = MC(Q)$$

In words: marginal revenue equals marginal cost, and for a competitive firm, $MR(Q)$ is just the market price of output, P , by definition.

Graphically, this process looks like:

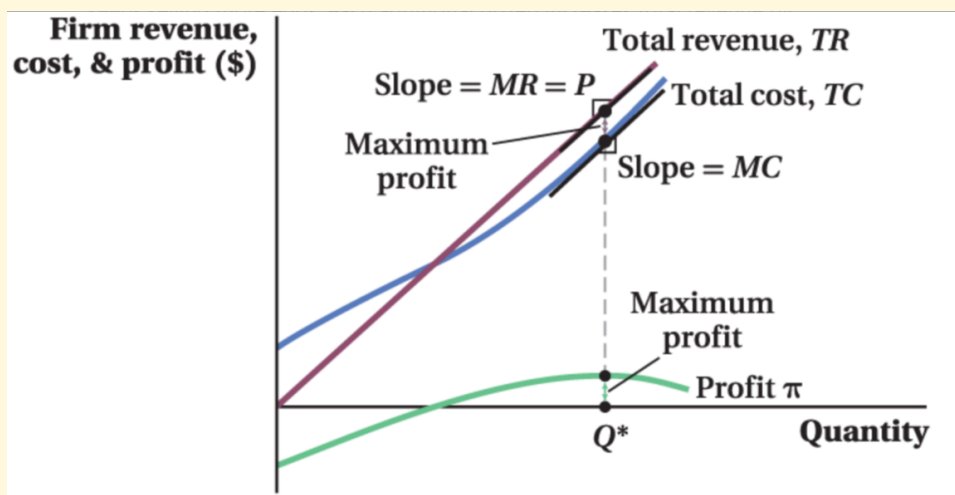


Figure 84: Maximizing profits

The profit-maximization condition for a competitive firm (conditional on cost-minimization) is

$$P = MC(Q^*)$$

A unique Q^* , representing a profit maximum, exists for this problem so long as $MC(Q)$ is (eventually) increasing with Q increases. Otherwise, the question will no longer make sense because the revenue will just be bigger than cost infinitely as Q goes infinitely. Also, we know for sure that in the short run, the marginal cost has to increase due to diminishing marginal return. However, in the long run, we are not sure, thus, we have this condition when finding the profit maximizing quantity.

Suppose we have an increasing marginal cost and conditional on cost-minimization, the company should keep producing until the profit maximum quantity is reached, or decrease the production to the optimal quantity.

In addition, profit maximization does not mean the firm will necessarily experience positive profit, but the profit will be the maximum profit at the current market price or marginal revenue in a competitive market.

Specifically, we can write profit as:

$$\pi = TR(Q) - TC(Q) = PQ - TC(Q) = PQ - ATC(Q)Q = (P - ATC)Q$$

The firm earns positive profit when the market price is above its average total cost of production and takes losses when price is below its average total cost.

We can visualize the situation with the follows:

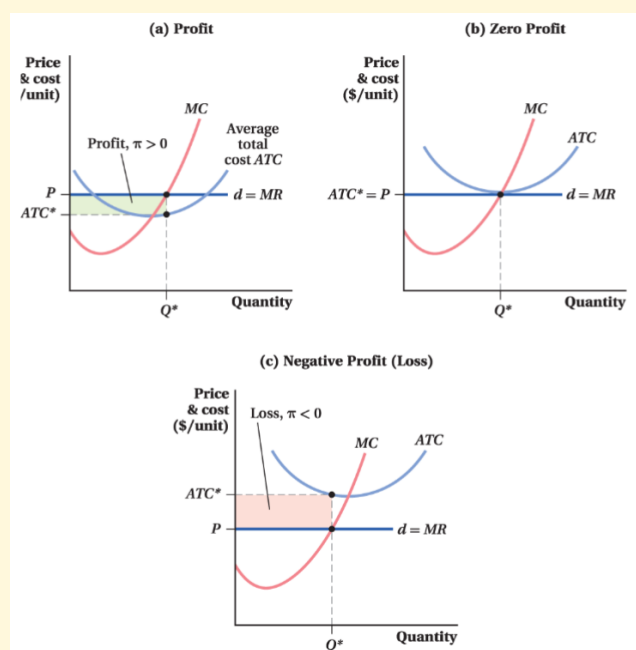


Figure 85: positive, zero and negative profit

Notice that the marginal cost curve intersect with the ATC curve at ATC's minimum, and ATC above MR means negative profit.

7.3 Competition in the short run

7.3.1 Make short run decision and short run supply curve

To understand firm behavior and equilibrium outcomes under competition, we need to distinguish what happens in the short run and the long run.

The profit maximization condition for output

$$P = MC(Q^*)$$

is the same for both long run and short run, except for finding the specific MC curve in long run and short run.

But the decision about whether to operate is different between the short and long runs.

In the short run, some (realistically) or all (Assumption in the model) costs attaching to the fixed factors of production, are sunk. As we already know, these costs cannot affect the short-run shut down decision. Because in the short run, doesn't matter what I do (shut down or continue), I have to bear the fixed costs. The fixed costs is also not included in the optimal quantity decision.

We have this table:

	Shut Down	Operate
Revenue	None	Some (TR)
Cost	Fixed (FC)	Fixed (FC) + Variable (VC)
Profit	$-FC$	$TR - FC - VC$

The firm should produce if $TR - FC - VC \geq -FC$. That is if $TR - VC \geq 0$. Otherwise it should shut down in the short run. We know that $TR = P \times Q^*$ and $VC = AVC(Q^*) \times Q^*$, and Q^* is the optimal quantity. Therefore, we can write the short run shut down decision as simply:

$$\text{Shut down if } P < AVC \text{ at } Q^*$$

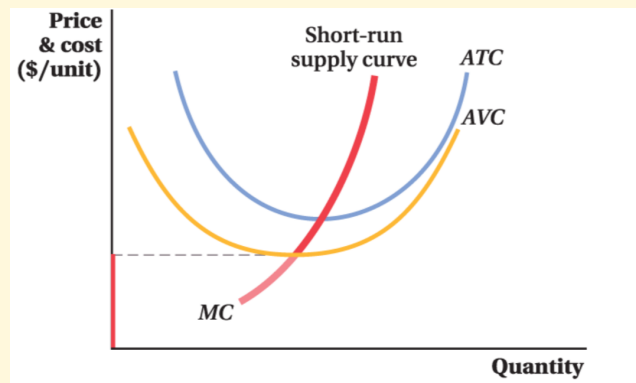


Figure 86: Shot down decision

Thus, the firm's short run supply curve in P/Q space is its $MC(Q)$ curve above the point where $AVC(Q)$ is at its minimum. At values of P lower than the minimum AVC , the firm produces zero.

If the price is above AVC , the firm will operate and making variable profits, if it is above ATC , then the firm is making economic profits.

Example: A firm has short run production function $Q = F(K, L) = K^{0.5}L^{0.4}$. If $W = 10, R = 2, K = \bar{K} = 4$, and the market price of output is $P = 4$, should this firm shut down in the short run?

Solution: The firm will shut down in the short run if $VC(Q^*) > PQ^*$, i.e. variable costs are greater than revenue at the profit-maximizing level of output. We already know P , so we need to find $VC(Q^*)$. So obviously we need to find Q^* . Q^* is the level of Q where $P = MC(Q)$. In the short run, $MC(Q) = \frac{W}{MP_L}$. (We need $\frac{1}{MP_L}$ units of L , our variable input, to produce one more unit of Q , and each unit of L costs W .) So we have

$$\frac{10}{2(0.4)L^{-0.6}} = 4 \rightarrow L^{0.6} = 0.32 \rightarrow L \approx 0.15$$

Then $Q^* = 2(0.15^{0.4}) \approx 0.94$. Now we ask: does revenue from Q^* cover variable costs? Revenue is $PQ^* \approx 3.74$. Variable costs are $WL = 1.50$. Since we are covering our variable costs, we should indeed produce in the short run.

Example: Should the firm in previous question produce in the long run?

Solution: We proceed as always. First find the optimal input ratio. We get

$$\frac{4}{5} \frac{K}{L} = \frac{10}{2} \rightarrow K = \frac{25}{4}L$$

Then use the production function to write:

$$Q = L^{0.4} \left(\frac{25}{4} \right)^{0.5} L^{0.5} = \frac{5}{2} L^{0.9}$$

Rearrange to isolate L :

$$L \approx 0.36Q^{1.11}$$

And from the optimality condition for L and K :

$$K = \frac{25}{4}L \approx 2.25Q^{1.11}$$

So we have $C(Q) = 2K + 10L = 4.5Q^{1.11} + 3.6Q^{1.11} = 8.1Q^{1.11}$. From here, we compute $MC(Q) = C'(Q) = 8.99Q^{0.11}$. So setting $MC(Q) = P$, we find $Q^* = 0.00062$. From here, $C(0.00062) = 8.1(0.00062)^{1.11} = 0.0024$. And $PQ^* = 0.0025$. So this firm should operate in the long run.

7.3.2 Short run market supply curve under competition

Under competition, all firms make supply decisions entirely independently of each other.

That is, a competitive firm makes its shut down and output decisions based on the market price and its own technology; not what other firms in the industry are doing.

An industry supply curve then indicates how much total output an industry supplies at any particular price.

We can use graphic approach to construct this industry supply curve with a stylized example:

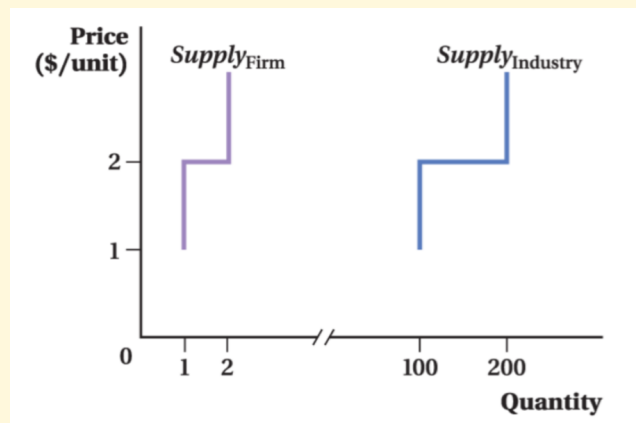


Figure 87: individual and aggregate supply curve

Here $Supply_{Firm}$ is the short-run supply curve of each firm in an industry with 100 firms. The short-run industry supply curve, $Supply_{Industry}$, is the horizontal sum of the individual firms' supply curves.

However, firms in the same industry don't need to be identical in the short run. This is because firms are likely to be operating with different levels of capital K in the short run, which they can't quickly adjust, and firms may be experimenting with different production processes, especially in young industries, to find the most cost-effective. This means firms have different cost structures and different prices at which they choose to shut down. Each firm's supply curve may be different, but market supply is still the horizontal sum of the individual supply curves.

A graphic example is:

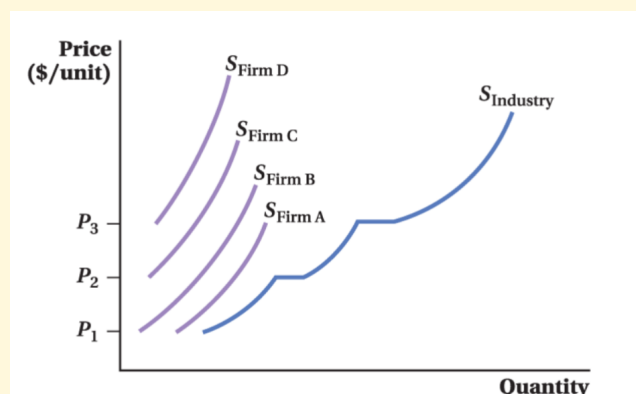


Figure 88: Different individual supply curve case

In a four-firm industry in which each firm faces different costs, these cost differences are reflected in their individual supply curves ($S_{Firm A}$, $S_{Firm B}$, $S_{Firm C}$, and $S_{Firm D}$). The industry supply curve, $S_{Industry}$, is the horizontal sum of the four individual firms' supply curves. At prices between P_1 and P_2 , only Firms A and B produce; at price P_2 , Firm C also supplies its product on the market; and at price P_3 , all four firms supply positive quantities.

7.3.3 Producer surplus in the short run

The firm produces to Q^* where $MC(Q^*) = P$.

At this level of output, $MR = MC$ and so the final unit of output yields no profit or surplus to the firm.

But at every other unit of $Q < Q^*$, $MR = P > MC(Q)$ and so the firm is earning variable profit on this unit of output.

The variable profit is therefore the area above the MC curve below the price. As we already know, since the MC curve above minimum AVC is the firm's supply curve, this is also producer surplus (PS) in the short run

Equivalently, PS and variable profit is the area between P and AVC out to Q^* .

Graphically, the PS looks like:

(a) At market price, a perfectly competitive firm produces Q^* . For each unit the firm produces below Q^* , the marginal cost MC is less than the market price, and the firm earns some producer surplus on that unit. As a result, total producer surplus is equal to the area below the demand curve and above MC .

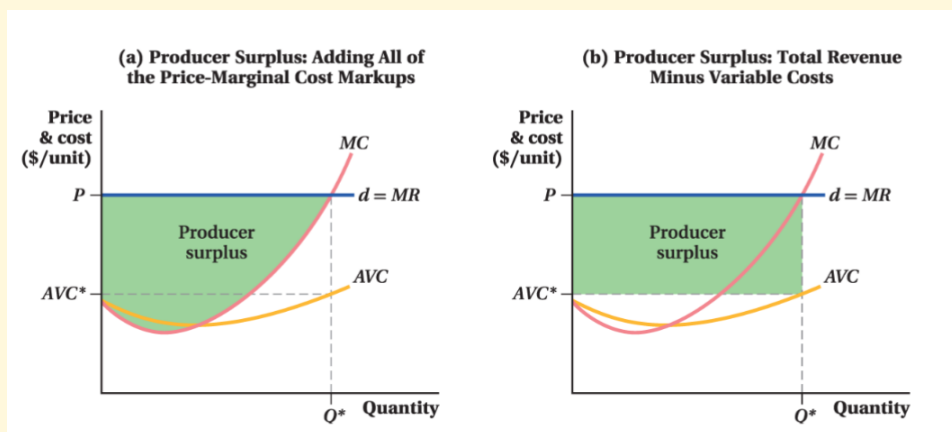


Figure 89: Producer surplus and Variable profit

(b) Producer surplus can also be calculated as a firm's total revenue minus its variable cost. A firm's total revenue is the entire rectangle with height P and length Q^* , and its variable cost is the rectangle with height AVC^* and length Q^* . Its producer surplus therefore is the area of the rectangle with height $(P - AVC^*)$ and length Q^* .

Producer surplus in the short run is not the same as profit, but it is equivalent to variable profit.

The firm produces in the short run so long as $PS \geq 0$ and shuts down otherwise.

7.3.4 Supply under competition in the long run

In the long run, a firm produces at Q^* where its long-run marginal cost ($LMC(Q^*)$) equals the market price P . Because all inputs, and costs, are variable in the long run, the firm's long run supply curve is the part of its LMC curve above minimum long run ATC ($LATC$).

At the industry level, the main difference between the short and the long run is that, in the long run, the number of firms producing in the industry can change.

To understand why this only happens in the long run, remember that K is fixed in the short run, which includes K being fixed at zero for firms not currently in the market.

In the long run, competitive industries are characterized by **free entry and exit**.

In the long run, we can assume that all existing firms in the industry are identical to each other. That is: they have the same cost structure and the same cost curves.

As new firms enter in search of profit, the total market supply increases, which means the (short run) supply curve shifts out from one "short run" to the next.

7.3.5 Long run competitive equilibrium

Equilibrium price falls until $P = ATC(Q^*)$ for each firm in the market. This is **long run competitive equilibrium** where all firms earn zero profits.

Graphically, a firm in the market makes profit in the long run when:

This "representative firm" in a competitive industry is earning profit in the long run. Therefore, this can't be an industry in equilibrium: new firms will enter.

Because new firm has entered the market, the aggregated market short run supply curve will shift:

New entrants shift the (short run) supply curve to the right; i.e. in the next short run, there are more firms in the industry. Entrance into the industry continues until P is pushed down to firms' $LATC$ at Q^* , which can only be at minimum $LATC$, which is the intersection with LMC .

This process is:

(a) The original long-run equilibrium for an industry is (P_1, Q_1) at the intersection of the long-run supply curve S_{LR} and the original demand curve D_1 . After a change in tastes, demand increases to D_2 , price increases to P_2 , and firms earn positive economic profits in the short run. In the long run, new firms enter the industry, shifting the short-run supply curve to S_2 until it reaches the long-run equilibrium price P_1 at the new equilibrium quantity Q_2 .

(b) At the long-run market price P_1 , the representative firm earns zero economic profit and produces quantity Q_1^* . When market demand rises, the market price increases to P_2 and the firm's output increases to Q_2^* . At this combination, the firm earns positive economic profit. As entry into the industry occurs, the price falls back to P_1 , and the firm reduces its output to Q_1^* . At this point, the firm earns zero economic profit.

Therefore the long run supply curve is horizontal at minimum $LATC$.

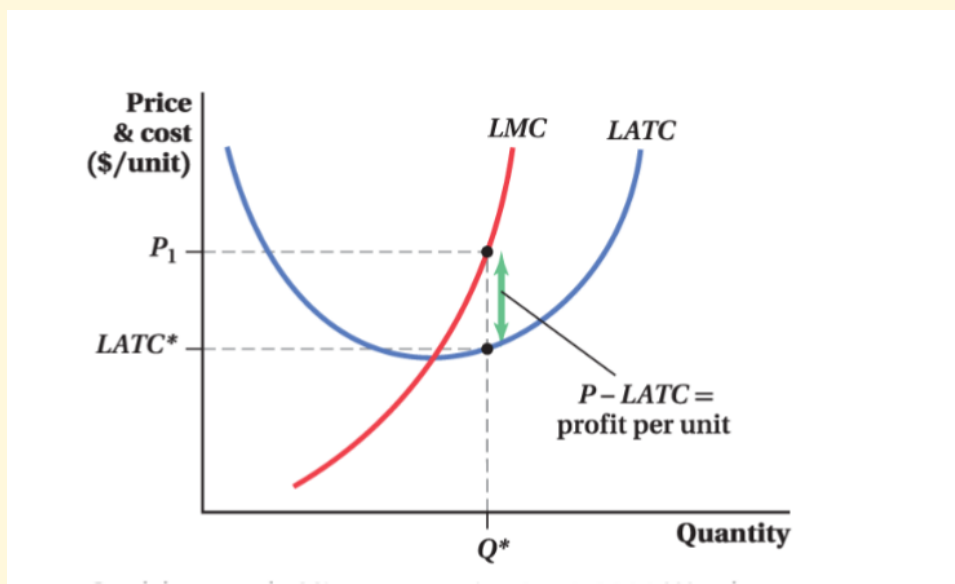


Figure 90: A firm in the market making profit in the long run

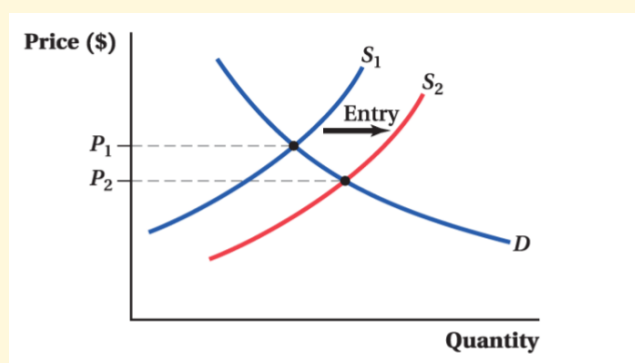


Figure 91: shift of market supply

7.3.6 Exit the market

The mirror images of free entry is of course free exit, which is equally important.

Suppose an industry experiences a negative shift in the market demand curve.

Or suppose too many firms enter an industry where profit is positive and push supply too far out.

In this case, P falls below the minimum LATC and firms take (economic) losses.

Some firms will respond by leaving the industry because doing so is relatively easy and their resources could be better deployed elsewhere.

Only where P equals the minimum LATC of firms in the industry can the equilibrium be stable, with no net entrance or exit in the long run.

7.3.7 Properties and implications of long run supply curve and more details

In a competitive market in the short run, if demand shifts out, we move along the supply curve to a new equilibrium with

1. Higher price
2. Higher quantity transacted

The long run is:

1. Firms respond by moving up their short run supply curves, hiring more variable factors and producing more output, i.e. above minimum LATC scale. And remember demand shock is only short run.
2. Now the existing firms will be earning positive short run profits.
3. In the long run, this attracts new entrants, until profits are dissipated.

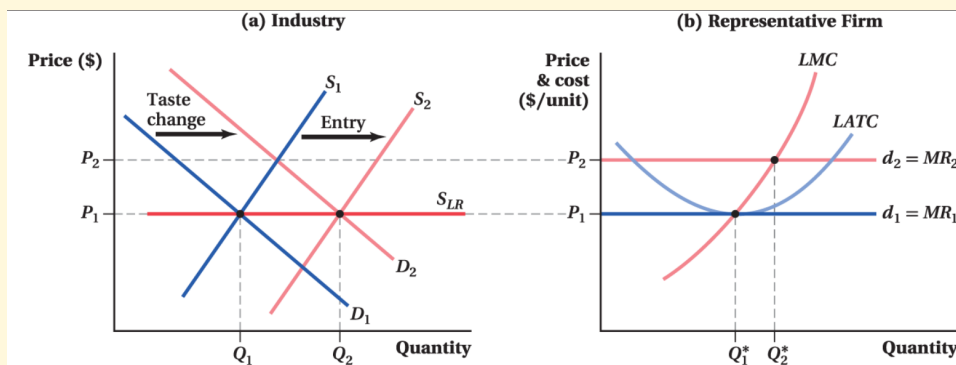


Figure 92: Long run supply curve and return to equilibrium

4. The existing firms scale back to their long run Q^* in order to avoid taking losses and equilibrium is restored at $P = \text{minimum } LATC$, which has not changed.
5. Thus, the price does not change, but the market quantity has changed forever.

Example: Consider a perfectly competitive market described by the following supply and demand functions:

$$Q = D(p) = a - bp$$

$$P = S^{-1}(Q) = c$$

1. This system of equations describes a long run competitive equilibrium. How do you know?

Solution: It describes a long run competitive equilibrium because the long run supply curve is horizontal at $c = P$. This is where each firm is operating at its minimum $LATC$.

2. What are P^* and Q^* ?

Solution: $P^* = c$ and $Q^* = a - bc$.

3. If the government decides to subsidize production on a per-unit basis (subsidy of some amount x per unit produced), what happens to consumer surplus, producer surplus, and their sum in the short and long runs?

Solution: In the short run, a per-unit subsidy x to producers effectively reduces their cost of production by x for each unit. Hence, the supply curve shifts downward by x . Since the price received by producers (net of subsidy) remains constant at c , the price they would accept from the market would reduce to $c - x$. Consumer surplus increases because the price decreases from p^* to $p^* - x$, and consumers enjoy more quantity at a lower price. Producer surplus also increases. Producers receive c per unit (as the market price plus subsidy), even though the market price is $c - x$. They also produce more, increasing total producer surplus. Both consumer and producer surplus increase, so total surplus (the sum of consumer and producer surplus) also increases in the short run.

8 Monopoly

8.1 Sources of market power

A (truly) competitive firm is a firm whose output is so small relative to the market in which it produces that it represents effectively 0% of the market output. At the other extreme of market structure is the monopolist: a firm that it represents 100% of market output for its good. In this chapter, we begin to look at firms' production choices when they have some ability to control the price at which their product sells.

Definition(Market power): A firm's ability to influence the market price of its product.

Definition(Monopoly): A market served by only one firm, and this firm is called a monopolist.

Definition(Barriers to entry): Factors that keep entrants out of a market despite the existence of a large producer surplus.

So, barriers to entry are factors that prevent or discourage potential competitors from entering a market and are crucial in the formation and maintenance of a monopoly. These barriers can be natural or artificial and vary widely depending on the industry.

8.1.1 Natural monopoly

Definition(Natural monopoly): A market in which it is efficient for a single firm to produce the entire industry output. This means when a single firm can supply the entire market with a particular good or service at a lower cost than any combination of two or more smaller firms could.

Example: Suppose a company could produce as large a quantity as it wants at a constant marginal cost of \$10 per unit and has a fixed cost of \$100. In this case, average total cost (total costs divided by output) declines across all quantities of output. In equation form,

$$TC = FC + VC = 100 + 10Q$$

so

$$ATC = \frac{TC}{Q} = \frac{100 + 10Q}{Q} = \frac{100}{Q} + 10$$

The larger the firm's quantity produced, Q , the lower is its average total cost. If all of the firms in an industry have this same cost structure, the lowest-total-cost way to serve industry demand is for one firm to produce everything. If more than one firm produces, the industry's average total cost of producing output is higher because each firm that operates has to pay the fixed cost of \$100 to produce anything at all. Having only one firm in the industry saves the replication of those fixed costs. That's why it is more cost-effective from society's perspective to have only one producer in the industry.

Characteristics of natural monopoly

- **High Fixed Costs:** Significant initial investment is required to enter the market. This might include the cost of building infrastructure, such as pipelines, cables, or networks, which do not change with the amount of service provided.
- **Economies of Scale:** As production increases, the average cost per unit decreases. In a natural monopoly, these economies of scale continue over such a large range of output that one firm can serve the entire market more cheaply than any smaller competitor.
- **Lower Marginal Costs Compared to Fixed Costs:** The cost of serving an additional customer is relatively low compared to the initial cost of setting up the business.

8.1.2 Switching costs

Definition(Switching costs): Switching costs are expenses or inconveniences that consumers must bear if they decide to switch from one product or service provider to another. These costs can act as a significant barrier to entry in various markets by discouraging customers from changing suppliers or brands, thus protecting the incumbents from new competitors.

Types of switching costs

- **Financial Costs:** These are direct costs involved in switching between products or services. For example, terminating a contract early might incur a fee, or there might be costs associated with setting up a new service, such as installation fees.
- **Time and Effort:** The time and effort required to learn how to use a new product or to integrate a

new service into one's daily routine can be significant. For instance, switching from one software system to another might require extensive employee training.

- **Psychological Costs:** These include the stress or uncertainty associated with using a new product or service. Brand loyalty and trust in a particular brand also play into psychological costs, as customers may feel a sense of risk when choosing a new provider.
- **Technical Compatibility and Integration:** In many cases, especially in technology and industrial markets, switching products might require additional changes to existing systems. For example, different software applications may not be compatible, or new machinery might require an overhaul of related systems.

Definition(Network goods): Network goods, also known as network externalities or network effects, refer to products or services whose value increases as more people use them. The classic examples include social media platforms, telecommunications services, and software ecosystems.

For network goods, a significant part of the product's value is derived from the number of users. Switching from one product to another often means losing access to that network and its associated benefits. For example, leaving a social media platform means losing connections, content, and interactions that are not transferable to other platforms. *Remark:* Switching costs don't need to be insurmountable to be effective. To reduce the threat of competition and give the incumbent firm some market power, switching costs only need to be high enough to make entry costly, not impossible.

8.1.3 Product differentiation

Definition(Product differentiation): Product differentiation is a strategy where firms attempt to make their products or services stand out as unique from those of their competitors. This differentiation can be based on various factors such as quality, features, design, branding, customer service, or technology. By differentiating their products, firms can reduce the substitutability between their offerings and those of their competitors, effectively decreasing the price sensitivity of consumers. This allows firms to exert a degree of market power, enabling them to command higher prices, attract more customers, or both.

8.1.4 Control key inputs

An absolute cost advantage occurs when a firm has the ability to produce a good or service at a lower absolute cost than its competitors. This advantage can derive from several factors, including more efficient production processes, lower labor costs, better terms from suppliers, or superior technology. One specific form of absolute cost advantage that can be particularly powerful as a source of market power is when a firm has a lower cost in obtaining a key input necessary for production compared to its competitors.

When a firm has access to a key input at a lower cost, it can produce its final product more cheaply than competitors. This allows the firm to either offer its product at a lower price, potentially capturing a larger market share, or maintain industry-standard pricing while enjoying higher profit margins.

If the cost advantage is significant and the input is critical to the industry, this advantage can act as a barrier to entry for other firms. Potential competitors may find it unprofitable to enter the market if they cannot access the key input at a competitive cost. This protects the advantaged firm from new entrants and reduces the intensity of competition.

8.1.5 Government regulations

Governments often require firms to attain licenses to enter an industry. For instance: almost every city government requires restaurants selling alcohol to obtain a license from the city. Before the advent of ride share companies, taxi service in most cities was severely restricted by the need for drivers to obtain taxi medallions. Patent and copyright law often create time-limited monopolies in all sorts of industries. In the simple competition vs. monopoly framework, these sorts of public interventions

8.2 Profit maximization for monopoly

A firm with market power faces a downward sloping demand curve. A firm that faces a downward sloping demand curve can raise its price, at any price below the market choke price, without losing all of its demand.

Example: In this example we will derive the marginal revenue. Suppose the firm faces inverse demand curve

$$P = a - bQ^D$$

where a and b are positive constants. The firm will always sell $Q = Q^D$. So the firm's revenue is

$$TR = PQ = (a - bQ)Q = aQ - bQ^2$$

Suppose the firm increases Q by a small amount:

$$\frac{dTR}{dQ} = \frac{d(aQ - bQ^2)}{dQ} = a - 2bQ = MR(Q)$$

Here, the MR curve has twice the slope, in absolute value, of the demand curve. MR lies everywhere below the price when $Q > 0$.

To maximize its profit, a firm should choose its quantity where its marginal revenue equals its marginal cost:

$$MR = MC$$

1. **Step 1:** Derive the marginal revenue curve from the demand curve.
2. **Step 2:** Find the output quantity at which marginal revenue equals marginal cost. This is the firm's profit-maximizing quantity of output.
3. **Step 3:** Determine the profit-maximizing price by locating the point on the demand curve at that optimal quantity level.

Example: Suppose a monopolistic firm has a production function

$$Q = F(K, L) = 10K^{0.5}L^{0.5}$$

Market (inverse) demand is given by

$$P = 150 - 0.25Q$$

1. What is Q^* for this monopolist in the short run if $K = 25$, $R = 10$ and $W = 10$?
2. What is Q^* for this monopolist in the long run if $R = 10$ and $W = 10$?

Solution: We need to find expressions for revenue and cost. We start with costs. In the short run, the production function becomes:

$$Q = f(L) = 10(5)L^{0.5}$$

Rearranging for L we get:

$$L = \frac{Q^2}{2500}$$

And total costs are $RK + WL$ or

$$TC = 250 + 10\left(\frac{Q^2}{2500}\right)$$

So marginal costs are:

$$MC(Q) = 0.008Q$$

Next, we need to find total revenue and marginal revenue. This is easy. We have

$$TR = PQ = (150 - 0.25Q)Q = 150Q - 0.25Q^2$$

So marginal revenue is

$$MR(Q) = 150 - 0.50Q$$

Combining our expressions, $MC(Q^*) = MR(Q^*)$ solves:

$$0.008Q^* = 150 - 0.50Q^*$$

Or $Q^* \approx 295$ in the short run when $K = 25$. Normally, finding $C(Q)$ in the long run takes some doing, but here it is easy since obviously our optimal input mix is $K = L = \frac{Q}{10}$. So we have

$$C(Q) = 10\frac{Q}{10} + 10\frac{Q}{10} = 2Q$$

And therefore $MC(Q) = 2$. Marginal revenue is the same as in the short run since demand does not change between the short and long runs. We have:

$$2 = 150 - 0.50Q^*$$

We get $Q^* = 296$ in the long run.

8.2.1 Lerner index and mark up

In the general case, the firm is constrained by its demand curve. If it chooses Q , P is a function of Q :

$$TR = P(Q)Q$$

Therefore:

$$MR = \frac{dTR}{dQ} = P(Q) + P'(Q)Q$$

Multiplying the second term by $\frac{P}{P}$ we can write:

$$MR = P(Q) + P(Q) \left(\frac{1}{E^D} \right) = P \left(1 + \frac{1}{E^D} \right)$$

Here, E^D is the elasticity of demand in the market. At the optimum, the monopolist sets $MR = MC$ which means

$$MC = P \left(1 + \frac{1}{E^D} \right) \rightarrow \frac{P - MC}{P} = -\frac{1}{E^D}$$

Definition(Mark up): The percentage of the firm's price that is greater than (or "marked up" from) its marginal cost.

Definition(Lerner index): The measurement of mark up is called Lerner index, which equals to $\frac{P - MC}{P}$.

8.3 Winners and losers from market power

Monopoly is obviously bad for consumers for the usual reasons:

1. Monopolists reduce total output in the industry, reducing transactions and the resulting gains to trade.
2. Monopolists raise prices above the competitive level, which consumers don't like.

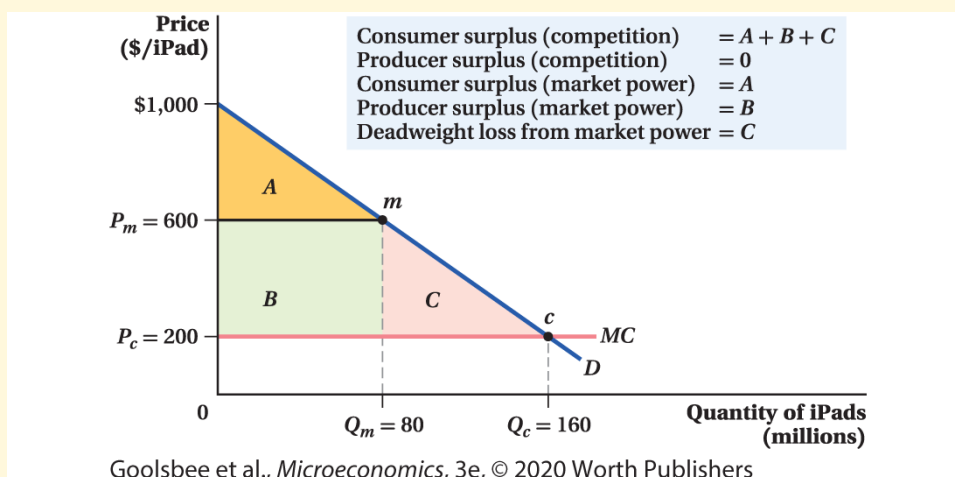


Figure 93: Transfer of surplus and DWL

Market power does not just transfer surplus from consumers to the monopolists, it also creates deadweight loss, also for the usual reasons. Therefore, an economy characterized by monopoly - more generally, by firms with price-setting power - will be inefficient.

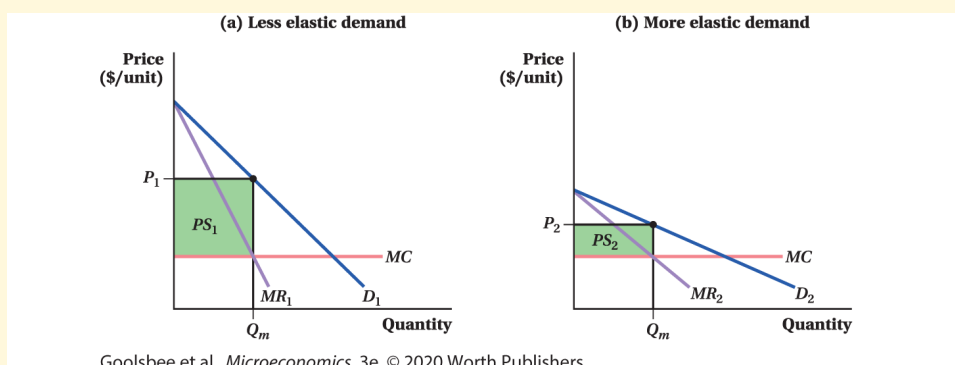


Figure 94: Elasticity and surplus

The more inelastic is demand in a monopolistic market, the worse the outcome for consumers will be.

9 Pricing strategy

In the previous chapters, we examined how firms with market power can generate positive economic profit by influencing the price at which their products or services are sold - but this is based on the assumption that firms must charge the same price to all customers.

In this chapter, we explore alternative pricing strategies and show that when a firm with market power can discriminate among customers, additional surplus (beyond that achieved by a single-price monopolist) can be generated.

9.1 Basics of pricing strategy

Definition(Pricing strategy): A firm's plan for setting the price of its product given the market conditions it faces and its desire to maximize profit.

Example: For a perfectly competitive firm, the pricing strategy is straightforward: charge the equilibrium market price and take zero economic profit in the long run.

Example: For firms with market power, strategies can become more complex.

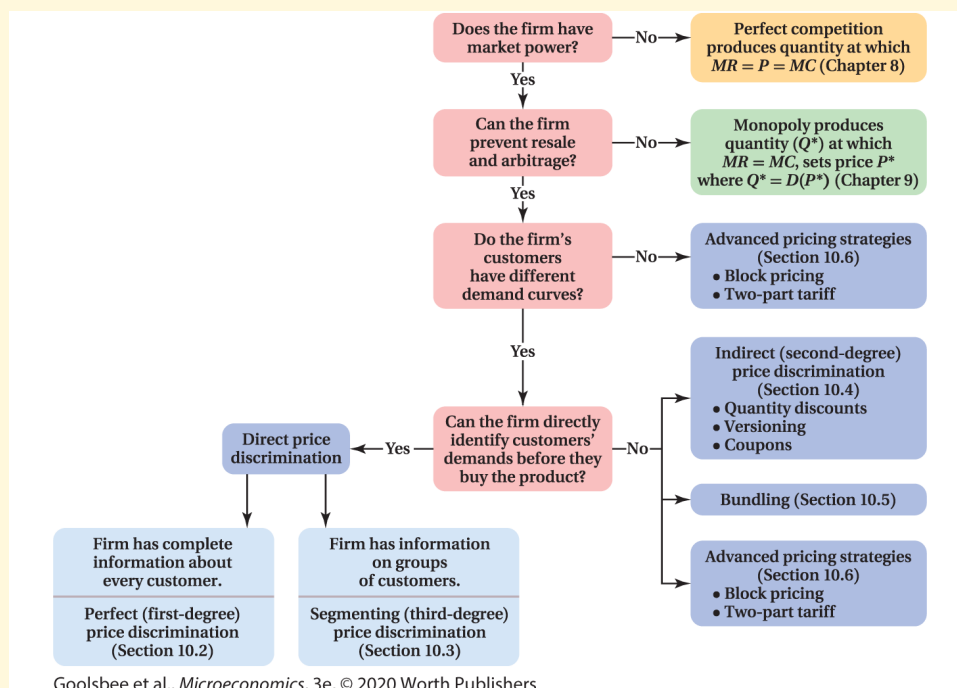
- For a single-price producer, the optimal strategy is to increase production until marginal revenue is equal to marginal cost, which yields maximum profit.
- However, some firms with market power are able to charge different prices to different customers (price discrimination).

Definition(Price discrimination): The practice of charging different prices to different consumers for the same product.

Pre-requisite of price discrimination

- The firm must have market power. A company must have market power to price discriminate. It's that simple.
- The firm must prevent resale and arbitrage. Arbitrage is the practice of reselling a product at a price higher than its original selling price. If this requirement is not met, customers subject to a lower price could simply purchase excess product and resell to those facing the higher price.

A firm's optimal pricing strategy is determined by its characteristics, its product, and its consumers. In particular, a firm takes into account its degree of market power, whether the product can be resold, and its knowledge of its customers' demand for the product.



Goolsbee et al., *Microeconomics*, 3e, © 2020 Worth Publishers

Figure 95: Different pricing strategies

9.2 Direct Price Discrimination I: Perfect (First-Degree) Price Discrimination

Definition(Perfect (First-Degree) Price Discrimination): A type of direct price discrimination in which a firm charges each customer exactly his willingness to pay.

Pre-requisite of perfect price discrimination

- The firm has market power and can prevent resale.
- The firm's customers have different demand curves.
- The firm has complete information about every customer and can identify each one's demand before purchase.

If a firm is able to observe characteristics of demand prior to purchase, it can increase producer surplus through direct price discrimination. It is a extreme case that the firm has complete information about customers, it can engage in perfect price discrimination.

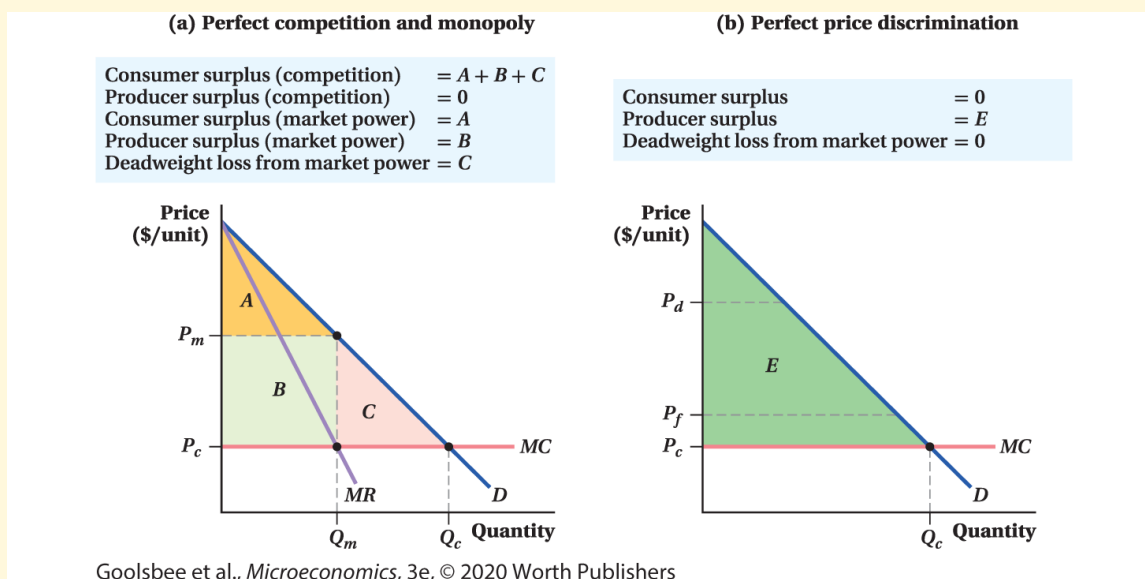


Figure 96: Surplus in perfect price discriminations

A competitive market will set price equal to marginal cost, producing Q_c and selling at a price of P_c . Consumers will receive consumer surplus equal to $A + B + C$ and the firm will earn zero producer surplus. A single-price monopoly will sell quantity Q_m at a price of P_m and receive producer surplus equal to B . Consumers will receive consumer surplus equal to A and the deadweight loss from market power will be area C .

If a firm with market power can identify each customer's demand curve, then it will charge each customer her willingness to pay and capture the entire surplus, E . For example, the firm will charge a customer willing to pay P_d exactly the price P_d and a customer willing to pay P_f the price P_f . The firm will sell up to the quantity Q_c , the perfectly competitive quantity where $P_c = MC$. There is no deadweight loss when a firm practices perfect price discrimination.

9.3 Direct Price Discrimination II: Segmenting (Third-Degree Price Discrimination)

Definition(Segmenting (Third-Degree Price Discrimination)): A type of direct price discrimination in which a firm charges different prices to different groups (segments) of customers based on the identifiable attributes of those groups.

Pre-requisite of segmenting

- The firm has market power and can prevent resale.
- The firm's customers have different demand curves.
- The firm can directly identify specific groups of customers with different price sensitivities (but not the demand of every individual customer) before purchase.

If a firm can directly identify groups that have different demands and charge different prices to each, it can essentially treat each group as a separate market. The firm then establishes its profit-maximizing quantity for each one of these "markets" where $MR = MC$ and sets the corresponding single-price profit-maximizing price according to each market's demand curve.

A firm following this pricing strategy will not earn as much producer surplus as one using perfect price discrimination (which would allow it to take the entire surplus from the market). However, it will earn more surplus than if it acted like a regular monopolist and charged the same price to everyone, because the strategy gives the firm some ability to charge a higher price to consumers with relatively inelastic demand and lower prices to consumers with relatively elastic demand.

Example: Consider an example with two consumer groups: The Ironman Cozumel 70.3 Triathlon is a prestigious annual race. Two types of consumers would like to enter the race: locals and people who fly in from somewhere else (usually the United States).

Traveling triathlon participants have demand

$$Q_T = 1,700 - 5P_T$$

while locals have demand

$$Q_L = 2,400 - 10P_L$$

As suggested, locals are more price-sensitive than travelers.

The marginal cost to the organizer of adding participants is assumed to be constant at \$100.

We follow the steps from previous chapters and first determine the marginal revenue curves for each group of participants.

Travelers	Locals
$Q_T = 1,700 - 5P_T$	$Q_L = 2,400 - 10P_L$
$P_T = 340 - 0.2Q_T$	$P_L = 240 - 0.1Q_L$
$MR_T = 340 - 0.4Q_T$	$MR_L = 240 - 0.2Q_L$

Setting marginal revenue equal to marginal cost for each group:

Travelers	Locals
$MR_T = MC$	$MR_L = MC$
$340 - 0.4Q_T = 100$	$240 - 0.2Q_L = 100$
$Q_T = 600$	$Q_L = 700$

And using the demand curves to solve for price: Travelers Locals

$$P_T = 340 - 0.2Q_T = \$220 \quad P_L = 240 - 0.1Q_L = \$170$$

Total producer surplus is the area below price but above marginal cost, summed over the two consumer groups:

- Travelers

$$\begin{aligned} PS_T &= (220 - 100) * 600 \\ &= \$72,000 \end{aligned}$$

- Locals

$$\begin{aligned} PS_L &= (170 - 100) * 700 \\ &= \$49,000 \end{aligned}$$

$$PS_{\text{combined}} = \$121,000$$

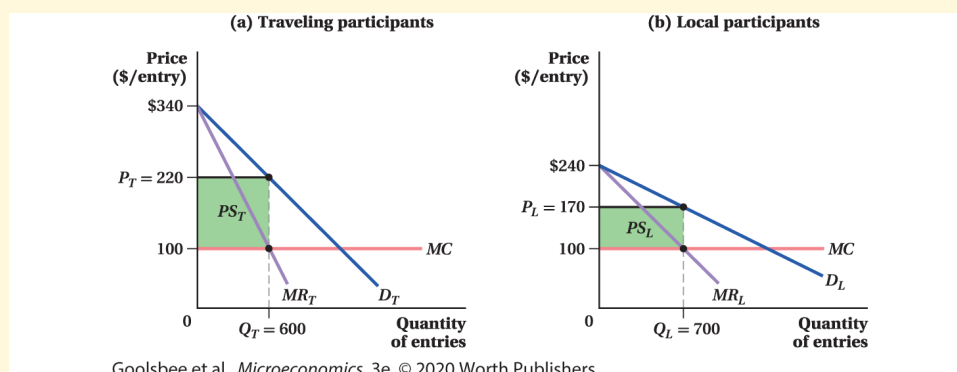


Figure 97: Graphic representation of the result

Example: Now we will make a comparison with the situation of regular monopolist with the same price for both segments.

First, find the equation for the market demand area to the right of the kink is

$$Q = 1,700 - 5P + 2,400 - 15P = 4,100 - 15P$$

Inverse demand is $P = \frac{4,100}{15} - \frac{Q}{15}$ and marginal revenue is $MR = \frac{4,100}{15} - \frac{2Q}{15}$. Setting marginal revenue equal to marginal cost yields the single-price equilibrium quantity

$$\begin{aligned} MR &= MC \\ \frac{4,100}{15} - \frac{2Q}{15} &= 100 \\ Q &= 1,300 \end{aligned}$$

And plugging in 1,300 to the market demand curve yields

$$P = \frac{4,100}{15} - \frac{1,300}{15} = \$186.67$$

As expected, the market price is slightly higher than the local price under segmentation but lower than the traveler price.

Producer surplus for the single-price monopolist is

$$PS = (186.67 - 100) * 1,300 = \$112,671$$

This is less than the \$121,000 achieved when the market can be segmented.

We can also use Lerner index to calculate the price for each segment. Because the standard market power pricing rule applies in each segment, it also means that the Lerner index, the basic markup formula we derived in previous chapters, applies in each market:

$$\begin{aligned} \frac{(P_1 - MC)}{P_1} &= -\frac{1}{E_1^D} \\ P_1 - MC &= -\frac{1}{E_1^D} \times P_1 \\ P_1 + \left(\frac{1}{E_1^D} \times P_1 \right) &= MC \\ P_1 \left(1 + \frac{1}{E_1^D} \right) &= MC \\ P_1 \left(\frac{E_1^D}{E_1^D} + \frac{1}{E_1^D} \right) &= MC \\ P_1 &= \left(\frac{E_1^D}{1 + E_1^D} \right) \times MC \end{aligned}$$

Likewise,

$$P_2 = \left(\frac{E_2^D}{1 + E_2^D} \right) \times MC$$

Now, we can compute the ratio of these prices:

$$\begin{aligned} \frac{P_1}{P_2} &= \frac{[E_1^D / (1 + E_1^D)] \times MC}{[E_2^D / (1 + E_2^D)] \times MC} \\ &= \frac{[E_1^D / (1 + E_1^D)]}{[E_2^D / (1 + E_2^D)]} \end{aligned}$$

Ways of segment customers

- **By customer characteristics:** such as age (e.g., senior citizen discounts) and gender (e.g., ladies night specials)
- **By past purchase behavior:** Repeat customers may be more price sensitive (e.g., software updates less expensive than initial purchase).

- **By location:** Based on local demand characteristics, prices are often different regionally (e.g., chain restaurants have higher prices in airports than other locations).
- **Over time:** Prices are higher when something is new as with games and hardcover books but, over time, the game will decrease in price and a paperback will be less expensive.

9.4 Indirect (Second-Degree) Price Discrimination

Definition(Indirect (Second-Degree) Price Discrimination): A pricing strategy in which customers pick among a variety of pricing choices offered by the firm.

Pre-requisite of indirect price discrimination

- The firm has market power and can prevent resale.
- The firm's customers have different demand curves.
- The firm cannot directly identify which customers have which type of demand before purchase.

Being able to directly observe a customer's elasticity before he buys (as required with direct price discrimination) is often difficult. A firm might know that its customers have different price sensitivities, but frequently can't tell which group any particular customer belongs to. Thus, we have the indirect price discrimination.

There are many different kinds of indirect price-discrimination techniques a company can use. The principle that underlies all of them, however, is the need to set up the pricing options and let the customers sort themselves into the "right" choice; that is, to purchase the option meant for their group rather than another option for a different group.

Definition(Incentive compatible): The requirement under an indirect price discrimination strategy that the price offered to each consumer group is chosen by that group.

9.4.1 Quantity discount

Definition(Quantity discount): A pricing strategy in which customers who buy larger quantities of a good pay a lower per-unit price.

Example: Suppose you are a pricing analyst for MegaDat Corporation, a firm that recently developed a new software program for data analysis. You have two types of clients who use your product. Type A's inverse demand for your software is $P = 120 - 10Q$, where Q represents users and P is in dollars per user. Type B's inverse demand is $P = 60 - 2Q$. Assume that your firm faces a constant marginal cost of \$20 per user to install and set up this software.

- If you can tell which type of buyer is buying the product before a purchase is made, what prices will you charge each type?
- Suppose instead that you cannot tell which type of buyer the client is until after the purchase. Suggest a possible way to use quantity discounts to have buyers self-select into the pricing scheme set up for them.
- Determine whether the pricing scheme you determined in part (b) is incentive compatible.

Solution: We will solve each question one by one:

- To maximize profit, set $MR = MC$ for each type. Therefore, we first need to solve for the marginal revenue curves for each type. Because we have linear inverse demand curves, we know that the MR curves will have the same vertical intercept but twice the slope. This means that $MR = 120 - 20Q$ for Type A buyers and $MR = 60 - 4Q$ for Type B buyers. Now set $MR = MC$ to find the profit maximizing quantity for each type:

For Type A:

$$\begin{aligned} 120 - 20Q_A &= 20 \\ 20Q_A &= 100 \\ Q_A^* &= 5 \end{aligned}$$

For Type B:

$$\begin{aligned} 60 - 4Q_B &= 20 \\ 4Q_B &= 40 \\ Q_B^* &= 10 \end{aligned}$$

At these quantities, the prices will be:

For Type A:

$$\begin{aligned} P_A^* &= 120 - 10Q_A^* \\ &= 120 - 10(5) \\ &= \$70 \end{aligned}$$

For Type B:

$$\begin{aligned} P_B^* &= 60 - 2Q_B^* \\ &= 60 - 2(10) \\ &= \$40 \end{aligned}$$

- The firm could charge \$70 per user for a package where the buyer can purchase any quantity she wishes and a price of \$40 for any buyer willing to purchase 10 or more units.
- This plan is incentive-compatible for Type B users. They are willing to continue to purchase $Q = 10$ at a price of \$40 each.

For a Type A consumer, we need to consider the amount of consumer surplus she receives under each scheme. We can do this with the help of a diagram showing the Type A demand curve and the two prices, \$70 and \$40. At a price of \$70, a Type A buyer would choose to purchase 5 units. Consumer surplus would equal area A , the area below the demand curve but above price.

If a Type A buyer were to opt to purchase the other package (10 units at a price of \$40 each), her consumer surplus would be the area above the price and below demand (area $A + B + C$), but she would also lose consumer surplus because she would be buying units that she values less than the price of \$40. This would be area L in the diagram.

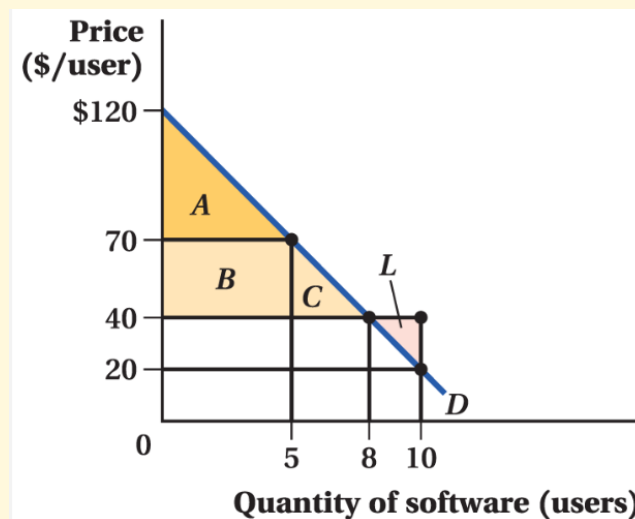


Figure 98: Type A surplus

Thus, opting for the quantity discount would change the Type A buyer's consumer surplus by area $B + \text{area } C - \text{area } L$. The \$40 10-unit package would be incentive compatible only if area $L > \text{area } B + \text{area } C$. Let's calculate those values:

$$\begin{aligned} \text{Area } B &= \text{base} \times \text{height} \\ &= (5)(\$70 - \$40) \\ &= (5)(\$30) \\ &= \$150 \end{aligned}$$

To calculate area C , we need to determine the base of the triangle. This means that we need to know the

quantity at which the Type A buyer's willingness to pay is exactly \$40 :

$$\begin{aligned}
 P &= 120 - 10Q \\
 40 &= 120 - 10Q \\
 10Q &= 80 \\
 Q &= 8 \\
 \text{Area } C &= \frac{1}{2} \times \text{base} \times \text{height} \\
 &= (0.5)(8 - 5)(\$70 - \$40) \\
 &= (0.5)(3)(\$30) \\
 &= \$45
 \end{aligned}$$

Therefore, area $B + \text{area } C = \$150 + \$45 = \$195$.

To calculate area L , we need to be able to determine the height of the triangle. To do so, we need the price at which a Type A buyer would be willing to purchase $Q = 10$ units:

$$\begin{aligned}
 P &= 120 - 10Q \\
 &= 120 - 10(10) \\
 &= 120 - 100 \\
 &= \$20 \\
 \text{Area } L &= \frac{1}{2} \times \text{base} \times \text{height} \\
 &= (0.5)(10 - 8)(\$40 - \$20) \\
 &= (0.5)(2)(\$20) \\
 &= \$20
 \end{aligned}$$

So, we know that area $B + \text{area } C = \$150 + \$45 = \$195$ and area $L = \$20$. Because area $A + \text{area } B > \text{area } L$, the \$40 10-unit pricing scheme is not incentive-compatible for Type A buyers. These buyers will want to receive the quantity discount and will purchase 10 units at a price of \$40 each. Thus, this pricing scheme would not be successful at making the buyers self-select into the pricing scheme established for their types.

9.4.2 Versioning

Definition (Versioning): A pricing strategy in which the firm offers different product options designed to attract different types of consumers.

To successfully implement versioning as a method of indirect price discrimination, it is not necessary for the marginal costs of the different product versions offered to consumers to be identical. The critical requirement is that the price markup should be higher for the product version targeted at the consumer segment exhibiting less price sensitivity (inelastic demand). This approach ensures higher revenue from those willing to pay more, regardless of the similarity in production costs between the versions.

Example: The table shows consumer valuations for two types of Callaway golf shoes.

Customer Type	C-Tech	Xtreme
Weekend Hacker	\$90	\$110
Club Champion	99	185

Suppose Callaway prices the C-Tech shoes at \$85 and the Xtreme shoes at \$125. Show that this pricing scheme is incentive-compatible.

Solution: The pricing scheme is incentive-compatible; customers purchase the version designed for their degree of price sensitivity. In particular, the weekend hacker will get more consumer surplus from the C-Tech shoes ($90 - 85$) than the Xtreme shoes ($110 - 125$). The club champion will get more consumer surplus from the Xtreme shoes ($185 - 125$) than the C-Tech shoes ($99 - 85$).

Example: With the same example above. Suppose Callaway prices the C-Tech shoes at \$70 and the Xtreme shoes at \$125. Show that this pricing scheme is NOT incentive-compatible.

Solution: The pricing scheme is also incentive-compatible. The price for the Xtreme version shoes is so low that the club champion will get more consumer surplus from the Xtreme version ($185 - 125$) than the lower-priced C-Tech version ($99 - 70$).

9.5 Bundling

Definition(Bundling): A pricing strategy in which the firm sells two or more products together at a single price. (p.

Pre-requisite of bundling

- A firm has market power and can prevent resale.
- A firm sells a second product and consumers' demand for that product is negatively correlated with their demand for the first product.

Example: Consider two consumers, Madison and Dakota, who are looking to subscribe to a cable service with two channels-ESPN and the truTV. First, consider positively correlated demand. (As demand for one channel increases across consumers, demand for the other increases as well.)

	ESPN	truTV	Bundle
Madison	\$9.00	\$1.00	\$10.00
Dakota	\$10.00	\$1.50	\$11.50

Without bundling, the company can charge a maximum of \$9 for ESPN and \$1 for the History Channel, for a total of \$20 in revenue, if they want both Madison and Dakota to subscribe.

With bundling, they can charge a maximum of \$10 for the bundle and get both customers.

No benefit to bundling with positive correlation.

Example: Now consider the same market, but with negatively correlated demand.

	ESPN	truTV	Bundle
Madison	\$9.00	\$1.50	\$10.50
Dakota	\$10.00	\$1.00	\$11.00

Once again, without bundling, the company can charge a maximum of \$9 for ESPN and \$1 for the History Channel, for a total of \$20 in revenue, if they want both Madison and Dakota to subscribe. However, with bundling, they can charge a maximum of \$10.50 for the bundle and keep both consumers, increasing total revenue to \$21 for the two consumers.

The reason why bundling works in the second scenario is the negative correlation between the two customers' willingness to pay. Dakota values one part of the bundle (ESPN) more than Madison, while Madison values truTV more than Dakota. If the cable company wants to sell to the entire market, it can only set a price equal to the smaller of the two customers' willingness to pay, whether pricing separately or as a bundle. In the first example with positively correlated demand (when Dakota had a higher willingness to pay for both channels), the lower of the customers' valuations for the bundle (\$10 per subscriber for Madison) is smaller by \$1.50 than the larger valuation (\$11.50 per month for Dakota) because it reflects Madison's lower valuations for both channels. Therefore, if the cable company wants to sell the channels as a bundle, it must offer Dakota a discount that embodies the fact that Madison has a lower willingness to pay for both channels. As a result, the cable company does no better than having sold the channels separately.

Definition(Mixed bundling): A type of bundling in which the firm simultaneously offers consumers the choice of buying two or more products separately or as a bundle.

Example: Consider a more complicate example with negative correlation valuation:

	ESPN (MC = \$6)	truTV (MC = \$1)	Bundle (MC = \$7)
Penny	\$12.00	\$0.50	\$12.50
Leonard	\$11.00	\$1.15	\$12.15
Raj	\$9.00	\$3.15	\$12.15
Sheldon	\$5.00	\$7.75	\$12.75

- Considering we use pure bundling. If the cable company only sells the bundle and for \$12.15 (the minimum valuation of the bundle across the customers), it will sell the bundle to all four customers. Subtracting costs, this will net a per-customer producer surplus of \$5.15 per month for a total of $(4 \times \$5.15)$, or \$20.60.
- Penny highly values ESPN but not truTV, and the opposite is true for Sheldon.

Selling only the bundle at \$12.15 ensures that all customers buy it, as it matches or exceeds their valuation for the combined channels, but this doesn't account for individual preferences and cost-effectiveness.

For Penny, supplying ESPN alone makes sense because her valuation for truTV (\$0.50) is below its cost (\$1). Similarly, Sheldon values ESPN (\$5.00) less than its cost (\$6), making it inefficient to supply him with ESPN.

Prices must be set so that buying a more preferred product individually is more appealing than buying a less preferred bundle, even if the bundle seems economically sensible at a glance. In Sheldon's case, setting the price of truTV alone at \$7.15 is critical because it keeps him from choosing the bundle (valued at \$12.75) that would give him more consumer surplus if truTV were priced at his full valuation of \$7.75. By pricing truTV at \$7.15, the company ensures Sheldon gains more surplus from buying just truTV than from purchasing the bundle. For Penny, pricing ESPN alone must be below the bundle price by at least her surplus from the bundle (35 cents), setting it at \$11.65.

By charging \$11.65 for ESPN to Penny and \$7.15 for truTV to Sheldon, and \$12.15 for the bundle to Leonard and Raj, the cable company can maximize its surplus.

Leonard and Raj would still opt for the bundle because their individual valuations for the channels do not justify buying them separately at the set prices.

Total producer surplus from this strategy would be \$22.10, higher than the \$20.60 from only selling the bundle, demonstrating the efficiency of mixed bundling.

Example: Fit Club, Inc. is a health club that offers two types of equipment: weight machines and a swimming pool. There are currently three customers (Abdul, Betty, and Chris), whose willingness to pay for using each type of equipment per month is listed in the table below:

	Willingness to Pay (per month)	
	WEIGHT MACHINES	INDOOR Pool
Abdul	\$60	\$50
Betty	\$50	\$125
Chris	\$25	\$140

The weight room and swimming pool each have a constant marginal cost of \$20 per month. In the case of the pool, the marginal cost is the price of the water and chemicals used, while the marginal cost of the weight machines is the cost of cleaning and maintaining them. Each customer is considering monthly access to each type of equipment, and the firm has to decide what type of membership package to offer the customers.

- What price will the firm charge for each product if it wishes to sell a health club membership to all three customers? What is the firm's producer surplus if it sells separate access to the weight room and the pool room at these prices?
- What price will the firm charge for a bundle of access to both the weight room and the swimming pool if it wishes to sell the bundle to all three customers? How much producer surplus will Fit Club, Inc. earn in this case?
- Suppose the firm is considering offering its customers a choice to either purchase access to the weight room and the swimming pool separately at a price of \$60 for the weight machine and \$140 for the pool, or to purchase a bundle at a price of \$175. Which option will each customer choose? How much producer surplus will Fit Club, Inc. earn in this situation?

Solution: We will solve each subquestion:

- To sell access to the weight machines to all three customers, the health club must charge a price no greater than \$25, the lowest willingness to pay of the customers (Chris). For the same reason, the price for the pool will be \$50.

At these prices, the firm's producer surplus for its sales of access to the weight machines will be

$$\begin{aligned}
 \text{Producer surplus for weight machine} &= (\text{price} - \text{marginal cost}) \times \text{quantity} \\
 &= (\$25 - \$20) \times 3 \\
 &= (\$5)(3) = \$15
 \end{aligned}$$

For access to the pool, producer surplus will be

$$\begin{aligned}
 \text{Producer surplus for the pool} &= (\$50 - \$20) \times 3 \\
 &= (\$30)(3) = \$90
 \end{aligned}$$

Total producer surplus will be $\$15 + \$90 = \$105$.

- To determine the price of the bundle, we need to calculate each buyer's willingness to pay for the bundle. This is done simply by summing the customers' willingness to pay for each product as shown in the table below:

	Willingness to Pay (per month)		
	WEIGHT MACHINES	INDOOR POOL	BUNDLE
Abdul	\$60	\$50	= \$60 + \$50 = \$110
Betty	\$50	\$125	= \$50 + \$125 = \$175
Chris	\$25	\$140	= \$25 + \$140 = \$165

So, the maximum price the health club can charge for its bundle (and still sell to all three buyers) is \$110. It will sell three bundles at this price. Therefore, its producer surplus will be

$$\begin{aligned}
 \text{Producer surplus for bundle} &= (\text{price} - \text{marginal cost}) \times \text{quantity} \\
 &= (\$110 - \$40) \times 3 \\
 &= (\$70)(3) = \$210
 \end{aligned}$$

- We need to compare each buyer's willingness to pay to the prices set for purchasing access to each room separately and the price of the bundle.

Abdul will only purchase a weight machine membership. His willingness to pay for the pool is below the price of \$140. The same is true for the bundle, which he values only at \$110. Therefore, the health club will only sell Abdul access to the weight machines.

Betty will not be willing to buy either membership separately, because her willingness to pay for each is below the set price. However, Betty's willingness to pay for the bundle (\$175) is exactly equal to the price, so she will purchase the bundle.

Chris will only purchase access to the indoor pool. His willingness to pay for weight machines is only \$25, far below the price of \$60. Likewise, Chris is willing to pay at most \$165 for the bundle. Thus, the health club will only be able to sell pool access to Chris.

Total producer surplus will therefore be

$$\begin{aligned}
 \text{Producer surplus for weight machines} &= (\text{price} - \text{marginal cost}) \times \text{quantity} \\
 &= (\$60 - \$20) \times 1 \\
 &= \$40 \\
 \text{Producer surplus for the pool} &= (\$140 - \$20) \times 1 \\
 &= \$120 \\
 \text{Producer surplus for bundle} &= (\$175 - \$40) \times 1 \\
 &= \$135
 \end{aligned}$$

Total producer surplus when the health club offers customers a choice of bundling or separate prices is $\$40 + \$120 + \$135 = \295 .

9.6 Advanced pricing strategy

Pre-requisite of advanced pricing strategy

- The firm has market power and can prevent resale.
- The firm's customers may have either identical or different demand curves.

9.6.1 Block pricing

Definition(Block pricing): The practice of reducing the price of a good when the customer buys more of it.

Example:

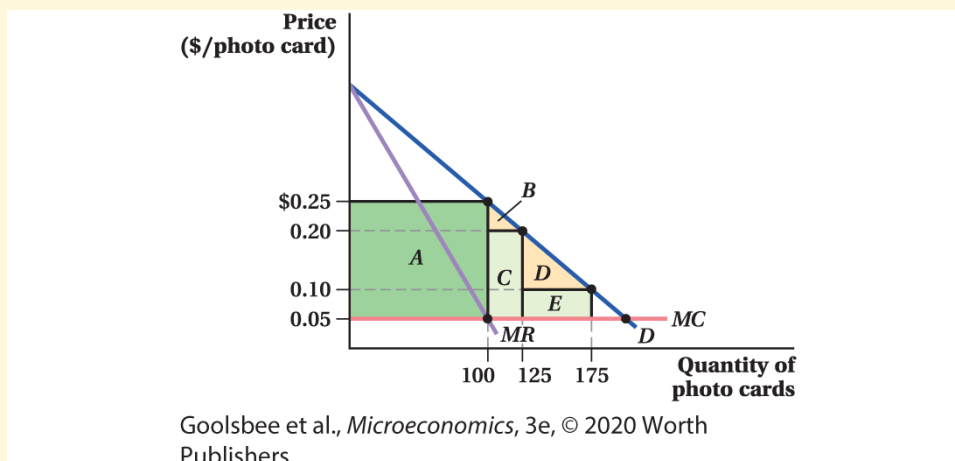


Figure 99: Selections of blocking price

D is the demand curve of an individual consumer of Walmart's photo cards. Under monopoly pricing, Walmart sells at the point on the demand curve corresponding to the quantity where $MR = MC$ ($Q^* = 100$ photo cards, $P^* = \$0.25$ per card). When Walmart can prevent resale, it can use a block pricing strategy instead. It could still sell the first 100 at a price of \$0.25 per card, while charging a lower price of \$0.20 each for the next 25 photos purchased (for a total quantity of 125 cards) and \$0.10 each for the next 50 cards (for a total of 175 cards). Producer surplus increases from area A to $A + C$ respectively, and consumer surplus increases by area D and area E respectively.

9.6.2 Two-part tariffs

Definition (Two-part Tariffs): A pricing strategy in which the payment has two components, a per-unit price and a fixed fee.

Example: Consider the market and a firm with market power. It has the demand for mobile phone service offered by the firm, the marginal revenue curve corresponding to demand, and the firm's constant marginal cost.

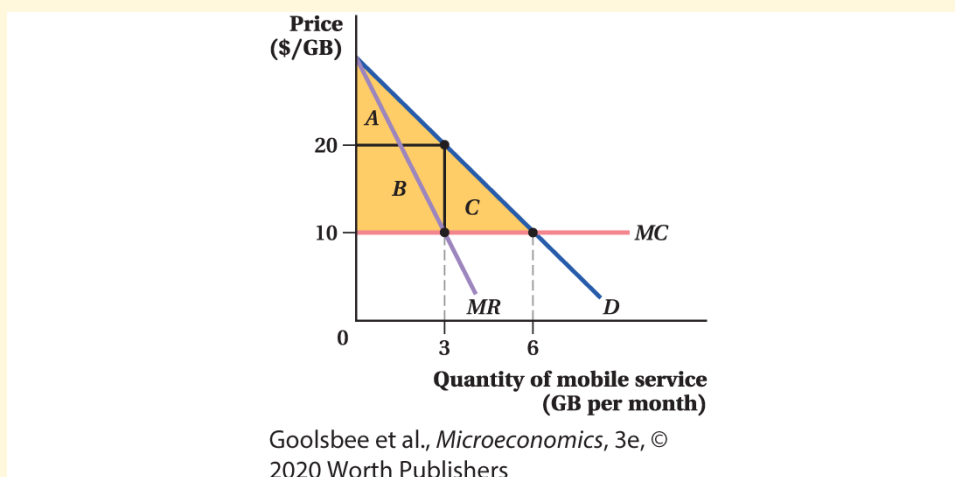


Figure 100: Two parts Tariffs surplus

As a single-price monopolist, a mobile phone service will sell 3 GB of mobile service per month at a price of \$20 per GB. Using a two-part tariff, however, the firm can increase its producer surplus from rectangle to the triangle. To do this, it will charge the per-unit price of \$10 per GB, where and set a fixed fee equal to the consumer's surplus at this quantity, the area A . Under this pricing scheme, the firm will sell 6 GB of mobile service per month.

Example: You have been hired as an intern at the Golden Eagle Country Club Golf Course. You have been assigned the task of creating the pricing scheme for the golf course, which typically charges an annual membership fee and a per-use cost to its customers. Each of your customers is estimated to have the following demand curve for rounds of golf per year:

If Golden Eagle can provide rounds of golf at a constant marginal cost of \$50 and charges that amount per round of golf, what is the most that members would be willing to pay for the annual membership fee?

Solution: This pricing scheme, with an annual membership fee and a per-unit price, is a two-part tariff. If the price per round of golf is set at then each member will want to play

With this knowledge, we can determine the maximum annual membership fee each customer is willing to pay. This will be equal to the amount of consumer surplus the customer will get from playing 50 rounds of golf each year at a price of \$50 per round.

To calculate consumer surplus, it is easiest to draw a diagram, plot the demand curve, and find the area of consumer surplus. To simplify matters, let's rearrange the demand function into an inverse demand function:

The vertical intercept is 60 and the consumer surplus is the area below the demand curve and above the price of \$50, area. We can calculate the area of triangle :

If the golf course set the price of a round of golf at \$50, the consumer would purchase 50 rounds per year. This gives the golfer a consumer surplus equal to \$250. Therefore, customers would be willing to pay up to \$250 for an annual membership.

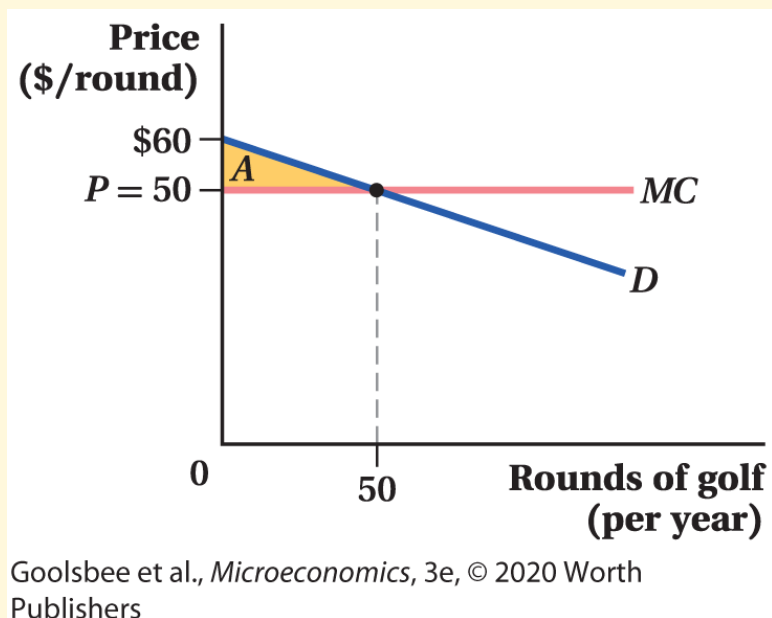


Figure 101: Surplus

10 Imperfect competition

In previous chapters, we studied the two ends of the market power spectrum: perfect competition and monopoly. Between these two ends of the spectrum are lots of industries that are neither perfectly competitive nor monopolistic.

Definition(Imperfect competition): The industry structure between perfect competition and monopoly.

Definition(Oligopoly): A market structure characterized by competition among a small number of firms.

Definition(Monopolistic competition): A type of imperfect competition where a large number of firms have some market power, but each makes zero economic profit in the long run.

10.1 Equilibrium in oligopoly

It is important to examine what equilibrium means in an oligopoly. Under perfect competition or monopoly, short-run equilibrium refers to a price-quantity combination that results in a market clearing. However, it is more complicated in oligopoly. In an oligopolistic industry, each company's actions influences what the other companies want to do. To determine an outcome when no firm wants to change its decision, we must determine more than just a price and quantity for the industry as a whole. This sort of equilibrium is called Nash equilibrium.

Definition(Nash equilibrium): An equilibrium in which each firm is doing its best conditional on the actions taken by other firms.

10.2 Oligopoly with Identical Goods: Collusion and Cartels

Definition(Collusion): Economic behavior in which all the firms in an oligopoly coordinate their production and pricing decisions to collectively act as a monopoly to gain monopoly profits to be split among themselves.

Definition(Cartel): The organization formed when firms collude.

Assumptions on this model

- Firms make identical products.
- Industry firms agree to coordinate their quantity and pricing decisions, and no firm deviates from the agreement even if breaking it is in the firm's best self-interest.

If the companies in an oligopoly can successfully collude, figuring out the oligopoly equilibrium is easy. The firms act collectively as a single monopolist would, and the industry equilibrium is the monopoly equilibrium (output is the level for which $MR = MC$ and the price is determined by the demand curve. However, this action is illegal in most countries. Collusion is very instable because people have incentive to cheat.

Example: Consider two firms, A and B, producing an identical product. Inverse demand is $P = 20 - Q$, and marginal cost is $MC = \$4$. If the firms collude, they will produce the monopoly output. Equate marginal revenue and marginal cost:

$$MR = 20 - 2Q = MC$$

$$20 - 2Q = 4$$

$$Q = 8$$

The monopoly price will be \$12, and total profits will be \$64. Assuming firms split production, each will produce 4 units, and each firm will earn \$32 in profit.

Example: With the same example, what happens if Firm A decides to produce 5 units instead of 4? Total production is 9 units instead of 8, and total industry profit will fall. Given inverse demand $P = 20 - Q$, the new price will be \$11, and total profits will be \$63. However, Firm A has increased individual profit:

$$\text{Profit}_A = (P - c) * Q_A$$

$$\text{Profit}_A = (11 - 4)5 = \$35$$

And Firm B has reduced profit:

$$\text{Profit}_B = (P - c) * Q_B$$

$$\text{Profit}_B = (11 - 4)4 = \$28$$

This incentive to cheat makes it difficult to maintain collusive agreements.

Thus, increasing the number of firms in the cartel also makes holding to the agreed production level more difficult. And having more firms in a cartel reduces the damages suffered by any one firm that continues to abide by the agreement because profit losses caused by cheating are spread out across more firms. Note that every firm in the cartel has an incentive to cheat, making it difficult to persuade any one firm to collude in the first place.

Ways to make collusion easier

- Making it easy to detect and punish cheaters.
- Little variation in marginal costs across producers; since the goal is to produce at lowest cost, it is difficult to share profits if production costs vary greatly across cartel members.
- Long time horizon makes defection more costly, as future monopoly profits are given more weight.

10.3 Oligopoly with Identical Goods: Bertrand Competition

Definition(Bertrand competition): Oligopoly model in which each firm chooses the price of its product.

Model assumptions

- Firms make identical products.
- Firms compete by choosing the price at which they sell their products.
- Firms set their prices simultaneously.

Suppose two firms, Target and Walmart, are selling Nintendo Switches.

- Products are identical; assume marginal cost is identical.
- Total quantity purchased is Q . Price at Walmart is P_W ; price at Target is P_T .

Demand for Switches at Walmart Demand for Switches at Target

$$\begin{aligned} Q, & \text{ if } P_W < P_T \\ \frac{Q}{2}, & \text{ if } P_W = P_T \\ 0, & \text{ if } P_W > P_T \end{aligned}$$

$$\begin{aligned} & \text{if } P_T < P_W \\ \frac{Q}{2}, & \text{ if } P_T = P_W \\ 0, & \text{ if } P_T > P_W \end{aligned}$$

The only way to sell Switches is to match or beat your competitor. What should Target do if Walmart lowers the price of the Nintendo Switch to less than Target's? Target is left with two options if it still wants to sell the Nintendo Switch.

- it can match Walmart, so that the market is shared equally.
- it can undercut Walmart, so that all consumers purchase from Target.

And vice versa. This incentive for undercutting would only stop once the price each store expects the other to charge falls to the level of the stores' marginal costs. At that point, cutting prices further would let a store gain the entire market, but that store would be selling every Switch at a loss. Thus, even though competition is imperfect, in Bertrand competition, market equilibrium is identical to perfect competition and price equals marginal cost.

Example: The inverse market demand curve for protocol droids is $P = 4,000 - 2Q$, where Q is the quantity of protocol droids and P is the market price. Protocol droids can be produced at a constant marginal cost of \$1,000, and all protocol droids are identical. Suppose the market for protocol droids is served by two firms that are engaged in Bertrand competition. What are the market output and price level?

Solution:

$$P = MC = \$1,000$$

To find the market output, plug \$1,000 into the inverse demand curve:

$$\begin{aligned} 1,000 &= 4,000 - 2Q \\ Q &= 1,500 \end{aligned}$$

10.4 Oligopoly with Identical Goods: Cournot Competition

Definition(Cournot Competition): Oligopoly model in which each firm chooses its production quantity.

Cournot competition

- Firms sell identical products.
- Firms compete by choosing a quantity to produce.

- All goods sell for the same price - the market price, which is determined by the sum of the quantities produced by all the firms in the market.
- Firms choose quantities simultaneously.

Example: Assume only two countries, **Saudi Arabia** and **Iran**, supply oil to the world. Each has a constant marginal cost of \$20 per barrel. Inverse demand is given by: $P = 200 - 3Q$; $Q = q_{SA} + q_I$. Solving for the equilibrium in this model is similar to the monopoly case, except Q is the sum of quantities. Rewriting the inverse demand curve,

$$P = 200 - 3Q = 200 - 3(q_{SA} + q_I)$$

$$P = 200 - 3q_{SA} - 3q_I$$

For Saudi Arabia: $MR_{SA} = 200 - 6q_{SA} - 3q_I$ because $TR_{SA} = P \times q_{SA}$, and $MR_{SA} = \frac{d}{dq_{SA}} (P \times q_{SA})$. Solving for Saudi Arabia's profit-maximizing output:

$$MR_{SA} = MC$$

$$200 - 6q_{SA} - 3q_I = 20$$

$$q_{SA} = 30 - 0.5q_I$$

Similarly, Iran's profit-maximizing output is:

$$q_I = 30 - 0.5q_{SA}$$

$q_{SA} = 30 - 0.5q_I$ and $q_I = 30 - 0.5q_{SA}$ are called Reaction curve for Saudi Arabia and Iran respectively. In Cournot competition, this is the firm's best production response to its competitor's possible quantity choices.

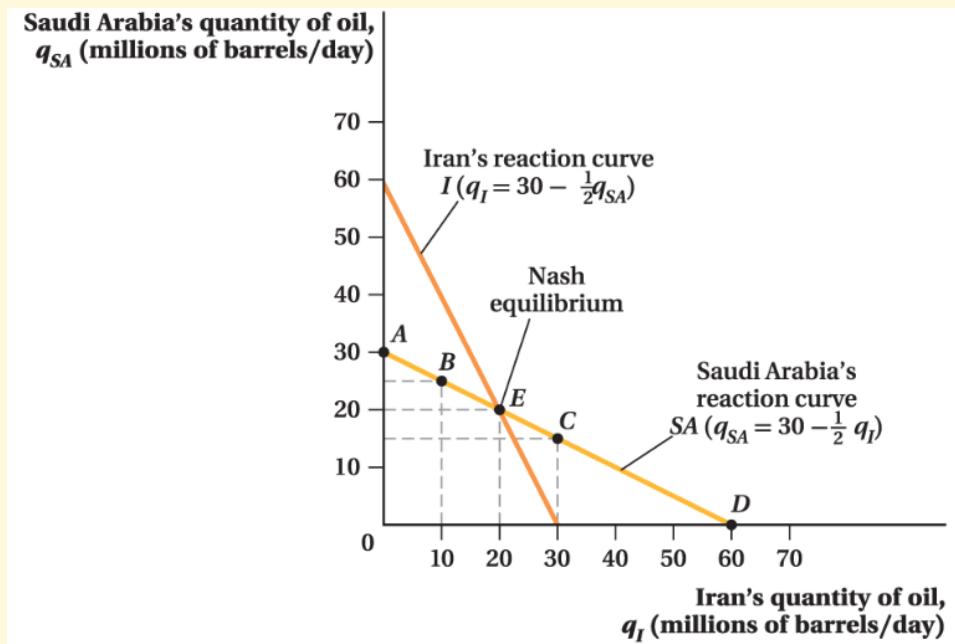


Figure 102: Nash equilibrium

$$q_{SA} = 30 - 0.5(30 - 0.5q_{SA}) = 30 - 15 + 0.25q_{SA}$$

$$q_{SA} = 20 \text{ million}$$

Saudi Arabia's equilibrium output is 20 million barrels per day. Since Iran and Saudi Arabia have identical production costs, Iran will also produce 20 million barrels per day, and the market price will be:

$$P = 200 - 3q_{SA} - 3q_I = 200 - 3(20) - 3(20) = \$80 \text{ per barrel}$$

Finally, we can compute the profit earned by Saudi Arabia:

$$\pi_{SA} = q_{SA} * (P - \$20) = 20 \text{ million} * (\$80 - \$20) = \$1.2 \text{ billion}$$

and Iran:

$$\pi_I = q_I * (P - \$20) = 20 \text{ million} * (\$80 - \$20) = \$1.2 \text{ billion}$$

Total output is 40 million barrels of oil per day, and total profit is \$2.4 billion.

Definition(Reaction curve): Here we will formally define reaction curve. A function that relates a firm's best response to its competitor's possible actions. In Cournot competition, this is the firm's best production response to its competitor's possible quantity choices.

Definition(Residue demand curve): In Cournot competition, the demand remaining for a firm's output given competitor firms' production quantities.

Definition(Marginal revenue residue curve): A marginal revenue curve corresponding to a residual demand curve.

Example: Using the same context as the previous example. If the Saudis expect Iran to produce 10 million barrels per day, they face the inverse demand curve:

$$P = 200 - 3q_{SA} - 3q_I = 200 - 3q_{SA} - 3(10) = 170 - 3q_{SA}$$

This leftover demand is the residual demand curve. And the residue marginal revenue curve would be $MR_{SA} = 170 - 6q_{SA}$. And the optimal quantity would be $MR_{SA} = MC_{SA} = 20 \Rightarrow q_{SA} = 25$ in response to the expectation of Iran producing 10 million barrels per day.

Remark: The approach of two firms in Cournot competition extends to the case of multiple firms.

10.4.1 Cournot compare to collusion and Bertrand oligopoly

We have the following result using the same example:

Oligopoly Structure	Total Output (million bpd)	Price (\$ per barrel)	Industry Profit (per day)
Collusion	30	\$110	\$2.7 billion
Bertrand (identical products)	60	20	0
Cournot	40	80	2.4 billion

To summarize then in terms of total industry output, the lowest is the collusive monopoly outcome, followed by Cournot, then Bertrand:

$$Q_m < Q_c < Q_b$$

The order is the opposite for prices, with Bertrand prices the lowest and the collusive price the highest:

$$P_b < P_c < P_m$$

Similarly, profit is lowest in the Bertrand case (at zero), highest under collusion, with Cournot in the middle:

$$\pi_b < \pi_c < \pi_m$$

Remark: In general, as the number of firms increases in Cournot, market outcomes still fall between the monopoly and perfectly competitive cases, but outcomes will approach the perfectly competitive case. And more competitors mean higher industry output, lower market price, and lower industry profit.

10.5 Oligopoly with Identical Goods but with a First-Mover: Stackelberg Competition

Definition(Stackelberg Competition): Oligopoly model in which firms make production decisions sequentially.

Definition(Frist-mover advantage): The advantage gained in a Stackelberg competition by the initial firm in setting its production quantity.

Definition(Stackelberg leader): The firm that moves first is called the Stackelberg leader.

Assumptions of Stackelberg competition

- Firms sell identical products.
- Firms compete by choosing a quantity to produce.
- All goods sell for the same price (which is determined by the sum total of quantities produced by all the firms combined).
- Firms do not choose quantities simultaneously. One firm chooses its quantity first. The next firm observes this and then chooses its quantity.

Example: Consider the same Saudi Arabia and Iran example. The market inverse demand for oil was $P = 200 - 3Q$, and both countries had a constant marginal cost of \$20 per barrel. Each firm produced where marginal revenue equaled marginal cost:

$$MR_{SA} = 200 - 6q_{SA} - 3q_I = 20$$

$$MR_I = 200 - 6q_I - 3q_{SA} = 20$$

In Cournot competition, we rearranged these equations to solve for each country's reaction curve:

$$q_{SA} = 30 - 0.5q_I$$

$$q_I = 30 - 0.5q_{SA}$$

Now suppose Saudi Arabia is a Stackelberg leader. Iran's incentives remain the same: for any quantity Saudi Arabia chooses to produce, Iran's reaction function describes the optimal response. Importantly, Saudi Arabia realizes Iran will do this before it makes its first move. This means we must substitute Iran's reaction curve into the inverse demand curve.

Substitute Iran's reaction curve into the inverse demand curve and solve for the optimal output for Saudi Arabia.

$$P = 200 - 3q_{SA} - 3q_I$$

$$P = 200 - 3q_{SA} - 3(30 - 0.5q_{SA})$$

$$P = 110 - 1.5q_{SA}$$

$$MR_{SA} = 110 - 3q_{SA} = 20$$

$$q_{SA} = 30$$

Plug this in to Iran's reaction function:

$$q_I = 30 - 0.5(30) = 15$$

Market price is $P = 200 - 3(45) = \$65$ per barrel, and profit for each country is:

Saudi Arabia	Iran
$\pi_{SA} = q_{SA}(P - 20) = 30(65 - 20)$	$\pi_I = q_I(P - 20) = 15(65 - 20)$
$\pi_{SA} = \$1,350,000,000/\text{day}$	$\pi_I = \$675,000,000/\text{day}$

Saudi Arabia makes slightly more (by \$150 million) than the Cournot equilibrium of \$1.2 billion per day as a result of holding first-mover advantage.

10.6 Oligopoly with Differentiated Goods: Bertrand Competition

Often a more realistic description of an industry is a set of firms that make similar but not identical products. We call this market differentiated goods market.

Definition(Differentiated goods market): A market in which multiple varieties of a common product type are available.

Assumptions of the models

- Firms do not sell identical products. They sell differentiated products, meaning consumers do not view them as perfect substitutes.
- Each firm chooses the price at which it sells its product.
- Firms set prices simultaneously.

Example: Suppose there are two snowboard manufacturers, Burton and K2. Their products are substitutes but not perfect substitutes in this differentiated goods market.

This product differentiation means that each firm faces its own demand curve, and each product's price has a different effect on the firm's demand curve. So:

<u>Burton</u>	<u>K2</u>
$q_B = 900 - 2p_B + p_K$	$q_K = 900 - 2p_K + p_B$

For simplicity, assume the marginal cost of producing snowboards is zero. Burton and K2 set their price so that marginal revenue is equal to its marginal costs. And we have the price reaction curve:

<u>Burton</u>	<u>K2</u>
$MR_B = 900 - 4p_B + p_K = 0$	$MR_K = 900 - 4p_K + p_B = 0$
$4p_B = 900 + p_K$	$4p_K = 900 + p_B$
$p_B = 225 + 0.25p_K$	$p_K = 225 + 0.25p_B$

To find the equilibrium prices, plug one company's reaction curve in to the other's:

$$\begin{aligned} p_B &= 225 + 0.25p_K \\ p_B &= 225 + 0.25(225 + 0.25p_B) \\ 0.9375p_B &= 281.25 \\ p_B &= \$300 \end{aligned}$$

Plugging this price in to K2's reaction curve yields K2's equilibrium price.

$$P_K = 225 + 0.25(300) = \$300$$

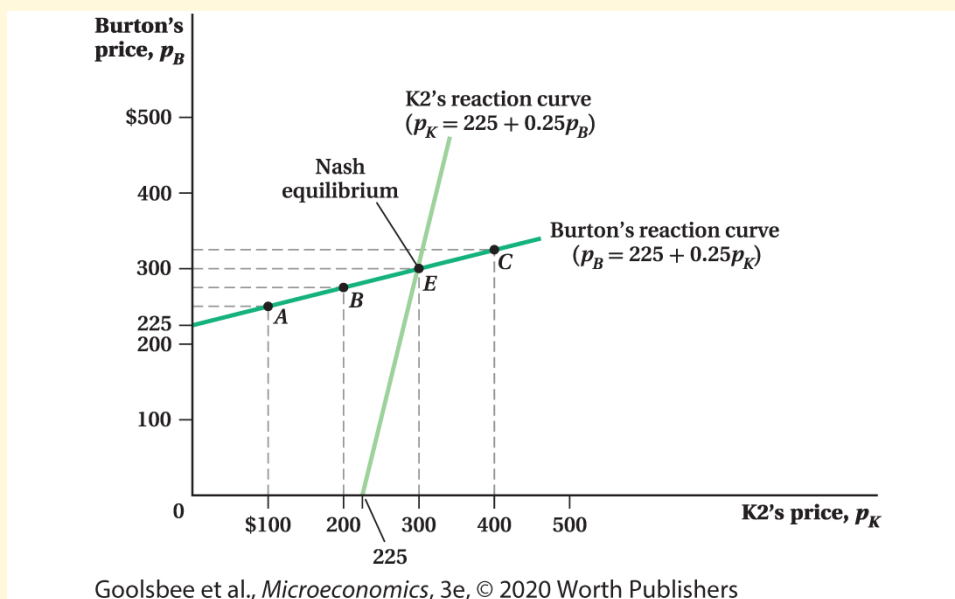


Figure 103: Nash equilibrium

10.7 Monopolistic Competition

Definition(Monopolistic competition): A market structure characterized by many firms selling a differentiated product with no barriers to entry.

Model assumptions

- Industry firms sell differentiated products that consumers do not view as perfect substitutes.
- Other firms' choices affect a firm's residual demand curve, but the firm ignores any strategic interactions between its own quantity or price choice and that of its competitors.
- There is free entry into the market.

Example: To understand how equilibrium is reached in a monopolistically competitive market, first examine how free entry affects noncompetitive market outcomes.

Consider a small town with a single fast-food burger restaurant, which is an effective monopolist.

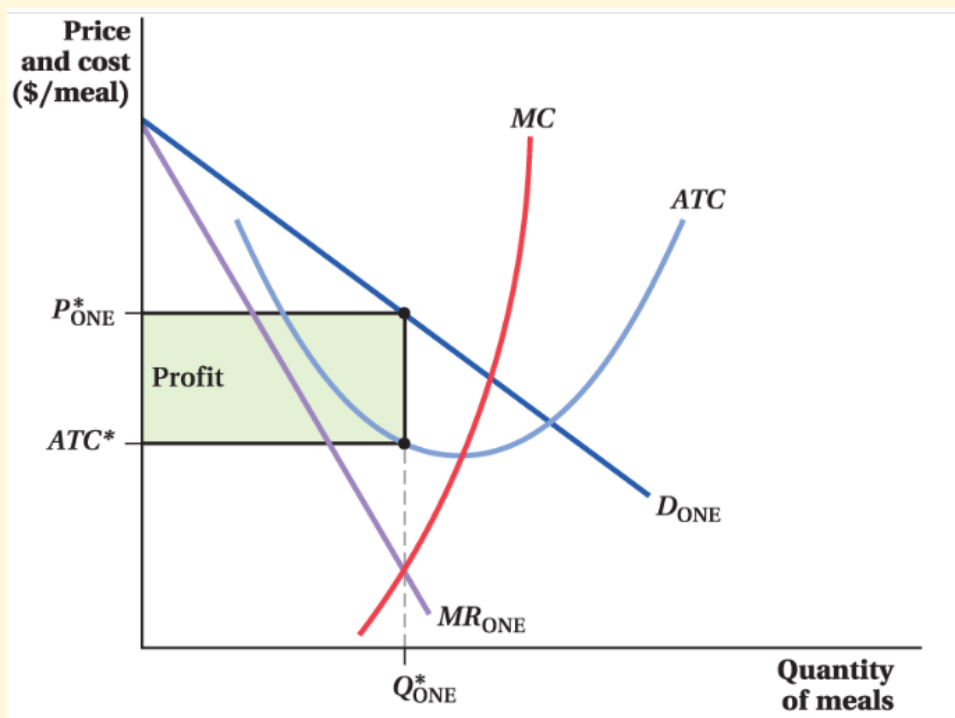
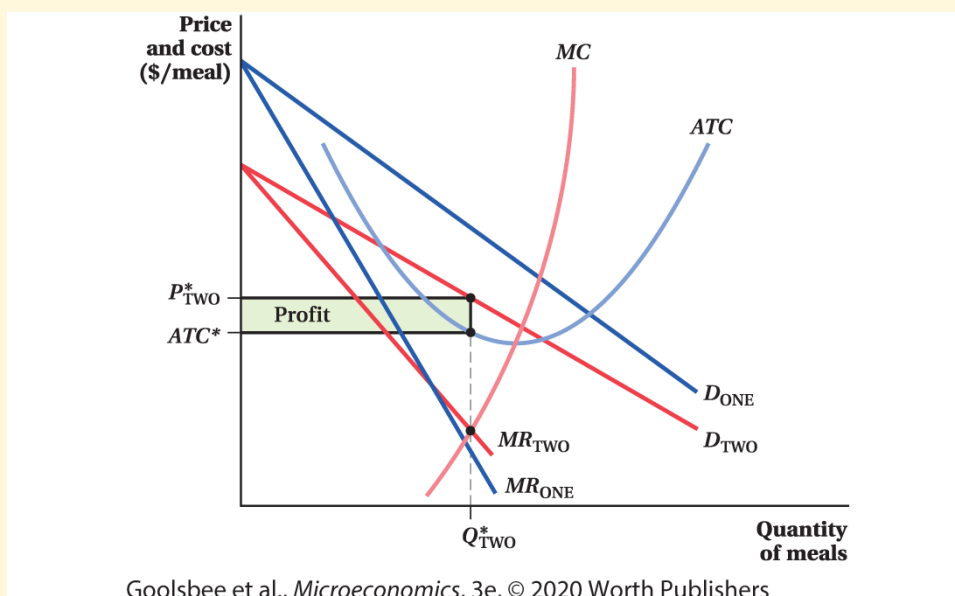


Figure 104: When there is only one restaurant

A monopolist restaurant has demand D_{ONE} , marginal revenue MR_{ONE} , average total cost ATC , and marginal cost MC . The restaurant produces where marginal revenue equals marginal cost, at quantity Q_{ONE}^* and average total cost ATC^* , multiplied by Q_{ONE}^* .

Now, suppose a second firm notices the profitability of operating a fast-food restaurant in this town since there is no barrier to entry.



Goolsbee et al., *Microeconomics*, 3e, © 2020 Worth Publishers

Figure 105: Now we have two restaurants

When a second restaurant enters the market, the original restaurant's demand curve shifts left from D_{ONE} to the more elastic residual demand curve D_{TWO} , and the marginal revenue curve MR_{ONE} shifts to MR_{TWO} . The restaurant now sells quantity Q_{TWO}^* at price P_{TWO}^* and earns profit represented by the shaded rectangle.

Just as with perfect competition, entry will continue to occur until economic profit is equal to zero. Entry will occur until demand is tangent with the average total cost curve.

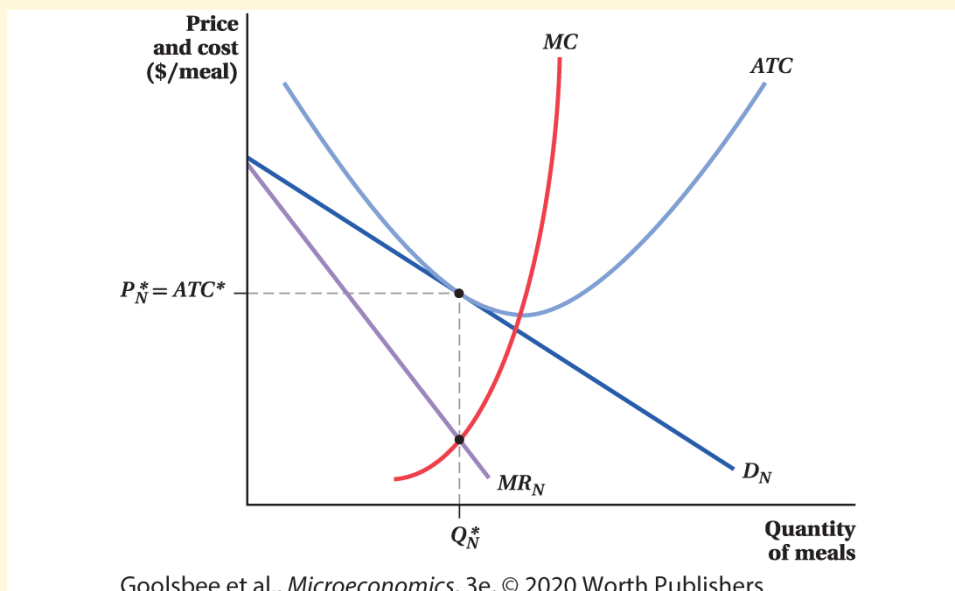


Figure 106: Now there is no more entry

In a monopolistically competitive market with N firms, firms face long-run demand D_N , marginal revenue MR_N , marginal costs MC , and average total cost ATC . At the long-run equilibrium, the firm's quantity is Q_N^* , price P_N^* is equal to average cost ATC^* , and each firm earns zero economic profit.

Example: Sticky Stuff produces cases of taffy in a monopolistically competitive market. The inverse demand curve for its product is $P = 50 - Q$, where Q is in thousands of cases per year and P is dollars per case. Sticky Stuff can produce each case of taffy at a constant marginal cost of \$10 per case and has no fixed cost. Its total cost curve is therefore $TC = 10Q$.

- To maximize profit, how many cases of taffy should Sticky Stuff produce each month? What price will Sticky Stuff charge for a case of taffy?
- How much profit will Sticky Stuff earn each year?
- In reality, firms in monopolistic competition generally face fixed costs in the short run. Given the information above, what would Sticky Stuff's fixed costs have to be in order for this industry to be in long-run equilibrium? Explain.

Solution:

- Sticky Stuff maximizes its profit by producing where $MR = MC$. Since the demand curve is linear, we know that the MR curve will be linear with twice the slope. Therefore, $MR = 50 - 2Q$. Setting $MR = MC$, we get

$$\begin{aligned} 50 - 2Q &= 10 \\ Q &= 20 \end{aligned}$$

Sticky Stuff should produce 20,000 cases of taffy each year.

- We can find the price Sticky Stuff will charge by substituting the quantity into the demand curve:

$$P = 50 - Q = 50 - 20 = \$30 \text{ per case}$$

- Total revenue for Sticky Stuff will be $TR = P \times Q = \$30 \times 20,000 = \$600,000$. Total cost will be $TC = 10Q = (10 \times 20,000) = \$200,000$. Therefore, Sticky Stuff will earn an annual profit of $\pi = TR - TC = \$600,000 - \$200,000 = \$400,000$.
- Long-run equilibrium occurs when firms have no incentive to enter or exit. Therefore, firms must be earning zero economic profit. From (c), we know that Sticky Stuff is earning a profit of \$400,000. In order for its profit to be zero, Sticky Stuff must face annual fixed costs equal to \$400,000.

11 Game theory

Definition (Game theory): The study of strategic interactions among two or more economic actors.

Definition (Strategic decisions): An action made based on the anticipation of others' actions. These actions are studied in game theory.

Definition (Simultaneous games): Games in which participants choose their actions at the same time without knowing their opponents' strategies.

Definition (Repeated game): A series in which the players play the same simultaneous game over and over.

Definition (Sequential game): Games where one player moves first and other players observe this action before making their decisions.

11.1 Definition of the game

Every game, no matter how simple or complex, shares three common elements: players, strategies, and payoffs.

Definition (Player): A participant in an economic game who must decide on actions based on the actions of others.

Definition (Strategy): A player's plan of action for a game.

Definition (Payoff): The outcomes the players receive from playing the game.

Definition (Optimal strategy): The action that has the highest expected payoff. Predicting behavior in games relies on finding the optimal strategy for the different players. This can be difficult because a particular strategy may be optimal for a player if her opponent chooses one action, but not optimal if her opponent chooses another action (this is another way of saying her optimal strategy must be a best response to the other player's chosen strategy).

Definition (Dominant strategy): A winning strategy for a player, regardless of her opponents' strategies. If there is a dominant strategy, all other strategies are dominated.

Definition (Dominated strategy): A losing strategy for a player, regardless of her opponents' strategies.

Definition (Payoff matrix): A table that lists the players, strategies, and payoffs of an economic game.

11.2 Nash Equilibrium in Simultaneous Games

When individuals or firms engaged in strategic competition have dominant strategies, determining equilibrium is easy. This is because each participant will naturally choose their dominant strategy, leading directly to an equilibrium state.

However, in most games, a Nash equilibrium only requires that an action is the best thing a player can do given the action the opponent happens to be taking. Therefore, a game can have a Nash equilibrium even if it has no dominated or dominant strategies.

Definition (Normal form): The common organization of an economic game into its players, strategies, and the payoffs in a payoff matrix.

The first step to finding a Nash equilibrium is to understand elements of the game using the normal form. Then we use The Check Method.

- Call our players Row and Column, and assume each has two actions available. What should Row do?
- If Column chooses action A, what is the best action for Row? Check off the payoff for that action.
- If Column chooses action B, what is the best action for Row? Check off that option.
- Repeat these steps for Column.
- If one of the boxes has two check marks, that is the Nash equilibrium.

Example: Consider the following example:

		Column	
		A	B
Row	A	✓ 100, 50	✓ 125, 100 ✓
	B	50, 100 ✓	75, 75

First, we examine Row's choices. If Column chooses action A, Row is better off choosing A ($100 > 50$). If Column chooses action B, Row is once again better off choosing action A ($125 > 75$). Therefore, we place check marks next to both of Row's payoffs from choosing A.

Next, we consider Column's choices. If Row chooses action A, Column is better off choosing B ($100 > 50$). If Row chooses action B, however, Column is better off choosing A ($100 > 75$). We place check marks next to these payoffs. Note that there is only one box in the payoff matrix in which two check marks appear: Row selecting A and Column choosing B. This is the Nash equilibrium of the game.

Using the checking method, the column or row a particular player never plays or plays all the time, it is the dominated strategy or dominant strategy.

11.2.1 Multiple equilibrium

Predicting outcomes of a game is relatively straightforward when there is only one Nash equilibrium. What if there is more than one? Considering the following example:

Example:

	Disney's Opening Date Choice: May	Disney's Opening Date Choice: December	Disney's Opening Date Choice: March
Warner Brothers' Opening Date Choice: May	100, 100	✓ 600, 400 ✓	✓ 600, 200
Warner Brothers' Opening Date Choice: December	✓ 400, 600 ✓	0, 0	400, 200
Warner Brothers' Opening Date Choice: March	200, 600 ✓	200, 400	−100, −100

Now we've identified each firm's best response to any possible strategy its competitor might pursue. Notice that a March opening is a dominated strategy for both players. That is, it is never optimal to open in March regardless of what the competitor might do. So, we can ignore any outcomes that involve either firm opening in March. This simplifies the game:

	Disney's Opening Date Choice: May	Disney's Opening Date Choice: December
Warner Brothers' Opening Date Choice: May	100, 100	✓ 600, 400 ✓
Warner Brothers' Opening Date Choice: December	✓ 400, 600 ✓	0, 0

Neither firm has an incentive to deviate from these two release date combinations. With the tools presented thus far, there is no way to determine which equilibrium occurs.

11.2.2 Mixed strategies

Definition(Pure strategy): A strategy in which the player chooses a particular action with certainty.

So far, we are only dealing with games that has a pure strategy. But, sometimes, there are other ways. And thus we introduce mixed strategies.

Definition(Mixed strategy): A strategy in which the player randomizes her actions according to a probability distribution.

Essentially, instead of consistently selecting one particular action every time the game is played (a pure strategy), the player randomly selects an action based on predetermined probabilities.

Example: Consider the case of a player taking a penalty kick in soccer. The payoff matrix for this game:

	Goalie: Left	Goalie: Right
Kicker: Left	0, 1 ✓	✓ 1, 0
Kicker: Right	✓ 1, 0	0, 1 ✓

There is no single strategy combination has two checks. So, there is no pure strategy equilibrium. Here, however, we will identify a mixed-strategy equilibrium by allowing for probabilistic selection of strategies.

Suppose the probability that the Goalie chooses "Left" is p . Then the probability that the Goalie chooses "Right" is $1 - p$.

- The Kicker's payoff for choosing "Left" is $0 \times p + 1 \times (1 - p) = 1 - p$.
- The Kicker's payoff for choosing "Right" is $1 \times p + 0 \times (1 - p) = p$.

To make the Kicker indifferent, set $1 - p = p$, leading to $p = 0.5$.

Suppose the probability that the Kicker chooses "Left" is q . Then the probability that the Kicker chooses "Right" is $1 - q$.

- The Goalie's payoff for choosing "Left" is $1 \times q + 0 \times (1 - q) = q$.
- The Goalie's payoff for choosing "Right" is $0 \times q + 1 \times (1 - q) = 1 - q$.

To make the Goalie indifferent, set $q = 1 - q$, leading to $q = 0.5$. Both players should randomize their choices, each selecting their strategies with a probability of 0.5. Hence, the mixed strategy Nash Equilibrium for both the Kicker and the Goalie is to choose each side with a probability of 0.5 :

- Kicker's Strategy: Kick Left with probability 0.5, Kick Right with probability 0.5.
- Goalie's Strategy: Dive Left with probability 0.5, Dive Right with probability 0.5.

Example: Consider the following game:

		Firm 2	
		Strategy A	Strategy B
Firm 1	Strategy A	8, 8	18, 18
	Strategy B	20, 20	15, 15

Find the mixed-strategy Nash equilibrium.

Solution: Let q be the probability that Firm 2 plays Strategy A.

- Firm 1's expected payoff for playing Strategy A is: $8q + 18(1 - q)$.
- Firm 1's expected payoff for playing Strategy B is: $20q + 15(1 - q)$.

For Firm 1 to be indifferent between playing Strategy A and Strategy B:

$$\begin{aligned}
 8q + 18(1 - q) &= 20q + 15(1 - q) \\
 8q + 18 - 18q &= 20q + 15 - 15q \\
 -10q + 18 &= 5q + 15 \\
 15q &= 3 \\
 q &= \frac{3}{15} = \frac{1}{5}
 \end{aligned}$$

Let p be the probability that Firm 1 plays Strategy A.

- Firm 2's expected payoff for playing Strategy A is: $8p + 20(1 - p)$.
- Firm 2's expected payoff for playing Strategy B is: $18p + 15(1 - p)$.

For Firm 2 to be indifferent between playing Strategy A and Strategy B:

$$\begin{aligned}
 8p + 20(1 - p) &= 18p + 15(1 - p) \\
 8p + 20 - 20p &= 18p + 15 - 15p \\
 -12p + 20 &= 3p + 15 \\
 15p &= 5 \\
 p &= \frac{5}{15} = \frac{1}{3}
 \end{aligned}$$

- Firm 1 plays Strategy A with probability $\frac{1}{3}$ and Strategy B with probability $\frac{2}{3}$.
- Firm 2 plays Strategy A with probability $\frac{1}{5}$ and Strategy B with probability $\frac{4}{5}$.

11.2.3 The Maximin Strategy

The idea of a Nash equilibrium is based on rational decision making by players. What if players make systematic errors? (In other words, they don't appear to be optimizing their own payoffs.) And a rational player facing less than rational opponents may find a more conservative strategy to be appealing.

Definition (Maximin strategy): A strategy in which the player minimizes her exposure to loss.

Example: Consider the payoff matrix:

	Firm B: High	Firm B: Low
Firm A: High	(10, 10)	(5, 15)
Firm A: Low	(15, 5)	(8, 8)

- Firm A will evaluate its worst-case outcomes for each of its strategies:
 - High Price: The worst outcome is if Firm A sets a high price and Firm B sets a low price, resulting in profits of 5 for Firm A.

- Low Price: The worst outcome is if both firms set low prices, resulting in profits of 8 for Firm A.

Therefore, Firm A's maximin strategy is to set a Low Price, ensuring a minimum profit of 8 .

- Firm B follows the same reasoning:

- High Price: The worst outcome is if Firm B sets a high price while Firm A sets a low price, resulting in profits of 5 for Firm B.
- Low Price: The worst outcome is if both firms set low prices, resulting in profits of 8 for Firm B.

Thus, Firm B's maximin strategy is also to set a Low Price, ensuring a minimum profit of 8 .

11.3 Repeated games

11.3.1 Finite repeated game

Definition(Backwards induction): The process of solving a multistep game by first solving the last step and then working backward.

Example: Suppose Kia and Hyundai are considering offering 10 -year full-coverage warranties for new cars. While offer-ing warranties is expensive, if one firm does offer a warranty, it will be very difficult for the other to compete without them. Assume the payoffs for this game are as follows (in millions of dollars per year).

	Offer	Don't offer
Offer	150,150	400,50
Don't offer	50,400	200,200

The Nash equilibrium occurs when both firms offer a warranty for game that only repeat once.

Suppose the game is repeated three times. There would be no change in the players' behavior. In the third period, both firms offer warranties because that is the Nash equilibrium. Knowing this, players will opt to offer warranties in both the first and second periods as well.

11.3.2 Infinitely repeated game

Infinitely repeated games sometimes sustain cooperation; however, they are more difficult to analyze.

Assume the two companies plan to continue to release Wonder Woman and Black Panther movies indefinitely. Warner Bros. will not advertise in the first period and will continue not to advertise as long as Disney does not advertise either; if Disney advertises, Warner Bros. will start advertising and continue to do so forever. And Disney does the same.

Consider the following single-period payoff matrix for the two firms. Payoffs are assumed to remain constant over time:

	Disney: Advertise	Disney: Don't Advertise
Warner Brothers: Advertise	250, 250	550, -80
Warner Brothers: Don't Advertise	-80, 550	320, 320

If both firms cooperate, the lower right box can be sustained, and periodic payoffs will be \$70 million greater for each firm than if both advertise. Consider Warner Bros.' payoffs under the two strategies of cooperating (no advertising) or defecting (advertising). Payoff from Defecting

$$550 + dx(250) + d^2x(250) + d^3x(250) \dots$$

Payoff from Cooperating

$$320 + dx(320) + d^2x(320) + d^3x320 \dots$$

If the payoff from cooperation is higher than the payoff from defection, cooperation (no advertising) can be sustained as a Nash equilibrium, or

$$320 + dx(320) + d^2x(320) + d^3x(320) + \dots > 550 + dx(250) + d^2x(250) + d^3x(250) \dots$$

Simplifying,

$$70x(d + d^2 + d^3 + \dots) > 230$$

$$(d + d^2 + d^3 + \dots) > 23/7$$

Using the mathematical identity for an infinite series when $0 \leq d < 1$ and substituting into the condition for a Nash equilibrium:

$$d + d^2 + d^3 = \dots = \frac{d}{(1-d)}$$

$$\frac{d}{(1-d)} > 23/7$$

$$d > 0.77$$

So long as Disney and Warner Bros. view a dollar earned in the next period to be worth at least as much as \$0.77 today, the cooperative Nash equilibrium (both choose Do not Advertise) can be sustained.

Definition(Grim trigger): A strategy in which cooperative play ends permanently when one player cheats. And it is the strategy we used in the example above.

Definition(Tit-for-tat): A strategy in which the player mimics her opponent's prior-period action in each round.

11.4 Sequential games

In many games, players make decisions one at a time. These types of games are called sequential games. One player moves first and other players observe this action before making their decisions. And we will use decision tree to represent this game.

Definition(Decision tree): Representation of a sequential game that shows both the choice and timing of players' actions. It is also called a extensive form.

Example: Two firms, GamesRUs (GRU) and PlayThings Incorporated (PTI), are considering a new television advertising campaign for their Christmas gift registries. Because television advertising is expensive, each firm earns greater profit (\$50 million) when it does not advertise. If both choose to advertise, very few new customers are gained and each firm earns only \$30 million profit. However, if one firm advertises while the other does not, the firm choosing to advertise gets the majority of customers, earning a profit of \$70 million, while its competitor only earns \$20 million profit.

If this game is played sequentially and GRU makes its decision before PTI, what will the outcome be?

Solution:

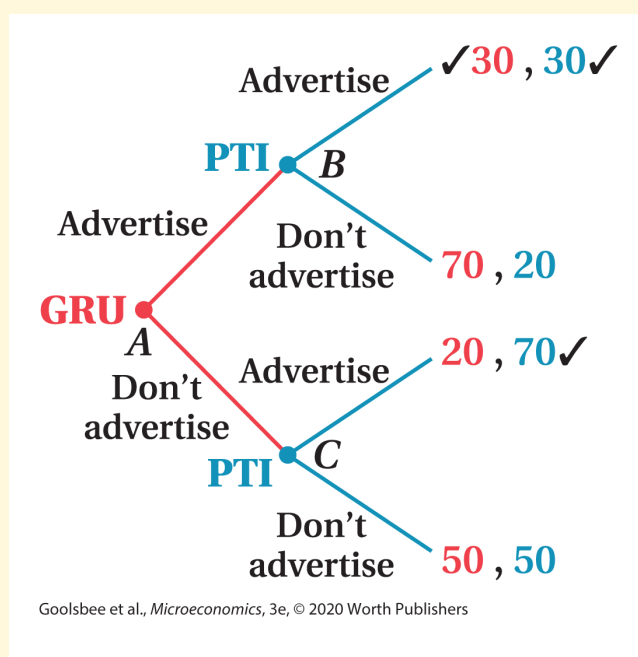


Figure 107: Decision tree

Using backward induction, we can see that, when GRU chooses to advertise, PTI will also choose to advertise (because \$30 million > \$20 million). If GRU chooses not to advertise, PTI will still choose to advertise (because \$70 million > \$50 million). Therefore, if GRU understands how PTI will respond to its strategy, it will also choose to advertise (because \$30 million > \$20 million).

Example: With the same example. Create a table showing the normal form of this game. List all Nash equilibria. Is there a first-mover advantage in this case? Explain.

Solution: We have the normal form

	ADVERTISE	DON'T ADVERTISE
ADVERTISE	30, 30	70, 20
DON'T ADVERTISE	20, 70	50, 50

	ADVERTISE	DON'T ADVERTISE
ADVERTISE	✓ 30, 30 ✓	✓ 70, 20
DON'T ADVERTISE	20, 70 ✓	50, 50

Because advertising is a dominant strategy for both firms, the outcome of this game is found in the upper-left portion of the payoff matrix and each firm earns a profit of \$30 million. This is the Nash equilibrium because neither firm has an incentive to change its strategy given the strategy of the other. And No first-mover advantage exists in this case because both firms have dominant strategies. A dominant strategy is the best strategy to follow no matter what your opponent does. Therefore, it is irrelevant if one firm makes its decision before the other firm; each firm will always choose its dominant strategy: to advertise.

Example: Consider another type of sequential game with a different structure. Suppose that Danica and Patrick are two contestants on a game show called "Play or Pass." They flip a coin to decide who goes first, and Danica wins. The rules are simple: Danica decides if she wants to play or pass. If Danica chooses "Play," the game ends and she receives a payoff of 1, and Patrick receives a payoff of -10. However, if Danica chooses "Pass," then Patrick gets a turn. If Patrick chooses "Play," he receives a payoff of 12 and Danica receives a payoff of 0. If Patrick chooses "Pass," however, both players receive a payoff of 11.

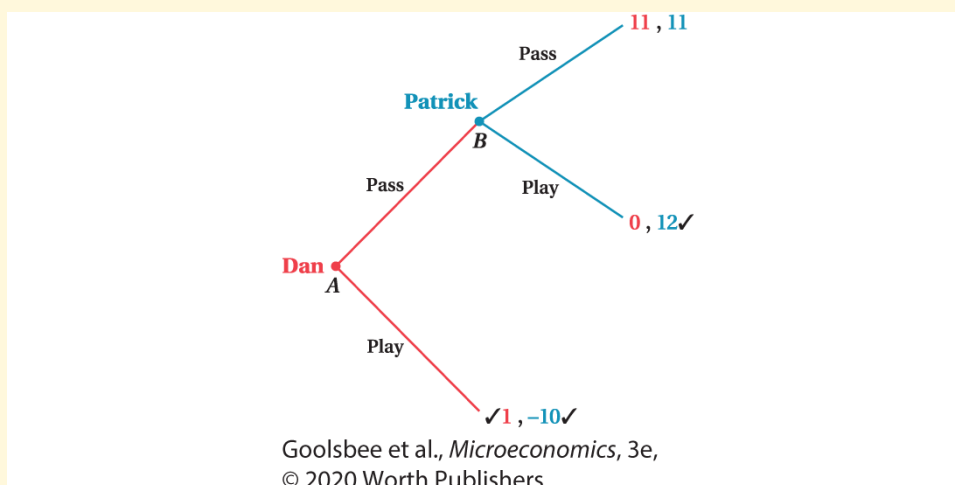


Figure 108: Play or pass game

If Danica chooses to play, the game ends. If she chooses to pass, Patrick moves. However, Patrick's best response is to play. Therefore, Danica will always choose to play.

11.5 Strategic Moves, Credibility, and Commitment

There's something not very satisfying about the equilibrium of the play or pass bargaining game. There was a big payoff for both players (11, 11) available later in the game, yet because each player followed his own self-interest, the equilibrium ended up with one player receiving something small (Danica earned a payoff of 1) and the other really getting hurt (Patrick lost 10). In this section, we discuss ways in which players can avoid such poor outcomes in certain kinds of sequential games.

Thus, the key is for player to use strategic moves.

Definition(Strategic moves): An action taken early in a game that favorably influences the ultimate outcome of the game.

11.5.1 Side payments

Definition(Side payments): A type of bribe that influences the outcome of a strategic game. It is a simple strategic moves.

Here we will return to the pass or play game before. Danica has no incentive to pass because Patrick will always play, leaving Danica with nothing. Danica might offer Patrick a deal to pass rather than play. Suppose Danica promises to pass and give Patrick a side payment if Patrick will pass, too. Take the payment to be 2, this works because:

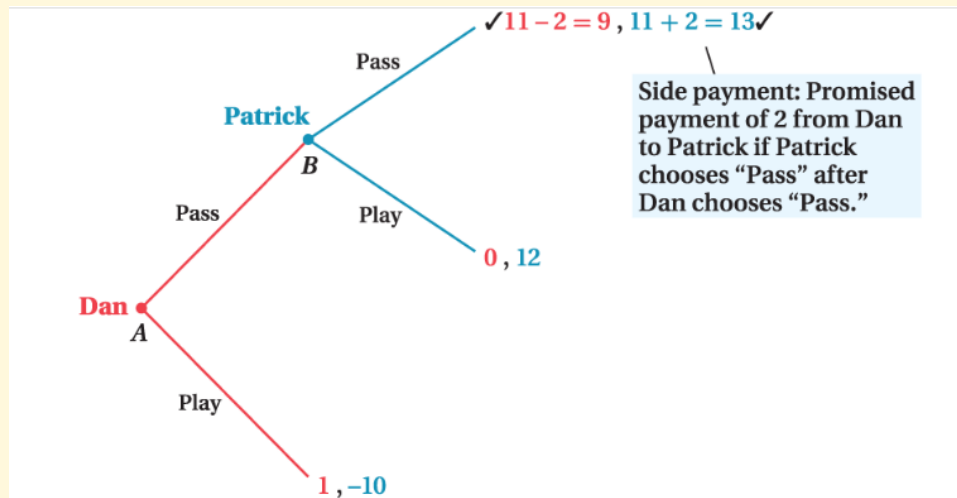


Figure 109: Side payment offered

Somewhat amazingly, just having the ability to make side payments can sometimes work, even if no money ever changes hands. Suppose, for example, that Patrick proposes the side payment this time. Before the game starts, he promises Danica that if he chooses "Play" at node *B*, he will give Danica a payment of 2 as a punishment to himself for hurting her. This promised punishment reduces Patrick's payoffs from "Play" at node *B* from 12 to 10 and changes his node *B* best option from "Play" to "Pass." Realizing that the side payment punishment makes "Pass" Patrick's best option, Danica now faces the following node *A* options: Choose "Play" and receive 1, or choose "Pass" and receive 11. Danica now chooses "Pass." Patrick responds by choosing "Pass" at node *B*, and both earn 11:

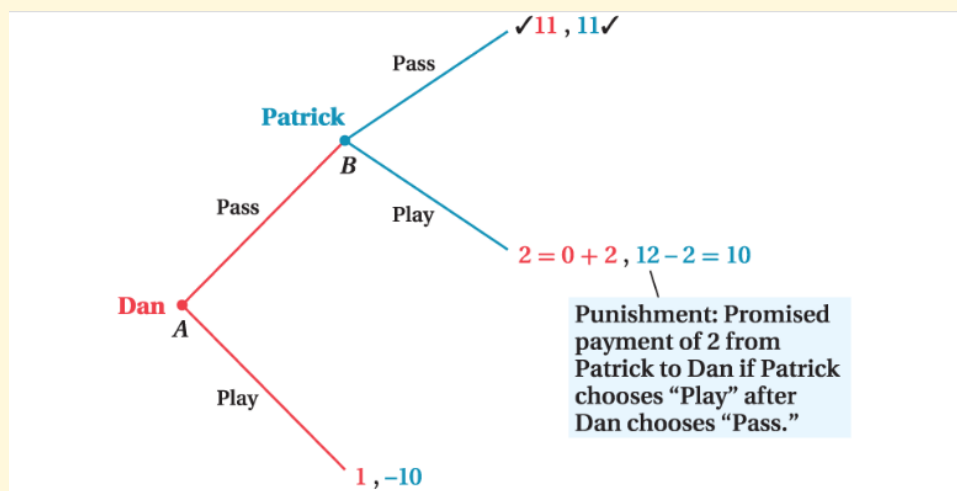


Figure 110: Side payment as punishment

11.5.2 Commitments

Side payments sound like a good idea, but in real life there may be incentives to renege.

Definition(Noncredible threats.): A threat made in a game that is not rational for the player to actually follow through on and, as such, is an empty threat.

Definition(Credible commitment): A choice (or a restriction of choices) that guarantees a player will take a particular future action if certain conditions occur that transform the game.

Example: MagicPill Inc. has developed a new wonder drug for curing obesity that has been approved by the Food and Drug Administration. If the drug is released for sale, a competitor, GenDrug, will attempt to copy the formula and steal all of MagicPill's customers by offering the wonder drug at a lower price. (Assume there are no patent laws at this time.) The extensive form of the game is shown below (payoffs represent profits in millions of dollars):

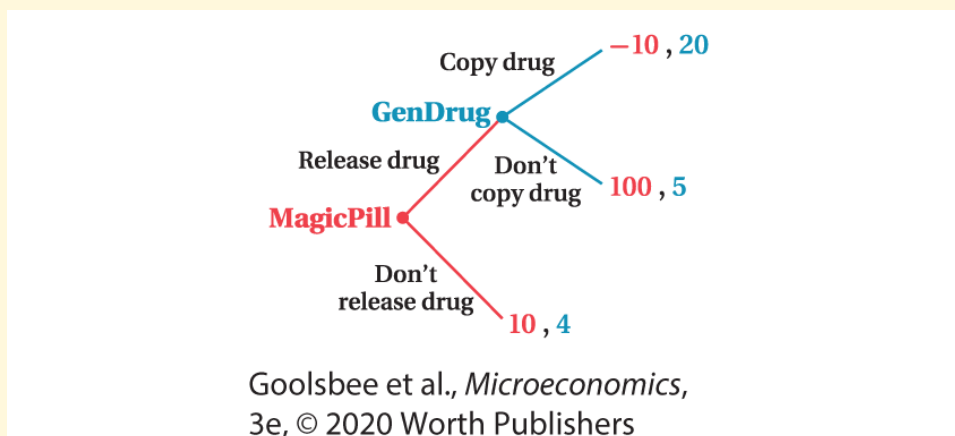


Figure 111: Extensive form

- Should MagicPill release this new wonder drug for sale? Explain.
- Would your answer to (a) change if GenDrug promised not to copy the new drug? Explain.
- Would your answer to (a) change if GenDrug signed a contract with MagicPill promising to pay \$10 million if it copies the drug? Explain.
- How would your answer to (a) change if patent laws protect MagicPill's exclusive right to produce its new wonder drug?

Solution:

- No. MagicPill will not release the drug. Using backward induction, we can see that if the drug is released, GenDrug will choose to copy it (because it can earn \$20 million profit rather than \$5 million). Knowing this, MagicPill is better off not releasing the drug (because it can earn \$10 million rather than losing \$10 million).
- No. GenDrug's promise would not be credible. The incentive (\$15 million additional profit) is large enough that MagicPill cannot believe the promise by GenDrug.
- No. The payment of \$10 million by GenDrug will not change GenDrug's incentive for copying the drug (\$20 million −\$10 million > \$5 million). Furthermore, the payment of \$10 million would not be enough to induce MagicPill to release the drug (−\$10 million +\$10 million < \$10 million).
- Yes. If the patent prohibited GenDrug from copying the wonder drug, we can ignore the "Copy Drug" option in the game. In this case, MagicPill will want to release the drug because \$100 million > \$10 million.

11.5.3 Entry Deterrence: Credibility Applied

A common application of strategic moves in microeconomics is firms deterring other firms from entering an industry.

Consider the market for tablet computers. Assume Apple is the only company making tablets but Samsung is thinking about entering the market.

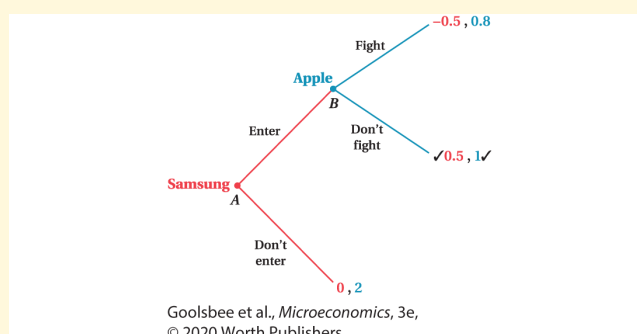


Figure 112: Samsung and Apple

Apple has promised to fight a price war if Samsung enters the tablet market. Is this a credible threat? If we look at the payouts for Apple, we can determine this is NOT a credible threat, as the payoff for not fighting is higher. Therefore, Samsung will choose to enter, and Apple will choose not to fight.

Rather than threatening just a price war, Apple could invest in excess capacity in anticipation of entry to lower its profits, putting itself in a better position for a price war. Apple has invested in excess capacity, lowering its cost of engaging in a price war. Is Apple's threat of a price war credible? Knowing this, Samsung will choose not to enter, and that will represent the Nash equilibrium.

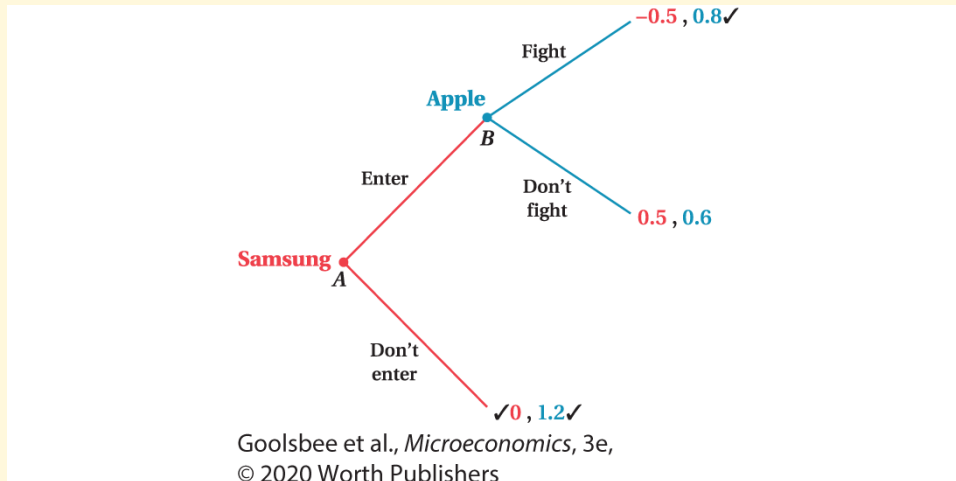


Figure 113: Apple invest in extra capital

12 Factor markets

Definition(Factor market): Markets for inputs such as labor, intermediate goods, and land (called factors of production) that are used to produce outputs.

In the previous chapters, we saw how firms choose the optimal mix of labor and capital inputs to produce their desired output, given the prices of labor and capital and the firm's production function. Now, we will examine how these prices are determined in factor markets.

12.1 Demand in a Perfectly Competitive Factor Market

The labor market is one of the most important factor markets. In most economies, about 60% to 70% of the money spent on inputs is paid to workers. The demand side of the labor market comprises all firms that would like to hire workers. And we examine the demand for labor for a single firm. Then we sum across all firms to obtain the market demand for labor.

Here, we assume all labor units, or workers, are the same. And firms choose the amount of labor to hire in the short run.

Definition(Marginal revenue product of labor MRP_L): $MRP_L = MP_L \times MR$, which is the marginal product of labor times the marginal revenue.

Firm's cost of hiring that additional unit of labor is the wage (set by the market). In a perfectly competitive market, the firm can hire as much labor as it would like at the market wage W .

Labor's marginal revenue product falls as more labor is used. This is because a firm's production exhibits diminishing marginal returns to labor (and capital). As production increases, the MP_L falls and the MR is falls or stay the same, causing the MRP_L to fall in both cases of market power and perfectly competitive market.

At relatively low levels of hiring, the MRP_L is high because the MP_L is large. When $MRP_L > W$, firms will want to hire more labor, as the benefit is greater than the cost. As hiring increases, the MP_L falls, causing the MRP_L to fall. A firm will not hire past the point where the cost (W) exceeds the benefit (MRP_L). It will not hire additional workers when $MRP_L < W$. Thus the optimal amount of labor is employed when $MRP_L = W$.

Definition(Choke wage): The wage at which a firm won't want to hire any labor.

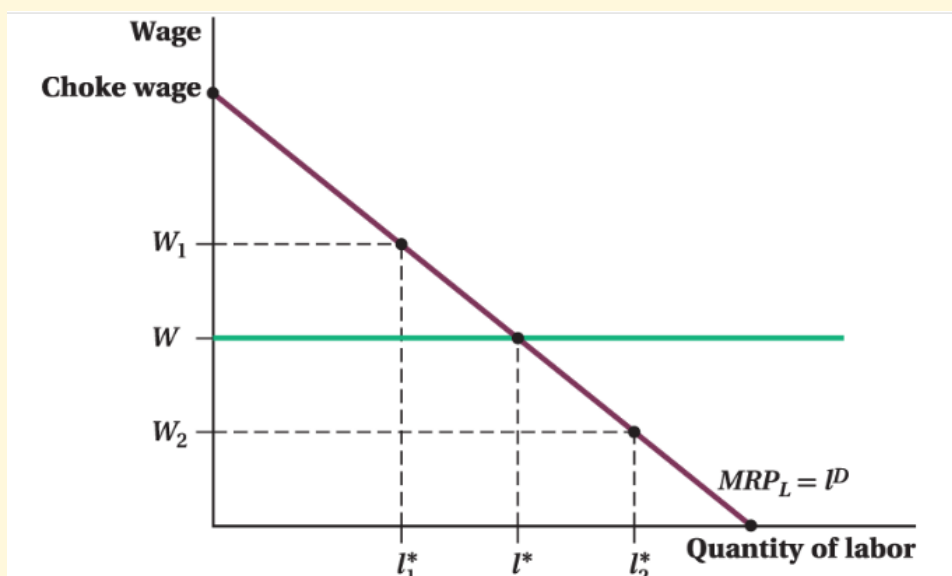


Figure 114: Optimal labor hired

Because the MRP_L curve shows the quantity of labor that a firm will hire at any given market wage, it is also the firm's labor demand curve. At market wage W , firm hires l^* units of labor. If the market wage were higher, like at W_1 , firm would hire l_1^* instead. At a lower wage like W_2 , firm hires a larger quantity, l_2^* .

There is another interpretation of the optimal hiring condition that can be seen if we break MRP_L into its two components, MP_L and MR :

$$MRP_L = MP_L \times MR = W$$

Rearranging this condition gives

$$MR = \frac{W}{MP_L}$$

Definition(Derived demand): A demand for one product (such as labor) that arises from the demand for another product (a firm's output). This also means labor demand is a derived demand.

Example: Victoria's Tours, Inc. is a travel company that offers guided tours of nearby mountain biking trails. Its marginal revenue product of labor is given by $MRP_L = 1,000 - 40l$, where l is the number of tour-guide-weeks it hires (a tour-guide-week is the amount of work done by one tour guide working one full-time week) and MRP_L is measured in dollars per tour-guide-week. The going market wage in the city where Victoria's Tours is located is \$600 per tour-guide-week.

- What is the optimal amount of labor for Victoria's Tours to hire?
- Above what market wage would Victoria's Tours not want to hire anyone?
- What is the most labor Victoria's Tours would ever hire given its current marginal revenue product ?

Solution:

- Victoria's Tours' optimal amount of labor is that which equates the marginal revenue product of labor and the wage:

$$\begin{aligned} MRP_L &= W \\ 1,000 - 40l &= 600 \\ 400 &= 40l \\ l^* &= 10 \end{aligned}$$

Victoria's Tours' optimal quantity of labor is 10 tour-guide-weeks.

- Victoria Tours will not want to hire any workers at all if the wage is higher than the marginal revenue product of the first unit of labor. That (choke) wage, which we label W_{choke} , is given by setting l in the MRP_L equation to zero:

$$\begin{aligned} W_{choke} &= 1,000 - 40(0) \\ W_{choke} &= 1,000 \end{aligned}$$

If the wage rises to or above \$1,000 per tour-guide-week, Victoria's Tours will not demand any labor.

- Given the MRP_L curve, there is an amount of labor at which the marginal revenue product becomes zero. Any more hiring, even if the wage is zero, will only reduce revenue. This maximum amount of hiring, l_{max} , sets $MRP_L = 0$:

$$\begin{aligned} 1,000 - 40l_{max} &= 0 \\ 1,000 &= 40l_{max} \\ l_{max} &= 25 \end{aligned}$$

The most labor Victoria's Tours would hire with its current MRP_L curve is 25 tour guide-weeks.

12.1.1 Shifts in the Firm's Labor Demand Curve

In general, forces that influence either the marginal product of labor or the marginal revenue will shift the labor demand curve (MRP_L).

- **Marginal Product of Labor:** Depends on both the production function and the firm's amount of capital. For example: An increase in total factor productivity (technical change) will shift the MPL outward, which would shift the labor demand curve outward.
- **Marginal Revenue:** Relationship between MR and shifts in the MRPL is the result of the labor demand being a derived demand. For example: If the demand for the firm's product falls, the price will fall, which will cause the marginal revenue to fall along with it.

12.1.2 Market labour demand curve

The market demand for labor is the horizontal sum of all the individual firms' labor demand curves. To compute it, we pick a wage level and add up all firms' quantity of labor demanded at that wage to get market quantity demanded at that wage. Then we repeat the process for every possible wage. The market labor demand curve must slope down because every individual firm's labor demand curve slopes down.

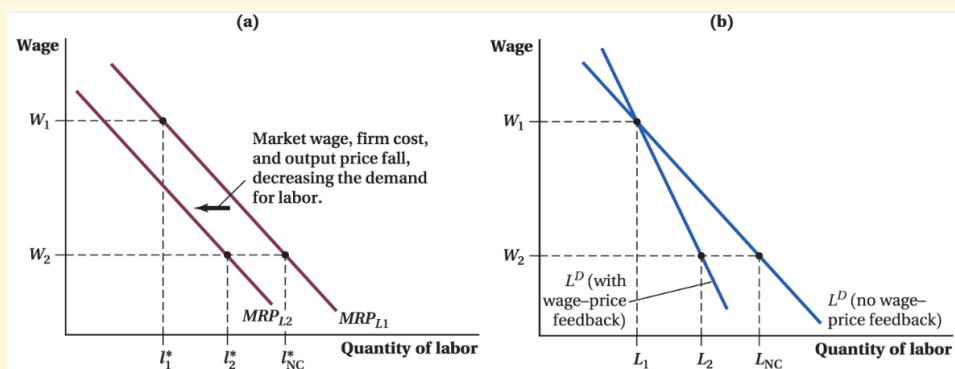


Figure 115: Market and individual labour demand

In panel a, which shows outcomes for an industry firm, the firm has a labor demand curve of MRP_{L1} at the initial market wage and output price. Given market wage W_1 , it hires l_1^* units of labor. A fall in the market wage from W_1 to W_2 reduces all industry firms' costs and thus the output price. The lower price reduces marginal revenue and shifts in the firm's labor demand curve to MRP_{L2} , implying that at W_2 the firm hires l_2^* instead of l_{NC}^* . At the industry level in panel b, which sums the responses of firms like the one shown in panel a, this feedback between the market wage and the output price implies that a wage drop from W_1 to W_2 increases the industry's quantity of labor demanded from L_1 to only L_2 instead of the no-feedback increase of L_1 to L_{NC} .

Therefore, firms' total response to a change in the market wage will be smaller in size than it would if there weren't a connection between the market wage and the output price.

13 Supply in a Perfectly Competitive Factor Market

In a perfectly competitive market, the firm can hire as much labor as it wants at the market wage. It faces a perfectly elastic supply of labor at the market wage. All shifts in the firm's labor demand show up as changes in quantity of labor demanded rather than as wage changes.

The supply of labor available to firms depends on the willingness of people to work. Willingness to work depends on age, family situation, health, and so on.

We will focus on the effect of the wage on the quantity of labor supplied while holding all other things constant. A key to grasping how wage affects people's labor supply is to understand how they think about leisure.

Assume people view leisure as a good, such as a shirt or a car. If leisure is a good, what is its price? Leisure takes time, so the price is what that person could be doing with that time. We assume that a person who isn't working is at leisure. By choosing leisure instead of work, they are giving up income, so the price of leisure can be thought of as an individual's wage.

Example: Assume Marissa can choose either work or leisure. The benefit of working is increased ability to buy and consume goods. By choosing leisure, she gives up that increase. We can think of her labor supply as a choice between consuming leisure and consuming the goods and services that can be bought with her income from work. In this example, we treat everything she can buy as one good, and call this consumption, with its quantity measured in dollars. The relative price between leisure and consumption is wage.

Marissa gets utility from both leisure and consumption. She can work only 24 hours in a day. If she doesn't work at all, she earns nothing and spends 24 hours at leisure. If she works constantly, she earns $W * 24$ and has no leisure. For each hour she works, she gives up W , so the slope of her budget constraint will be equal to $-W$.

Marissa maximizes her utility subject to a budget constraint for which the relative price of leisure is her wage.

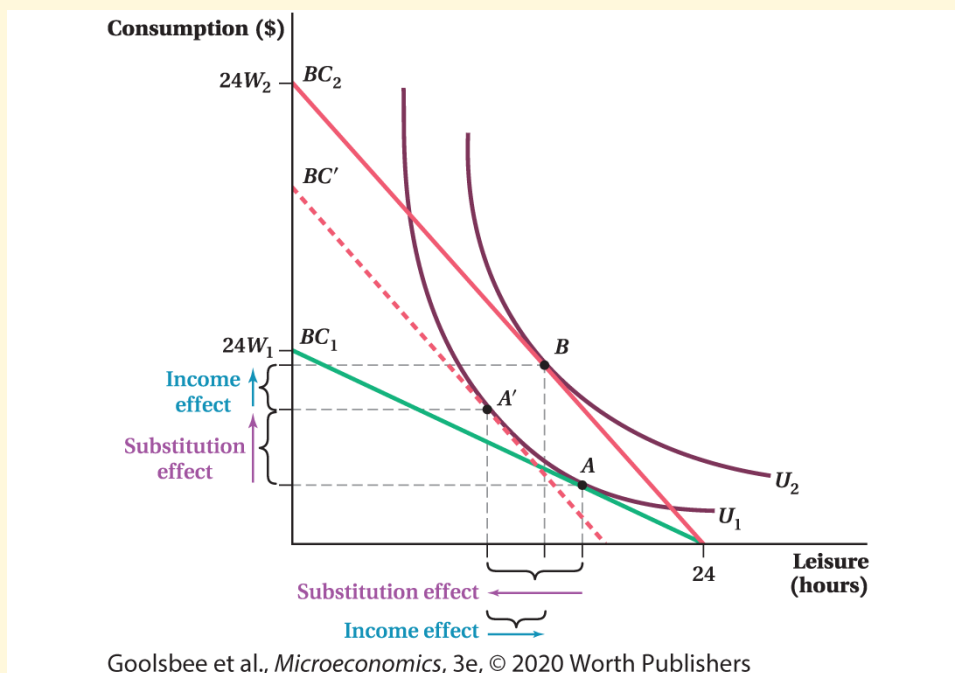


Figure 116: Substitution and income effect

An increase in the wage from W_1 to W_2 shifts the budget constraint from BC_1 to BC_2 , and the optimal consumption-leisure bundle changes from A to B . The substitution effect of the wage increase, which decreases leisure and increases consumption because leisure has become relatively more costly, accounts for the shift from bundle A to A' , the tangency of U_1 and BC' . The shift from A' to B is the income effect.

- **When wages increase:** The substitution effect tends to make leisure more costly in terms of foregone earnings, leading to more hours worked and less leisure. However, the income effect allows a person to feel wealthier, potentially leading them to enjoy more leisure since they can now afford it. The overall impact on labor supply (hours worked) and leisure depends on the relative strength of these two effects.
- **When wages decrease:** The substitution effect makes leisure cheaper (as the opportunity cost of leisure is lower), so individuals might choose more leisure over work. Conversely, the income effect from a wage decrease means overall lower income, potentially forcing the individual to work more hours to maintain their standard of living, despite the lower wage rate.

Therefore, if the income effect is large enough, you could see an individual work less as wage increases. Normally, we might expect that as wages increase, people will choose to work more hours because the reward for working (i.e., the wage) is higher. This is driven by the substitution effect, where labor is substituted for leisure because leisure becomes relatively more expensive. However, beyond a certain wage level, the labor supply curve can start to bend backward, meaning that as wages continue to rise, individuals might choose to work fewer hours. This happens when the income effect outweighs the substitution effect.

- When wages initially increase from a lower level, the substitution effect dominates. Since the opportunity cost of leisure is higher (every hour not worked means more money foregone), people are incentivized to work more.
- As wages continue to rise and people accumulate more income, their basic needs and some of their higher wants are increasingly met. They reach a level of financial comfort and security.
- At higher wage levels, the income effect can become stronger than the substitution effect. Here, individuals might feel that they have sufficient income to maintain their desired standard of living and start valuing leisure more. Essentially, they can afford to enjoy more leisure time without significantly compromising their lifestyle.
- As a result, at these higher wage rates, individuals may opt to work fewer hours despite the potential to earn more per hour. This choice reflects a preference for leisure over additional income, which causes the labor supply curve to bend backward.

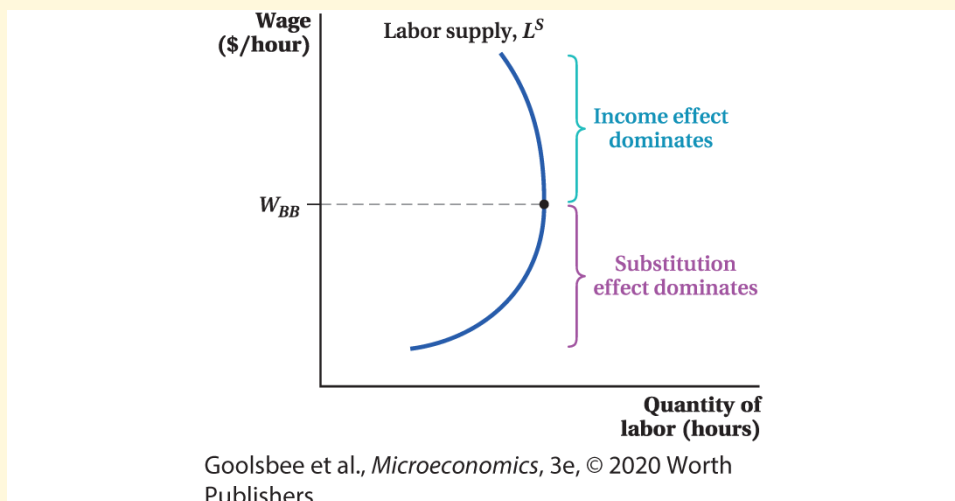


Figure 117: Backwards labor supply curve

13.0.1 Market level labor supply

The market-level supply curve shows the total amount of labor all workers are willing to supply at every possible wage. Horizontal sum of every worker's labor supply curve. To construct it, simply fix wage and add each worker's quantity of labor supplied at that wage. Once this is done for every possible wage, the market-level labor supply is the result. While some individuals can have a backward-bending portion of their labor supply curve, research suggests that, at the market level, the substitution effect dominates the income effect. Thus, the market labor supply curve is upward sloping.

13.1 Labor Market Equilibrium

The labor market reaches equilibrium when the wage equates the quantity of labor demanded by firms and the quantity of labor supplied by workers. Each firm takes the wage as given and hires labor until its marginal revenue product of labor equals wage. The MRP_L varies across firms because of differing production functions, amounts of capital, and demand for their output. The market equilibrium is the sum of all firms' hiring (L^*) at the equilibrium wage (W^*).

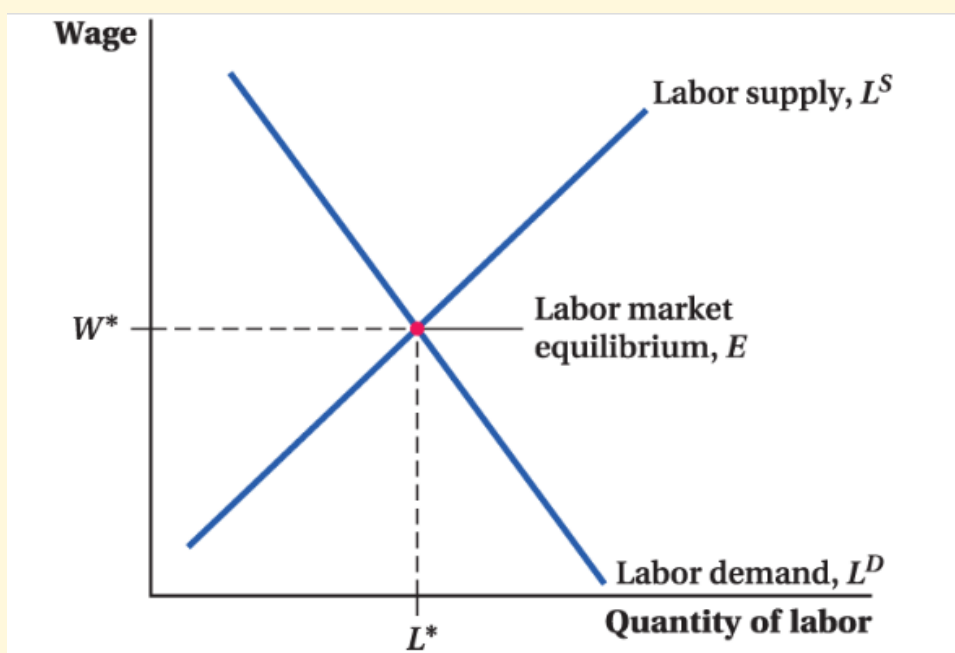


Figure 118: Labor market equilibrium

The labor market equilibrium wage, W^* , equates the market quantity demanded (as determined by the labor demand curve L^D) and the market quantity supplied (as determined by the labor supply curve L^S). The sum of all

firms' labor hiring when the wage is W^* will be L^* .

13.2 The Labor Market in the Long Run

Our labor market analysis up to this point has been for the short run when the firm's level of capital is fixed. Let's look at what happens when the firm can change its capital level.

Suppose there is a permanent drop in the market equilibrium wage.

- In the short run when capital is fixed: When the wage drops, the firm hires more labor until the MRP_L equals the lower wage.
- In the long run when we can change capital: The increase in labor from the short-run increase the marginal product of capital. The increase in the marginal product of capital gives the firm an incentive to employ more capital. This shifts the MRP_L outward, and the firm hires more labor.

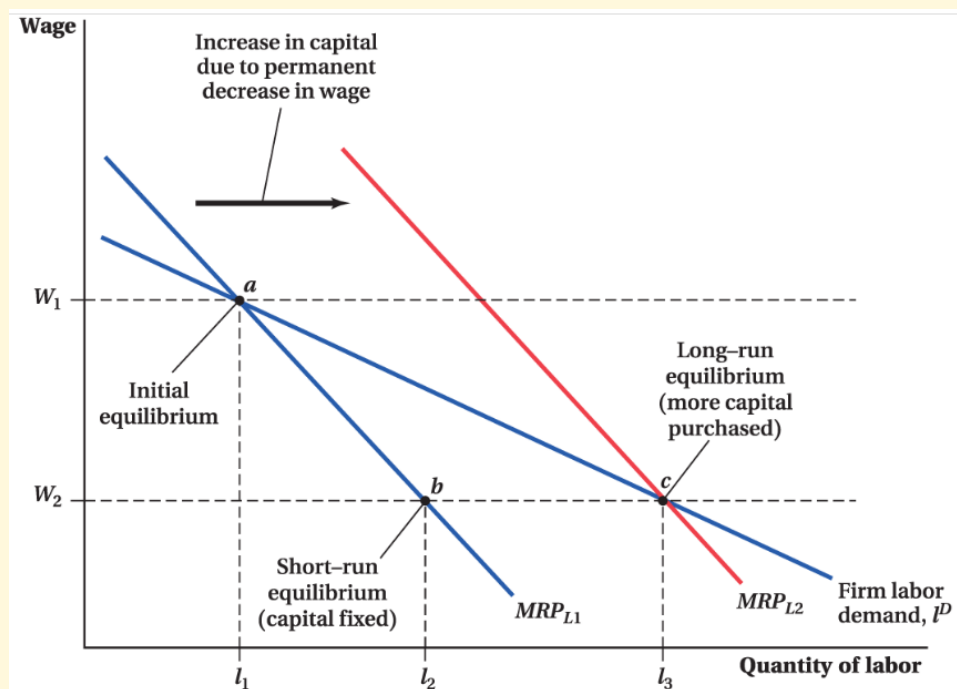


Figure 119: Short run and long run labor demand

A firm facing a market wage W_1 and having a short-run demand for labor given by MRP_{L1} initially hires labor quantity l_1 (point a). The short-run effect of a permanent drop in the market equilibrium wage from W_1 to W_2 will cause the firm to hire more labor until MRP_{L1} equals W_2 . This is labor quantity l_2 (point b). In the long run, the increase in labor hired to l_2 raises the firm's marginal product of capital, which leads the firm to purchase more capital and, in turn, raises the firm's marginal revenue product of labor. This shifts out the firm's short-run labor demand curve to MRP_{L2} , and the firm hires additional labor until $W_2 = MRP_{L2}$. This is labor quantity l_3 (point c). The long-run labor demand curve, l^D , runs through points a and c .

Long run demand curve is flatter than the short run because firms have these options in the long run, their demand for labor becomes more elastic. This means a change in wages leads to a proportionally larger change in the quantity of labor demanded compared to the short run. As a result, the long-run labor demand curve is flatter, indicating that firms are more responsive to changes in wage rates when they have more time to adjust their production processes and input usage.

13.3 Other Perfectly Competitive Factor Markets

We have looked specifically at labor markets as our example of factor markets. But there are other important factor markets, too. Capital is one of the more prominent of these.

13.3.1 Demand in other factor market

The demand for any factor is similar to that for labor.

- A firm's demand for any factor is given by the marginal revenue product (MRP_f) of that factor.
- A factor's MRP falls as more of it is bought.
- The optimal quantity of a factor is purchased by a firm such that its MRP equals the market price.
- Total market demand is the horizontal quantity of each firm's individual demand.

13.3.2 Supply in other factor market

The supply of a factor comes from producers or owners of those factors.

Definition(Intermediate goods): A product that is made specifically to be used as a factor input into the production of another good.

The supply of an intermediate good is just as for other products. As price increases, producers of the factor are willing and able to supply more of the good, and vice versa.

The equilibrium price is where quantity demanded equals quantity supplied.

13.3.3 Unimproved lands

Definition(Unimproved lands): The physical space on which economic activity (agriculture, housing, manufacturing, etc.) takes place.

The difference is that unlike most other factors, the supply of unimproved land is close to fixed. The land area of the earth is what it is (with sea levels rising, it may even be shrinking).

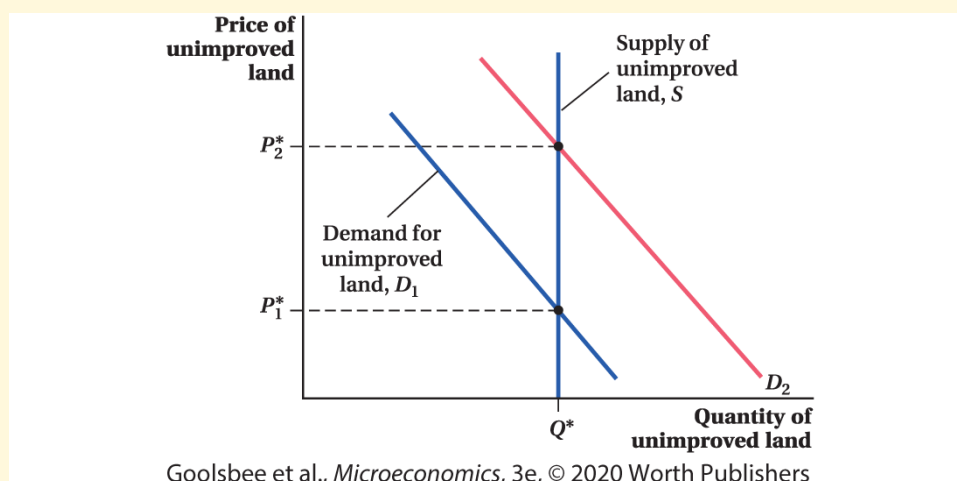


Figure 120: Supply and demand of unimproved lands

The supply of unimproved land is fixed at Q^* . Therefore, any changes in demand end up changing only the price of land. For example, an increase in the demand for unimproved land from D_1 to D_2 raises the equilibrium price of land from P_1^* to P_2^* with no change in quantity.

Note that the supply of **land for any particular use**, however, is not fixed. If the prices of crops rise, for example, you can expect people to begin to cultivate some land that wasn't formerly farmed. Likewise, if crop prices fall, some farmland will be transformed into other uses such as housing or industry. And, if the demand to live in an area rises, land that was once a farm or logging forest will be converted to housing, commercial, or industrial use.

13.4 Imperfectly Competitive Factor Markets: Monopsony, a Monopoly in Factor Demand

We've restricted our attention so far to perfectly competitive factor markets. What happens when factor demanders or suppliers are not price takers?

Definition(Monoposy power): When a buyer's choice of how much of a product to buy affects the market price of that product.

A monopsonist is a buyer of inputs with market power. Because it is such a large buyer, the monopsonist faces a dilemma in that it influences price as it demands more product. In other words, it faces an upward-sloping supply curve instead of horizontal supply curve.

Definition(Marginal expenditure): The incremental amount spent to buy one more unit of labor. This is constant and equal to the market price for firms without market power.

For monopsonists, demand is large enough to require the market price to rise as they purchase more. Marginal expenditure equals the price plus the incremental price it has to pay for all other previously purchased units.

Example: Initially, Syncrude hires 1,000 employees at \$50,000 each.

Syncrude's payroll = \$50 million

If it hired 1,100 employees, the added demand would push the wage up to \$60,000.

Syncrude's new payroll = \$66 million

Marginal expenditure of 100 workers = \$16 million

We can mathematically express marginal expenditure for a monopsonist as the price plus the amount that prices would change from increasing quantity times the quantity:

$$ME = P + (\Delta P / \Delta Q) \times Q$$

Notice that for a buyer who faces a flat supply curve - that is, a buyer who has no monopsony power and therefore takes the factor's purchase price as given - $\Delta P / \Delta Q$ is zero and $ME = P$, as we saw earlier.

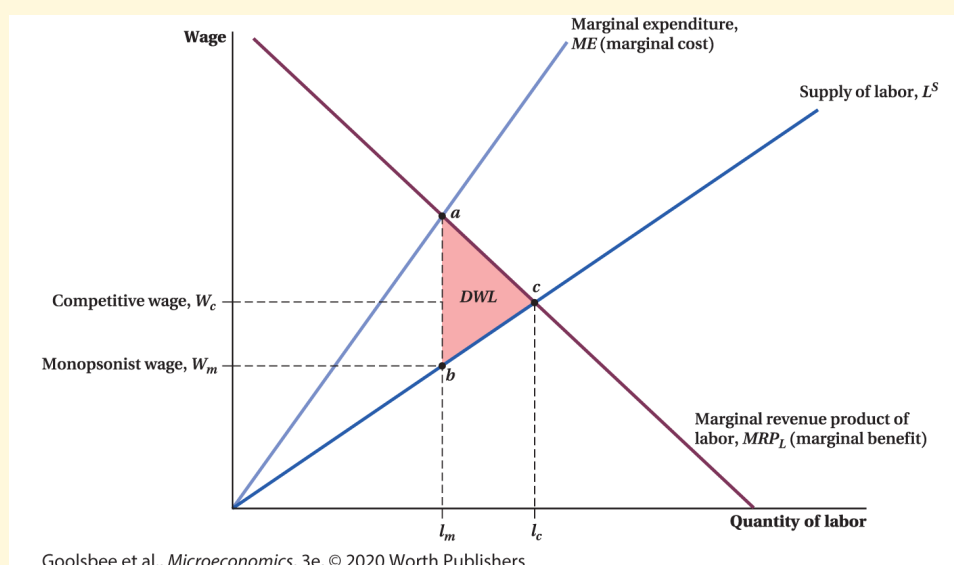


Figure 121: A Monopsonist's Hiring Decision

A monopsonist faces an upward-sloping supply curve L^S , implying that its marginal expenditure curve, ME , is above the supply curve. It hires labor quantity L_M at point a , where its marginal expenditure (its marginal cost of labor) equals its MRP_L (its marginal benefit of labor). It pays a wage W_m at point b , the height of the supply curve at that quantity of labor L_M . A set of perfectly competitive firms facing the same supply curve would hire the labor quantity L_C and pay a wage W_c . The deadweight loss of monopsony, area DWL , arises because there will be some workers who would work for wages that are lower than the marginal revenue product they create for the firm, but are not hired because doing so would raise overall wages too much. The deadweight loss is due to the labor that is not sold in the monopsonist market, which would be sold in the competitive market. This lost social surplus equals DWL , the area between the MRP_L and labor supply curves for labor units between L_M and L_C .

Example: Richland Uranium Mining operates a mine in a remote area. Because of its location, it has monopsony power in the labor market. Its marginal revenue product of labor curve is $MRP_L = 800 - 10L$, where L is the total number of miners it hires and MRP_L is measured in thousands of dollars per miner. The labor supply curve of local miners is $W = 10L - 100$, where W is the wage.

- What is Richland's marginal expenditure curve for labor?
- What quantity of labor does it want to hire, and what wage will it pay?
- What would the quantity of labor and wage be if Richland's marginal revenue product curve belonged to a perfectly competitive set of mines?

Solution:

- Because the labor supply curve is linear, we know the marginal expenditure curve has twice the slope of the labor supply curve:

$$ME = 20L - 100$$

- Richland will want to hire the quantity of labor that equates its marginal revenue product of labor with its marginal expenditure:

$$\begin{aligned}MRP_L &= ME \\800 - 10L &= 20L - 100 \\30L &= 900 \\L_m &= 30\end{aligned}$$

Richland's optimal amount of labor to hire is 30 miners.

To find the wage it must pay, we substitute the quantity of labor into the labor supply curve:

$$\begin{aligned}W_m &= 10L_m - 100 \\W &= 10(30) - 100 = 300 - 100 \\W_m &= 200\end{aligned}$$

Richland must pay a wage of \$200,000 per miner to hire 30 miners.

- A competitive market would hire labor until MRP_L equaled the wage from the labor supply curve. The competitive quantity of labor would therefore be

$$\begin{aligned}MRP_L &= W \\800 - 10L &= 10L - 100 \\20L &= 900 \\L_c &= 45\end{aligned}$$

A competitive set of mines would hire 45 miners. The wage can be found using the supply curve:

$$\begin{aligned}W_c &= 10L_c - 100 \\W &= 10(45) - 100 = 450 - 100 \\W_c &= 350\end{aligned}$$

The wage in a competitive market would be \$350,000 per miner.

13.5 Imperfectly Competitive Factor Markets: Monopoly in Factor Supply

These concepts apply to factor markets in which sellers have market power.

Example: Labor unions are the most studied holders of market power in factor markets. Unions coordinate labor supply actions of their members so that these workers act together as a monopoly seller of labor. The union's coordination gives it market power, so the demand for its members' labor is downward-sloping.

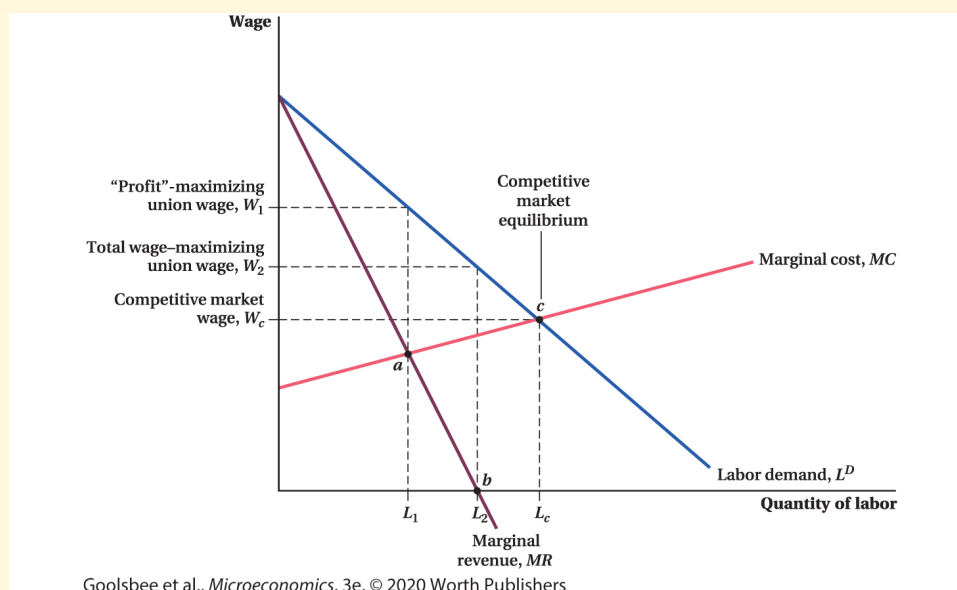


Figure 122: Union labor supply decision

If, like a typical profit-maximizing seller with market power, the union aims to maximize total "profit" for union members, it will supply the quantity of labor L_1 at point a , where the marginal cost curve and marginal revenue curve intersect. At this quantity, the wage will be W_1 , the wage indicated by the labor demand curve, L^D . If the union instead aims to maximize total wage earnings, it would sell at point b , where the union's marginal revenue equals 0. At this point, the quantity of labor supplied would be L_2 and the wage would equal W_2 . Point c represents the competitive market equilibrium, where the labor demand and marginal cost curves intersect. The competitive market wage is W_C and quantity of labor is L_C . Thus, regardless of the union's goal, it will supply a smaller quantity of labor to the market at a higher wage than in the competitive equilibrium.

13.6 Bilateral Monopoly

Definition(Bilateral Monopoly): The market structure that exists when there is major concentration of market power on both sides of a market.

Models of market power discussed previously assume that only one side of the market has pricing power. In this case, it is uncertain which model should be used to figure out what will happen in a factor market or what either party should do. Anything between a monopoly outcome and a monopsony outcome could be observed.

14 General equilibrium

Definition(Partial equilibrium analysis): Determination of the equilibrium in a particular market that assumes there are no cross-market spillovers.

Definition(General equilibrium analysis): The study of market behavior that accounts for cross-market influences and is concerned with conditions present when all markets are simultaneously in equilibrium.

Our analysis of markets, thus far, has been focused on individual markets; this is called partial equilibrium analysis. In this chapter, we introduce general equilibrium analysis.

14.1 General Equilibrium Effects in Action

General equilibrium analysis has two parts. One part describes the mechanics of market interactions and illustrates how various market features affect the size and direction of equilibrium effects in all these markets. This branch of general equilibrium analysis describes markets as they are. The other part asks whether economy-wide market equilibria are efficient or equitable (and how to define those terms).

We start with the first approach: how general equilibrium effects work in markets and what market features influence these mechanisms.

Since the oil crisis of the 1970 's, the U.S. government has subsidized the production of ethanol, and most ethanol is made from corn in the U.S. This mandate increased demand for corn, hence the price. However, the prices of wheat, rice, and soybeans also increased during this period. As demand for corn increases, the price increases. As the price increases, consumers demand more of substitute goods. For instance, wheat-based cereals versus corn-based. As demand for other commodities increases, the demand for corn increases further. These effects can be depicted graphically:

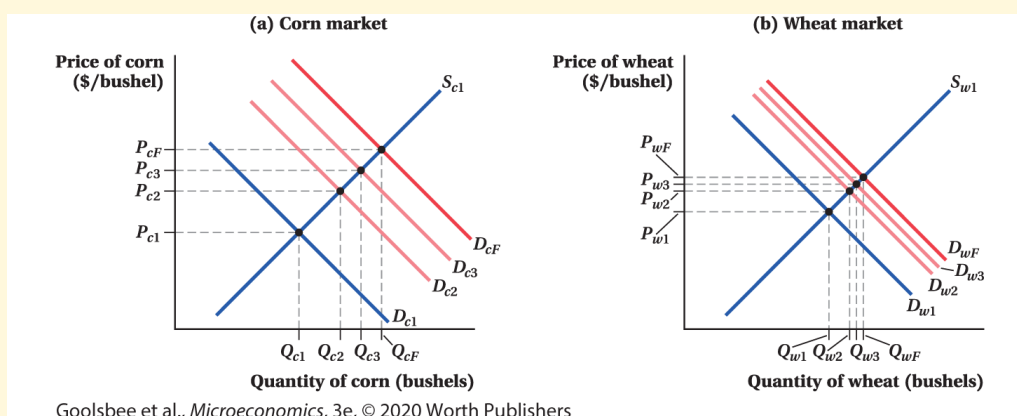


Figure 123: General Equilibrium Effects in Corn and Wheat Markets

- Before an ethanol-from-corn mandate, the corn market is in equilibrium at (Q_{c1}, P_{c1}) , where the initial demand curve D_{c1} and supply curve S_{c1} intersect. The direct effect of the mandate shifts demand out to D_{c2} . Because the mandate also increases the price of wheat, however, the demand for corn continues to shift out until general equilibrium is reached at (Q_{cF}, P_{cF}) , where D_{cF} intersects S_{c1} .
- Wheat, a substitute good for corn, is at an initial equilibrium of (Q_{w1}, P_{w1}) , where the initial demand curve D_{w1} and supply curve S_{w1} intersect. When the mandate increases corn prices, the demand for wheat increases to D_{w2} . The subsequent increases in corn prices continue to shift out the demand for wheat until general equilibrium is reached at (Q_{wF}, P_{wF}) , where D_{wF} intersects S_{w1} .

We can also analyze the supply side links of the industry. As demand for corn increases, the price increases. As the price of corn increases, farmers will shift land to corn production from other crops, reducing the supply of other commodities and increasing prices. As in the case of demand-side spillovers, this increase in price in other commodities will lead farmers to shift land and other resources back to those commodities. These effects can also be depicted graphically:

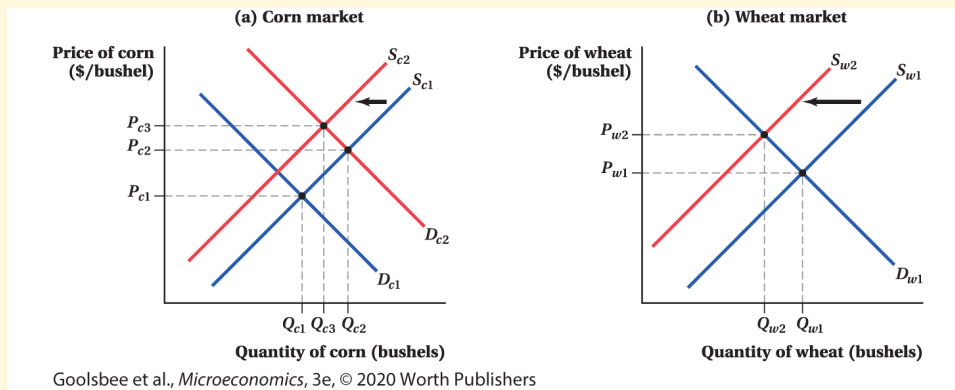


Figure 124: Supply-Side Input Links across Industries

- The increase in the demand for corn shifts demand from D_{c1} to D_{c2} . As a result, the quantity of corn increases from Q_{c1} to Q_{c2} , and the price of corn increases from P_{c1} to P_{c2} .
- The increased demand for corn causes farmers to shift some of their production from wheat to corn. Consequently, the supply of wheat shifts in from S_{w1} to S_{w2} . The quantity of wheat decreases from Q_{w1} to Q_{w2} , and the price of wheat increases from P_{w1} to P_{w2} . The resulting increase in the wheat price feeds back, in turn, to the corn market, shifting the supply of corn from S_{c1} to S_{c2} , reducing the quantity of corn to Q_{c3} and raising its price to P_{c3} , because of increases in the price of inputs into corn production.

Let's put some specific numbers on the types of processes discussed above to get a better feel for analyzing general equilibrium effects. For simplicity, assume there are only two goods in the economy, corn and wheat. Demand for corn is given by: $Q_c^d = 20 - P_c + P_w$, where P_c is the price of corn and P_w the price of wheat (each in dollars per bushel). Quantities are measured in millions of bushels. Demand for wheat is given by: $Q_w^d = 20 - P_w + P_c$. The supply of each is equal to price, $Q_c^s = P_c$; $Q_w^s = P_w$.

In a two-good general equilibrium market, we must solve for the pair of prices that clear the two markets. This means demand must equal supply in each market.

- First, consider the market for wheat:

$$\begin{aligned} Q_w^d &= Q_w^s \\ 20 - P_w + P_c &= P_w \\ P_w &= 10 + \frac{P_c}{2} \end{aligned}$$

- Doing the same for corn yields:

$$\begin{aligned} Q_c^d &= Q_c^s \\ 20 - P_c + P_w &= P_c \\ P_c &= 10 + \frac{P_w}{2} \end{aligned}$$

To solve for the equilibrium prices, substitute the equation for the wheat price:

$$\begin{aligned} P_c &= 10 + \frac{10 + \frac{P_c}{2}}{2} = 15 + \frac{P_c}{4} \\ \frac{3P_c}{4} &= 15 \\ P_c &= \$20 \end{aligned}$$

Substituting the price of corn into the price equation for wheat:

$$P_w = 10 + \frac{P_c}{2} = 10 + \frac{20}{2} = \$20$$

Using the supply or demand equation, plug in the prices:

$$\begin{aligned} Q_c^s &= 20 - P_c + P_w = 20 - 20 + 20 = 20 && \text{million} \\ Q_w^s &= 20 - P_c + P_w = 20 - 20 + 20 = 20 && \text{million} \end{aligned}$$

Take example of demand substitution effects. Consider how the renewable fuel mandate affects this market. Assume the mandate increases demand for corn by 12 million bushels at every price, so that the new demand for corn is given by:

$$Q_c^d = 32 - P_c + P_w$$

To determine the new equilibrium price and quantity in each market, we go through the same process as previously. Equating demand and supply in the corn market yields:

$$\begin{aligned} Q_c^d &= Q_c^s \\ 32 - P_c + P_w &= P_c \\ P_c &= 16 + \frac{P_w}{2} \end{aligned}$$

The equilibrium condition for the wheat market is the same as before:

$$P_w = 10 + \frac{P_c}{2}$$

Substituting in the equilibrium condition for the corn market:

$$\begin{aligned} P_w &= 10 + \frac{16 + \frac{P_w}{2}}{2} \\ P_w &= 18 + \frac{P_w}{4} \\ \frac{3P_w}{4} &= 18 \\ P_w &= \$24 \end{aligned}$$

And the price of corn is:

$$P_c = 16 + \frac{P_w}{2} = 16 + \frac{24}{2} = \$28$$

Finally, the new equilibrium quantities are:

$$\begin{aligned} Q_c^S &= 32 - P_c + P_w = 32 - 28 + 24 = 28 \text{ million} \\ Q_w^S &= 20 - P_w + P_c = 20 - 24 + 28 = 24 \text{ million} \end{aligned}$$

The increase in demand for corn has led to an increase in the equilibrium price and quantity of both corn and wheat. Now, consider the possibility that there are supply-side links. Demand for corn is given by:

$$Q_c^d = 20 - P_c$$

where P_c is the price of corn and P_w the price of wheat (each in dollars per bushel). Quantities are measured in millions of bushels. Demand for wheat is given by:

$$Q_w^d = 20 - P_w$$

The supply of each is related to the supply of the other. The supply of corn is given by:

$$Q_c^s = 2P_c - P_w$$

The supply of wheat is given by:

$$Q_w^s = 2P_w - P_c$$

These equations capture the notion, for example, that as agricultural production shifts to corn, scarce resources are reallocated from wheat to corn, raising the marginal cost of production for wheat.

Setting supply equal to demand in the wheat market yields:

$$\begin{aligned} Q_c^D &= Q_c^S \\ 20 - P_c &= 2P_c - P_w \\ P_w &= \frac{20}{3} + \frac{P_c}{3} \end{aligned}$$

And for corn,

$$\begin{aligned} Q_w^D &= Q_w^S \\ 20 - P_w &= 2P_w - P_c \\ P_c &= \frac{20}{3} + \frac{P_w}{3} \end{aligned}$$

Solving for equilibrium prices yields a price of \$10 in each market, with 10 million bushels of corn and wheat supplied. Suppose again, there is a 12-million-bushel increase in the quantity of corn demanded at every price, so that:

$$Q_c^d = 32 - P_c$$

Going through the same process as before, we find that the equilibrium condition for the wheat market stays the same but the equilibrium condition for the corn market changes to:

$$P_c = \frac{32}{3} + \frac{P_w}{3}$$

The process of finding the equilibrium once more yields: $P_w = \$11.50$ $P_c = \$14.50$ and

$$Q_w^S = 8,500,000$$

$$Q_c^S = 17,500,00$$

Example: Apple cider and orange juice are substitutes. Suppose that the demand for cider is given by $Q_c = 20 - P_c + 0.5P_o$, and that the demand for orange juice is given by $Q_o = 20 - P_o + 0.5P_c$, where Q_c and Q_o are measured in millions of gallons and P_c and P_o are the prices per gallons. The supplies of cider and orange juice are given by $Q_c = P_c$ and $Q_o = P_o$, respectively.

- Solve for the general equilibrium prices and quantities of apple cider and orange juice.
- Suppose that the demand for cider falls by 5 units at every price so that $Q_c = 15 - P_c + 0.5P_o$. Calculate the new general equilibrium prices and quantities of cider and juice.

Solution:

- We first need to solve for the price of apple cider as a function of orange juice by setting quantity demanded and quantity supplied equal in the cider market:

$$20 - P_c + 0.5P_o = P_c$$

$$2P_c = 20 + 0.5P_o$$

$$P_c = 10 + 0.25P_o$$

Then we follow the same step for the orange juice market so that the price of juice is expressed as a function of the price of cider:

$$20 - P_o + 0.5P_c = P_o$$

$$2P_o = 20 + 0.5P_c$$

$$P_o = 10 + 0.25P_c$$

To solve for P_c , we can substitute in the equation we just derived for P_o :

$$P_c = 10 + 0.25P_o = 10 + 0.25[10 + 0.25P_c] = 10 + 2.5 + 0.0625P_c$$

$$0.9375P_c = 12.5$$

$$P_c = \$13.33$$

This means that P_o is

$$P_o = 10 + 0.25P_c = 10 + 0.25(13.33) = 10 + 3.33 = \$13.33$$

Because $Q_c = P_c$ and $Q_o = P_o$, $Q_c = 13.33$ million gallons and $Q_o = 13.33$ million gallons.

- b. When the demand for cider falls, both the markets for apple cider and orange juice will be affected. We will need to follow the same steps used in previous part to solve for the new equilibrium prices and quantities. First, we solve for the price of cider as a function of the price of juice:

$$15 - P_c + 0.5P_o = P_c$$

$$2P_c = 15 + 0.5P_o$$

$$P_c = 7.5 + 0.25P_o$$

Because the supply and demand for juice are not affected initially, we know from previous part that

$$P_o = 10 + 0.25P_c$$

Therefore, we can substitute P_o into the equation for P_c :

$$\begin{aligned} P_c &= 7.5 + 0.25P_o = 7.5 + 0.25 [10 + 0.25P_c] = 7.5 + 2.5 + 0.0625P_c \\ 0.9375P_c &= 10 \\ P_c &= 10.67 \end{aligned}$$

Substituting for P_c and solving for P_o , we get

$$P_o = 10 + 0.25P_c = 10 + 0.25(10.67) = 10 + 2.67 = 12.67$$

Because $Q_c = P_c$ and $Q_o = P_o$, $Q_c = 10.67$ million gallons and $Q_o = 12.67$ million gallons.

14.2 General Equilibrium: Equity and Efficiency

The second branch of general equilibrium analysis is an important tool for evaluating performance of policies, particularly to identify effects on specific sectors of the economy. Often, policies and projects are evaluated with the use of a social welfare function.

Definition(Social welfare function): A mathematical function that combines the utility levels of the individuals in a society to obtain a single overall measure of an economy's performance.

That way, we can compare various market outcomes (i.e., distributions of utilities across all individuals in the economy) to one another. If one outcome has a higher value than another according to the social welfare function, the higher-valued outcome is considered more desirable. The ranking of various market outcomes depends on the particular form of the social welfare function one chooses to use to evaluate outcomes in the first place.

Definition(Utilitarian social welfare function): A mathematical function that computes society's welfare as the sum of every individual's welfare.

Definition(Rawlsian social welfare function): A mathematical function that computes society's welfare as the welfare of the worst-off individual.

Draw backs of social utility functions

While social welfare functions are useful for analyzing the distributional consequences of policies and projects, there are many complicating factors:

- Choosing a social welfare function is ultimately subjective and different functions might give varied answers about what makes for desirable outcomes.
- Furthermore, it is difficult to mathematically combine individuals' utility levels.

The difficulties with trying to use social welfare functions to evaluate how well markets are working have led economists to use a criterion that everyone can understand and agree on: Pareto efficiency.

Definition(Pareto efficiency): An economic allocation of goods in which the goods cannot be reallocated without making at least one individual worse off. This means the economy does not have any free lunch.

Remark: Note that Pareto efficiency doesn't imply it is fair or that it maximizes some social welfare function. Pareto-efficient allocations can exist even with large differences in individuals' utility levels.

Remark: When certain conditions are met, competitive market allocations are Pareto-efficient and consumer and producer surplus are maximized; and no alternative allocation can lead to gains to one person without being more than offset by losses to another. If a particular allocation is not Pareto-efficient, there may be opportunities for improvements through market intervention or the removal of market impediments.

We have three conditions for markets that result in Pareto-efficient allocations:

- **Definition(Exchange efficiency):** A Pareto-efficient allocation of a set of goods across consumers. No one can be made better off with respect to consumption without harming another.
- **Definition(Input efficiency):** A Pareto-efficient allocation of inputs across producers. No firm or individual can be made better off with an alternative input allocation without harming another.
- **Definition(Output efficiency):** A mix of outputs that simultaneously supports exchange and input efficiency. The mix and amount of goods that the economy produces cannot be changed without making some consumer or producer worse off.

14.2.1 Efficiency in Markets: Exchange Efficiency

We introduce an extremely handy tool for analyzing market efficiency named Edgeworth box.

Definition (Edgeworth box): Graph of an economy with two economic actors and two goods that is used to analyze market efficiency.

Example: Suppose there are two consumers, Jerry and Elaine, and two types of goods, bowls of cereal and pancakes.

- A total of 10 bowls of cereal and 8 pancakes are available.
- We want to determine the Pareto-efficient ways these two goods can be split between Jerry and Elaine.

We can use an Edgeworth box to depict various allocations of cereal and pancakes:

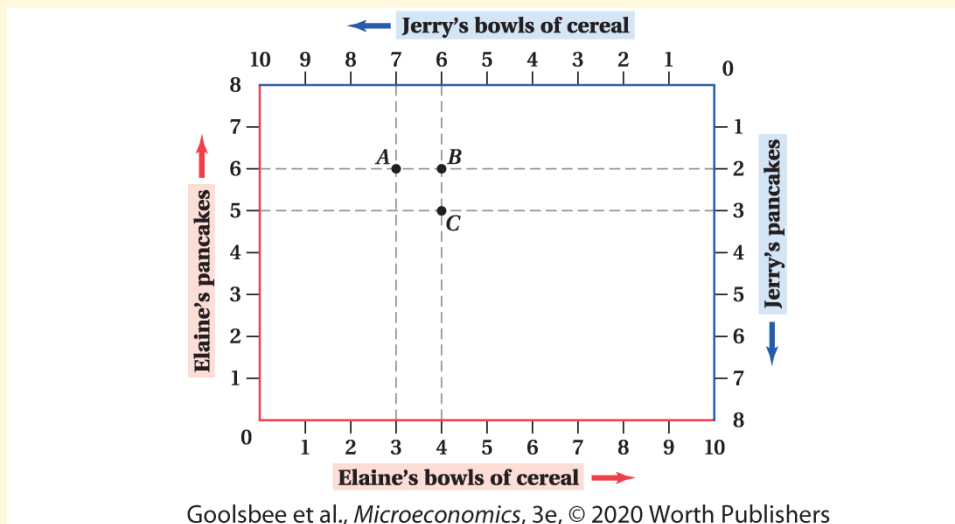


Figure 125: A Consumption Edgeworth Box

- If Jerry consumes 7 bowls of cereal and 2 pancakes, Elaine consumes 3 bowls of cereal and 6 pancakes (point A).
- If Jerry consumes one fewer bowl of cereal at point B, then Elaine's consumption of cereal increases by 1 bowl to 4 bowls of cereal (point B).
- At point C, Jerry eats one more pancake, decreasing Elaine's consumption of pancakes by 1, to 5 pancakes (point C).

To determine whether a particular allocation is efficient, we will use the analysis of utility curve. Analogous to the single-consumer case, Jerry and Elaine are assumed to gain utility from consuming pancakes and cereal. We represent utility with indifference curves. Indifference curves are bowed in toward the origin. And complete preferences means every possible allocation can be ranked for both Jerry and Elaine:

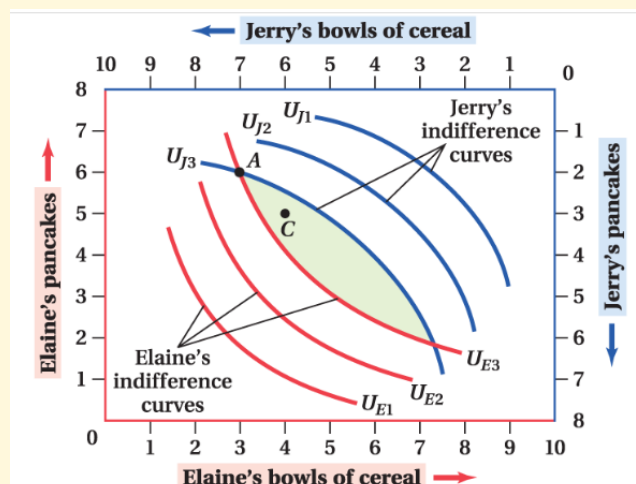


Figure 126: Edgeworth Box with Two Sets of Indifference Curves

As indifference curves move away from the respective origins, they represent greater levels of utility. Point *A* is an inefficient allocation because any point in the shaded area, including point *C*, will yield both Jerry and Elaine higher utilities. How might Jerry and Elaine move from the less desirable allocation *A* to the more desirable *C*? At allocation *A*, Jerry's marginal utility of consuming cereal is less than Elaine's, and Elaine's marginal utility of consuming pancakes is less than Jerry's. Therefore, Jerry should be willing to trade some cereal for pancakes, and Elaine should be willing to trade some pancakes for cereal. A Pareto-efficient allocation of pancakes and bowls of cereal occurs at a tangency between Jerry's and Elaine's indifference curves. In this case, the Pareto-efficient allocation is point *D*, where Elaine's indifference curve U_{E5} is tangent to Jerry's indifference curve U_{J5} . Here, Jerry and Elaine each consume 5 bowls of cereal and 4 pancakes.

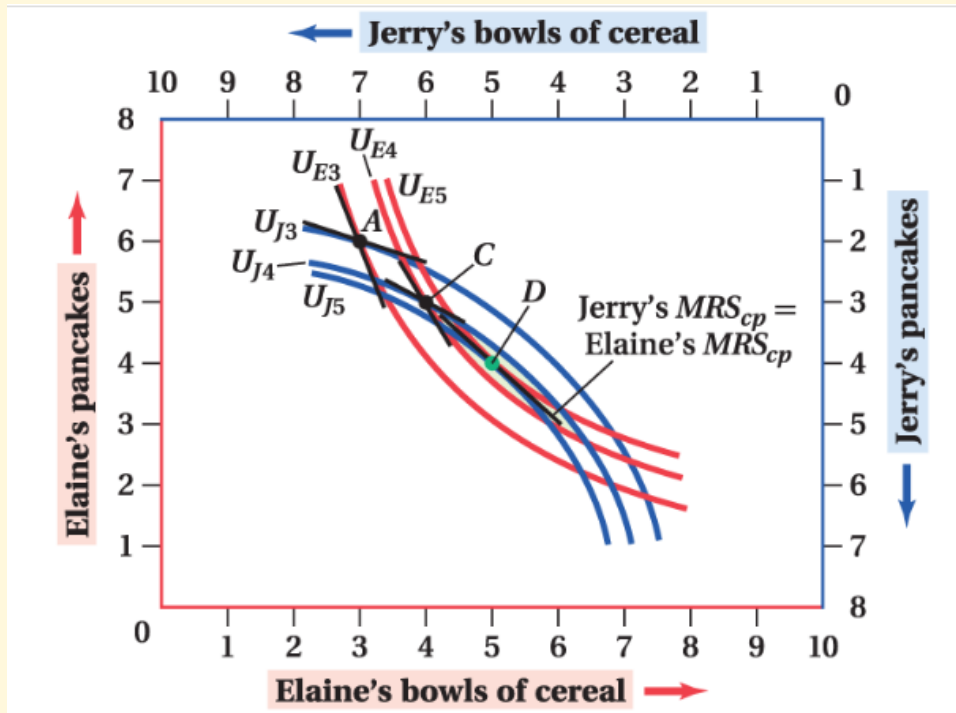


Figure 127: Closing In on Pareto Efficiency for Elaine and Jerry

Mathematically, for an allocation to be efficient, the marginal rate of substitution must be the same for each consumer:

$$MRS_{CP}^{\text{Jerry}} = MRS_{CP}^{\text{Elaine}}$$

$$\frac{MU_C^{\text{Jerry}}}{MU_P^{\text{Jerry}}} = \frac{MU_C^{\text{Elaine}}}{MU_P^{\text{Elaine}}}$$

In Elaine and Jerry's case, then, we know that they will both consume amounts where $MRS_{CP} = \frac{P_C}{P_P}$. We also just saw that Elaine's and Jerry's MRS_{CP} will be equal in a Pareto-efficient allocation. Thus, it must be that Pareto efficiency in a market implies

$$\text{Elaine's } MRS_{CP} = \text{Jerry's } MRS_{CP} = \frac{P_C}{P_P}$$

The key condition of exchange efficiency that consumers have equal marginal rates of substitution - that is, their indifference curves are tangent to one another - is true at more than one location in this Edgeworth box.

Definition(Consumption contract curve): Curve that shows all possible Pareto-efficient allocations of goods across consumers.

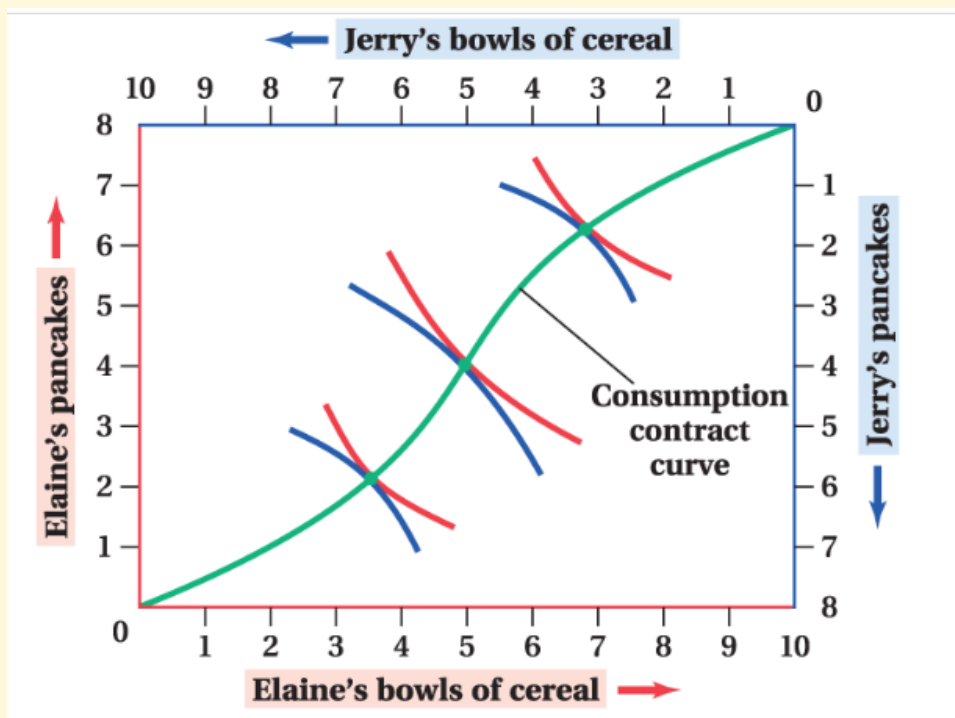


Figure 128: A Consumption Contract Curve

Here the consumption contract curve connects every point of tangency between Jerry's and Elaine's indifference curves for bowls of cereal and pancakes. Each point on the contract curve represents a Pareto-efficient allocation of bowls of cereal and pancakes between Jerry and Elaine.

14.2.2 Efficiency in Markets: Input Efficiency

Exchange efficiency shows what conditions must hold on the demand side of a market for the market to be efficient. Now we will deal with the supply side. Since production inputs (capital, labor) are scarce, input efficiency refers to the idea that we are allocating our inputs to the most valued production activities.

We can use the same tools as we did in the previous section, namely, a production Edgeworth box.

Example: Consider again the example of pancakes (produced by Pancake, Inc.), and cereal (produced by Cereal, Inc.). Assume each production process uses labor and capital and that a total of 20 labor inputs and 12 capital inputs are available for use.

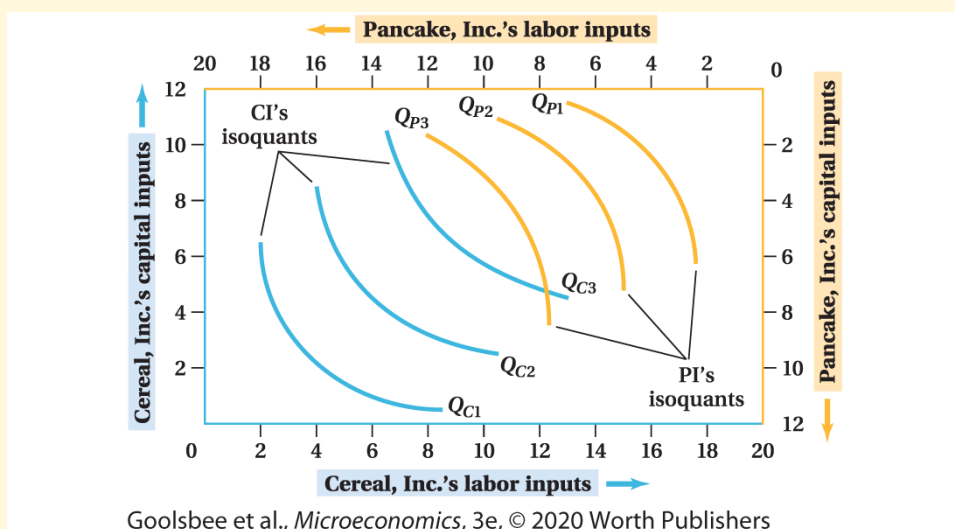


Figure 129: A Production Edgeworth Box

An Edgeworth box can be used to determine the efficient input allocation between two firms. In this case, Cereal, Inc.'s and Pancake, Inc.'s labor inputs are on the horizontal axes and their capital inputs are shown on the vertical

axes. Q_{C1} , Q_{C2} , and Q_{C3} are examples of Cereal, Inc.'s isoquants, and Q_{P1} , Q_{P2} , and Q_{P3} are examples of Pancake, Inc.'s isoquants.

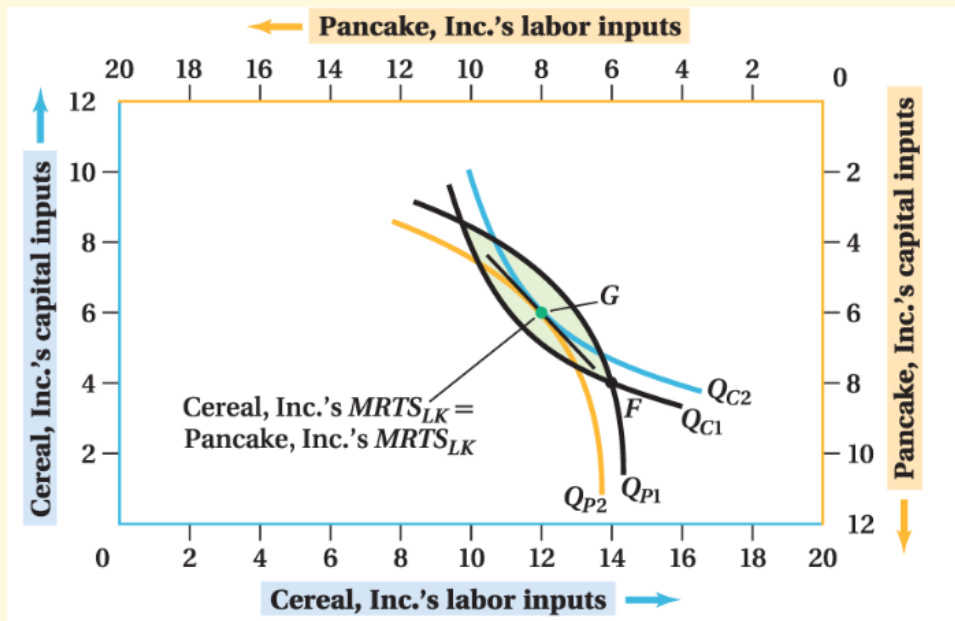


Figure 130: Edgeworth Box with Two Sets of Isoquants

The isoquants for the two firms are bowed outward from the origin because of our assumption of a diminishing marginal rate of technical substitution (MRTS). MRTS, the slope of the isoquant, indicates the rate at which two inputs can be traded off while maintaining a constant level of production (the ratio of the inputs' marginal products).

$$MRTS_{LK} = \frac{MP_L}{MP_K}$$

Just as with the exchange efficiency example, input efficiency is achieved when the isoquants for the two firms are tangent, or

$$CI's \text{ } MRTS_{LK} = PI's \text{ } MRTS_{LK}$$

$$\frac{MP_L^{CI}}{MP_K^{CI}} = \frac{MP_L^{PI}}{MP_K^{PI}}$$

Definition(Production contract curve): Curve that shows all Pareto-efficient allocations of inputs across producers.

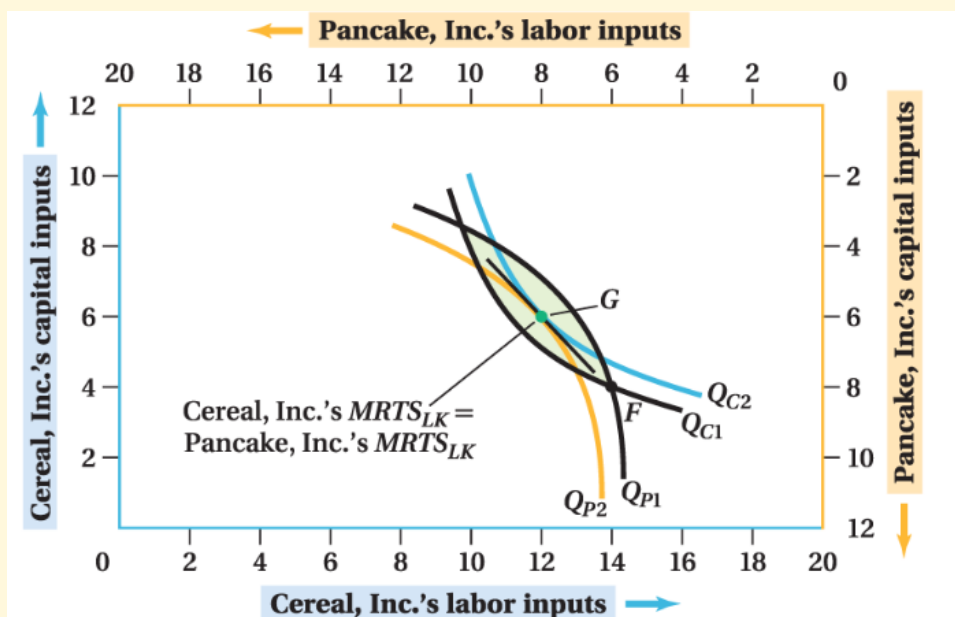


Figure 131: Closing in on pareto efficiency of inputs

The production contract curve connects every point of tangency between Pancake, Inc.'s and Cereal, Inc.'s isoquants for capital and labor. Each point on the production contract curve represents a Paretoefficient input allocation. At points *H* and *I*, relatively little cereal is produced, while relatively more pancakes are produced. At point *J*, no pancakes are made, while Cereal, Inc. produces the maximum amount of cereal given the available inputs.

The production contract curve generates a useful idea. Think about what various input allocations across the two firms imply with regard to the tradeoffs between the output of either good. As we move from point *H* to point *J*, we are producing fewer pancakes and more cereal.

Definition(Production possibilities frontier (PPF).): Curve that connects all possible efficient output combinations of two goods.

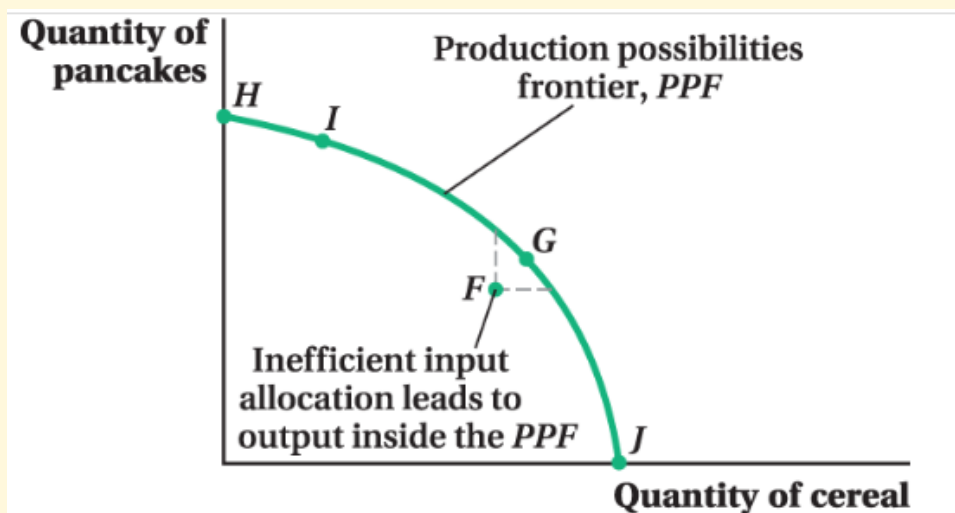


Figure 132: A Production Possibilities Frontier

The production possibilities frontier (PPF) plots all possible output combinations of cereal and pancakes that can be made if inputs are allocated efficiently. Points *H*, *I*, *G*, and *J* all lead to efficient outputs. Point *F*, which lies within the PPF, results from an inefficient allocation of inputs across producers.

14.2.3 Efficiency in Markets: Output Efficiency

Exchange efficiency and input efficiency helped develop consumption and production contract curves. Understanding the requirements for output efficiency requires us to see the inherent tradeoff between the production of various goods. We describe this tradeoff first and then show how it ties together the three efficiency conditions.

Definition(Marginal rate of transformation (MRT)): The tradeoff between the production of any two goods along the PPF.

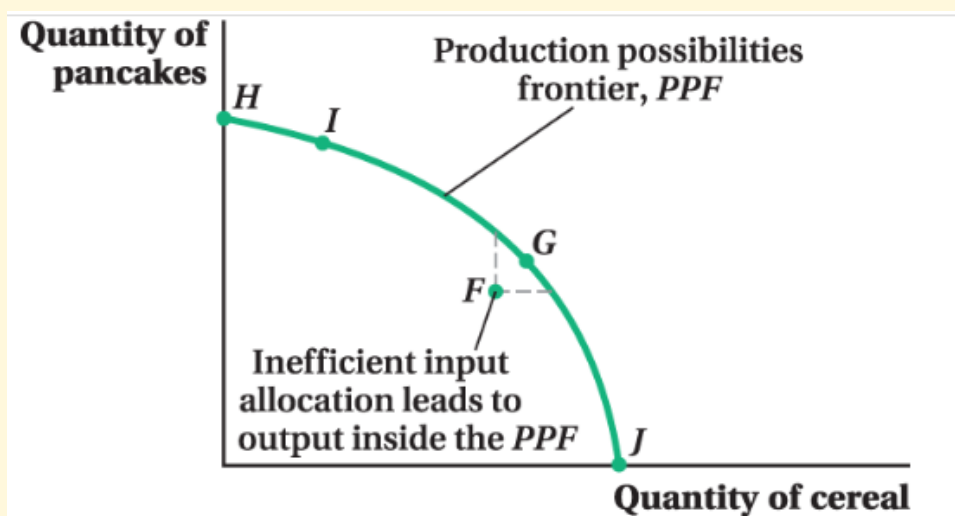


Figure 133: A Production Possibilities Frontier

- At the beginning of the PPF (when producing near zero of one good and maximum of the other), the resources

used to increase production of the first good are those that are least suited for the production of the second good. Hence, only a small amount of the second good needs to be sacrificed to increase production of the first. This makes the MRT (opportunity cost) small.

- As production shifts towards more and more of the first good, the resources now being reallocated are increasingly better suited for producing the second good. Therefore, the cost in terms of foregone production of the second good rises. The slope of the PPF becomes steeper, indicating a higher MRT.
- Towards the end of the PPF, where nearly all resources are devoted to the first good, reallocating any further resources (which are now highly efficient at producing the second good) requires giving up large amounts of the second good to increase production of the first by just a little. Thus, the MRT becomes very large.

The MRT is equivalent to the ratio of the marginal products of labor for the two products.

$$MRT_{PC} = \frac{MP_L^{PI}}{MP_L^{CI}}$$

Remember, the PPF represents all of the production combinations whose marginal rate of technical substitution is equal for the two products.

$$CI's \quad MRTS_{LK} = PI's \quad MRTS_{LK} \rightarrow \frac{MP_L^{CI}}{MP_K^{CI}} = \frac{MP_L^{PI}}{MP_K^{PI}}$$

This means that the *MRT* is also equal to the ratio of the marginal products of capital.

$$\frac{MP_L^{CI}}{MP_L^{PI}} = \frac{MP_K^{CI}}{MP_K^{PI}}$$

Output efficiency exists when the tradeoffs on the consumption and production sides of an economy are equal.

- The tradeoff on the consumption side is the marginal rate of substitution (MRS).
- The tradeoff on the production side is the marginal rate of transformation (MRT).

Output efficiency requires that $MRS = MRT$.

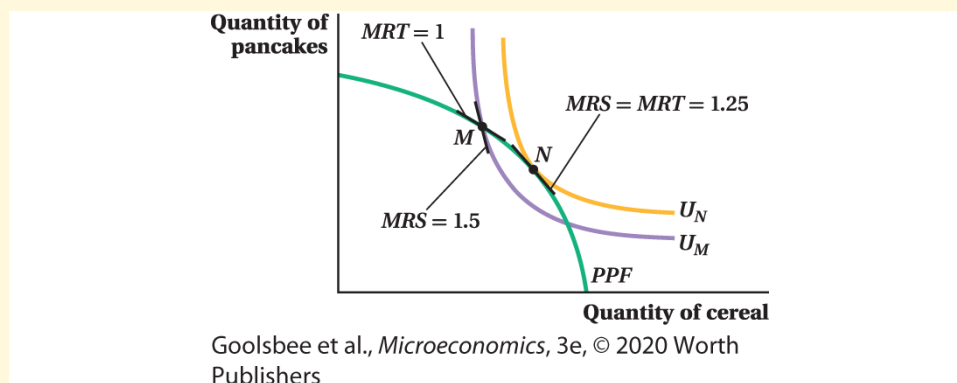


Figure 134: Achieving Output Efficiency

Output efficiency occurs at the tangency between the consumers' indifference curves and the production possibilities frontier, where the marginal rate of substitution MRS equals the marginal rate of transformation MRT. Point *M* shows a possible output combination of pancakes and cereal. Because it lies on the intersection of the PPF and the indifference curve U_M - not at the tangency - *M* is an inefficient output combination. *N*, which is located at the tangency between the PPF and U_N where $MRS = MRT = 1.25$, is an efficient output allocation.

Example: In Ecoland, producers make both televisions and clocks. At current production levels, the marginal cost of producing a clock is \$50 and the marginal cost of producing a television is \$150.

- How many clocks must Ecoland give up if it wishes to produce another television?
- What is the marginal rate of transformation from clocks to televisions in Ecoland?
- At current production levels, consumers are willing to give up 2 clocks to obtain another television. Is Ecoland achieving output efficiency? Explain.

Solution:

- Because the marginal cost of producing a television is 3 times that of producing a clock, Ecoland must give up 3 clocks for each television it produces.
- The marginal rate of transformation (MRT) measures how much of one output must be given up to obtain one more unit of another. Therefore, the MRT from clocks to televisions is 3.
- Because consumers are only willing to trade 2 clocks for a television, the marginal rate of substitution (MRS) is 2. However, output efficiency requires that $MRS = MRT$. Therefore, Ecoland is not achieving output efficiency.

14.3 Markets, Efficiency, and the Welfare Theorems

Our discussion has suggested that the simple solution to inefficiency on the exchange, input, and output margins is to shift the allocation of inputs or outputs. However, this extremely simple model assumes the following:

- Only two inputs, two outputs, two firms, and two consumers.
- Perfect information.

The real economy is vastly more complicated. And it is likely impossible for a central planner to efficiently allocate inputs and outputs across firms and consumers.

With exchange efficiency, if consumers maximize their utility (taking prices as given), they will have the same MRS. Thus, consumers will reach exchange efficiency without market intervention.

For any given set of input prices, profit-maximizing firms will choose their input mix so that their MRTS equals the input-price ratio. Firms will achieve input efficiency without market intervention.

The final link, output efficiency, ties these two conditions together. Output efficiency's $MRS = MRT$ condition sets the ratio of goods' prices equal to their marginal costs of production. If the goods are produced by a perfectly competitive industry, price will equal marginal cost; therefore, the price and cost ratios are equal, satisfying output efficiency while preserving input and exchange efficiency. Thus a decentralized competitive markets can achieve all three efficiency conditions.

Thus, we have the first welfare theorem:

First Welfare theorem

Theorem stating that perfectly competitive markets in general equilibrium distribute resources with rational customers in a Pareto-efficient way.

While the initial allocation of inputs and goods across consumers and firms does not affect the efficiency of competitive markets, it does affect the distribution of final goods and income across consumers and firms. This leads to the Second Welfare Theorem.

Second Welfare theorem

Theorem stating that any given Pareto-efficient equilibrium can be achieved by choosing the right initial allocation of goods.

If equity is a concern, society can reallocate some inputs across society and still not sacrifice the efficiency of markets. Nevertheless, redistribution can lead to efficiency losses. If the amounts of redistribution are affected by the choices of firms and individuals, efficiency losses will occur. To maintain efficiency, redistribution should take the form of a lump-sum transfer.

Definition(Lump sum transfer): Transfer to or from an individual where the size of the transfer cannot be affected in any way by the individual's choices.

15 Assymmetric information

Our analysis of markets, thus far, has assumed that participants have complete information in market transactions.

Definition(Complete information): A situation where all market participants know everything important for making economic decisions.

In reality, many market transactions occur under conditions of incomplete information, in which one or more parties to a market cannot determine with certainty all of the important attributes of a market or a particular good.

This is called asymmetric information.

Definition(Assymmetric information): A situation where there is an imbalance of information across participants in an economic transaction.

15.1 The Lemons Problem and Adverse Selection

15.1.1 The Lemons problem

A common manifestation of asymmetric information in markets is the lemons problem.

Definition(Lemons problem): An asymmetric information problem that occurs when a seller knows more about the quality of the good he is selling than does the buyer.

Example: Consider the market for a used car from a private-party seller. Assume there are two types of used cars-good ones (plums) and bad ones (lemons). Half of the cars available are plums and half are lemons. Potential buyers value plums at \$10,000 but place no value on lemons. Sellers value plums at \$8,000 and also have no value for lemons.

- First, think about market outcomes if quality attributes are observable to both sellers and buyers.
Since sellers value plums at \$8,000 and buyers value plums at \$10,000, the half of used cars that are plums will sell at prices between these two values.
Both parties are better off following the trade.
Buyers value the cars more than former owners, and former owners are happier with the money.
Lemons have no value, so none will be sold.
- Now, consider what happens when sellers know whether their offering is a plum or a lemon but buyers do not.
Buyers know 50% of cars are lemons. Therefore, buyers recognize that the probability that they will purchase a lemon is 50.
Since buyers value plums at \$10,000 and lemons at \$0 and since there is a 50% chance that a given car is a lemon, the most a buyer is willing to pay is:

$$EV = (\$10,000 * 0.50) + (\$0 * 0.50) = \$5,000$$

Any buyer who pays more is worse off (in expectation) from engaging in the trade.

Now, think about the owner of a plum who is considering whether to sell.

The seller values his car at \$8,000 but recognizes that, because buyers can't know for sure whether or not his car is a plum, he could never get more than \$5,000.

Therefore, he does not offer the car for sale.

The lemons problem is that market transactions fail to occur because of asymmetric information; this a market failure.

Mechanisms That Mitigate Lemons Problems

The lemons problem destroys economic value by preventing beneficial exchanges. In response, institutions to mitigate information asymmetries have developed. These can work in three ways:

- Address the information asymmetry directly by allowing buyers to observe quality characteristics before a transaction takes place.
- Punish sellers who misrepresent their lemons as plums.
- Use incentives to increase the number of plums brought to market.

Example: Suppose that consumers value a high-quality used laptop computer at a price of \$400, while they value a low-quality used laptop at \$100. The supply of high-quality laptops is $Q_H = P_H - 100$, while the supply of low-quality laptops is $Q_L = 2P_L - 50$. Potential buyers cannot tell the difference between high-quality and low-quality laptops when purchasing one.

- Assume that buyers believe there is a 50% probability that a used laptop will be of high quality. What would be the price that buyers are willing to pay for any used laptop?
- If the price determined in (a) is offered in the market for used laptops, how many high-quality laptops will be made available in the market? How many low-quality laptops will be available in the market? Are buyers correct in their assumption that 50% of the used laptops available for sale are of high quality? Explain.
- What would you expect to happen over time as information about the true odds of buying a high-quality used laptop becomes known? Explain.

Solution:

- If buyers expect that 50% of the used laptops available are of high quality (meaning that the other 50% are of low quality), then the expected value of a laptop is equal to

$$0.50 \times \$400 + 0.50 \times \$100 = \$200 + \$50 = \$250$$

Therefore, \$250 would be the most that buyers are willing to pay for a used laptop.

- If the price of a used laptop is \$250, then the quantity supplied of high-quality laptops is $Q_H = P_H - 100 = 250 - 100 = 150$. The quantity supplied of low-quality used laptops is $Q_L = 2P_L - 50 = 2(250) - 50 = 500 - 50 = 450$. Therefore, there will be 600 used laptops for sale (150 of high quality and 450 of low quality). The probability of buying a used laptop of high quality is not 50%, but actually equal to $150/600 = 0.25$ or 25%. Because of asymmetric information, buyers are not willing to pay a very high price for a used laptop. Therefore, the owners of high-quality laptops will be reluctant to sell them, while the owners of low-quality laptops will be eager to sell them. This changes the proportion available in the market.
- Over time, buyers will adjust their expected value of used laptops. This will further reduce the price buyers are willing to pay. Owners of high-quality used laptops will be even less inclined to sell them, reducing even more the proportion of high-quality used laptops available. Ultimately, it is possible that the market could end up with only low-quality used laptops available.

15.1.2 Adverse selection

Definition(Adverse selection): A situation where there are stronger incentives for "bad" types of a product to be involved in a transaction than "good" types of the product.

The existence of quality differences is not by itself a market failure; instead, it is a lack of information. This leads to a market price that provides no incentive for plums to be offered for sale but every incentive for lemons to be offered for sale. Information asymmetries hurt not only those with little information but also those with more information. Both sides lose because a lack of complete information prevents trades. In this example, buyers and owners of plums are both hurt despite the owners having complete information.

In some cases of adverse selection, it is possible for buyers to have more information than the seller. People with higher health risks are more likely to purchase health insurance, which can lead to a pool of insured individuals that is sicker than the general population, increasing insurance premiums and possibly driving healthier people away from purchasing insurance.

Mitigating Adverse Selection in Insurance

A number of mechanisms have emerged to deal with adverse selection in insurance markets.

- **Group policies:**
Tying insurance to employment removes the link between the individual's riskiness and the decision to purchase insurance.
Pooling individuals reduces the effect of any given poor-risk person.
- **Screening:** Detailed questionnaires, health exams, driving records, and so on
- **Denying coverage:** Insurers try to deny coverage to individuals with certain risk factors or pre-existing conditions. This was the reasoning behind the protections provided in the ACA (Affordable Care Act).

15.2 Moral Hazard

Definition(Moral hazard): A situation that arises when one party in an economic transaction cannot observe another party's behavior.

When quality is difficult to observe, a party to a transaction may have a financial incentive to engage in fraud. In insurance markets, adverse selection refers to the problem of deciding who to insure and at what price. Moral hazard refers to the effect of being insured on the behavior of an individual. Knowing you are insured may make you more willing to take risks, since part of your risk is being borne by a third party.

Example: Consider comprehensive auto insurance. Comprehensive insurance compensates drivers in the event their vehicle is stolen (among other things). How might drivers' behavior change if they know they are covered by a comprehensive insurance policy? The coverage may lead them not to try to prevent theft.

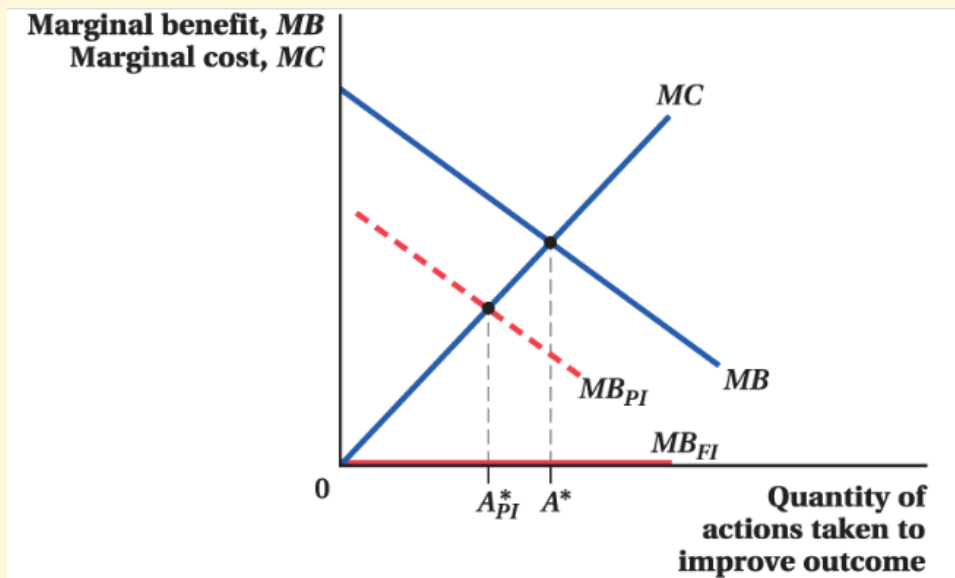


Figure 135: Moral hazard in the insurance market

In a market without insurance, a potential policyholder would take actions to improve the potential for a good outcome up to A^* , where the marginal benefit of further action, MB , equals the marginal cost of further action, MC . If instead the policy offers full insurance, MB shifts to MB_{FI} , and the policyholder does not take any action (makes zero effort) to make the good outcome more likely. If the policy offers partial insurance, MB shifts to MB_{PI} , and the policyholder takes action level A_{PI}^* .

Example: Anastasia and Katherine own a café. Because of their equipment and business, they run a risk of loss due to small kitchen fires. This risk can be mitigated by taking precautions such as purchasing fire extinguishers or by increasing the training and awareness of the café's employees. Assume that the marginal cost of these precautions can be represented by $MC = 80 + 8A$, where A is equal to the actions taken to mitigate the risk of a fire. Likewise, the marginal benefit of these precautions is $MB = 100 - 2A$.

- If the café has no insurance, what would be the optimal level of precautions for Anastasia and Katherine to take?
- Suppose the café has insurance that reduces the marginal benefit of taking precautions to $MB = 90 - 4A$. What happens to the optimal level of precautions? Explain why this is the case.

Solution:

- With no insurance, the optimal level of precautionary actions would occur where $MB = MC$:

$$100 - 2A = 80 + 8A$$

$$10A = 20$$

$$A = 2$$

- Once the insurance is in place, the marginal benefit of taking precautions falls to $MB = 90 - 4A$. The optimal level of precautions also falls:

$$MB = MC$$

$$90 - 4A = 80 + 8A$$

$$12A = 10$$

$$A = 0.83$$

The optimal level of precautionary actions falls when insurance is available. If a fire occurs, the cafe's owners will experience a smaller loss as a result of the insurance coverage. Therefore, the owners' incentives to try to prevent a loss are reduced.

Example: Another common instance of moral hazard occurs between borrowers and lenders in financial markets. Borrowers' liability may be limited (e.g., in bankruptcy proceedings), meaning they may be willing to take unjustified risks with borrowed funds.

The recent financial crisis has highlighted the issue of "too big to fail." Implicit guarantees to large financial institutions may increase risk taking. Employer-employee relationships often are a source of moral hazard. Inability to observe all of employees' activities provides an opportunity and incentive for employees to shirk.

Lessening Moral Hazard

Market mechanisms have developed to diminish the effects of moral hazard.

- Insurance policies often mandate actions to be taken by the insured.
Commercial property insurance often requires working smoke detectors and regularly inspected fire extinguishers.
- Policies can be structured to encourage good behavior.
Some auto insurance deductibles fall after each accident-free year.
Life insurance premiums usually fall when efforts are made to improve health (e.g., quit smoking).

15.3 Asymmetric Information in Principal-Agent Relationships

Definition(Principle-agent relationships): Economic transactions that feature information asymmetry between a principal and her hired agent, whose actions the principal cannot fully observe. This means involves one party (the principal) hiring a second party (the agent) to perform a task and being unable to completely observe the effort of the agent

Remark: Information asymmetry is insufficient to develop a principal-agent problem. There must also be a misalignment between the incentives and preferences of the principal and the agent. For example, an employer wants her employees to work as hard as possible, but employees want to work as little as possible without being fired.

Example: Consider a mall kiosk selling mobile phones. The kiosk is owned by Selena (the principal), who hires a single employee (Joe, the agent) to staff the kiosk.

If Joe works hard, the kiosk earns a daily profit of \$1,000 with 80% probability and \$500 with 20% probability (expected profit of \$900). If Joe does not work hard, the probabilities are reversed (expected profit of \$600). Joe does not like to work hard and requires at least \$150 per day to do so. Selena cannot observe Joe's effort.

Selena would be happy to pay Joe a flat rate of \$150 per day to work hard, but as she cannot observe his effort (and profits are uncertain), she cannot be sure that he will indeed, follow through.

Rather than paying Joe a flat rate, Selena can instead pay Joe a wage that varies with the profits of the kiosk.

For example, Joe will be paid \$255 if daily profits are high (\$1,000) and \$0 when profits are low (\$500). If Joe works hard, the 80% probability of high profits yields an expected wage of \$204 and a surplus of \$54. (Remember, he also suffers an effort cost of \$150.) If Joe does not work hard, the 20% probability of high profits yields an expected wage of \$51, with no cost to him because he need not exert any effort.

Under this compensation plan, Joe prefers working hard to being lazy.

Paying \$255 to Joe for high-profit days yields an expected wage of \$204.

This is less than the increase in expected profits associated with Joe working hard (\$300), so she will prefer this compensation plan. Expected profits are significantly higher than under the fixed compensation plan of \$96 per day.

15.4 Signaling to Solve Asymmetric Information Problems

One major result of principal-agent problems is that good agents or products are not identifiable and, therefore, cannot command full value.

Definition(Signaling): A solution to the problem of asymmetric information in which the knowledgeable party alerts the other party to an unobservable characteristic of the good.

Definition(Signal): A costly action taken by an economic actor to indicate something that would otherwise be difficult to observe.

Often, economic actors will attempt to communicate their quality via a signal. To signal high quality credibly, a signal must be less costly for high-quality agents than low-quality agents. In the warranty example, sending the signal of providing a warranty is less costly for producers of high-quality products than for producers of low-quality

products. This difference makes it more likely that high-quality products will have a warranty, allowing consumers to infer something about a product's quality, which would be otherwise unobservable at the time of purchase.

Example: Education - specifically the granting of degrees - is a classic case of signaling. Highly productive workers like to let employers know they are, indeed, highly productive so they might receive a job offer and/or a higher wage. However, simply telling an employer that you are productive is cheap talk. A college degree, however, signals that you are, indeed, productive. College is difficult. It takes time, a lot of money, and an ability to learn and apply new information. These are the same attributes that often make employees productive. The fact that a third party (an accredited university) has issued a degree is a credible signal of productivity.

Example: For the same example, we can use numbers to make this example clearer.

Suppose there are two types of workers: high-productivity and low productivity.

Each year of higher education costs high-productivity workers \$25,000, including psychic costs of going to class, finishing assignments, studying for exams, and so on.

Each year of higher education costs low-productivity workers \$50,000 because of their higher psychic costs.

$$C_H = \$25,000y$$

$$C_L = \$50,000y$$

Over a lifetime, high-productivity workers produce \$250,000 worth of value to employers, whereas low-productivity workers produce \$125,000 of value, regardless of education.

Thus, employers are willing to pay high-productivity workers up to \$125,000 more in wages. However, employers must be able to tell them apart from the low-productivity workers.

Suppose employers view a four-year degree as a signal of high productivity and are willing to pay workers with a degree \$125,000 more.

For this to be effective, it must be incentive-compatible.

High-productivity workers must find it advantageous to use college as a signal, and low-productivity workers must find a college degree to be not worth the effort.

To determine whether the strategy is effective, compute the net benefits for each type of worker; ignore discounting.

$$NB_H = \text{Benefit} - (C_H * 4 \text{ years}) = 125,000 - 100,000 = \$25,000$$

$$NB_L = \text{Benefit} - (C_L * 4 \text{ years}) = 125,000 - 200,000 = \$ - 75,000$$

High-productivity workers earn \$25,000 more in surplus with a college degree, while low-productivity workers lose \$75,000.

Therefore, this is an effective strategy, as the low-productivity workers will choose not to go to college.

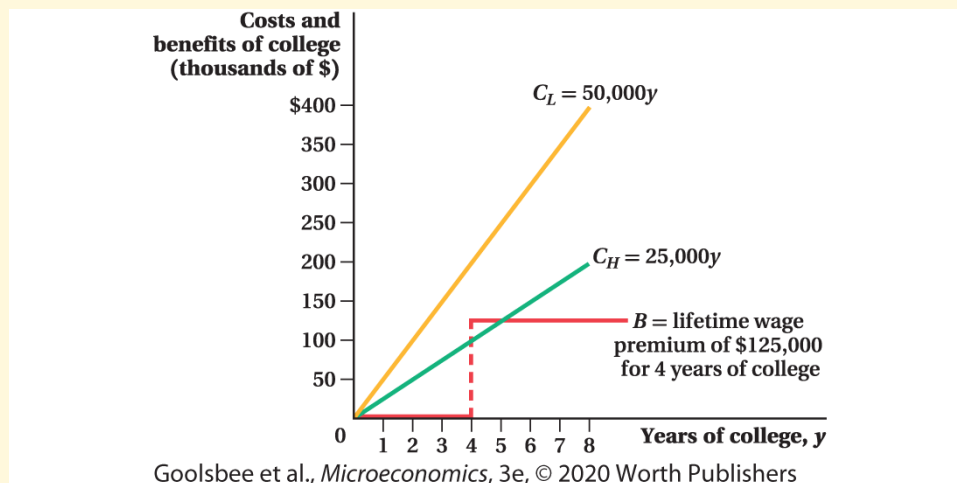


Figure 136: Education as a Signal on the Job Market

16 Externalities and Public Goods

16.1 Externalities

Definition(Externalities): A cost or benefit that affects a third party not directly involved in an economic transaction.

Definition(Negative externality): A cost imposed on a party not directly involved in a transaction

Definition(Positive externality): A benefit conferred on a party not directly involved in a transaction.

Definition(External marginal cost): The cost imposed on a third party when an additional unit of a good is produced or consumed.

Definition(External marginal benefits): The benefit conferred on a third party when an additional unit of a good is produced or consumed.

When there are externalities, the social costs and social benefits will differ from the private costs and private benefits.

Definition(Social costs): The cost of an economic transaction to society as a whole, equal to the private cost plus the external cost.

Definition(Social benefits): The benefit of an economic transaction to society as a whole, equal to the private benefit plus the external benefit.

16.1.1 Negative externalities

Negative externalities occur when a market transaction imposes an external cost on society.

Consider the example of a coal-fired power plant. The power plant produces electricity, which is good, but in the process, it releases pollutants into the air: particulate matter, nitrogen oxides (NO_x), and sulfur dioxide (SO₂). These pollutants both directly and indirectly damage human and environmental health, leading to welfare losses.

The costs of operating the plant are borne by the plant, but the health effects are external costs, borne by society. Consider a competitive market for electricity.

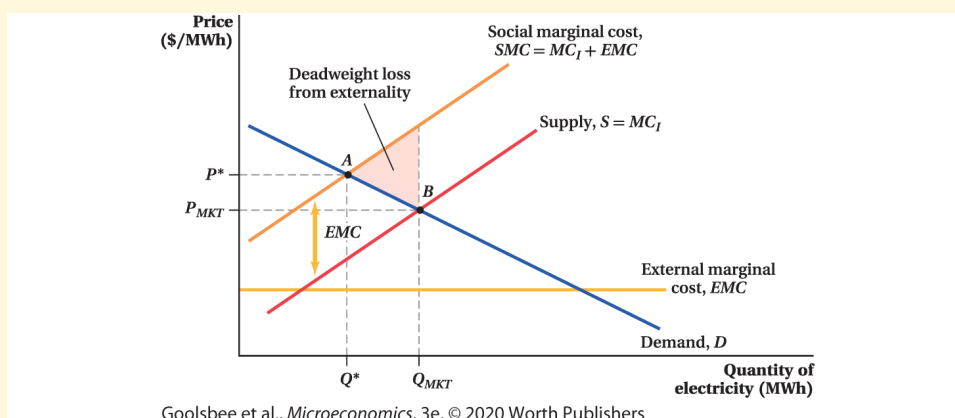


Figure 137: Negative Externalities in a Competitive Electricity Market

The social marginal cost of electricity (SMC) equals the industry's private marginal cost (MC_I) plus the external marginal cost (EMC). The socially optimal quantity of electricity, Q^* , is found at point A , the intersection of the social marginal cost curve (SMC) and D . In a competitive market, however, production occurs at point B (Q_{MKT}, P_{MKT}), where $MC_I = S = D$. Because the industry does not take into account the external marginal cost (EMC), it produces a quantity of electricity Q_{MKT} that is higher than the socially optimal quantity (Q^*). This results in a deadweight loss equal to the shaded triangle. The deadweight loss represents the social cost associated with the competitive outcome in the presence of an externality. The magnitude of the deadweight loss depends on two factors: the size of the externality and electricity that wouldn't be bought if the price reflected the true social cost.

Example: Assume that notebook paper is sold in a perfectly competitive industry. The industry short-run supply curve (or marginal cost curve) is $P = MC = 2Q$, where Q is measured in millions of reams per year. The inverse demand for notebook paper is $P = 40 - 8Q$.

- Find the equilibrium market price and quantity sold.
- Suppose that, in their production processes, paper manufacturers have been dumping waste in nearby streams. The external marginal cost is estimated to be \$0.50 for each ream produced. Calculate the socially optimal level of output and price for the paper industry.

Solution:

- First, rearrange the inverse supply and demand equations to put them in terms of quantity supplied and quantity demanded:

$$\begin{aligned} P &= 2Q & P &= 40 - 8Q \\ Q^S &= 0.5P & 8Q &= 40 - P \\ & & &= 5 - 0.125P \end{aligned}$$

Market equilibrium occurs where $Q^D = Q^S$:

$$\begin{aligned} 0.5P &= 5 - 0.125P \\ 0.625P &= 5 \\ P &= \$8 \end{aligned}$$

When $P = \$8$,

$$\begin{aligned} Q^S &= 0.5P = 0.5(8) = 4 \\ Q^D &= 5 - 0.125P = 5 - 0.125(8) = 5 - 1 = 4 \end{aligned}$$

Paper will sell for \$8 per ream, and 4 million reams will be sold each year.

- The social marginal cost is equal to the industry marginal cost plus the external marginal cost:

$$\begin{aligned} SMC &= MC + EMC \\ &= 2Q + 0.50 \end{aligned}$$

To find the socially optimal price and quantity, we equate the social marginal cost with the inverse demand:

$$\begin{aligned} 2Q + 0.50 &= 40 - 8Q \\ 10Q &= 39.50 \\ Q^* &= 3.95 \end{aligned}$$

The socially optimal level of output is 3.95 million reams of paper each year. The socially optimal price can be found by substituting the optimal quantity into either the social marginal cost or inverse demand curves:

$$\begin{aligned} P^* &= 2Q + 0.50 = 2(3.95) + 0.50 = \$7.90 + 0.50 = \$8.40 \\ &= 40 - 8Q = 40 - 8(3.95) = 40 - 31.60 = \$8.40 \end{aligned}$$

Therefore, the socially optimal price is \$8.40 per ream.

16.1.2 Positive externality

Positive externalities exist when an economic activity has a spillover benefit enjoyed by third parties. Marginal social benefit of an economic activity higher than the private marginal benefit (i.e., the demand curve)

The classic example of a positive externality is education. Education is associated with private benefits, including higher lifetime earnings and many others. Education is also associated with broader social benefits, such as an increase in overall entrepreneurial activity, higher incomes, and a faster pace of technology growth.

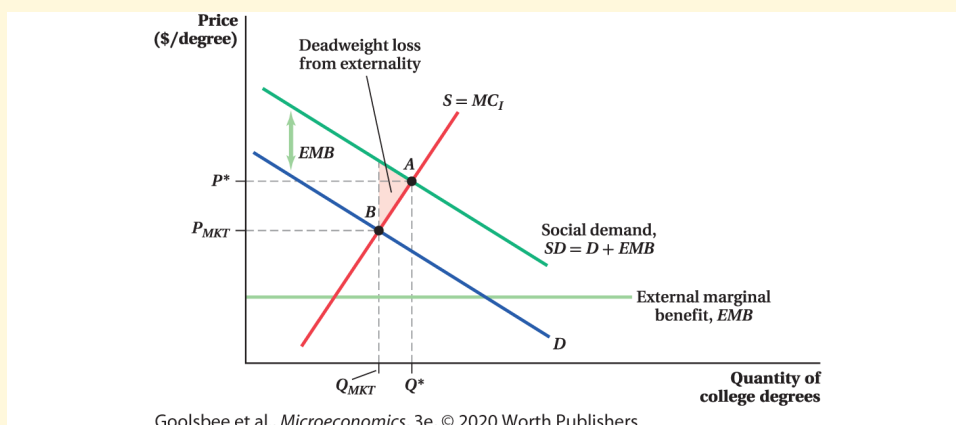


Figure 138: Positive Externalities in the Market for College Degrees

The social demand for college degrees (SD) equals the private marginal benefit curve (D) plus the external marginal benefit (EMB). The socially optimal number of college degrees, Q^* , is found at point A , the intersection of the marginal cost curve $S = MC$ and SD . In an unregulated market for college degrees, production occurs at point $B (Q_{MKT}, P_{MKT})$, where $D = S = MC$. Because the market does not take into account the external marginal benefit (EMB), it ends up producing fewer college degrees Q_{MKT} than the socially optimal quantity (Q^*), resulting in a deadweight loss equal to the shaded triangle.

16.2 Correcting externalities

Competitive markets with externalities produce more or less than is socially efficient. A number of market interventions can help correct externalities.

- Some work through their effect on prices (e.g., taxes).
- Some target the quantity produced and consumed.

Example: To understand the mechanisms, we will focus on the case of the negative externality from pollution. Before considering how to correct for an externality, we need to consider how the government chooses its efficiency target.

Definition (Effective level of pollution): The level of emissions necessary to produce the efficient quantity of the good tied to the externality.

Thus the goal is to determine the efficient level of pollution. We can also think about the efficient amount of pollution as the level that balances its costs and benefits.

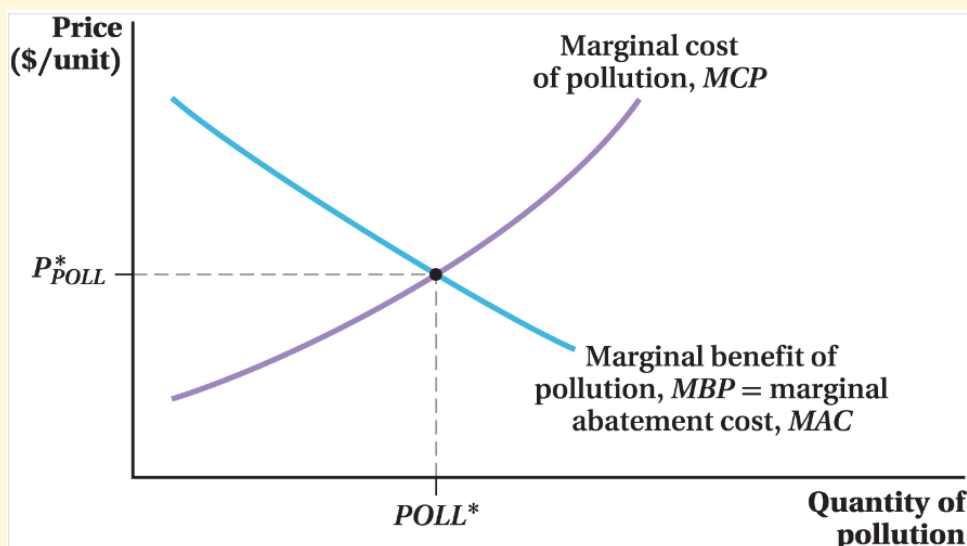


Figure 139: Effective level of pollution

For the marginal cost of pollution (MCP) curve, it slopes upwards. This is because at low levels of pollution, the damage associated with an additional unit of pollution is relatively low, at higher levels of pollution, health effects become more severe, and additional units of pollution are more costly to society.

The marginal benefit of pollution (MB_P), on the other hand, is likely to be high at low levels because some positive level of pollution is necessary to make goods. But the marginal benefit will fall as pollution output increases (and as output of the good increases).

Definition (Marginal abatement cost): The cost of reducing emissions by 1 unit.

Every unit of pollution a producer cuts entails a lost marginal benefit - giving up the sale of something that they could make (but would generate that extra unit of pollution). Thus, $MBP = MAC$.

The efficient level of pollution (Q_{poll}) occurs where the marginal cost of pollution MC_P equals the marginal benefit or marginal abatement cost of pollution ($MBP = MAC = P^*_{poll}$).

This represents a hypothetical market for pollution, with a resulting optimal price and quantity placed on pollution. Since this market doesn't exist in the real world, what can be done to induce private parties to produce and consume at the socially efficient level?

16.2.1 Using Prices to Correct Externalities

Consider changing the price of goods produced in the presence of externalities so that the social and private benefits and costs align.

- For a negative externality, levy a tax on production or consumption.
- For a positive externality, subsidize production or consumption.

Definition(Pigouvian tax): A type of tax levied on an activity that raises a good's price to take into account the external marginal costs imposed by a negative externality.

Example: Consider the same power plant example.

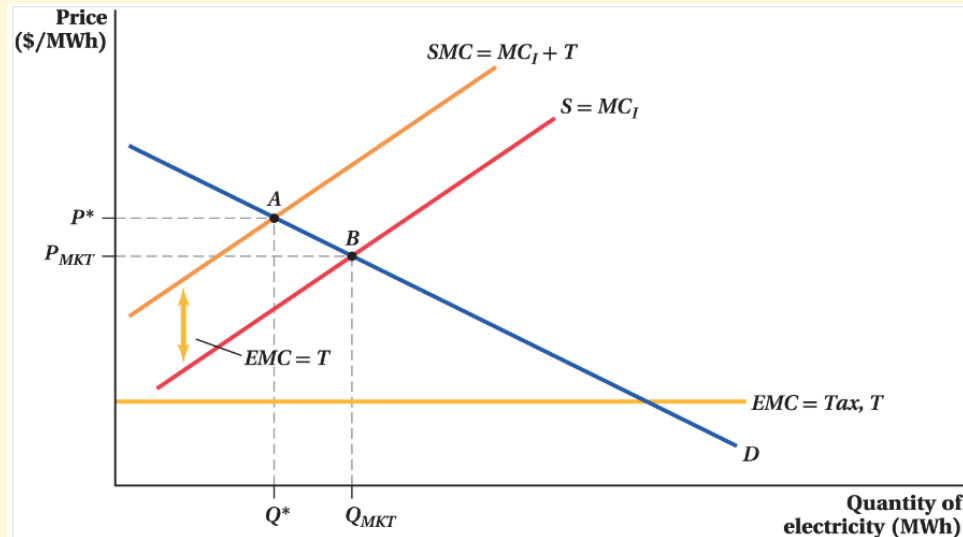


Figure 140: A Pigouvian Tax Corrects for a Negative Externality

In an unregulated market, the power industry overproduces quantity Q_{MKT} at price P_{MKT} (point B). A Pigouvian tax(T) equal to the external marginal cost EMC shifts the supply curve (S) up from marginal cost $S = MC_I$ to the social marginal cost curve (SMC). Now, the industry produces at point A, where SMC intersects demand (D), and supplies the socially efficient quantity Q^* MWh at price P^* .

Definition(Pigouvian subsidy): A subsidy paid for an activity that can be used to decrease a good's price (paid by the buyer) to take into account the external marginal benefits.

Pigouvian subsidy serves to equate the external marginal benefit imposed by an externality.

Example: Assume that notebook paper is sold in a perfectly competitive industry. The industry short-run supply curve (or marginal cost curve) is $P = MC = 2Q$, where Q is measured in millions of reams per year. The inverse demand for notebook paper is $P = 40 - 8Q$. Suppose the government places a \$0.50 tax on each ream of paper sold.

- What price would buyers pay, and what price would sellers receive (net of the tax)?
- How many reams of paper would be sold?

Solution:

▪

$$Q^D = 5 - 0.125P_b \quad \text{and} \quad Q^S = 0.5P_s$$

Because $P_b = P_s + 0.50$, we can substitute this expression into the demand curve for P_b :

$$\begin{aligned} Q^D &= 5 - 0.125P_b = 5 - 0.125(P_s + 0.50) \\ &= 5 - 0.125P_s - 0.0625 = 4.9375 - 0.125P_s \end{aligned}$$

Now that the supply and demand equations are both written in terms of P_s , we can solve $Q^D = Q^S$:

$$\begin{aligned} 4.9375 - 0.125P_s &= 0.5P_s \\ 0.625P_s &= 4.9375 \\ P_s &= \$7.90 \\ P_b = P_s + 0.50 &= \$7.90 + \$0.50 = \$8.40 \end{aligned}$$

- To solve for the quantity sold, we can substitute the buyer's price ($P_b = \$8.40$) into the demand curve or the seller's price ($P_s = \$7.90$) into the supply curve:

$$Q^D = 5 - 0.125P_b = 5 - 0.125(8.4) = 5 - 1.05 = 3.95$$

$$Q^S = 0.5P_s = 0.5(7.9) = 3.95$$

This is the socially optimal quantity.

16.2.2 Quantity Mechanisms to Correct Externalities

Quantity-based interventions have the same goal as taxes or subsidies—to move a market with externalities toward the efficient outcome.

Definition(Quota): A regulation mandating that the production or consumption of a certain quantity of a good or externality be limited (negative externality) or required (positive externality)

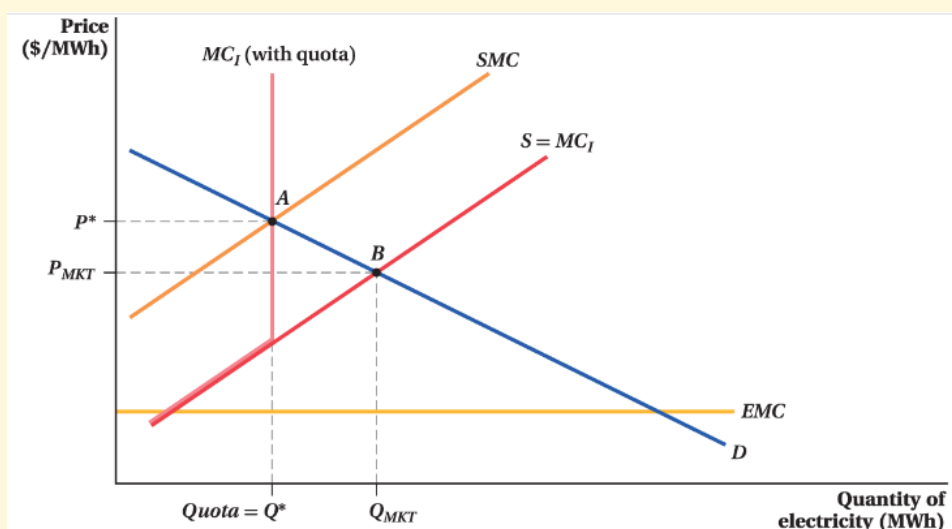


Figure 141: The Effects of a Quota on a Market with a Negative Externality

In an unregulated market, the power industry overproduces the quantity of electricity (Q_{MKT}) at price P_{MKT} (point B). When the government enacts a quota limiting production to Q^* , the private marginal cost curve MC_I becomes vertical at Q^* , intersecting the social marginal cost SMC and demand D at the socially optimal quantity Q^* and price P^* (point A).

16.2.3 Price-Based versus Quantity-Based Interventions with Uncertainty

With perfect information, price and quantity instruments are equally effective in controlling externalities. Without perfect information, significant problems face regulators.

- The costs of controlling externalities (marginal abatement costs) are often difficult to ascertain ex ante.
- The benefits of controlling externalities (marginal pollution costs) are similarly difficult to estimate—in the case of pollution, valuing changes to human health, ecosystems, and such.

When there is uncertainty in marginal abatement costs, price and quantity mechanisms are not equivalent.

We saw above that, as a matter of principle, quantity and price mechanisms can be equally effective at addressing externalities. In reality, though, there are situations in which one type of market intervention might be preferred to the other.

A particularly important situation in which one mechanism might be preferred to the other is when the optimal level of the externality is not fully known.

The marginal benefits and the marginal costs of pollution, for example, are difficult to measure and often shift with changes in technologies, input costs, and new discoveries about how damaging pollution is to human health. As a result, efforts to intervene in markets with externalities are generally inexact, so any Pigouvian taxes or quantity mandates that regulators set are not exactly right. Not being exactly right is always costly, but it can be even more costly if the wrong intervention is used.

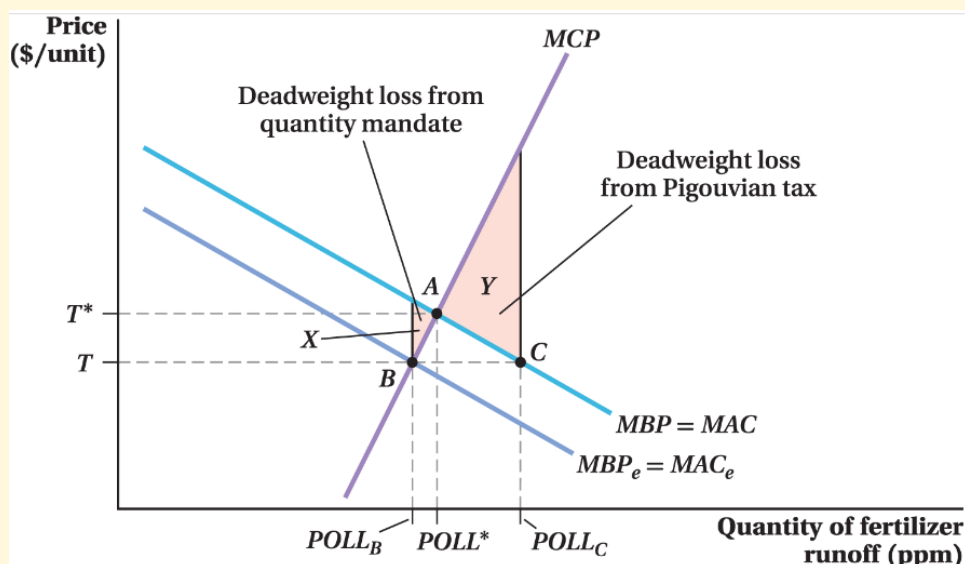


Figure 142: When Quantity Mechanisms Are Preferable to Price Mechanisms

$MB_P = MAC$ and MC_P are the marginal benefit (marginal abatement cost) and marginal cost of pollution curves for farmers' fertilizer use. The efficient level of water pollution is quantity Q_{poll}^* of fertilizer runoff. If government regulators incorrectly estimate farmers' marginal abatement costs at $POLL_C$, a Pigouvian tax would increase pollution to $Q_{poll_C} > Q_{poll}^*$. **Because MAC is flat relative to MC_P** , the quantity-based intervention is preferable to the price mechanism, as seen by comparing the resulting deadweight losses from the two interventions ($X < Y$).

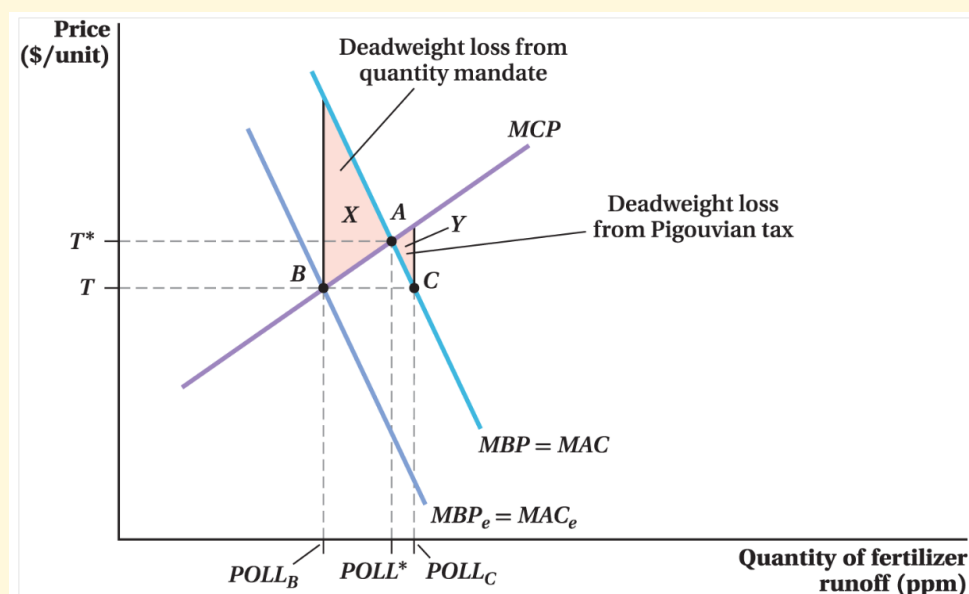


Figure 143: When Price Mechanisms Are Preferable to Quantity Mechanisms

If government regulators incorrectly estimate farmers' marginal abatement costs at $MAC_e < MAC$, the quantity-based intervention would reduce pollution to $Q_{poll_B} < Q_{poll}^*$, while the Pigouvian tax would increase pollution to $Q_{poll_C} > Q_{poll}^*$. **Because MAC is steep relative to the marginal cost of pollution MC_P** , the price mechanism is preferable to the quantity mechanism, as seen by comparing the resulting deadweight losses from the two interventions ($Y < X$).

Therefore, when regulators are uncertain about the optimal level of an externality, the choice of quantity- or price-based market interventions should depend on what the regulator perceives are the relative sensitivities of the marginal cost of pollution and the marginal abatement cost (or marginal benefit of pollution).

16.2.4 Tradable permits

In response to these challenges, many regulatory agencies issue tradable permits to control pollution.

Definition(Tradable permits): A government-issued permit that allows a firm to emit a certain amount of pollution during production and that can be traded to other firms.

The tradable permit has the following mechanics:

- A regulatory authority sets a cap on the total amount of a particular pollutant that can be emitted. This cap is determined based on environmental, health, economic, and political considerations, and is typically set to lower pollution levels over time.
- The total amount of allowable emissions under the cap is divided into permits, where each permit allows the holder to emit a certain amount of the pollutant. The initial allocation of permits can be done through an auction or for free, based on historical emissions levels.
- Once distributed, these permits can be bought and sold on a secondary market. Companies that can reduce emissions at a lower cost can sell their excess permits to companies for whom reducing emissions is more expensive. This creates a financial incentive for companies to innovate and reduce their emissions.

In a competitive tradable permits market:

- The demand for permits is driven by companies needing to comply with emission regulations. A firm's demand for permits depends on its abatement costs (the cost of reducing emissions) and its production needs. Companies with higher abatement costs will demand more permits, as purchasing permits may be cheaper than reducing emissions.
- The supply of permits is fixed by the cap and does not change. The total quantity of permits supplied is determined by the environmental targets set by the regulatory authorities.
- Market will arrive at a price per permit that equates the marginal cost of pollution abatement across all firms. equates the marginal cost of pollution abatement across all firms. If the permit price was above a firm's marginal abatement cost (alternatively, its marginal benefit of polluting), the company could make a profit by reducing its pollution by an extra unit and selling the permit for that extra pollution to someone else. If permit prices were instead below its marginal benefit of pollution, the firm could raise its profits by buying a permit and going ahead with polluting that extra unit.

16.3 The Coase Theorem

Under certain circumstances, it may be possible to reach the efficient market outcome through private negotiation.

The Coase theorem

States that if negotiation costs are low enough, negotiation among market participants will lead to the efficient market outcome regardless of who holds the property rights.

Example: Green Acres Fertilizer Company is located near Lucy's Dry Cleaning Service. In its production process, Green Acres emits noxious odors that are absorbed by the clothing that Lucy is cleaning. The result is that Lucy has lost many customers over time. Lucy estimates that the odors cost her business \$10,000 per year. Green Acres can eliminate its odors by altering its production process at a cost of \$12,000 per year.

- If Green Acres has the right to emit the odors, what will the socially optimal outcome be? How will it be reached? Will any money change hands?
- If Lucy has the right to odor-free air, what will the socially optimal outcome be? How will it be reached? Will any money change hands?

Solution:

- The socially optimal outcome will occur when Green Acres emits the odor. The cost of eliminating the emissions (\$12,000 per year) is greater than the external cost of the odor (\$10,000 per year). If Green Acres has the right to emit the odors, it will continue to do so. Lucy does not value clean air enough to purchase that right from Green Acres, so no money will change hands.
- The socially optimal outcome will still be for Green Acres to emit the odor. The optimal outcome is determined by the relative values that Green Acres and Lucy place on the resource (air). Because Green Acres values the air more highly, it should use the resource and emit the odor. However, because Lucy has a right to odor-free air, Green Acres will have to purchase the right to emit odors from her. Assuming that this transaction can be done without cost, Green Acres will have to pay Lucy between \$10,000 and \$12,000 for that right.

16.4 Public goods

Definition(Public goods): A good that is accessible to anyone who wants to consume it, and that remains just as valuable to a consumer even as other people consume it.

There is another type of good for which markets can fail to deliver the socially optimal level of output.

Public goods have two important properties:

- They are nonexcludable, meaning that consumers cannot be prevented from consuming the good.
- They are nonrival, meaning that one person's consumption of the good does not diminish another consumers enjoyment of the same good.

Using the definitions of nonexcludable and nonrival, we can define four categories of goods:

	Excludable Individuals can be kept from consuming.	Nonexcludable Individuals cannot be kept from consuming.
Rival One individual's consumption affects another's consumption.	Private good: tacos, gasoline, paper	Common resource: shared property, fisheries, interstate highways
Nonrival One individual's consumption has no effect on another's consumption.	Club good: satellite TV services, private parks, movies	Public good: fireworks display, mosquito abatement, national defense

Market efficiency requires that a good be produced until the marginal benefits of production are equal to the marginal costs. Since public goods are nonrival, the marginal benefit of a given level of provision is the vertical sum of all individuals' marginal benefit curves. Total marginal benefit is defined as the vertical sum of the marginal benefit curves of all of a public good's many customers. This is shown by the equation $MB_T = \sum MB_i$ where i indicates each individual. The cost side is the same as with any other good.

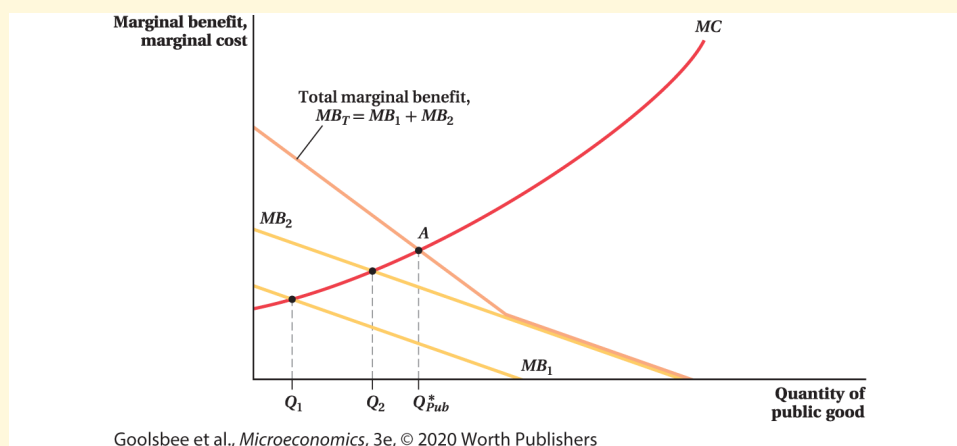


Figure 144: Efficiency in the Market for a Public Good

The two consumers of a public good have marginal benefit curves MB_1 and MB_2 . At the efficient point, both consumers consume Q^*_{Pub} of the public good, where their total marginal benefit curve (MB_T) intersects the marginal cost curve (MC). If the two consumers privately bought the good, each would consume less than the optimal level at quantities Q_1 and Q_2 , where MB_1 and MB_2 each intersect MC .

There are two reasons markets will underprovide public goods:

- If individuals purchase the goods themselves, they will do so only until private marginal benefit equals marginal cost. The efficient level occurs when the total marginal benefits equal marginal cost.
- A second problem results from the nonexcludability of public goods.

This is called the free-rider problem: a source of inefficiency resulting from individuals consuming a public good or service without paying for it.

Example: Jasmine and Casey are neighbors in a rural area. They are considering the joint installation of a large fountain near their joint property line so that each can enjoy its beauty and also improve the value of their property. Jasmine's marginal benefit from the fountain is $MB_J = 70 - Q$, where Q measures the diameter of the fountain (in feet). Casey's marginal benefit from the fountain can be represented as $MB_C = 40 - 2Q$. Assume that the marginal cost of producing the fountain is constant and equal to \$80 per foot (in diameter).

- Find an equation to represent the total marginal benefit of the fountain.
- What is the socially optimal size of the fountain?
- Show that if either Jasmine or Casey had to build a fountain by themselves, neither would find even the smallest fountain worth building.

Solution:

- The total marginal benefit is the vertical summation of Jasmine's and Casey's individual marginal benefit curves:

$$MB_T = MB_J + MB_C = (70 - Q) + (40 - 2Q) = 110 - 3Q$$

- The socially optimal size of the fountain occurs where MB_T equals the marginal cost of producing the fountain:

$$\begin{aligned} MB_T &= MC \\ 110 - 3Q &= 80 \\ 3Q &= 30 \\ Q &= 10 \end{aligned}$$

The optimal size of the fountain is 10 feet in diameter.

- Jasmine's marginal benefit is $MB_J = 70 - Q$. If the smallest possible fountain was zero feet in diameter ($Q = 0$), Jasmine's marginal benefit would only be \$70, while the marginal cost would be \$80. Therefore, Jasmine would not find even the smallest fountain to be worth the cost. Likewise, the marginal benefit of a zero-foot fountain for Casey will be $MB_C = 40 - 2Q = \$40$, which is also less than the \$80 marginal cost.

There are ways to solve the free-rider problem:

- The power to tax is often used to provide public goods such as national defense, air traffic control, and weather forecasting.
- Groups of private citizens can form to provide public goods (e.g., homeowner associations).
- People do not always act as free riders. In many cases, individuals have complex motivations that extend beyond the simplistic assumptions of rational self-interest typically used in economic theories.

Definition(Tragedy of the commons): The dilemma that common resources create when everyone has free access and the resource is used more intensively than it would be if privately owned, leading to a decline in its value for everyone. Tragedy of the commons is a special form of a negative externality.

Many of the solutions for negative externalities can serve as remedies for the tragedy of the commons:

- Pigouvian taxes: charge individuals for the external damage they do to the common resource
- Quotas: quantity restrictions on the rate at which common resources can be extracted (limits on the amount of fishing, hunting, logging, for example)
- Defining property rights and facilitating negotiation among those who share the common resource as laid out in the Coase theorem.